



Product Technology and Research Compendium

An integrated view of the proposed hardware, software and services for plate observation, traceable measurement and antimicrobial susceptibility research.

Prepared for technical partners, laboratories and controlled investor review.

AMReye.AI develops from a practical measurement priority: an endpoint disk diffusion reader with controlled imaging, spatial calibration, quality assessment, versioned interpretation and mandatory human review. The broader programme investigates multimodal observation, laboratory automation and predictive models.

This compendium describes product intent, engineering choices and the evidence required to advance them. It does not represent a clinically validated or cleared diagnostic device. Literature findings belong to their original studies; proposed capabilities require their own evaluation.

14 September 2026

Controlled external review edition

F / FRONT MATTER

Purpose and scope of controlled review

This compendium presents a proposed product platform for laboratory observation, measurement and research. Its readers should be able to determine what a module is intended to do, what physical or computational principle supports it, what evidence is missing and what would have to be demonstrated before deployment.

Scope of the review

The central development priority is Reader V1, an endpoint disk diffusion imaging and measurement system. The wider portfolio includes integrated instruments, research imaging, automation, software and services. These share recordkeeping and engineering principles, but their intended uses and evidence requirements differ. A research imaging result should not acquire a clinical meaning simply because it appears in the same interface as a laboratory record.

Table F.1. Purpose and scope of controlled review

Review perspective	Questions to resolve
Laboratory	Is the workflow useful, reviewable and compatible with the laboratory quality system?
Technical partner	Are interfaces, measurement assumptions and qualification responsibilities sufficiently explicit?
Research partner	Can a hypothesis be investigated with representative observations and an independent comparator?
Investor	Does the development sequence reduce uncertainty before expanding commitments?

Distribution and interpretation

This edition is intended for controlled external review. The product concepts, diagrams and interfaces describe engineering intent. They are not instructions for patient treatment, evidence of regulatory clearance or confirmation that every proposed module has been built. Clinical use would require an appropriate intended use, completed validation and applicable authorization.

Financial projections, fundraising scenarios and proposed selling prices are outside the book. A dated Reader V1 candidate bill of materials appears at the end to make one engineering configuration inspectable. Component prices are purchasing observations or explicitly identified estimates, not a validated cost of a finished medical device.

F / FRONT MATTER

The product proposition

AMReye.AI aims to make laboratory observations easier to acquire consistently, inspect critically and preserve with their context. The first useful unit of value is a trustworthy measurement record: an image associated with the correct plate, a valid calibration, a reproducible measurement and a review decision that can be reconstructed.

A platform built around observation

A plate contains more information than a final diameter alone. Its appearance, temporal development and measurement conditions can support quality assessment and research. Multiple sensing methods may add complementary information, but each added channel also introduces calibration, maintenance and interpretation work. The platform therefore treats integration as a question of demonstrated information value.

Table F.2. The product proposition

Product element	Proposed contribution
Instruments	Acquire controlled images and, in later configurations, complementary sensor observations.
Software	Maintain identities, assess quality, measure features, apply eligible rules and support review.
Services	Establish calibration, competency, maintenance, integration and reproducible research support.
Research platform	Investigate temporal models, richer sensing and selected early response workflows.

What success would look like

A successful endpoint reader would reduce avoidable measurement and transcription variability while preserving laboratory oversight. A successful research platform would produce observations that other investigators can interpret, compare and challenge. A successful deployment would remain usable when connectivity is interrupted and would make failed or uncertain results visible.

The common aim is better evidence at the point where a laboratory makes a decision. Speed, automation and AI are valuable when they improve that evidence or reduce a verified operational burden. They are not substitutes for method suitability, traceability or biological validation.

F / FRONT MATTER

Reading the maturity of a capability

Capability maturity is expressed by what has been demonstrated, not by the sophistication of its name. The same module can have a working software demonstration, an unqualified measurement chain and no clinical validation. Those are different states and should remain distinguishable.

Table F.3. Reading the maturity of a capability

Term in this book	Meaning for a reviewer
Established principle	The physical, biological or computational principle is supported independently; its integration here still needs evaluation.
Proposed engineering	A defined design intention with requirements and a verification route; completion is not implied.
Research target	An outcome whose feasibility or generalizability remains to be tested.
External study result	A result reported by the cited investigators under their own conditions.
Validated product claim	A bounded claim requiring evidence from the actual product configuration and intended workflow.

Read the scope with the result

A reported agreement percentage is incomplete without the tested population, comparator, denominator, excluded cases and uncertainty. A short assay time is incomplete without its clock origin. A simulation is incomplete without the last real observation and the horizon over which it predicts.

No clinical accuracy, sub-two-hour performance, resistance-gene identification from plate images or zero-error operation is established for AMReye.AI by this compendium. Those ambitions are translated into testable research and engineering requirements. The near-term product path prioritizes controlled endpoint measurement and human review; broader capabilities advance through separate evidence gates.

This vocabulary should help technical partners compare proposals fairly. It also protects useful exploratory ideas from being discarded merely because they are not ready for clinical use. Research potential and readiness for deployment are separate assessments.

F / FRONT MATTER

How the platform fits laboratory work

A laboratory result emerges from a chain of activities. Specimen collection, transport, processing, organism identification, susceptibility testing, review and communication can each contribute delay or uncertainty. An instrument can improve a particular step without controlling the whole chain.

Table F.4. How the platform fits laboratory work

Workflow stage	Proposed AMReye.AI relationship
Specimen and case registration	Receive only the identity and contextual fields required for the permitted workflow.
Laboratory preparation	Preserve assay metadata and relevant quality records; preparation remains method governed.
Plate observation	Acquire a controlled image and verify identity, calibration and readability.
Measurement and interpretation	Preserve measured values; apply a separately versioned eligible ruleset.
Human review	Display source evidence, uncertainty, edits and unresolved discrepancies.
Reporting and follow-up	Export the reviewed state, retain acknowledgements and preserve corrections.

Measure the contribution at the right boundary

Reader acquisition time, analysis latency and total laboratory turnaround should be reported separately. A fast image measurement cannot recover time already spent in specimen transport or biological growth. Similarly, an early research signal may not yet be a reportable susceptibility result.

The proposed software should make these boundaries visible through distinct timestamps. Where a timestamp is missing, elapsed time should be unavailable rather than inferred silently. Partners can then identify the stage that actually limits their service and assess whether a reader, workflow integration or a different assay would address it.

F / FRONT MATTER

Contents by part

The main parts progress from product purpose to engineering, software, research applications and the evidence required for delivery. Page numbers refer to this controlled review edition.

[I Purpose and product portfolio • page 11](#)

[II Sensing and engineering • page 35](#)

[III AI software and data • page 83](#)

[IV Digital twins and simulations • page 133](#)

[V Applications and impact • page 165](#)

[VI Validation safety and security • page 197](#)

[VII Scientific evidence • page 229](#)

[VIII Development and delivery • page 249](#)

[Appendices Interfaces examples glossary and references • page 271](#)

[Concluding Reader V1 bill of materials • page 293](#)

Suggested reading paths

Laboratory reviewers can begin with Reader V1, routine disk diffusion, human review and measurement validation. Engineering partners can move directly to sensing, interfaces and the development sequence. Research collaborators should connect a candidate modality to its evidence synthesis, application boundary and validation plan.

Investors should read the product portfolio together with development gates and service requirements. The breadth of the platform is an opportunity map; it is not a promise that every proposed family will be delivered in one instrument or one development cycle.

F / FRONT MATTER

Contents by technical theme

Use these links to reach the principal catalogues and questions. The PDF bookmarks provide finer navigation, and the cross-reference index connects common terms to related topics.

[Reader V1 scope and product boundary](#)

[Candidate sensing catalogue with 40 technology entries](#)

[Optical geometry and measurement uncertainty](#)

[PIMS and plate record management](#)

[Proposed record interfaces](#)

[AI catalogue with 40 capability entries](#)

[Genomic repositories and provenance](#)

[Requirements for a predictive digital twin](#)

[Plate loss and continuity of the digital record](#)

[Heteroresistance and population diversity](#)

[UTI and women's health workflows](#)

[Goa and a regional research pathway](#)

[Sensor failure and common causes](#)

[Validation of early results](#)

[Data protection and purpose limitation](#)

[Comparator families and differentiation](#)

[Custom models and staged silicon strategy](#)

[Reader V1 components and dated INR price basis](#)

Technology descriptions explain the measurement or computational choice. Exact candidate component identities appear in the concluding BOM, where compatibility and availability can be assessed together.

F / FRONT MATTER

Abbreviations for microbiology and measurement

Abbreviations are expanded at first use where practical. The definitions below keep similar-looking biological and measurement terms distinct.

Table F.5. Abbreviations for microbiology and measurement

Abbreviation	Meaning in this compendium
AMR	Antimicrobial resistance.
AST	Antimicrobial susceptibility testing.
MIC	Minimum inhibitory concentration, defined by the applicable testing method.
BMD	Broth microdilution, a method family used for quantitative susceptibility testing.
QC	Quality control of a method, instrument or process, with the object identified.
RAST	Rapid AST within a specified method and eligible conditions; EUCAST uses this term for its defined pathway.
ATU	Area of technical uncertainty in applicable EUCAST interpretation contexts.
CFU	Colony-forming unit, an operational estimate dependent on the counting method.
UTI	Urinary tract infection.
MRSA	Methicillin-resistant Staphylococcus aureus.
PCR	Polymerase chain reaction; distinguish real-time PCR from reverse-transcription PCR when discussing a molecular result.
OD	Optical density or absorbance-related measurement, with wavelength and optical geometry stated.
RH	Relative humidity, normally expressed as a percentage.
FOV	Field of view at a specified object plane.
SNR	Signal-to-noise ratio with its calculation and units or logarithmic convention stated.
MTF	Modulation transfer function, describing contrast transfer across spatial frequencies.

The presence of an abbreviation in this glossary does not establish product capability. For example, MIC prediction is a research track; a reviewed disk-zone diameter should not be presented as a measured MIC. Standards terminology must be interpreted within the selected method and document version. [\[R01\]](#) [\[R02\]](#)

F / FRONT MATTER

Abbreviations for software and engineering

Software and engineering terms can obscure responsibility when used loosely. Here, each term refers to a defined function or record, rather than a claim that a particular architecture has been implemented.

Table F.6. Abbreviations for software and engineering

Abbreviation	Meaning in this compendium
AI	Artificial intelligence, including task-specific learned models.
ML	Machine learning, a subset of computational methods whose parameters are learned from data.
PIMS	Plate Information Management System.
BAHU	Bio-Automation and Human User Interface, the proposed operator-facing orchestration concept.
LIS	Laboratory Information System.
LIMS	Laboratory Information Management System.
API	Application programming interface with an explicit input and output contract.
OOD	Out of distribution, relative to a declared training or validation domain.
VVUQ	Verification, validation and uncertainty quantification of computational models.
RTD	Resistance temperature detector.
RTOS	Real-time operating system, where deterministic scheduling is needed.
BOM	Bill of materials for a specified configuration.
SBOM	Software bill of materials describing software dependencies.
QMS	Quality management system.
EMC	Electromagnetic compatibility.
OEM	Original equipment manufacturer or an associated integration relationship.
UTC	Coordinated Universal Time, suitable for unambiguous stored event times.
IST	Indian Standard Time, UTC plus five hours and thirty minutes.

“Edge” means computation close to the instrument; “cloud” means managed remote computing services. Neither word alone specifies security, reliability or performance. “Digital twin” is reserved for a model-linked representation whose relationship to observations and uncertainty is explicit.

F / FRONT MATTER

Questions this compendium answers

The book is organized around decisions that a serious reviewer should be able to examine. Each answer connects the proposed benefit to a mechanism and an evidence requirement.

Table F.7. Questions this compendium answers

Question	Where the answer is developed
Why does the platform matter?	Part I connects measurement, workflow and product value.
What are the product families?	Part I distinguishes instruments, software and services.
What can sensors actually measure?	Part II describes signals, units, calibration and confounders.
Where does AI add value?	Part III compares each capability with a conventional baseline.
What happens when observations stop?	Part IV preserves the final observation and distinguishes forecasts from measurements.
How could laboratories and communities benefit?	Part V examines individual workflows and their feasibility boundaries.
How will errors be detected and managed?	Part VI specifies evaluation, disagreement, abstention and recovery.
What is already known from research?	Part VII separates external findings from original development questions.
How can the product be built and supported?	Part VIII connects stages, suppliers, production and service.
What information crosses an interface?	The appendices define records, status fields and provenance.
What does one reference configuration require?	The final eight BOM pages identify candidates, prices and gaps.

An unanswered question should lead to a defined research or engineering task. It should not be concealed by a broad capability label. The strongest product proposition is one whose limitations are specific enough to be tested and whose useful functions remain clear within those limits.

The laboratory problem AMReye.AI addresses

Laboratories need results that can be trusted, explained and delivered through a workable process. Manual observation can preserve valuable expert judgement, but it can also leave measurement context dispersed across images, worksheets and information systems. Automation is useful when it improves a defined part of that process without removing the evidence a reviewer needs.

Three practical gaps

The first gap is repeatable acquisition: lighting, pose and scale can vary between images. The second is traceable interpretation: a measured quantity, the applicable standards rule and the reviewer's decision may not be stored as separate objects. The third is continuity: an image or observation series may become difficult to retrieve after a correction, instrument change or research reanalysis.

AMReye.AI proposes a connected measurement workflow around those gaps. The wider AMR problem provides the public health motivation, but a reader's contribution must be demonstrated through laboratory outcomes rather than inferred from global burden figures. WHO identifies AMR as a major global health challenge; that does not establish the impact of a particular device. [\[R06\]](#)

Table I.1. The laboratory problem AMReye.AI addresses

Practical problem	Proposed product response	Evidence to seek
Acquisition variability	Controlled lighting, pose checks and calibration	Repeatability across operators and conditions
Unclear measurement history	Original observations linked to derived measurements	Complete reconstruction of a reviewed result
Difficult discrepancy review	Visible quality flags and correction history	Review accuracy, workload and usability
Fragmented records	PIMS and controlled exports	Identity integrity and successful receiving-system reconciliation

The initial product should earn its place by solving these bounded problems. More sensing or computing should be added when it addresses a documented residual limitation. This sequence makes the programme testable: partners can evaluate a useful measurement workflow while the more ambitious research tracks develop their own evidence.

Antimicrobial susceptibility as a measurement problem

Susceptibility testing converts a biological response under specified conditions into an interpretable laboratory result. The measured signal may be a zone diameter, a growth endpoint or another method-defined quantity. Interpretation depends on the assay, organism, antimicrobial agent and applicable standards context. A visually convincing measurement can still be ineligible for interpretation.

Separate the layers of the result

An image contains optical intensities. A measurement algorithm estimates a physical feature from those intensities. A standards engine applies eligible rules to the measured feature. A qualified reviewer considers the evidence and resolves exceptions. Combining these layers into a single opaque score makes it harder to identify the source of an error.

The proposed reader should retain the boundaries between them. For example, a changed breakpoint table can justify a new interpretation of an unchanged diameter. A corrected plate association may invalidate both the measurement context and the prior interpretation. Those changes have different meanings and should not be represented by the same generic “updated” status. [\[R01\]](#) [\[R02\]](#)

Table I.2. Antimicrobial susceptibility as a measurement problem

Layer	Representative object	What it does not establish alone
Observation	Original plate image and capture conditions	A calibrated length
Measurement	Zone diameter in millimetres with uncertainty	Method eligibility or a clinical category
Interpretation	Ruleset-qualified category and explanation	Approval to release the record
Review	Accepted, corrected, rejected or deferred decision	A new biological measurement

A measurement-centred design also makes conventional methods easier to compare with learned methods. The relevant question is whether a method produces a sufficiently reliable estimate under the declared conditions, including difficult and unreadable cases. Model complexity is secondary to that result.

From specimen to reviewed laboratory result

The proposed record follows the laboratory's workflow without pretending to control every upstream activity. A case or study identifier is associated with a plate, observations are acquired, measurements are made and a reviewer determines the disposition. Each transition has a responsible actor, a timestamp and a reason when the normal path is interrupted.

Preserve meaningful states

“Captured,” “measured,” “interpreted” and “reviewed” should be distinct states. A capture can be technically successful but unreadable. A measurable zone can lack an eligible interpretation rule. A technically consistent analysis can still be rejected because the plate identity is unresolved. These are useful distinctions for both laboratory work and subsequent evaluation.

Proposed path from identity to a reviewed export



AMReye.AI · conceptual diagram

Figure I.1. Proposed workflow states. A failed transition remains visible and does not silently advance.

Make interruption visible

When a capture fails, the absence of an image is itself a recorded event. When an export is not acknowledged, the record remains pending reconciliation. When a reviewer changes a measurement, the original estimate remains retrievable. None of these situations should silently advance the record to a completed state.

This workflow model supports clear performance measures. Acquisition success, readable coverage, review time and export reliability can be assessed separately. It also provides a stable foundation for later automation: additional hardware can produce observations within the same record structure without redefining what a reviewed result means.

What an endpoint reader can contribute

An endpoint reader can improve the consistency and traceability of reading a plate that has already reached the appropriate method-defined endpoint. It can assist with image acquisition, disk localization, zone measurement, quality checks, overlay presentation and record export. Its value depends on the accuracy and usability of those functions in the intended laboratory setting.

A bounded and useful contribution

The reader should first establish that it can measure the permitted plate types reliably. That requires controlled geometry, calibration, readable images and a clear response to ambiguous edges. The software can then help a reviewer compare an overlay with the original plate image and identify cases that need another observation or manual handling.

The approach fits a measurement workflow already governed by method-specific standards. It does not shorten the underlying biological endpoint merely by computing the diameter quickly. EUCAST distinguishes its routine disk diffusion method from its rapid AST pathway; the product should preserve that distinction in eligibility checks and claims. [\[R03\]](#) [\[R04\]](#)

Table I.3. What an endpoint reader can contribute

Proposed function	Useful output
Standardized acquisition	A reproducible image with capture conditions and quality state.
Calibrated measurement	A diameter estimate tied to a valid spatial calibration.
Review overlay	Inspectable boundaries, disk centres and uncertainty indications.
Rule application	An eligible interpretation linked to the exact ruleset version.
Record preservation	Original image, analysis, reviewer decision and export lineage.

An endpoint reader should also have a useful failure mode. It should identify when the available image cannot support the requested measurement, preserve the evidence and allow the laboratory to use its established alternative workflow. The aim is a reliable contribution to laboratory work, including reliable recognition of the limits of that contribution.

Reader V1 intended development scope

Reader V1 is the near-term development priority: a controlled imaging and measurement workflow for endpoint disk diffusion plates, with mandatory human review. The reference configuration at the end of this book makes one candidate optical and computing arrangement concrete. It is a development configuration, not a declaration of a finished commercial enclosure or validated clinical performance.

Proposed minimum functional scope

The first release should associate each image with the correct plate, verify acquisition readiness, preserve the original image and calibration context, estimate permitted zone measurements and support review. Standards interpretation should be independent of the image algorithm and should refuse unsupported combinations or incomplete context.

Table I.4. Reader V1 intended development scope

Included development objective	Required boundary
Controlled endpoint imaging	Permitted plate geometry, pose and lighting are declared and verified.
Spatial measurement	Calibration validity and measurement uncertainty are visible.
Readability assessment	Failed, ambiguous and unsupported images are retained as such.
Standards-assisted review	Exact ruleset and method context accompany the interpretation.
Traceable correction	Original estimates and subsequent reviewer changes remain linked.
Export	Only the intended reviewed record state is released.

Evidence needed to advance

Bench measurements should establish the optical measurement chain before biological agreement is evaluated. Representative plate studies should include difficult cases, independent review and prespecified handling of unreadable results. Workflow studies should then assess whether the interface helps users without encouraging inappropriate reliance on the automated estimate.

Rapid AST, MIC estimation, genomic inference and predictive twins remain separate research tracks. They may reuse selected infrastructure, but success in Reader V1 does not validate their biological conclusions. This separation keeps the initial product objective sufficiently narrow to verify while preserving the broader platform vision.

The integrated instrument research family

An integrated instrument could bring controlled observation, environmental monitoring, incubation-related functions and material handling into a coordinated laboratory workstation. Its purpose would be to make a longer biological process observable and reproducible. The appropriate physical arrangement remains open and should follow the selected assays, service requirements and risk assessment.

Integration creates new responsibilities

Imaging a plate at one endpoint is different from maintaining conditions over time. A time-series instrument must control or measure relevant environmental variables, preserve time origins, detect interruptions and distinguish instrument faults from biological changes. Added motion or liquid handling introduces further requirements for identity, containment, cleaning and mechanical safety.

Table I.5. The integrated instrument research family

Candidate module	Research or engineering question
Controlled incubation	Can conditions be measured and maintained within the selected method's requirements?
Scheduled imaging	Does repeated observation add useful information without disturbing the process?
Plate queue and indexing	Can identity and pose remain reliable through every transfer?
Broth microdilution integration	Can a separately qualified assay produce direct MIC results with appropriate controls?
Atmosphere monitoring	Which measurements are relevant to the intended organisms and assays?
Research sensing	Does each added channel provide independent information worth its burden?

Keep assay boundaries intact

A direct MIC method, a disk diffusion reader and a research growth model are distinct functions even if they share an instrument housing. Each needs its own eligibility rules, controls and evidence. A research atmosphere or specialized optical mode should not be enabled within a clinical workflow merely because the hardware can support it.

The development decision is therefore modular: select a useful combination, define the interfaces and demonstrate the integrated behaviour. A broad catalogue describes possibilities across the portfolio, not a requirement to install every module in one device.

The imaging and microscopy family

The imaging family extends from whole-plate observation to selected microscopic or spectroscopic investigations. Different questions demand different fields of view, spatial resolution, contrast mechanisms and temporal sampling. A single camera arrangement is unlikely to provide the best measurement for every scale.

Choose the scale from the question

Whole-plate imaging supports spatial context, zone measurements and colony distribution. Higher magnification can support cell-level or microcolony research, but reduces the area observed at one time and increases the burden of focusing, positioning and data handling. Spectral and phase-sensitive approaches may reveal information that ordinary colour images do not, while introducing their own calibration and interpretation requirements.

Table I.6. The imaging and microscopy family

Observation scale	Candidate use	Main qualification issue
Whole plate	Zone geometry, readability and spatial overview	Distortion, scale and edge interpretation
Local colony region	Texture, boundary development and unusual growth	Selection bias and registration to the parent plate
Microscopic field	Cell or microcolony research	Focus, tracking, sampling representativeness and reference labels
Spectral measurement	Chemical or metabolic research signals	Background, spectral calibration and assay-specific meaning

Preserve context across scales

A high-magnification field should retain its position within the parent specimen or plate. A reviewer needs to know whether the field was selected manually, by a prespecified sampling pattern or by an algorithm attracted to an unusual feature. Otherwise, a striking image can be mistaken for a representative observation.

Published single-cell AST research illustrates that specialized imaging can support short assay windows in selected experimental settings. It does not show that whole-plate imaging automatically has the same timing or clinical scope. [\[R11\]](#) The proposed imaging family should therefore qualify each scale and workflow on its own terms, with shared provenance rather than interchangeable claims.

The automation and robotics family

Automation should remove a demonstrated source of variation or workload. Candidate functions include plate loading, indexing, scheduled observation, barcode association and transfer between qualified modules. More advanced liquid handling and robotic workflows belong to separately assessed development tracks.

Make every physical action accountable

A motion command should have a permitted starting state, a target state, feedback and a defined response when completion is uncertain. The software record should describe what the instrument observed about the action, not merely what it requested. A command to move a plate is not evidence that the plate arrived in the correct position.

Table I.7. The automation and robotics family

Automation function	Feedback needed	Failure response
Plate loading	Identity, presence and permitted orientation	Stop association and request resolution
Indexing	Position or pose confirmation	Hold capture and verify the mechanism
Scheduled observation	Clock, readiness and capture acknowledgement	Record the missed observation explicitly
Module transfer	Source and destination custody confirmation	Quarantine the ambiguous record state
Research liquid handling	Qualified volume and path verification	Apply the assay-specific interruption procedure

Separate safety from conversational control

Operator assistance may explain a sequence or collect an annotation, but a learned language model should not become the sole authority for safety-critical motion or environmental control. Interlocks, command authorization and deterministic limits should remain effective when the interface or cloud service fails.

The strongest early automation candidate is often the simplest one that makes a repeated operation more consistent. The development sequence should measure operator time, identity integrity, recovery burden and maintenance demand alongside mechanical cycle time. A faster mechanism that creates more ambiguous plate associations would not satisfy the product objective.

The research instrumentation family

Research instrumentation should support investigators who need reproducible observations and transparent access to data. Its outputs may include raw images, calibrated measurements, temporal features and sensor traces. The value lies in the quality and interpretability of those observations, rather than in assigning a clinical category before the relevant assay has been validated.

A configurable but bounded environment

Research configurations can expose additional optical modes, sampling schedules and analysis options. Those choices should be versioned and recorded. An investigator should be able to reconstruct the configuration that produced a dataset and distinguish an intentional experimental change from an unrecorded instrument drift.

Table I.8. The research instrumentation family

Research function	Proposed deliverable
Time-resolved phenomics	Registered observations with biological and instrument time origins.
Multimodal comparison	Synchronized channels with units, calibration and missingness states.
Algorithm development	Original observations, annotation provenance and declared dataset splits.
Simulation research	Model inputs, parameter assumptions, outputs and uncertainty.
Reproducibility support	Exportable metadata, versioned configurations and analysis manifests.

Protect comparability

Open access to settings is useful only when changes remain interpretable. For example, an exposure adjustment can alter image intensity without indicating a biological change. A different illumination mode can reveal another feature while breaking direct comparability with the prior series. The research interface should make these transitions explicit.

Partnerships should define who owns observations, who can reuse them, what de-identification is required and which results may be published. Research use is not a blanket exemption from privacy, biosafety or instrument safety responsibilities. A well-designed research product gives investigators the information needed to meet those responsibilities and to evaluate whether its measurements answer their question.

Plate information management software

Plate information management is a proposed software product in its own right. Laboratories and research groups may need reliable plate records even when their images come from different instruments. PIMS should provide a stable identity and provenance model across those acquisition sources.

Product functions

The software should organize plate associations, observations, analysis versions, quality decisions and review states. Search should answer operational questions such as which records need review, which images used a superseded calibration and which exports remain unacknowledged. Retrieval should not depend on a staff member remembering a file naming convention.

Table I.9. Plate information management software

User need	Proposed PIMS function
Find a plate record	Search permitted identifiers and metadata under role-based access.
Explain a result	Follow the chain from original observation to review and export.
Correct an association	Create an authorized reassociation event with a reason and preserved history.
Reanalyse observations	Attach a new analysis version without replacing the original data.
Manage retention	Apply data-class policies and preserve required audit evidence.

Product boundaries

PIMS should integrate with the laboratory’s authoritative patient and order systems rather than create unnecessary duplicate identity stores. The appropriate source of truth for each field must be agreed during integration. A research deployment may use study identifiers and require no patient details at all.

The initial product value can be evaluated through successful retrieval, identity consistency, correction completeness and reviewer usability. Predictive features can be added later without changing the meaning of the original observation record. The detailed PIMS chapter and interface appendices define the proposed record relationships and expected handling of measured, derived, imputed, simulated and unavailable values.

Analysis and decision support software

Analysis software converts observations into candidate measurements and reviewable evidence. Decision support adds context that helps a qualified user understand the result. These functions should be distinguished from autonomous clinical decision-making and from the deterministic application of an eligible standards rule.

A layered analysis product

The proposed software combines conventional image processing, task-specific learned models and explicit quality logic. A geometric method may be sufficient for one measurement, while a learned model may help with variable appearance or ambiguous segmentation. Every AI capability should be compared with a simpler baseline and judged by its effect on the intended workflow.

Table I.10. Analysis and decision support software

Software element	Proposed output	User control
Image quality assessment	Specific readiness or readability findings	Inspect evidence and reacquire where appropriate
Measurement assistance	Boundary and calibrated quantity	Accept, adjust, reject or defer
Contextual interpretation	Eligible rule result and source version	Confirm context and resolve exceptions
Research inference	Bounded prediction with uncertainty	Treat as research and inspect supporting inputs
Evidence retrieval	Source-linked material	Verify relevance before use

Keep explanations operational

An explanation should identify what affected the result and what the user can do next. A coloured overlay may help locate a boundary, but it does not establish that the algorithm used a biologically valid feature. A confidence score should have a defined empirical meaning and should be evaluated on representative data. [\[R21\]](#) [\[R22\]](#)

The product should make uncertainty actionable through review, additional observation or abstention. It should also preserve a record of overrides so that developers can investigate recurring weaknesses. Human review is most useful when the interface exposes the evidence needed for independent judgement, rather than simply asking the user to confirm an automated answer.

Laboratory connectivity products

Connectivity products should move laboratory information with its meaning intact. The challenge is not only transmitting a value; it is preserving identity, units, method, status, version and correction semantics across systems that may use different conventions.

Define a mapping before an interface

Each integration needs a field-level agreement between the sending and receiving systems. The agreement should identify authoritative identifiers, required terminology, permitted result states and acknowledgement behaviour. Common messaging standards can support that agreement, but they do not remove the need for site-specific mapping and verification.

Table I.11. Laboratory connectivity products

Information class	Integration requirement
Order and identity	Unambiguous association and handling of duplicate or changed identifiers.
Measured quantity	Units, method and precision retained without silent conversion.
Interpretation	Standards family, version and eligibility context preserved where supported.
Review status	Preliminary, reviewed, rejected and corrected states mapped explicitly.
Correction	Receiving system can identify the superseded record and the replacement.
Acknowledgement	Delivery, acceptance and application processing states are distinguished.

Design for disagreement between systems

A successful network request does not prove that a result reached the correct record. The product should reconcile acknowledgements and expose unresolved exports. Duplicate submissions should be idempotent where the receiving interface supports that behaviour. When a destination cannot represent a required field, the limitation should be resolved through the approved workflow rather than by dropping the field silently.

Connectivity can be offered as adapters, supported integrations and partner interfaces. Each should carry a compatibility statement and a conformance test set. Changes in a laboratory information system, terminology dictionary or result schema should trigger a review of the integration evidence, even when the instrument’s measurement software has not changed.

Fleet and instrument management services

Fleet services should help a laboratory understand the condition and configuration of its instruments. A useful fleet view reports calibration validity, software versions, storage health, outstanding faults and synchronization state. It should not require central access to every patient-linked image to provide those functions.

A proposed service model

Routine telemetry should contain the minimum information needed for instrument support. A service operator may need a fault code, configuration identifier and diagnostic trace, while the laboratory retains control of clinical records. Access to a specific image for troubleshooting should be a separate, authorized action with an audit trail.

The service can coordinate signed software packages, staged updates and rollback. Instruments in the middle of a run should retain their active qualified configuration. A fleet update should not create a mixture of rules or models within one result record.

Table I.12. Fleet and instrument management services

Service measure	Why it matters
Calibration status by instrument	Prevents an apparently healthy device from using an expired or invalid measurement basis.
Update success and rollback rate	Reveals whether change delivery itself creates operational risk.
Unresolved fault duration	Measures time during which a laboratory may need an alternative workflow.
Synchronization backlog	Distinguishes local work completed from records available centrally.
Restore verification	Demonstrates recovery of usable records, not merely existence of a backup file.

Contract the responsibilities

The laboratory, instrument provider and hosting provider should agree who can authorize remote actions, who responds to alerts and how incidents are escalated. Service availability targets should be negotiated against the deployment context and demonstrated operating data. This compendium does not assign a contractual uptime promise.

A fleet service becomes valuable when it reduces uncertainty about instrument readiness and recovery. A dashboard full of nominally green indicators is inadequate if it cannot reveal stale telemetry, failed checks or a disconnected instrument.

I / PURPOSE AND PRODUCT PORTFOLIO

Calibration and measurement services

Calibration services connect instrument output to a defined measurement basis. For an imaging reader, the central task is spatial calibration at the relevant object plane, supported by checks of field dependence, stability and configuration control. Other sensing configurations require their own reference materials, transfer functions and uncertainty budgets.

Deliver a usable calibration record

A service report should identify the instrument and configuration, the reference used, the measured residuals, the permitted operating range and the conditions that invalidate the calibration. A coefficient file without this context is insufficient for laboratory review or later investigation.

The proposed service should also distinguish calibration from adjustment. Calibration characterizes a relationship; adjustment changes the instrument. After an adjustment, the resulting measurement performance needs to be established again. Routine checks can detect departure from an accepted state, but they should not be labelled a full calibration unless they fulfil that function. [\[R16\]](#)

Table I.13. Calibration and measurement services

Service output	Proposed content
Calibration certificate or report	Instrument identity, method, references, conditions, results and uncertainty basis.
Machine-readable calibration	Coefficients, valid region, version and relationship to the signed report.
Invalidation rules	Lens, focus, mounting, image pipeline or other changes that require review.
Routine verification guidance	Appropriate checks and escalation when results exceed the accepted range.
As-found and as-left records	Condition before service and condition after any intervention.

Make serviceability a design input

Reference targets need accessible placement, stable mounting and protection from damage. The instrument should allow a service engineer to verify the measurement plane without dismantling unrelated assemblies. Calibration effort, required expertise and replacement logistics should therefore influence the design early, rather than being treated as an after-sales detail.

Maintenance and repair services

Maintenance services preserve a qualified operating state and restore it after faults. They include inspection, cleaning-compatible component care, replacement, software recovery and requalification where a change can affect measurement. The scope should be specific to the instrument configuration and the laboratory’s use conditions.

Classify the intervention by its effect

A replacement storage device may affect data integrity and recovery. A replacement light source may affect exposure, spectral response and contrast. A camera or lens change can alter the spatial measurement chain. These interventions should not share a generic “service complete” test.

Table I.14. Maintenance and repair services

Intervention	Evidence before return to use
Lighting replacement	Illumination uniformity, stability and image quality checks.
Camera or lens work	Configuration identity and appropriate spatial requalification.
Cooling or power repair	Stable operation under the declared load and fault monitoring.
Storage replacement	Verified software state, record restoration and write integrity.
Mechanical adjustment	Pose, alignment and any affected calibration checks.

Support recovery at the laboratory

The service offer should define how users report a fault, what information is collected, when remote diagnosis is permitted and when a physical intervention is required. Spare-part planning should consider lead time, interchangeability and the evidence needed after substitution. A replacement that fits mechanically may still change the measurement behaviour.

Maintenance records should link to instrument configuration history so that later analysis can identify observations acquired before and after a change. This also supports investigation of recurring faults. The measure of a successful service is restoration of a verified, usable function with its records intact, not simply replacement of the component that first appeared suspicious.

Training and competency services

Training should enable users to recognize valid operation, uncertain evidence and situations requiring an alternative workflow. It should develop the ability to challenge the software when appropriate. Familiarity with the interface alone is not evidence of measurement or review competency.

Build training around realistic decisions

A proposed training programme should include plate identity resolution, capture readiness, calibration status, overlay inspection, correction, rejection and export reconciliation. Difficult examples should be included deliberately: glare, unclear edges, incomplete context and disagreement between the visible image and the automated estimate.

Competency evaluation should ask users to make decisions on cases they have not already memorized. The evaluation should preserve the distinction between learning the interface and demonstrating the judgement required for the laboratory’s intended role.

Table I.15. Training and competency services

User role	Competency focus
Acquisition operator	Identity, readiness, failed capture and permitted recovery actions.
Reviewer	Independent evidence inspection, uncertainty and justified corrections.
Laboratory administrator	Access control, workflow configuration and exception monitoring.
Service engineer	Configuration integrity, calibration impact and return-to-use evidence.
Research investigator	Experimental metadata, sampling bias and measured versus simulated values.

Keep training current

A substantial change in overlays, quality flags, rules interpretation or export behaviour may require targeted retraining. The change should be assessed for its effect on user decisions rather than simply whether a button moved. Training records should identify the product version and the relevant tasks.

The service can include guided demonstrations, practice cases, assessed exercises and refresher material. Claims that training reduces errors or review time should be supported by evaluation, including whether users become excessively trusting of the automated output. Human review works only when the human has the information, time and competence to perform it.

Research and data partnership services

Research support services can help partners turn a broad idea into a reproducible investigation. The useful deliverable is a defined question, a suitable observation plan, a provenance-preserving dataset and an analysis that distinguishes evidence from interpretation. Scientific conclusions remain the responsibility of the investigators and their review process.

Proposed support functions

AMReye.AI could support instrument configuration, acquisition metadata, annotation workflows, data quality review, interface development and reproducible analysis packaging. Partners may also need help comparing candidate sensing approaches against a conventional baseline before committing to a larger study.

Table I.16. Research and data partnership services

Partnership stage	Concrete output
Question definition	A bounded hypothesis and a measurable outcome.
Feasibility assessment	Required signals, confounders, reference method and expected limitations.
Dataset preparation	Data dictionary, provenance, access controls and annotation instructions.
Analysis support	Reproducible processing with declared assumptions and exclusions.
Publication support	Traceable figures, method descriptions and separation of original from external evidence.

Agree the evidence and rights in advance

Data-sharing terms should address permitted reuse, confidentiality, publication review, attribution and handling of derived models. A partner’s access to an instrument should not imply permission to reuse identifiable records for unrelated model training. Similarly, a promising preliminary result should not become a public product claim before the relevant evidence review.

Research support should be evaluated by reproducibility and usefulness: whether another analyst can understand the dataset, reconstruct the processing and identify the limits of the conclusion. A large volume of generated analysis is not a substitute for an answer to the original research question. The service should make uncertainty explicit enough to guide the next experiment or engineering decision.

The laboratory user experience

The operator-facing experience should make the state of the plate and the state of the analysis immediately understandable. BAHU, the proposed Bio-Automation and Human User Interface concept, brings together acquisition, review and instrument orchestration while preserving the distinct responsibilities of each function.

Organize the interface around decisions

Before capture, the user needs identity and readiness information. During review, the user needs the original image, the measurement overlay, quality findings and the relevant interpretation context. After release, the user needs export status and a reliable way to issue or inspect a correction.

An interface should display “unavailable” differently from zero, “simulated” differently from observed and “pending review” differently from approved. These distinctions should appear in text as well as colour so that the workflow remains understandable under varied displays and accessibility needs.

Assistance without hidden authority

Voice annotation, touchless controls, mobile monitoring and augmented overlays are proposed interaction options. Their usefulness depends on the environment and on whether they reduce or increase distraction. Voice transcription should be reviewable before becoming part of a released record. A remote screen should not imply permission to control local hardware. An augmented overlay should retain registration and visibility checks.

Table I.17. The laboratory user experience

User question	Information the interface should expose
What am I reviewing?	Plate identity, observation time and original evidence.
Why is it flagged?	Specific quality issue and the affected measurement.
What changed?	Original value, revised value, reason and responsible actor.
What can I do next?	Permitted review, reacquisition, escalation or rejection action.

The design should be evaluated with representative users performing realistic tasks. A visually attractive interface can still mislead if it conceals uncertainty, gives excessive prominence to an automated category or makes rejection harder than acceptance.

The technical partner proposition

Technical partners need a clear account of the component they are being asked to develop, integrate or evaluate. The partnership proposition should define the interface, the operating conditions, the evidence expected and the responsibilities on either side. Broad platform language is insufficient for an engineering agreement.

Make the contribution inspectable

An optics partner may contribute a qualified imaging path and calibration method. An electronics partner may contribute a power and control assembly with verified fault behaviour. A software partner may contribute an adapter with a conformance suite. A laboratory partner may evaluate a defined workflow using an agreed comparator and review process.

Table I.18. The technical partner proposition

Partner contribution	Proposed acceptance evidence
Optical subsystem	Field coverage, image quality, stability and spatial measurement residuals.
Control subsystem	State transitions, interlocks, failure response and timing behaviour.
Data interface	Schema mapping, identity integrity, correction semantics and reconciliation.
Research modality	Measurable signal, confounder assessment and incremental information value.
Manufacturing process	Repeatable assembly, inspection records and configuration traceability.

Share interfaces before fixing form

The physical product remains malleable. Interfaces can nevertheless specify coordinate frames, timestamps, electrical limits, data types and quality states. This allows alternative layouts to be compared without losing the ability to assess their behaviour.

Partners should also agree how design changes are reviewed and how a supplied component’s evidence transfers to the integrated product. A component test can support an argument about the system, but it does not automatically establish system performance. A productive partnership makes that integration gap visible and assigns responsibility for closing it.

The responsible investment proposition

A responsible investment review should examine whether the programme is converting uncertainty into evidence in a sensible order. The breadth of the proposed platform is relevant, but it should be assessed alongside the narrower product path that can be built, evaluated and supported first.

Review the evidence behind the opportunity

For Reader V1, the essential questions concern measurement performance, laboratory usability, integration and serviceability. For an integrated instrument, the review expands to environmental control, handling, assay boundaries and production complexity. For predictive twins or genomic inference, the central uncertainty includes whether the available observations support the intended biological conclusion.

Table I.19. The responsible investment proposition

Review dimension	Evidence that would strengthen the case
Technical feasibility	Repeatable measurements and successful tests of declared failure modes.
Workflow value	Demonstrated usefulness to representative laboratory users.
Product readiness	Defined configuration, documentation, training and service process.
Regulatory path	Intended-use assessment and a credible evidence programme.
Scalability	Repeatable assembly, qualified substitutions and sustainable support requirements.

Avoid premature extrapolation

An attractive prototype demonstration is not a substitute for representative validation. An external paper does not supply product accuracy. A component BOM does not include every cost of a supported, authorized product. These distinctions make development milestones more meaningful and prevent a broad vision from obscuring the next decision.

The financial narrative is deliberately excluded from this compendium. The final BOM supports a concrete engineering discussion about one reference configuration. Investment decisions would additionally require separate, current commercial and financial diligence, based on verified commitments and an agreed product scope.

Lab in a box as a system concept

“Lab in a box” describes the integration of several laboratory functions into a coordinated instrument or system. It does not mean that all laboratory responsibilities are physically contained in a small enclosure. Specimen handling, quality management, review, biosafety and reporting can still depend on the surrounding laboratory.

Apply the term to functions

For AMReye.AI, the concept could include plate observation, selected environmental functions, scheduled capture, analysis, record management and later handling modules. The appropriate combination depends on the intended application. An endpoint reader and a research workstation may share software while needing very different hardware.

Table I.20. Lab in a box as a system concept

Functional boundary	Design question
Observation	Which signals must be acquired, at what scale and with what traceability?
Biological conditions	Which conditions are controlled, monitored or supplied by the laboratory?
Material handling	Which transfers are automated and how is custody confirmed?
Computation	Which functions must remain available locally and which can be remote?
Review and service	How can people inspect evidence, recover from faults and maintain calibration?

Integration should simplify the useful workflow

Combining modules may reduce handling, but it can also couple failures. A thermal disturbance may affect optics; a software update may affect scheduling; a mechanical fault may obstruct a plate needed for review. The design should therefore expose functional boundaries and preserve safe failure behaviour even when modules share a housing.

The term is useful when it helps explain a coherent workflow. It becomes misleading if it suggests universal assay coverage, independence from laboratory expertise or readiness that has not been demonstrated. The product should be described through the functions it has qualified and the environment in which those functions remain reliable.

Lab on a chip as an assay concept

“Lab on a chip” refers to miniaturized fluidic or analytical structures that perform selected laboratory operations. Such devices can position small samples near sensors, reduce reagent volumes or support parallel observations. Their suitability depends on the assay and on the complete path from sample preparation to interpretable output.

Miniaturization changes the measurement problem

At small scales, surface interactions, fluidic resistance, bubbles, evaporation and cell positioning can become central sources of variation. A chip that produces a useful signal in a research study may still require substantial work on sample compatibility, manufacturing consistency, storage and operator handling.

Published microfluidic single-cell imaging and impedance studies provide examples of rapid phenotypic research approaches. Their timing and performance belong to their particular methods and tested samples. They do not establish that a conventional plate reader can provide the same result. [\[R11\]](#) [\[R38\]](#)

Table I.21. Lab on a chip as an assay concept

Chip-level advantage to investigate	Corresponding development burden
Small sample volumes	Accurate delivery, losses to surfaces and representative sampling.
Close sensor coupling	Electrode or optical stability and assay compatibility.
Parallel test regions	Uniform exposure conditions and channel-to-channel comparability.
Rapid observation	A validated relationship between the early signal and the intended result.
Disposable formats	Manufacturing variation, shelf life and waste handling.

A separate product track

A chip-based assay could become a complementary research or product family. It should have its own intended use, consumable specification, reader interface and validation programme. Shared PIMS records can preserve provenance across formats, but the software should not collapse chip results and disk diffusion measurements into an interchangeable quantity.

The near-term question is whether a particular miniaturized format solves a defined assay limitation better than a simpler alternative. The terminology should describe that mechanism, not function as a general promise of speed or precision.

Portfolio dependencies and sequencing

The portfolio should advance through dependencies that can be tested. Shared infrastructure can reduce duplicated work, but evidence should transfer only where the relevant assumptions remain valid. A stable plate identity model may support several product families; an optical calibration for one camera arrangement usually does not.

Proposed dependency sequence

First establish reliable acquisition, spatial calibration and record lineage. Then evaluate endpoint measurement and human review. Add laboratory interfaces and service processes around the qualified workflow. Investigate time-series capture and additional modalities when a specific research question requires them. Integrated handling, predictive models and custom hardware should follow their own feasibility evidence.

Table I.22. Portfolio dependencies and sequencing

Shared element	Potential reuse	Reassessment needed
Identity and audit model	Across readers and research instruments	New custody events and data classes
User access model	Across software products	New roles, remote actions and sensitive data
Capture pipeline	Across related optical configurations	Sensor, lens, processing and timing changes
Measurement algorithm	Within a declared image domain	New media, geometry or appearance distributions
Standards engine	Across eligible measurement sources	Method compatibility and source-specific verification

Preserve optionality through interfaces

The platform should avoid making an early enclosure, processor or model version a permanent conceptual boundary. Stable interfaces allow alternative components to be evaluated against the same requirements. Optionality is useful when it enables a better verified design, rather than delaying all decisions indefinitely.

Portfolio reviews should ask which uncertainty the next stage resolves and which future choices it enables. A module that adds complexity without answering an important question should be deferred. This approach allows ambitious research to proceed while the first product remains focused enough to evaluate thoroughly.

How a proposed capability becomes a product

A proposed capability becomes a product when its purpose, configuration, performance limits and support responsibilities are sufficiently defined and evidenced for the intended use. A functioning demonstration is an important step, but it is not the final criterion.

A capability transition

Consider a proposed condensation detector. It begins as a hypothesis that image features can identify a condition that compromises reading. Development needs representative examples, a conventional baseline and a definition of the condition that should trigger action. Evaluation must then determine whether the detector catches relevant failures without unnecessarily rejecting useful observations. The interface must communicate the finding and support an appropriate recovery action.

The same logic applies to a sensor, an AI model or an integration adapter: define what failure means, measure performance in the intended context and make the resulting limitation operationally manageable.

Table I.23. How a proposed capability becomes a product

Transition	Evidence expected
Idea to feasibility	A measurable signal or behaviour linked to a useful question.
Feasibility to engineering	Defined interfaces, operating range and reproducible implementation.
Engineering to evaluation	Controlled configuration and prespecified assessment methods.
Evaluation to deployment	Acceptable evidence, user information, support and applicable authorization.
Deployment to sustained product	Monitoring, correction, maintenance and controlled change.

A disciplined claim boundary

The claim should describe the function and conditions actually evaluated. A detector that identifies one type of artifact does not establish universal image quality. An algorithm that performs well at one site does not establish transferability. A sensor that agrees with another sensor does not prove both are correct.

Product development therefore includes the ability to say when the function cannot support the requested decision. Explicit limitations, abstention and recovery are part of the product design, not merely qualifications added after the technology has been built.

II / SENSING AND ENGINEERING

Measurement architecture without a fixed enclosure

The proposed platform architecture should be described through functions and interfaces before it is expressed as an enclosure. Observation, environmental management, physical control, computation, review and data storage have different timing, failure and service requirements. Their separation can be logical, electrical or physical; no single compartment arrangement is prescribed here.

Define what crosses a boundary

An imaging subsystem should provide observations with configuration, timing and quality metadata. A control subsystem should expose permitted commands and confirmed states. An analysis subsystem should return derived values linked to exact inputs. These contracts let designers compare alternative mechanical layouts and computing arrangements without changing the meaning of a record.

Table II.1. Measurement architecture without a fixed enclosure

Boundary	Information or constraint to specify
Optical to mechanical	Object plane, usable field, permitted pose, focus range and vibration sensitivity.
Mechanical to control	Motion limits, position feedback, interlocks and uncertain-state behaviour.
Acquisition to analysis	Original data, exposure state, timestamps, calibration and quality flags.
Analysis to review	Quantity, units, uncertainty, eligible context and evidence overlay.
Local to remote	Data purpose, authorization, synchronization state and offline behaviour.

Design for coupling that cannot be eliminated

Heat from computation can change an optical assembly. A fan can introduce vibration. An illumination change can alter both image contrast and sample heating. Treating modules as separate boxes in a diagram does not remove these interactions. Integration testing should identify and quantify the couplings that matter to the selected measurement.

A candidate layout should therefore be assessed against measurement quality, access for service, contamination management, power, thermal behaviour and recovery. The strongest arrangement is the one that meets the chosen requirements with evidence. It need not be the most compact, the most sensor-rich or the one containing the newest processor.

Optical geometry and spatial calibration

An imaging instrument measures a physical object through an optical projection. A zone diameter expressed in pixels becomes a length only when the relationship between the image and the relevant plate plane is established. Resolution, repeatability and accuracy describe different properties; a dense pixel array cannot substitute for a calibrated measurement chain.

Define the measurement plane

The proposed reader should reference the plane at which zone boundaries are interpreted. A ruler attached to a raised lid can have a different scale from the agar surface under perspective imaging. Calibration therefore needs a traceable target at the intended plane, a declared working range and a check for local distortion across the field. Telecentric optics are a candidate for reducing magnification changes with depth, but their benefits must be weighed against field size, working distance and integration cost. [\[R15\]](#)

Carry geometry into uncertainty

For an ideal uniform scale, diameter d equals pixel span p multiplied by scale s in millimetres per pixel. A first order independent uncertainty approximation is $u(d)^2 = s^2u(p)^2 + p^2u(s)^2$. Correlated terms must be included when the same images or reference features influence both quantities. This approximation does not cover all lens distortion or ambiguous biological edges. [\[R16\]](#)

Table II.2. Optical geometry and spatial calibration

Source of uncertainty	Proposed engineering treatment
Scale target uncertainty	Store target identity, certificate or verification basis, and calibration date.
Position dependent distortion	Compare residuals across the usable field; restrict the valid region when necessary.
Plate height and tilt	Measure or constrain the permitted pose and verify the resulting scale error.
Zone edge ambiguity	Preserve boundary confidence and reviewer edits rather than hiding them in a single rounded number.
Configuration changes	Invalidate or requalify calibration after lens, focus, mount or image pipeline changes.

The acceptance decision should use a measurement uncertainty budget and independent target checks. A successful camera self-test establishes instrument readiness; biological agreement still requires a separate study against the appropriate reference workflow.

II / SENSING AND ENGINEERING

Illumination and image formation

Illumination is part of the measurement chain. A plate image records the combined effects of the light source, optical path, specimen, lens, sensor and processing pipeline. Changes in any of these can alter apparent contrast even when the underlying biological object is unchanged.

Control the useful dimensions of light

The proposed reader should qualify source geometry, intensity stability, spectral content and spatial uniformity. Reflected and transmitted light can answer different questions. Oblique illumination may reveal texture while making a boundary appear different from its appearance under a standard reading geometry. A multimode acquisition should therefore identify the mode in each observation rather than merge images without provenance.

Table II.3. Illumination and image formation

Lighting property	Measurement consequence
Spatial nonuniformity	Position-dependent brightness can bias threshold-based segmentation.
Temporal modulation	Exposure timing can interact with flicker and create variable intensities.
Spectral change	Pigment, medium and sensor responses can shift differently.
Specular reflection	Glare can erase visible boundaries or create false bright features.
Stray ambient light	An apparently identical setting can produce a different image between sites.

Preserve linearity where it matters

Automatic enhancement, tone mapping and local contrast processing can improve appearance while changing quantitative relationships. A measurement pipeline should declare which processing stages are permitted and retain enough original data to investigate their effect. Camera characterization methods can help quantify noise, response and dynamic range, but the complete illuminated object still requires application-specific evaluation. [\[R14\]](#)

The preferred illumination arrangement should be selected through comparisons on representative non-biological targets and appropriate plate images. Evaluate not only whether a feature becomes more visible, but whether the resulting measurement remains stable across location, exposure and the permitted range of plate appearance.

Environmental and mechanical stability

Mechanical and environmental stability determine whether an instrument can reproduce its measurement geometry over time. A stable image requires more than a rigid camera mount: the plate support, focus mechanism, lighting fixture and surrounding thermal conditions can all affect the acquired signal.

Identify slow and fast disturbances

Slow drift may follow temperature changes, material relaxation or repeated loading. Fast disturbances may arise from fans, motion stages, nearby equipment or contact with the bench. Their signatures differ, so the instrument should not rely on a single generic stability indicator.

Table II.4. Environmental and mechanical stability

Disturbance	Proposed observation or control
Plate tilt or height variation	Verify pose against the permitted geometry and assess scale sensitivity.
Focus drift	Track image quality on a stable reference and control adjustment history.
Vibration during exposure	Compare inertial or image-based evidence with blur and motion artifacts.
Thermal expansion	Characterize measurement residuals across the intended ambient range.
Condensation	Record optical obstruction and assess readability instead of assuming a corrective mode works.
Airflow disturbance	Separate electronics cooling requirements from the observation environment.

Qualify the assembled instrument

Component specifications do not fully describe the behaviour of the assembly. A low-noise camera mounted on a vibrating structure can still produce unreliable images. A stable temperature sensor at one location does not prove uniform conditions throughout a chamber.

The engineering programme should measure the relationship between disturbances and the quantities that matter to the workflow. If a disturbance has no material effect within the accepted range, it may need only monitoring. If it can invalidate measurement, the response should be explicit: prevent capture, flag affected observations or require requalification. The decision should follow measured sensitivity rather than the mere presence of a sensor.

Timing synchronization and acquisition control

Time is an essential coordinate in any repeated observation system. The instrument must distinguish the biological event used as a time origin, the start and end of an exposure, the arrival of data and the completion of analysis. These timestamps can differ substantially and should not be collapsed into one “result time.”

Synchronize the events that need comparison

For multimodal analysis, each observation should identify the interval over which it was measured. A gas sensor may average over seconds while a camera integrates light over milliseconds. Assigning both the same timestamp can create a false impression of simultaneity. Where sensor clocks differ, the synchronization method and estimated timing uncertainty should be recorded.

Table II.5. Timing synchronization and acquisition control

Time field	Proposed meaning
Biological origin	Declared assay event from which biological age is calculated.
Observation interval	Start and end of the actual measurement or exposure.
Receipt time	Time the record reached the acquisition or storage service.
Analysis time	Time the specified computation completed.
Review time	Time the authorized reviewer made the recorded decision.

Handle clock faults explicitly

A system restart, manual clock correction or loss of synchronization can disrupt ordering. Monotonic counters can help preserve local event sequence, while UTC timestamps support comparison across systems. Both may be useful, but neither should conceal an uncertain biological start time.

Acquisition scheduling should prioritize the observations required by the selected workflow. A missed scheduled image is a missing observation, not an invitation to substitute a generated image into the measured series. Later analysis can model the gap, provided that its status and uncertainty remain explicit. Timing integrity is therefore both an engineering requirement and a condition for credible growth modelling.

Sensor selection and information value

Sensor selection should begin with a question that the existing observations cannot answer adequately. The relevant comparison is the information gained after accounting for calibration, confounding, maintenance, power and integration. A sensor count is not a measure of diagnostic reliability.

Compare incremental value

A candidate channel should be assessed against the current baseline using the same records and a prespecified outcome. If it improves a model only because it identifies the acquisition site or experimental batch, it may fail when deployed elsewhere. Its value should persist under tests that separate the intended biological signal from instrument and workflow context.

Table II.6. Sensor selection and information value

Selection criterion	Question for the design review
Signal relevance	Is the measured quantity plausibly connected to the intended question?
Sensitivity and range	Can the channel resolve useful variation within the permitted operating conditions?
Independence	Does it add information or repeat the same failure mechanism?
Calibration burden	Can its response be verified and maintained in the deployment setting?
Integration burden	Does it disturb the sample, optical path, atmosphere or operator workflow?
Service and supply	Can equivalent parts and reference materials remain available?

Use staged experiments

Begin with instrument characterization and non-biological controls where possible. Then test the assay-specific relationship using suitable reference observations. Finally, evaluate whether the channel improves the complete decision process, including cases in which it is missing or disagrees with other inputs.

The outcome may be to select the sensor, restrict it to research, use it only for instrument quality checks or reject it. A channel that reliably detects a lighting fault may be highly valuable even if it contains no biological information. Conversely, an attractive biological correlation may not justify product integration if it cannot be calibrated or transferred reliably.

II / SENSING AND ENGINEERING

Combining modalities without counting votes

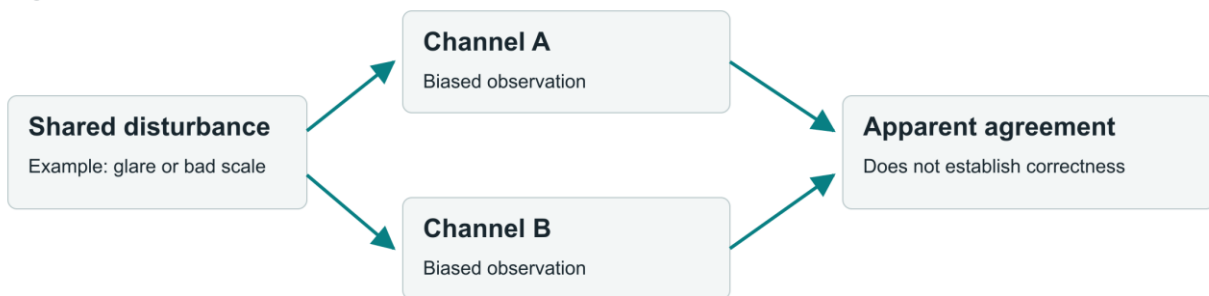
Multiple sensors can improve observability, but they do not guarantee independent evidence. Two cameras can share the same glare, two temperature sensors can share the same poor placement and several learned models can inherit the same biased training labels. Agreement is useful only when the reasons for possible agreement are understood.

Model disagreement as information

The proposed fusion system should preserve each channel's quality state and uncertainty before combining outputs. A disagreement may indicate a sensor fault, timing mismatch, domain shift or a real difference between the quantities being observed. The correct response depends on which of these explanations remains plausible.

A channel that fails its own quality check should be excluded from eligible fusion while its failure record remains. Different quantities require an explicit relationship model, not numerical voting. Independent disagreement should trigger checks of timing, calibration and assumptions. Shared disturbances require common-cause tests and independent references. A missing required channel permits only a separately evaluated degraded mode; otherwise the system should decline the result.

Agreement can survive a shared failure



AMReye.AI · conceptual diagram

Figure II.1. Shared disturbances can make multiple channels agree incorrectly.

A simple uncertainty example

For two equally precise estimates with variance σ^2 and correlation ρ , the variance of their average is $\sigma^2(1 + \rho)/2$. With zero correlation, averaging can reduce variance. With perfect positive correlation, it does not. This mathematical example concerns uncertainty in comparable estimates; it is not a diagnostic accuracy claim.

The zero-error ambition should therefore become measurable requirements for disagreement detection, fault containment, abstention and recovery. Evaluation should include failures that affect several channels together. The product should be able to explain why a combined result is eligible, which observations contributed and which conditions would invalidate the inference.

Using the candidate technology catalogue

The following forty entries form a candidate technology catalogue across the platform. They are not an installed sensor list. Some entries describe biological observation methods, some monitor the environment and others assess instrument readiness or control. Several optical entries are related acquisition modes; the catalogue also spans distinct thermal, electrical, chemical, acoustic, mechanical and magnetic principles.

Read each entry as a selection argument

Each entry identifies the measurable signal, useful question, calibration needs, confounders, integration burden and evidence boundary. A candidate can be well established for measuring a physical quantity while remaining unproven for an AMReye.AI biological application.

Table II.7. Using the candidate technology catalogue

Catalogue range	Principal role
S01 to S18	Optical imaging, spectroscopy and radiometric observations.
S19 to S30	Thermal, electrical, chemical and acoustic research measurements.
S31 to S35	Environmental and mechanical monitoring.
S36 to S40	Position, vibration, power and optical reference checks.

Use the catalogue to narrow choices

Selection should start with the intended assay and the unresolved measurement problem. An environmental channel may help identify an invalid observation without contributing directly to a biological model. A sophisticated spectroscopic channel may require a different sample format and belong in a research instrument rather than Reader V1.

The catalogue deliberately avoids current sensor model names and component generations. The concluding BOM identifies exact candidates for one Reader V1 configuration. That configuration uses only the components needed for its defined endpoint measurement workflow and does not attempt to implement the complete catalogue.

All application proposals require suitable qualification. A literature demonstration is cited where it supports a relevant principle or research precedent; it should not be read as evidence that the proposed AMReye.AI integration has achieved the same result.

S01 Visible reflectance imaging

S01 · Biological observation candidate · Core imaging pathway

Visible reflectance imaging measures light returned from the plate, medium, disks and growth under a defined illumination geometry. A colour sensor samples spectral bands rather than measuring a complete reflectance spectrum. Exposure, illumination spectrum and image processing influence the recorded values alongside the object itself.

Signal and useful question

The primary signal is pixel intensity in digital counts. Calibrated spatial features can be expressed in millimetres, while colour or texture features require their own definitions. This pathway supports plate overview, disk localization, zone review and selected morphology research. Camera characterization can inform response and noise assessment. [\[R14\]](#)

Table II.8. S01 Visible reflectance imaging

Qualification element	Proposed treatment
Calibration	Spatial target at the measurement plane; dark and uniform-field checks; colour reference when colour is analysed.
Confounders	Glare, medium colour, condensation, automatic white balance, lid markings and uneven lawn appearance.
Integration burden	Stable optics and lighting, clean optical access, controlled focus and sufficient data storage.
Selection criterion	Reliable contrast and spatial measurement across the intended plate domain, not maximum megapixel count.

What the channel can support

The observation can support a reviewed geometrical measurement when the image and calibration are eligible. It cannot independently establish organism identity, a resistance mechanism or a quantitative MIC. Those outputs require a suitable assay or separately validated inference.

The proposed evaluation should compare measurements across plate positions, lighting drift and repeated loading. Preserve original images so that a reviewer can distinguish an algorithmic boundary error from a limitation in the acquired evidence. A failed reflectance image should remain a failed observation even if another mode later provides a usable view.

S02 Transmitted light imaging

S02 · Biological observation candidate · Alternative optical geometry

Transmitted-light imaging observes light that passes through the plate and its contents. Attenuation depends on absorption, scattering, thickness and optical path. It can provide contrast different from reflected light, particularly where the medium and vessel permit a useful transmission path.

Signal and useful question

The camera records transmitted intensity in digital counts. A normalized transmission ratio is dimensionless when a suitable reference is available. The useful question is whether this geometry makes a relevant boundary or distribution more reproducible for the selected workflow. It should not be assumed to be interchangeable with the geometry specified by a standard reading method. [\[R03\]](#)

Table II.9. S02 Transmitted light imaging

Qualification element	Proposed treatment
Calibration	Uniform backlight reference, dark response and spatial calibration through the complete vessel path.
Confounders	Agar thickness, opaque media, bubbles, plate base curvature, labels and stray light.
Integration burden	Optical access below the plate, repeatable support and separation of direct from scattered light.
Selection criterion	Measurable improvement in the intended feature without unacceptable position or medium dependence.

Interpretation boundary

A darker region may reflect greater thickness or scattering rather than a proportional increase in biomass. Converting intensity into a biological quantity therefore requires an assay-specific calibration and a declared operating range.

A useful bench comparison would hold the object and camera geometry stable while examining how known optical disturbances affect the image. Subsequent plate evaluation should determine whether the transmitted view assists measurement or quality review. The software should retain mode identity so that a model cannot silently treat a transmitted image as a reflected one.

II / SENSING AND ENGINEERING

S03 Oblique dark field imaging

S03 · Biological observation candidate · Surface and scattering contrast

Oblique and dark-field arrangements reduce direct light reaching the detector and emphasize light scattered by the object. They can make edges, texture and small surface features conspicuous. The resulting contrast is strongly dependent on illumination angle, collection geometry and the object's scattering behaviour.

Signal and useful question

The measured signal is spatial intensity in digital counts, often interpreted through normalized contrast or texture descriptors. Candidate questions include whether an unusual surface feature is present and whether a boundary remains visible when conventional illumination is ambiguous. Such images should be treated as complementary observations rather than automatically preferred measurements.

Table II.10. S03 Oblique dark field imaging

Qualification element	Proposed treatment
Calibration	Fixed illumination geometry, repeatable reference texture and field uniformity assessment.
Confounders	Scratches, dust, meniscus edges, lid defects, plate tilt and specular reflections.
Integration burden	Additional lighting positions, optical baffling and mode-specific acquisition control.
Selection criterion	Incremental detectability with manageable false feature rates across vessel conditions.

Avoid a visibility trap

A bright feature in dark field is evidence of scattering, not proof of a colony or a resistant subpopulation. A scratch can be optically prominent while biologically irrelevant. The review workflow should retain the corresponding conventional image and the feature's location.

Qualification should include objects that challenge specificity: clean vessel surfaces, known surface defects and representative permitted plates. If the mode changes the apparent zone boundary, the effect must be evaluated against the selected measurement definition. Its most appropriate initial role may be research annotation or artifact review rather than the sole source of a reported diameter.

S04 Polarization resolved imaging

S04 · Biological observation candidate · Polarization contrast

Polarization-resolved imaging changes or measures the orientation state of light. Polarizers can suppress some specular reflections, while polarization-dependent signals may reveal optical structure. The benefit depends on the source, object, vessel and geometry; cross-polarization does not remove every reflection or scattering artifact.

Signal and useful question

Candidate outputs include intensity at declared polarizer angles and dimensionless polarization contrast. More complete measurements can estimate polarization state under a calibrated instrument model. The near-term question is whether the approach improves readability or artifact discrimination for a specified plate and lid material.

Table II.11. S04 Polarization resolved imaging

Qualification element	Proposed treatment
Calibration	Polarizer orientation, extinction behaviour, illumination stability and channel registration.
Confounders	Stress birefringence in plastics, depolarizing scattering, scratches, angle changes and reduced light throughput.
Integration burden	Additional optical elements, repeatable rotation or channel switching, exposure compensation.
Selection criterion	Better usable-image coverage without biasing the feature being measured.

Preserve the comparison

The same region should be examined with and without the polarization arrangement so that a gain in readability can be distinguished from a change in appearance. A detector trained on one vessel material may learn its polarization artifacts and fail on another.

This candidate should begin as an optical quality improvement investigation. It is not a direct measurement of antimicrobial response. If selected, the configuration and angle state belong in the capture record, and any calibration affected by the added optics must be requalified. A nominally sharper image is insufficient evidence unless the intended measurement becomes more reliable.

S05 Near infrared imaging

S05 · Biological observation candidate · Experimental spectral mode

Near-infrared imaging uses wavelengths beyond the main visible band, with a compatible detector and optical path. Absorption and scattering can differ from visible-light behaviour, so a feature may gain or lose contrast. Those differences do not imply that near-infrared light will reliably see through condensation or every plate material.

Signal and useful question

The recorded signal is wavelength-band-dependent image intensity. Spatial measurements remain in millimetres only after suitable calibration. The relevant research question is whether a defined near-infrared mode improves usable observation under particular media, lids or condensation conditions.

Table II.12. S05 Near infrared imaging

Qualification element	Proposed treatment
Calibration	Spectral response, source stability, dark response, focus and spatial calibration in the selected band.
Confounders	Water absorption, vessel transmission, focus shift, detector sensitivity and differences in medium contrast.
Integration burden	Compatible illumination, optics and filtering; assessment of optical exposure and heating.
Selection criterion	Verified improvement over visible controls across the declared disturbance range.

Test the claim directly

The evaluation should preserve paired visible and near-infrared observations and independently assess the disturbance being addressed. Improved appearance in a few images is insufficient to establish a general “fog vision” capability. The comparison should include failure cases in which the added mode makes an edge less visible or creates misleading contrast.

Near-infrared imaging remains a proposed research pathway. It should not supply a biological ground truth or replace a missing standard observation automatically. The software should identify the spectral mode and require a separately qualified analysis route before using it for a reported measurement.

S06 Ultraviolet excited autofluorescence

S06 · Biological observation candidate · Fluorescence research

Ultraviolet-excited autofluorescence observes light emitted after endogenous materials absorb excitation energy. The emitted signal can reflect several molecular and environmental contributions. The measured fluorescence is not a unique identifier of an organism or resistance state without an appropriately validated assay and analysis.

Signal and useful question

The detector measures emission intensity within declared wavelength bands, usually in counts or normalized relative units unless radiometrically calibrated. Candidate questions concern whether a reproducible fluorescence contrast exists and whether it adds information beyond ordinary imaging in a defined research workflow.

Table II.13. S06 Ultraviolet excited autofluorescence

Qualification element	Proposed treatment
Calibration	Excitation intensity, emission filters, detector response, background and reference fluorescence stability.
Confounders	Medium fluorescence, vessel fluorescence, photobleaching, exposure history and spectral overlap.
Integration burden	Filtered optical paths, excitation containment, suitable materials and exposure assessment.
Selection criterion	Reproducible signal separation with a credible control for non-biological background.

Treat the exposure as part of the experiment

Excitation can affect the specimen as well as reveal it. The research design should assess whether repeated observation changes the process being measured. The acquisition record should therefore retain exposure conditions and sequence history.

Initial use should focus on optical characterization and bounded research questions. Fluorescent reporters or added probes would constitute separate assay choices with their own validation and handling requirements. The product should not imply that a general fluorescence channel provides direct susceptibility, species identification or a clinically meaningful metabolic measurement merely because it produces a distinctive image.

S07 Fluorescence lifetime sensing

S07 · Biological observation candidate · Time-resolved fluorescence research

Fluorescence lifetime sensing measures the temporal decay of emission after excitation rather than relying only on brightness. A decay parameter can be influenced by the molecular environment and by mixtures of emitting species. The instrument response and analysis model are essential parts of the measurement. [\[R56\]](#)

Signal and useful question

Candidate outputs include photon arrival distributions, phase shifts or fitted decay times, commonly expressed in nanoseconds for relevant fluorophores. The proposed question is whether a stable time-resolved feature provides useful research information that intensity alone cannot supply.

Table II.14. S07 Fluorescence lifetime sensing

Qualification element	Proposed treatment
Calibration	Timing response, reference lifetime material, background and detector temporal behaviour.
Confounders	Mixed emitters, low photon counts, scattering, photobleaching and model-dependent fitting.
Integration burden	Pulsed or modulated excitation, suitable detection, timing electronics and substantial analysis complexity.
Selection criterion	Identifiable and reproducible decay parameters with an incremental benefit over simpler fluorescence measurements.

Do not overinterpret a fitted lifetime

A single fitted number can conceal a mixture of signals or an inadequate decay model. The research record should retain fitting assumptions, residuals and uncertainty. Comparisons across instruments require evidence that their timing responses and analysis methods produce comparable quantities.

This is an exploratory platform candidate, not a Reader V1 requirement. Its biological interpretation would need assay-specific studies and suitable controls. A technically successful lifetime measurement establishes that a temporal optical signal was measured; it does not establish that the signal identifies a pathogen, resistance mechanism or antimicrobial response. Selection should depend on a research question sufficiently important to justify the added optical and computational burden.

S08 Multispectral imaging

S08 · Biological observation candidate · Selected spectral bands

Multispectral imaging records a limited set of spectral bands chosen for a particular measurement question. The bands may be acquired sequentially or through parallel optical paths. Unlike a colour photograph, the acquisition can be designed around specific spectral regions, but its usefulness still depends on the sample, illumination and detector response. [\[R55\]](#)

Signal and useful question

The primary output is an image stack indexed by band, with calibrated or normalized intensity values. Ratios between bands are dimensionless but remain sensitive to how their signals were acquired. Candidate questions include whether spectral contrast improves artifact discrimination or characterizes a reproducible research phenotype.

Table II.15. S08 Multispectral imaging

Qualification element	Proposed treatment
Calibration	Band transmission, source spectrum, flat-field response and spatial registration.
Confounders	Motion between bands, exposure differences, spectral leakage, medium variation and detector drift.
Integration burden	Filters or multiple sources, synchronized switching, more data and a band-aware analysis pipeline.
Selection criterion	Each retained band adds reproducible information on held-out observations.

Select bands through evidence

Adding bands increases the number of possible correlations a model can learn. A band that appears predictive within one batch may encode lighting or medium differences. Selection should therefore be evaluated under site, batch and time separation, with a conventional baseline and an ablation analysis.

The resulting image stack is a richer observation, not a direct biological conclusion. The capture record should identify all band settings and mark incomplete stacks explicitly. If a required band fails, the system should use a separately evaluated reduced-band route or decline the inference rather than silently synthesize the missing measurement.

S09 Hyperspectral imaging

S09 · Biological observation candidate · Spectral research

Hyperspectral imaging samples a densely resolved spectral dimension together with spatial information. The result is a data cube whose coordinates describe position and wavelength. Acquisition may involve spatial or spectral scanning, so the cube may represent a sequence of observations rather than one instantaneous view. [\[R55\]](#)

Signal and useful question

The primary signal is intensity or calibrated reflectance as a function of wavelength and position. Spectral features can be described through band ratios, absorption features or learned representations. The research question is whether these features explain variation that a smaller set of selected bands cannot capture reliably.

Table II.16. S09 Hyperspectral imaging

Qualification element	Proposed treatment
Calibration	Wavelength accuracy, spectral response, dark reference, spatial registration and illumination reference.
Confounders	Spectral mixing, sample movement, stray light, medium composition and preprocessing choices.
Integration burden	Specialized optics, longer or more complex acquisition, large datasets and spectral quality control.
Selection criterion	Reproducible incremental information relative to multispectral and ordinary imaging baselines.

Control dimensionality and leakage

Many correlated spectral channels can make a small dataset appear highly separable. Model evaluation should therefore separate sites, batches and acquisition sessions, and feature selection must occur within the training process. A spectrum from one region should not be allowed to reveal the identity of the same plate in a test set.

Hyperspectral observation is an exploratory research option. A useful result could be a justified reduction to a few robust bands rather than a permanent requirement for a full spectral instrument. Biological interpretation and any susceptibility claim would need independent assay-specific evidence; a visually distinctive spectral map is not sufficient.

II / SENSING AND ENGINEERING

S10 Raman spectroscopy

S10 · Biological observation candidate · Vibrational spectroscopy

Raman spectroscopy measures inelastically scattered light whose frequency shift relates to molecular vibrational behaviour. Biological spectra can contain overlapping contributions from many constituents. The measured fingerprint depends on the optical system, sample state and background, as well as on biological composition.

Signal and useful question

The output is intensity against Raman shift, conventionally expressed in inverse centimetres. Candidate research questions include whether spectra provide reproducible phenotypic information and whether an assay-specific response can be detected. Published work has investigated Raman observations with machine learning for microbial identification and susceptibility research; its results do not validate this proposed integration. [\[R44\]](#)

Table II.17. S10 Raman spectroscopy

Qualification element	Proposed treatment
Calibration	Wavenumber reference, spectral response, optical alignment and background characterization.
Confounders	Fluorescence, substrate signal, medium residues, sample heterogeneity and laser-induced changes.
Integration burden	Spectrometer, stable excitation, optical containment, sensitive detection and often local sampling.
Selection criterion	Reproducible biological information after controlling substrate and acquisition effects.

Preserve what was actually measured

A spectrum taken from a selected microscopic region does not automatically represent the whole plate. The record should retain sampling location, selection method and any preparation required. A model should not report a measured gene merely because a spectral pattern correlates with a genotype in its training set.

This candidate belongs to a dedicated research track. The first decision is whether the intended question requires molecularly sensitive information and whether Raman provides it more usefully than a simpler method. Assay transfer, representative sampling and independent validation are likely to matter as much as spectral resolution.

S11 Infrared absorption spectroscopy

S11 · Biological observation candidate · Infrared spectroscopy

Infrared absorption spectroscopy measures wavelength-dependent attenuation associated with molecular vibrations. A spectrum can reveal chemical composition and changes in a sample, but the optical path and measurement format strongly influence the signal. Water and substrate contributions can dominate biologically interesting features.

Signal and useful question

The output is absorbance or transmittance against wavenumber, commonly in inverse centimetres. Candidate questions concern whether a reproducible chemical pattern can be measured in a permitted sample format and whether it relates to a defined research phenotype. It should not be assumed that an ordinary closed culture plate offers a suitable infrared transmission path.

Table II.18. S11 Infrared absorption spectroscopy

Qualification element	Proposed treatment
Calibration	Wavenumber accuracy, background spectrum, optical path and instrument response.
Confounders	Water absorption, vessel material, sample thickness, contact pressure and baseline correction.
Integration burden	Suitable infrared optics and detector; potentially a separate sample interface or preparation route.
Selection criterion	Stable spectral information that survives realistic sample and instrument variation.

Keep assay format explicit

Transmission, reflection and attenuated-total-reflection arrangements do not produce interchangeable observations. A contact measurement can require a different handling workflow from non-contact plate imaging. The product description should identify that difference before discussing automation or throughput.

Infrared spectroscopy is an exploratory candidate across the broader platform. The proposed qualification should compare repeated observations of reference materials and then assess the chosen biological question with appropriate controls. A classifier’s apparent accuracy should be challenged for dependence on medium, preparation batch or instrument background. No clinical identity or susceptibility function is established by including this modality in the catalogue.

S12 Optical coherence tomography

S12 · Biological observation candidate · Depth-resolved optical research

Optical coherence tomography uses interferometric information to resolve optical path differences within a sample. It can produce depth profiles and cross-sectional representations of scattering structures. The relationship between optical path length and physical depth depends on refractive properties and the reconstruction model. [\[R57\]](#)

Signal and useful question

Candidate outputs include depth-dependent backscatter and reconstructed structural coordinates, with depth expressed in micrometres or millimetres under declared assumptions. Research questions could concern surface structure, thickness or subsurface morphology that a two-dimensional plate image cannot resolve.

Table II.19. S12 Optical coherence tomography

Qualification element	Proposed treatment
Calibration	Axial and lateral scales, reference surfaces, sensitivity range and reconstruction settings.
Confounders	Multiple scattering, refractive-index variation, vessel reflections, shadowing and motion.
Integration burden	Interferometric optics, scanning or specialized detection, stable alignment and substantial data processing.
Selection criterion	Depth information that is accurate enough and relevant enough to justify the added system complexity.

Distinguish structure from biological meaning

A reconstructed surface is a model-dependent measurement, not a direct resistance result. Features may arise from the medium or vessel rather than the growth being studied. The observation should therefore remain linked to a conventional overview and to the region selected for depth imaging.

This modality is a future research option, not a requirement for endpoint Reader V1. A development study should begin with known structural references and evaluate how vessel and medium effects alter reconstruction. Only then should it assess the intended biological application and the additional value relative to simpler focus, stereo or surface-imaging methods.

S13 Quantitative phase imaging

S13 · Biological observation candidate · Quantitative optical phase

Quantitative phase imaging estimates changes in the phase of transmitted light caused by optical path differences. For a suitable sample, the signal relates to both thickness and refractive-index variation. Separating those contributions or converting them into another biological quantity requires additional assumptions and calibration.

Signal and useful question

The primary output is phase in radians or optical path difference in length units. Candidate research questions include cell or microcolony growth, morphology and temporal changes that produce weak intensity contrast. Dry-mass estimates, where scientifically justified, are derived quantities rather than direct balance measurements.

Table II.20. S13 Quantitative phase imaging

Qualification element	Proposed treatment
Calibration	Phase or path-length reference, background correction, spatial scale and reconstruction verification.
Confounders	Phase wrapping, multiple scattering, refractive-index changes, drift and overlapping structures.
Integration burden	Interferometric or computational phase optics, stable illumination and a suitable thin-sample geometry.
Selection criterion	Reproducible phase information with a defensible link to the intended research quantity.

Respect the sample geometry

A method suitable for thin cells in a controlled optical path may not transfer to a thick, scattering agar plate. The proposed platform should treat cell-level phase research as a separate acquisition route with its own sample and field-selection records.

The evaluation should distinguish numerical reconstruction error from uncertainty in the biological interpretation. A phase map may be repeatable while an inferred mass is biased by an incorrect refractive increment. This candidate should be selected only where its assumptions can be tested and where the resulting measurement answers a question that ordinary intensity imaging cannot address adequately.

II / SENSING AND ENGINEERING

S14 Laser speckle activity sensing

S14 · Biological observation candidate · Dynamic scattering research

Laser speckle imaging records interference patterns produced when coherent light is scattered. Temporal changes in those patterns can reflect movement or other changes in the scattering scene. They can also arise from vibration, source instability or optical-path changes unrelated to biology.

Signal and useful question

Candidate outputs include speckle contrast, correlation decay and other dimensionless temporal descriptors. Published plate-related studies have investigated speckle observations for early antibacterial-effect research. These are method-specific precedents, not established AMReye.AI performance. [\[R42\]](#) [\[R43\]](#)

Table II.21. S14 Laser speckle activity sensing

Qualification element	Proposed treatment
Calibration	Stable scattering references, exposure timing, optical alignment and instrument-motion characterization.
Confounders	Vibration, evaporation, thermal drift, motion of the vessel and changes in illumination coherence.
Integration burden	Coherent source, optical containment, mechanically stable acquisition and time-series analysis.
Selection criterion	A biological signal that remains distinguishable from instrument and environmental dynamics.

Compare against an instrument-only explanation

An early change in a speckle statistic should be challenged with controls that can reveal non-biological motion. The analysis should also preserve the relationship between exposure duration, sampling interval and the reported temporal feature.

This is an exploratory sensing pathway. Its first product contribution might be an additional research observation rather than an early clinical category. Any future susceptibility inference would need a defined clock origin, organism and drug scope, comparator and treatment of unreadable or inconsistent cases. A shorter detection window in an external study cannot be transferred to a different optical assembly by analogy alone.

S15 Elastic light scattering

S15 · Biological observation candidate · Angular scattering

Elastic light scattering measures light redirected by an object without a change in photon frequency. The angular and intensity distribution depends on size, shape, refractive properties and the measurement geometry. In a heterogeneous biological sample, several structures can contribute to the same pattern.

Signal and useful question

The instrument may record intensity against angle or a two-dimensional scattering pattern. Normalized descriptors are generally dimensionless; absolute intensity requires a calibrated optical response. Candidate questions concern whether a stable pattern can characterize a sample or flag a change that deserves further observation.

Table II.22. S15 Elastic light scattering

Qualification element	Proposed treatment
Calibration	Angular geometry, source stability, detector response and reference scattering objects.
Confounders	Vessel curvature, multiple scattering, sample concentration, alignment and stray reflections.
Integration burden	Controlled source-detector geometry, baffling and a reproducible sample presentation.
Selection criterion	Pattern stability and useful discrimination after vessel and concentration effects are controlled.

Avoid one-to-one biological assumptions

A scattering pattern is not uniquely determined by organism identity or susceptibility. Different combinations of shape, concentration and refractive properties can produce similar observations. A learned classifier may exploit stable laboratory artifacts rather than the desired biological variation.

The proposed evaluation should include repeated measurements after repositioning and across permitted sample containers. Where a biological inference is investigated, it should be compared with a suitable reference and tested outside the acquisition batches used for development. The initial role is research characterization; a direct clinical conclusion would require substantially more evidence than pattern separation on a small dataset.

S16 Nephelometric turbidity sensing

S16 · Preparation quality candidate · Nephelometric measurement

Nephelometry measures light scattered by suspended material at a defined observation geometry. It can support a turbidity comparison used in inoculum preparation workflows. Its output depends on particle properties and instrument geometry, so it should not be treated as a universal cell counter.

Signal and useful question

The output may be an instrument response, a calibrated turbidity unit or a comparison with an applicable McFarland reference. The useful question is whether the preparation falls within the method’s permitted turbidity condition. A McFarland value and a viable count in CFU per millilitre are not interchangeable measurements.

Table II.23. S16 Nephelometric turbidity sensing

Qualification element	Proposed treatment
Calibration	Appropriate turbidity references, vessel path, detector response and repeatability checks.
Confounders	Cell size, clumping, non-cell particles, bubbles, tube scratches and settling.
Integration burden	Defined sample holder, optical geometry, cleanable surfaces and preparation-record linkage.
Selection criterion	Reliable method-relevant comparison across the permitted sample and vessel range.

Keep preparation QC distinct from final AST

A preparation check can reduce one source of assay variation, but it does not establish that the rest of the assay was performed correctly. The record should retain the turbidity observation, its reference basis and any action taken, rather than storing only a generic “QC passed” flag.

An integrated nephelometric pathway is a proposed module. It should be evaluated independently and then as part of the selected laboratory workflow. If an organism or sample type produces an atypical optical relationship, the system should preserve the limitation rather than convert the response into an unsupported precise viable-cell concentration. [\[R03\]](#)

S17 Absorbance and optical density

S17 · Biological observation candidate · Optical attenuation

Absorbance-related measurements compare incident and transmitted light. A common expression is $A = -\log_{10}(I/I_0)$, where I is the sample signal and I_0 is an appropriate reference. In microbial suspensions, scattering can contribute strongly, so an optical-density reading is not necessarily a pure molecular absorbance measurement.

Signal and useful question

The output is dimensionless absorbance or optical density at a declared wavelength and path length. Candidate uses include tracking a reproducible optical change in a research assay or checking a preparation against a qualified reference range. Converting the reading into cell concentration requires a separate sample-specific relationship.

Table II.24. S17 Absorbance and optical density

Qualification element	Proposed treatment
Calibration	Blank reference, optical path, detector linearity and response across the intended range.
Confounders	Multiple scattering, coloured media, sedimentation, bubbles, path-length variation and stray light.
Integration burden	Stable source and detector geometry, reproducible vessel positioning and reference management.
Selection criterion	Useful dynamic range and repeatability under the actual sample format.

Define the valid range

At high attenuation, stray light and detector limits can distort the response. At low attenuation, reference noise can dominate. A growth curve should therefore identify observations outside the qualified range rather than treating every recorded number as equally informative.

This candidate is most naturally associated with a defined liquid or optical-path assay. Its interpretation should not be imported into whole-plate imaging without evidence. The platform should preserve wavelength, path and preprocessing metadata so that researchers can distinguish biological trends from differences in optical configuration.

S18 Thermal infrared radiometry

S18 · Environmental and research candidate · Surface radiometry

Thermal infrared radiometry estimates temperature-related information from emitted radiation. The estimate depends on emissivity, reflected background radiation, optical transmission and the detector calibration. A thermal camera observes an accessible radiating surface; it does not automatically measure the temperature of material hidden beneath a lid.

Signal and useful question

Raw signals are detector responses, while calibrated outputs may be radiance or apparent surface temperature in degrees Celsius or kelvin. Candidate questions include whether there are spatial thermal gradients, whether an assembly is heating unevenly or whether a process has departed from its qualified conditions.

Table II.25. S18 Thermal infrared radiometry

Qualification element	Proposed treatment
Calibration	Traceable temperature reference, emissivity assumptions, optical path and spatial response.
Confounders	Reflections, changing emissivity, vessel transmission, viewing angle and ambient background.
Integration burden	Appropriate infrared optics, unobstructed view and comparison with contact references.
Selection criterion	Surface-temperature information that cannot be obtained adequately with simpler placed sensors.

Do not confuse thermal structure with metabolic heat

A warmer region can result from lighting, airflow or electronics. Demonstrating a biological heat signal requires a much more controlled heat-flow measurement than observing apparent surface temperature. The instrument should not label a thermal image a biomass or susceptibility map without an independently validated relationship.

The proposed initial use is engineering characterization and environmental research. A thermal observation can help locate a fault or gradient, but its relationship to the plate’s relevant internal conditions must be tested. The uncertainty and emissivity assumptions should remain available to anyone interpreting the image.

S19 Microcalorimetry

S19 · Biological observation candidate · Heat-flow research

Microcalorimetry measures small heat flows associated with physical, chemical or biological processes. In a controlled assay, changes in heat production can provide a time-resolved signal. The measured heat is the sum of relevant processes in the measurement volume and is not inherently specific to microbial growth.

Signal and useful question

The primary output is thermal power, typically expressed in watts at an appropriate scale, and its time integral is energy in joules. Published work has investigated isothermal microcalorimetry for antimicrobial susceptibility research in defined sample contexts. [\[R45\]](#)

Table II.26. S19 Microcalorimetry

Qualification element	Proposed treatment
Calibration	Electrical or other suitable heat reference, baseline stability and thermal response time.
Confounders	Environmental drift, equilibration, chemical reactions, vessel differences and changing heat transfer.
Integration burden	Strong thermal control, sensitive differential measurement and a carefully defined sample interface.
Selection criterion	A reproducible assay signal with acceptable detection time and baseline uncertainty.

Preserve the distinction from temperature sensing

A temperature sensor measures a local thermal state. A calorimeter estimates heat flow through a calibrated thermal relationship. Adding several temperature sensors does not automatically create a qualified calorimeter.

This candidate belongs to a dedicated research instrument or assay track. The first evaluation should establish baseline and response behaviour before asking a biological question. Any future susceptibility result would require a method-specific interpretation and comparison with a suitable reference. The cumulative heat curve should remain linked to its raw baseline, model assumptions and actual observation intervals, particularly when a run is interrupted.

S20 Electrical impedance spectroscopy

S20 · Biological observation candidate · Frequency-dependent electrical response

Electrical impedance spectroscopy measures the relationship between an applied alternating signal and the resulting electrical response across frequency. Resistance, capacitance and interfacial effects can contribute to the complex impedance. In biological systems, the measured response may involve cells, medium, electrodes and geometry simultaneously.

Signal and useful question

Outputs include impedance magnitude in ohms, phase in degrees or radians, and fitted model parameters. Published AST research has used several electrical measurement formats with different response mechanisms and timing constraints. Their results should be interpreted within those specific assays. [\[R38\]](#) [\[R40\]](#) [\[R41\]](#)

Table II.27. S20 Electrical impedance spectroscopy

Qualification element	Proposed treatment
Calibration	Open, short and suitable reference loads; frequency response; electrode and geometry characterization.
Confounders	Medium conductivity, temperature, electrode polarization, fouling, bubbles and contact variation.
Integration burden	Analog front end, controlled electrodes or microfluidics, shielding and electrical sample compatibility.
Selection criterion	A stable signal linked to the intended response beyond medium and electrode changes.

Explain the fitted circuit

Equivalent-circuit parameters are model-based estimates. Different circuits may fit similar data, so a numerical fit should not be assigned a biological meaning without supporting evidence. Retain the measured spectrum, fitting range, residuals and parameter uncertainty.

This is a separate assay research pathway rather than an automatic add-on to endpoint plate reading. Evaluation should compare the electrical result with an appropriate reference and assess whether the relevant antimicrobial mechanism produces a measurable response within the proposed window. Short electrical acquisition time alone is not total time to a reportable result.

S21 Electrochemical pH sensing

S21 · Chemical observation candidate · Electrochemical measurement

Electrochemical pH sensing estimates hydrogen-ion activity through an electrode system and a reference potential. The response depends on temperature and on the condition of the sensing and reference interfaces. pH is logarithmic and dimensionless; it is not a direct measure of acid concentration in every sample.

Signal and useful question

The instrument records an electrical potential, typically in millivolts, and converts it to pH using a qualified calibration. Candidate uses include monitoring a defined liquid research environment or identifying a chemical change that may affect an assay. The measured location must be explicit because a local probe does not describe every region of a plate.

Table II.28. S21 Electrochemical pH sensing

Qualification element	Proposed treatment
Calibration	Appropriate reference buffers, temperature record, response slope and offset.
Confounders	Reference-junction effects, fouling, ionic composition, temperature, drift and slow equilibration.
Integration burden	Sample contact, cleaning or disposable interfaces, stable reference and electrical isolation.
Selection criterion	Reliable local chemical information that matters to the chosen assay or control question.

Interpretation boundary

A pH change can have biological or non-biological causes. Buffer capacity, gas exchange and chemical reactions can alter the relationship between biological activity and the observed value. The system should not label a pH trend as a growth rate or susceptibility result without an independently established model.

This candidate is most suitable for a dedicated research or environmental-monitoring module. Qualification should separate electrode performance from the biological hypothesis. Original voltage, temperature, calibration and derived pH should remain linked so that an investigator can evaluate whether a trend reflects the sample or a changing electrode response.

S22 Optical dissolved oxygen sensing

S22 · Chemical and environmental candidate · Optical oxygen measurement

Optical dissolved-oxygen sensing commonly uses oxygen-dependent quenching of a luminescent indicator. The instrument measures intensity or temporal behaviour and converts it through a calibrated relationship. The sensing element observes its local environment and can respond differently as temperature and the surrounding matrix change.

Signal and useful question

Outputs may include luminescence phase or lifetime and derived oxygen partial pressure, concentration or percentage saturation. These quantities should not be interchanged without the required conditions and conversion model. Candidate questions include whether a defined research environment remains within its intended oxygen state.

Table II.29. S22 Optical dissolved oxygen sensing

Qualification element	Proposed treatment
Calibration	Suitable oxygen references, temperature compensation and indicator-response characterization.
Confounders	Temperature, salinity or matrix effects, indicator ageing, optical background and diffusion lag.
Integration burden	Compatible sensing spot or probe, optical access, sample-contact assessment and local-position records.
Selection criterion	A useful local oxygen measurement with an understood response time and maintenance burden.

Link the signal to its location

An oxygen reading near a surface may differ from conditions deeper in a sample. A declining value can reflect transfer limitations as well as consumption. The research model should therefore state the sampled volume or location and distinguish observation from inferred bulk behaviour.

This is a proposed monitoring and research channel. It does not independently establish organism identity or antimicrobial susceptibility. If used in fusion, its time constant and measurement interval should accompany the value so that it is not treated as an instantaneous observation synchronized perfectly with a camera exposure.

S23 Electrochemical redox potential

S23 · Chemical observation candidate · Redox environment

Oxidation-reduction potential measures an electrical potential associated with the combined electrochemical environment at a sensing electrode relative to a reference. In complex media, the value can reflect several redox couples and surface processes. It is not generally a selective measurement of one molecule or pathway.

Signal and useful question

The primary quantity is potential in millivolts, with the reference electrode and temperature identified. A candidate use is monitoring whether a defined research environment changes in a way that merits further investigation or affects assay comparability.

Table II.30. S23 Electrochemical redox potential

Qualification element	Proposed treatment
Calibration	Suitable reference solutions, reference-electrode identity and response stability checks.
Confounders	Electrode fouling, dissolved gases, medium chemistry, reference drift and slow stabilization.
Integration burden	Contact electrode, cleanable or disposable interface and high-input-impedance measurement circuitry.
Selection criterion	Reproducible environmental information with a defensible relationship to the selected research question.

Avoid mechanistic overreach

Two samples can have similar redox potentials while differing substantially in composition or biological state. Conversely, an electrode-surface change can produce a shift without a meaningful change in the bulk sample. A model should therefore not translate the potential directly into a metabolic identity or a resistance mechanism.

The proposed evaluation should document baseline drift, response time and effects of the permitted medium range. If the channel is used as a quality indicator, define the condition it is intended to detect and the consequence of a missed or false alarm. Its strongest initial role may be contextual monitoring rather than a primary biological inference.

S24 Capacitive dielectric sensing

S24 · Physical observation candidate · Dielectric response

Capacitive dielectric sensing measures how an electric field interacts with the materials between or near electrodes. The response depends on permittivity, geometry, frequency and losses. Water content and container position can strongly influence a biological sample measurement.

Signal and useful question

Candidate outputs include capacitance in farads at an appropriate scale, conductance and derived dielectric descriptors. Potential uses include detecting presence, monitoring a reproducible physical change or investigating whether a sample’s electrical properties contain useful research information.

Table II.31. S24 Capacitive dielectric sensing

Qualification element	Proposed treatment
Calibration	Reference capacitors or materials, electrode geometry, frequency response and shielding.
Confounders	Humidity, temperature, vessel wall thickness, position, parasitic capacitance and sample conductivity.
Integration burden	Stable electrodes, guarded circuitry, mechanical repeatability and a clear contact or non-contact interface.
Selection criterion	Signal changes that remain interpretable after geometry and environmental effects are controlled.

Separate presence from biology

A reliable detector of a plate or liquid level can be useful even if it provides no biological information. That engineering function should be evaluated independently from a proposed growth or composition inference. Counting the same channel as both a presence check and a biological sensor does not create two independent measurements.

This candidate requires a carefully defined operating geometry. Qualification should determine how much signal change is caused by normal placement variation before attributing a similar change to the sample. Where a derived dielectric feature is used in AI, the original measurement settings and quality state should remain available for review and reanalysis.

II / SENSING AND ENGINEERING

S25 Metal oxide gas sensing

S25 · Chemical observation candidate · Broad gas-response research

Metal-oxide gas sensors change electrical resistance in response to interactions at a heated sensing surface. Their response can depend on several gases, humidity, temperature, exposure history and the heater programme. A broad gas-response pattern is not a species-specific chemical analysis. Manufacturer documentation supports the underlying gas-sensing capability, not a microbial diagnostic claim. [\[R17\]](#) [\[R18\]](#)

Signal and useful question

The primary output is resistance or conductance with associated operating conditions. A derived feature may summarize a response over time. The research question is whether controlled headspace observations add reproducible information after medium, cleaning agents and environmental conditions are accounted for.

Table II.32. S25 Metal oxide gas sensing

Qualification element	Proposed treatment
Calibration	Defined reference exposures, baseline history, environmental compensation and sensor-to-sensor comparison.
Confounders	Humidity, medium volatiles, carryover, cleaning chemicals, ageing and mixed gas responses.
Integration burden	Controlled sampling path, exposure timing, recovery management and separation from electronics heat.
Selection criterion	Transferable information beyond environmental or batch identity, with manageable drift.

Bound the “AI nose” concept

The informal phrase describes pattern analysis of gas responses. It should not imply a measured resistance gene, a unique pathogen identity or a validated AST result. Several sensors exposed to the same uncontrolled headspace can agree for the same wrong reason.

This is a research-only biological inference candidate. A useful development outcome might instead be detection of an instrument contamination event or a changed sampling condition. The intended function should be selected from evidence rather than inferred from the sensor’s ability to produce a complex response curve.

S26 Photoacoustic gas spectroscopy

S26 · Chemical observation candidate · Photoacoustic gas measurement

Photoacoustic spectroscopy converts absorbed, modulated optical energy into a pressure-related acoustic signal. With a suitable wavelength and measurement cell, the response can support gas analysis. Selectivity depends on the optical absorption being addressed, the gas mixture and the instrument model.

Signal and useful question

The primary signal is an acoustic or pressure-related response, converted to a concentration or partial-pressure estimate through calibration. Candidate uses include monitoring a relevant gas in a controlled research environment where the measurement adds more specificity than a broad gas-response sensor.

Table II.33. S26 Photoacoustic gas spectroscopy

Qualification element	Proposed treatment
Calibration	Reference gas mixtures, optical power, pressure, temperature and acoustic-cell response.
Confounders	Overlapping absorption, water vapour, background acoustics, leaks and sampling-path memory.
Integration burden	Modulated source, optical cell, acoustic detection, gas routing and environmental correction.
Selection criterion	Required gas selectivity and sensitivity with a qualified sampling path.

Qualify the sampling chain

A precise sensor cannot correct an unrepresentative gas sample. The record should identify where gas was sampled, the flow or diffusion arrangement and the response interval. A delayed signal should not be assigned to the current plate state without a model of that delay.

This modality is a candidate for specialized monitoring or research instruments. Even a well-qualified gas concentration does not identify the biological source of that gas by itself. A biological interpretation requires an appropriate assay and controls that distinguish sample activity from medium, background and instrument contributions.

S27 Nondispersive infrared carbon dioxide sensing

S27 · Environmental and research candidate · Infrared gas absorption

Nondispersive infrared gas sensing measures attenuation in selected infrared bands associated with a target gas, commonly using a reference channel or other compensation. Carbon dioxide monitoring can provide useful environmental context, but the concentration observed depends on sampling location, ventilation and background sources.

Signal and useful question

The output may be gas concentration in parts per million or volume fraction, with pressure and temperature conditions relevant to the conversion. Candidate questions include whether an instrument environment remains within its declared range and whether a research headspace changes reproducibly.

Table II.34. S27 Nondispersive infrared carbon dioxide sensing

Qualification element	Proposed treatment
Calibration	Suitable gas references, optical baseline, pressure and temperature compensation.
Confounders	Humidity, optical contamination, leaks, human breath, background drift and automatic baseline assumptions.
Integration burden	Representative gas access, response-time characterization and protection of the optical path.
Selection criterion	Stable environmental measurement with a sampling arrangement appropriate to the intended function.

Avoid inappropriate baseline correction

Some compensation strategies assume that the sensor periodically sees a known background condition. That assumption may be invalid in a controlled chamber or specialized research setting. The selected device and software configuration should therefore be assessed together.

Carbon dioxide observations may support environmental quality checks or a bounded research model. They do not independently measure viable count or susceptibility. If a biological interpretation is investigated, the model should account for transport, volume, exchange and non-biological sources rather than treating concentration as a direct growth curve.

S28 Surface acoustic wave mass sensing

S28 · Surface observation candidate · Acoustic-wave research

Surface acoustic wave devices propagate an acoustic disturbance along a material surface. Changes in the surface environment can alter frequency, phase, velocity or attenuation. A functional coating or sample contact can make the response sensitive to mass loading or material properties, but also introduces additional sources of variation.

Signal and useful question

Outputs include frequency shift in hertz, phase and attenuation. Candidate research questions concern whether a controlled surface interaction produces a reproducible signal that is useful for the selected assay or environmental measurement.

Table II.35. S28 Surface acoustic wave mass sensing

Qualification element	Proposed treatment
Calibration	Frequency and phase reference, temperature response, baseline coating and controlled loading.
Confounders	Temperature, humidity, viscosity, coating ageing, nonspecific adsorption and mechanical stress.
Integration burden	Specialized transducer, radio-frequency electronics, surface preparation and a qualified sample interface.
Selection criterion	A stable response with a defensible separation between intended loading and environmental effects.

Treat selectivity as an assay property

A transducer does not become biologically selective merely because it is highly sensitive. If a surface chemistry is used to create selectivity, that chemistry, its stability and its cross-reactions become part of the assay that must be evaluated.

This is a future research candidate with substantial integration uncertainty. It is not proposed as a direct replacement for whole-plate imaging. The first development task would be to establish a reproducible physical response on controlled reference materials, followed by an assessment of whether the intended biological question can be addressed without confounding the signal with surface or fluid properties.

S29 Quartz crystal microbalance sensing

S29 · Surface observation candidate · Resonant mass and viscoelastic response

A quartz crystal microbalance detects changes in a resonator’s behaviour when its surface environment changes. Under restricted thin, rigid loading assumptions, frequency shifts can be related to mass per area. In liquids or soft biological layers, viscoelastic and hydrodynamic effects can make a simple mass conversion inappropriate.

Signal and useful question

The primary outputs are resonant frequency and, where measured, dissipation or related damping quantities. A mass-per-area estimate is derived under an explicit model. Candidate research uses concern controlled surface attachment or material changes rather than weighing an entire culture plate.

Table II.36. S29 Quartz crystal microbalance sensing

Qualification element	Proposed treatment
Calibration	Frequency reference, clean-surface baseline, temperature stability and suitable loading models.
Confounders	Liquid viscosity, density, soft-film behaviour, nonspecific adsorption and surface history.
Integration burden	Resonator interface, stable fluid or contact conditions, surface management and specialized electronics.
Selection criterion	A reproducible response whose model assumptions remain credible in the selected sample format.

Retain the measured response

The record should preserve frequency and damping data even when the analysis reports an inferred mass. A reviewer should be able to see which model was used and whether the data were consistent with its assumptions. An apparent mass change should not be labelled biomass without an independent basis.

The modality is an exploratory research option. Its qualification should first distinguish surface loading from changes in the surrounding fluid. Clinical susceptibility or organism identification would require a separate assay and validation programme, and neither follows from the resonator’s nominal mass sensitivity alone.

S30 Acoustic resonance and attenuation

S30 · Physical observation candidate · Acoustic material response

Acoustic resonance and attenuation measurements examine how sound propagates through or interacts with a material system. The observed response depends on geometry, density, compressibility, viscosity and interfaces. In a plate or fluid assembly, vessel properties can contribute strongly to the signal.

Signal and useful question

Candidate outputs include resonant frequency in hertz, sound velocity in metres per second, attenuation and dimensionless quality factor. Potential research questions include whether a reproducible physical change can be detected and separated from the mechanical properties of the holder and vessel.

Table II.37. S30 Acoustic resonance and attenuation

Qualification element	Proposed treatment
Calibration	Known mechanical or fluid references, geometry, coupling and frequency response.
Confounders	Temperature, bubbles, contact pressure, vessel variation, mounting and background vibration.
Integration burden	Transducers, coupling arrangement, excitation control and stable mechanical interfaces.
Selection criterion	A signal linked to the intended physical change with acceptable sensitivity to placement.

Establish a useful baseline first

An assembly may have many resonances unrelated to the sample. Development should characterize the empty holder and permitted vessels before interpreting changes in a biological context. The analysis should retain the excitation and detection conditions so that a spectral shift can be investigated later.

This is a future research candidate, with no AMReye.AI assay evidence established here. It may prove more useful for checking fill state, mechanical coupling or instrument integrity than for biological inference. Selection should follow the demonstrated function rather than the general sensitivity of acoustic methods.

S31 Resistance temperature measurement

S31 · Environmental monitoring candidate · Resistance thermometry

Resistance temperature measurement uses a material's predictable resistance-temperature relationship. The complete result depends on the sensor, excitation, wiring, reference and conversion model. Four-wire arrangements can reduce lead-resistance effects, but they do not remove self-heating, placement or calibration uncertainty. [\[R20\]](#)

Signal and useful question

The instrument measures resistance in ohms and derives temperature in degrees Celsius or kelvin. Useful questions include whether the relevant observation environment is stable, whether electronics are overheating and whether a process experienced an excursion.

Table II.38. S31 Resistance temperature measurement

Qualification element	Proposed treatment
Calibration	Traceable temperature references across the intended range and complete signal-chain verification.
Confounders	Self-heating, lead mismatch, thermal contact, response lag, gradients and electrical noise.
Integration burden	Suitable placement, wiring, analog measurement and protection from damage or contamination.
Selection criterion	Temperature uncertainty and response time appropriate to the monitored location.

Location defines meaning

A sensor attached to a wall measures a different thermal environment from a sensor near the plate or on a processor. The record should identify location and mounting, and any inference about the sample temperature should be supported by a transfer study.

For Reader V1, temperature monitoring can support instrument characterization and fault detection. Integrated incubation would require a broader spatial and temporal assessment. Multiple agreeing sensors do not demonstrate uniformity if they are all placed in the same unrepresentative region. Qualification should include the conditions and positions that could reveal a meaningful gradient.

S32 Capacitive relative humidity sensing

S32 · Environmental monitoring candidate · Capacitive humidity measurement

A capacitive humidity sensor uses a material whose dielectric response changes with water vapour exposure. The instrument converts that response into relative humidity through calibration and compensation. Relative humidity depends on temperature and describes a local gas condition, not the total water content of a plate.

Signal and useful question

The derived output is relative humidity in percent, accompanied by temperature and sensor quality status. Candidate uses include detecting environmental changes that may affect condensation, evaporation or instrument reliability.

Table II.39. S32 Capacitive relative humidity sensing

Qualification element	Proposed treatment
Calibration	Suitable humidity references, temperature dependence and response-time checks.
Confounders	Condensation, chemical exposure, contamination, hysteresis, placement and slow recovery after saturation.
Integration burden	Representative airflow, protection that does not create excessive lag and a serviceable mounting.
Selection criterion	Stable local measurement under the intended humidity and contamination conditions.

Condensation is a separate event

A high humidity reading does not establish that droplets are present on the optical path, and a local sensor may miss condensation elsewhere. Image-based obstruction checks and environmental monitoring can complement each other, but they answer different questions.

The proposed system should retain sensor saturation or recovery states rather than displaying an apparently precise value throughout a compromised interval. If humidity is used in a biological model, the analysis should account for the sensor’s location and lag. Its most defensible initial role is environmental context and instrument quality assessment, with any growth-related inference evaluated separately.

S33 Pressure measurement

S33 · Environmental and control candidate · Pressure measurement

Pressure sensors convert mechanical deformation or another pressure-dependent effect into an electrical signal. Absolute, gauge and differential measurements use different reference conditions. A value without its reference type can be misleading even when its numerical units are correct.

Signal and useful question

The output is pressure in pascals or a declared multiple. Candidate functions include monitoring a gas path, detecting an obstruction or assessing a pressure difference across a relevant assembly. The selected function determines the required range, response time and reference.

Table II.40. S33 Pressure measurement

Qualification element	Proposed treatment
Calibration	Traceable pressure reference, zero, span, temperature behaviour and line integrity.
Confounders	Leaks, blocked tubing, condensate, offset drift, mounting stress and pressure-port placement.
Integration burden	Compatible ports, tubing, seals, protective interfaces and accessible maintenance.
Selection criterion	Reliable detection of the intended condition with a defined fault response.

Pressure alone does not prove containment

A pressure difference can support an engineering assessment, but it does not independently establish airflow direction everywhere, filter integrity or biological containment. Those claims require the relevant system-level evaluation.

The instrument should distinguish a measured process change from a failed sensor line. Plausibility checks, maintenance records and independent observations can help resolve that difference. If a pressure measurement is required for a control decision, the response to unavailable or implausible data should be specified before the control function is enabled. The initial catalogue role is engineering monitoring rather than a direct biological measurement.

S34 Airflow measurement

S34 · Environmental and control candidate · Flow measurement

Airflow can be measured through several physical approaches, including thermal transfer and pressure-related methods. Each responds to a defined flow path and fluid condition. A local reading should not be assumed to describe circulation throughout an enclosure.

Signal and useful question

Outputs may be volumetric flow in litres per minute, mass flow or local velocity in metres per second. Temperature, pressure and gas composition may be needed to interpret or convert the quantity. Candidate uses include monitoring cooling, detecting a restricted path or characterizing a research gas-sampling arrangement.

Table II.41. S34 Airflow measurement

Qualification element	Proposed treatment
Calibration	Suitable flow reference, gas conditions, path geometry and response across the intended range.
Confounders	Leaks, turbulence, fouling, gas composition, temperature and flow-profile changes.
Integration burden	Defined duct or channel, low unwanted restriction and service access.
Selection criterion	A flow quantity relevant to the function and robust to expected path changes.

Connect flow to the consequence

The product should identify why the measurement matters. Reduced electronics cooling may require throttling or shutdown. A changed research sampling flow may invalidate the time relationship between a plate and a gas observation. Those responses have different priorities and should not share an unexplained generic alarm.

Flow measurement is a proposed engineering channel. Its presence does not establish sterile airflow or safe containment. Qualification should evaluate the complete path and the relevant failure modes, including disconnected tubing or a sensor that reports plausible flow while the intended sampling route is bypassed.

S35 Strain gauge weighing

S35 · Mechanical monitoring candidate · Gravimetric measurement

Strain-gauge weighing measures deformation of a load-bearing element and converts it to force or mass under a calibrated model. A plate assembly includes the vessel, medium, water and any biological material. Changes in those contributions are not automatically separable.

Signal and useful question

The primary electrical response is a bridge signal, with derived load in newtons or mass in grams. Candidate functions include plate presence, gross fill checks, evaporation research and mechanical monitoring. A whole-plate mass change should not be labelled microbial biomass without a separately established relationship.

Table II.42. S35 Strain gauge weighing

Qualification element	Proposed treatment
Calibration	Traceable masses, zero, span, loading position, hysteresis and drift.
Confounders	Evaporation, condensation, handling forces, thermal drift, cable forces and mechanical creep.
Integration burden	Stable support, isolation from motion and a reproducible load path.
Selection criterion	Useful change resolution under actual environmental and mechanical conditions.

Compare signal size with the disturbance

Nominal converter resolution is not the same as usable mass resolution. An instrument may display many decimal places while drift and handling variation dominate the change of interest. The evaluation should report repeatability and stability over the intended observation interval.

The proposed initial role is engineering and research context. Presence detection or evaporation tracking may be useful without making any biological inference. If mass data are included in fusion, preserve taring events and mechanical configuration changes so that an apparent discontinuity is not interpreted as a sudden biological event.

S36 Optical distance and presence sensing

S36 · Instrument-control candidate · Optical ranging and presence

Optical distance sensing estimates range through time-of-flight, triangulation or another declared optical principle. A coarse multizone sensor can support presence, position or clearance checks. Its spatial sampling and ranging uncertainty should not be confused with microscopic three-dimensional imaging. Manufacturer multizone ranging documentation supports an engineering presence use, not colony topography. [\[R19\]](#)

Signal and useful question

Outputs include distance in millimetres, return strength and measurement-validity indicators. Candidate functions include checking that a plate or mechanism occupies a permitted region and detecting gross deviations before capture or motion.

Table II.43. S36 Optical distance and presence sensing

Qualification element	Proposed treatment
Calibration	Known distances, target materials, ambient-light conditions and alignment.
Confounders	Transparent surfaces, multiple reflections, low return signal, contamination and object angle.
Integration burden	Clear optical path, mounting stability and quality-aware control logic.
Selection criterion	Reliable detection of the intended mechanical state under permitted target conditions.

Use validity information

A numerical range can be unreliable when the return signal is weak or when several surfaces contribute. The control interface should preserve validity flags and avoid converting an invalid range into a confident “plate present” state.

This candidate supports instrument readiness and motion checks. If finer shape information is needed for research, a separately qualified optical method should be considered. The proposed evaluation should challenge the sensor with empty holders, permitted vessels, misalignment and reflective or transparent surfaces, then verify that the system responds appropriately when distance is unavailable or ambiguous.

S37 Magnetic position encoding

S37 · Instrument-control candidate · Magnetic position measurement

Magnetic position encoders infer displacement or rotation from a magnetic field and its relationship to a sensing element. They can provide feedback without an exposed optical scale. Their performance depends on magnet geometry, alignment, field conditions and the conversion model.

Signal and useful question

Outputs include angle in degrees or radians, position in length units and status information. Candidate functions include verifying mechanism position, indexing a plate holder or detecting whether a shutter has reached a permitted state.

Table II.44. S37 Magnetic position encoding

Qualification element	Proposed treatment
Calibration	Known positions, magnet alignment, full travel behaviour and repeatability.
Confounders	External magnetic fields, temperature, magnet variation, mechanical backlash and mounting shifts.
Integration burden	Stable magnet-sensor geometry, control interface and independent limit or interlock strategy where required.
Selection criterion	Feedback accuracy and fault detectability appropriate to the controlled mechanism.

Measure the mechanism rather than only the motor

An encoder on a motor shaft may report commanded rotation while a coupling slips or a downstream mechanism remains displaced. The relevant feedback location should follow the physical state that the system needs to know.

This is a proposed control-sensing option, not a biological modality. Qualification should include plausible failures such as misalignment, lost feedback or mechanical play. The software should distinguish estimated, commanded and confirmed positions. A safety-relevant action should not proceed merely because the last remembered position was acceptable before a restart or service intervention.

S38 Inertial vibration measurement

S38 · Instrument-monitoring candidate · Inertial measurement

An accelerometer measures specific force along defined axes; related inertial sensors may measure angular motion. These channels can help characterize vibration, handling shocks and movement during acquisition. Their output depends on mounting, bandwidth, orientation and the influence of gravity.

Signal and useful question

Acceleration is expressed in metres per second squared or in a declared multiple of standard gravity. Frequency-domain descriptors can summarize vibration over an observation interval. The useful question is whether mechanical disturbance is large enough to compromise the selected image or instrument state.

Table II.45. S38 Inertial vibration measurement

Qualification element	Proposed treatment
Calibration	Axis scale, offset, alignment, frequency response and timestamp accuracy.
Confounders	Mounting resonance, temperature, aliasing, gravity projection and electrical noise.
Integration burden	Rigid representative placement, synchronized sampling and signal processing.
Selection criterion	Demonstrated relationship between the monitored disturbance and measurement quality.

Link vibration to an optical consequence

A vibration signal should not trigger an arbitrary rejection threshold without evidence. The qualification should compare inertial observations with image blur, registration error or another relevant outcome. Some measured vibration may be harmless, while an unmonitored local motion may still affect the optics.

This channel is proposed for engineering characterization and readiness checks. It may also help investigate transport damage or recurring faults. The record should preserve the interval and bandwidth used for a summary, because a single “vibration score” can conceal whether a short shock occurred during the actual exposure or at an irrelevant time.

S39 Electrical power monitoring

S39 · Instrument-monitoring candidate · Electrical power measurement

Voltage and current monitoring can reveal supply instability, overload, abnormal consumption or a failing subsystem. Electrical power is derived from the appropriate voltage-current relationship, with waveform and measurement timing considered where relevant. A low reported average can conceal short disturbances that affect capture or storage.

Signal and useful question

Outputs include volts, amperes, watts and integrated energy in joules or watt-hours. Candidate functions include detecting undervoltage, verifying a load budget and relating faults to acquisition or processing activity.

Table II.46. S39 Electrical power monitoring

Qualification element	Proposed treatment
Calibration	Reference voltage and current, sense-path resistance, offset and sampling response.
Confounders	Transients, wiring drops, thermal drift, measurement bandwidth and shared ground effects.
Integration burden	Appropriate sensing location, protection, isolation where needed and event synchronization.
Selection criterion	Detection of disturbances that can affect the actual instrument function.

Preserve the affected operation

A power fault during storage may threaten record completeness, while a disturbance during exposure may alter an image. The system should associate the event with the relevant operation and verify recovery rather than merely clearing an alarm after the voltage returns.

This proposed channel supports reliability and service. It does not measure biological activity. Qualification should include controlled electrical load scenarios, safe shutdown and restart behaviour, and verification that a partial record cannot appear complete. Power monitoring is especially useful when paired with data-integrity checks that reveal whether an electrical event had a lasting consequence.

S40 Reference photodiode monitoring

S40 · Instrument-monitoring candidate · Optical reference measurement

A reference photodiode measures light at a defined location in the illumination path. With a suitable response calibration, it can help distinguish a changing source from a changing sample. Its observation is local and spectrally weighted by the detector response.

Signal and useful question

The primary quantity is photocurrent or converted voltage. Derived optical power or relative intensity requires a stated responsivity and geometry. Candidate uses include monitoring source stability, detecting a failed illumination channel and verifying exposure synchronization.

Table II.47. S40 Reference photodiode monitoring

Qualification element	Proposed treatment
Calibration	Detector responsivity, linearity, dark offset, amplifier gain and reference geometry.
Confounders	Detector ageing, temperature, contamination, stray light and spectral mismatch with the imaging path.
Integration burden	A representative pick-off location, stable optics and synchronized acquisition.
Selection criterion	Reliable detection of illumination changes relevant to image quality.

A source check is not a full-field check

The photodiode may report stable light while a diffuser shifts, a lens becomes contaminated or one part of the field darkens. It should complement, rather than replace, spatial image-quality verification. A reference channel also shares some failure mechanisms with the source it monitors, so independent checks remain useful.

This candidate is particularly relevant to a controlled imaging reader. The proposed evaluation should determine which faults the reference can detect, how quickly it detects them and which faults remain invisible. The result should be a specific quality-control function with an explicit limit, not a general assurance that every captured image is correctly illuminated.

The software system and its responsibilities

The proposed software system should assign clear responsibility for acquisition, physical control, analysis, review and record management. These functions have different failure consequences. A delayed research computation is not equivalent to a failed interlock, and an unavailable cloud service should not silently invalidate a locally completed observation.

Define responsibilities independently of deployment

Physical control should enforce permitted states and limits. Acquisition should preserve original observations and their context. Analysis should produce traceable derived values. PIMS should maintain identity and lineage. The operator interface should expose evidence and permitted actions. Remote services can support fleet management, collaboration and research computation when the data purpose and connection state are explicit.

Table III.1. The software system and its responsibilities

Responsibility	Proposed contract
Control	Authorized commands, confirmed states, fault handling and safe limits.
Capture	Immutable observation, timing, configuration, calibration and quality metadata.
Analysis	Exact input references, method version, output type and uncertainty.
Interpretation	Eligible method context and independently versioned standards rules.
Review	Responsible actor, disposition, edits, reasons and release state.
Remote services	Purpose-limited exchange, acknowledgements, update control and recovery.

BAHU and the shared research services

BAHU is the proposed operator-facing orchestration concept. The broader cloud or “Hive” concept describes shared services for authorized data processing, models and research collaboration. These names should not conceal the functions or imply a fixed number of software layers.

The software should be testable through these contracts even when individual components are replaced. A new model or processor should not change the meaning of a plate identifier or a reviewer correction. Stable semantics are a practical form of modularity: they allow innovation while preserving the evidence chain on which laboratory trust depends.

III / AI SOFTWARE AND DATA

PIMS as the plate record system

PIMS means **Plate Information Management System** in this compendium. Its central responsibility is to maintain the relationship between a physical plate, its observations, the analyses performed on those observations and the person who reviews the result. It should make that relationship inspectable without requiring a reviewer to understand the internal software implementation.

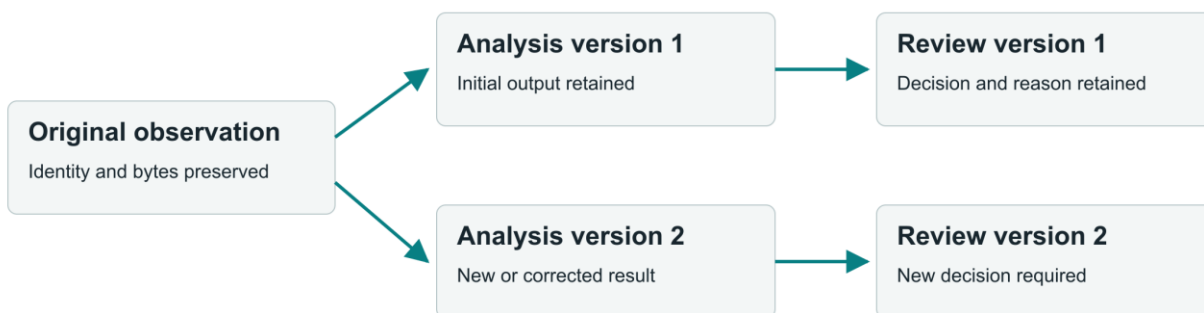
A plate record is more than an image folder

The proposed record begins with an immutable plate identifier and an association to the permitted laboratory case or study identifier. Each capture adds its own time, instrument configuration, calibration reference, image checksum and quality state. A repeated image is another observation of the plate, not a replacement for the original. Where the plate association is uncertain, the system should stop interpretation and request identity resolution.

Preserve the path from observation to decision

Analysis outputs reference their exact inputs. The standards interpretation records the ruleset version separately from the measurement algorithm. The reviewer can inspect the original image, measurement overlay, uncertainty and applicable interpretation context. A correction creates a new linked version with a reason; it does not erase the result that was initially presented.

One original observation can support distinct, preserved analyses



AMReye.AI · conceptual diagram

Figure III.1. Proposed PIMS lineage. A new analysis and review preserve the original evidence and earlier decisions.

PIMS is proposed product functionality. The field dictionaries in the appendices describe interfaces, not an assertion that a complete implementation exists. A minimum useful release should demonstrate identity integrity, reproducible retrieval and a complete correction trail before adding conversational or predictive features. Laboratory connectivity should export reviewed records through a controlled mapping rather than letting a receiving system infer missing context from file names.

Capture and observation interfaces

Capture and sensor interfaces should preserve what the instrument actually observed. Each record needs identity, timing, configuration, units where applicable and a quality state. The schema should make missing information explicit rather than depend on a reader to infer meaning from a filename or a blank cell.

Proposed capture contract

A capture record identifies the plate, the acquisition device, the observation interval and the original image. It references the calibration and optical configuration used. A checksum supports integrity checking, while a storage location identifies where authorized software can retrieve the original bytes. The checksum does not replace access control or establish biological truth.

Table III.2. Capture and observation interfaces

Record element	Required meaning
observation_id	Unique identity of the observation, stable across retrieval and reanalysis.
plate_id	Association to the physical plate or explicitly unresolved identity state.
observed_interval	Actual acquisition start and end, with clock basis and uncertainty when relevant.
configuration_id	Optical, sensor and acquisition settings needed to interpret the observation.
quality_state	Usable, failed, uncertain or another controlled state with reasons.
value_kind	Measured, derived, imputed, simulated or unavailable.

Proposed sensor contract

A sensor observation additionally identifies the channel, physical quantity, unit, sampling or averaging interval and calibration reference. Raw response and converted value should remain linked. A value derived through a compensation model should identify that model rather than appear to be an unprocessed measurement.

Failed acquisition should create an event with an unavailable value and a reason. It should not create a zero-valued measurement. These interfaces are specifications for development; their inclusion here does not claim that a complete instrument or data service already implements them.

III / AI SOFTWARE AND DATA

Calibration and measurement interfaces

Calibration and measurement interfaces should make the conversion from observation to quantity reproducible. The measurement record should identify the calibration that was valid for the acquisition and the analysis method that used it. Replacing a coefficient silently would make the historical result impossible to reconstruct.

Proposed calibration contract

A calibration record includes the instrument and configuration, reference identity, method, validity range, coefficients, residuals and uncertainty basis. It also records who authorized the calibration and which changes invalidate it. A calibration can be superseded without deleting its relationship to older observations.

Table III.3. Calibration and measurement interfaces

Measurement field	Proposed meaning
measurement_id	Unique identity for this derived measurement version.
input_observation_ids	Exact original observations used by the method.
calibration_id	Calibration and configuration relationship applied to the quantity.
quantity and unit	Named physical or analytical quantity with an explicit unit.
value and uncertainty	Estimate and uncertainty representation, including the basis or coverage convention.
geometry	Coordinates, boundary representation and coordinate frame where relevant.
method_version	Algorithm and processing configuration that produced the estimate.

Preserve corrections as new evidence

A reviewer-adjusted boundary produces a new measurement linked to the original estimate and the decision record. The system should preserve whether the change was manual, algorithmic reprocessing or a corrected calibration. Each has a different provenance.

For a zone diameter, a rounded display value should not be the only stored quantity. Preserve sufficient precision for the qualified measurement and interpretation process, while avoiding a display that implies more accuracy than the uncertainty supports. The data model should therefore separate stored numerical representation, reported precision and the uncertainty claim. [\[R16\]](#)

AI and interpretation interfaces

AI outputs and standards interpretations should remain separate records. A learned model estimates a feature or outcome from data; a standards engine applies eligible rules from a controlled source. Combining them into one unversioned label would obscure both the statistical uncertainty and the normative basis of the result.

Proposed AI output contract

An AI record identifies the task, model package, input observations, preprocessing, output type, confidence or uncertainty representation and domain checks. It should distinguish a segmentation, a quality flag, a research prediction and a retrieved explanation. These outputs have different meanings and different permitted uses.

Table III.4. AI and interpretation interfaces

Interpretation field	Proposed meaning
interpretation_id	Unique version of the rule-based interpretation.
measurement_id	Exact measurement and context being interpreted.
standards_family and version	Authoritative ruleset identity, including applicable corrections.
method_context	Assay, organism, antimicrobial and other required eligibility fields.
rule_reference	Specific applicable rule or an explicit reason why no rule applies.
interpretation_state	Eligible result, technical uncertainty, unsupported or unavailable.

Reject invalid combinations

The standards engine should not infer a missing organism or method from a plausible image. It should also avoid choosing whichever ruleset yields a more convenient category. The selected standards policy and its version should be explicit, and changes should create a new interpretation record. [\[R01\]](#) [\[R02\]](#)

A research prediction may be displayed alongside measured evidence only when its status remains clear. It should not overwrite the measured quantity or become a standards interpretation by renaming a field. These interfaces are proposed contracts intended to make such misuse detectable during development and review.

Review correction and export interfaces

Reviewer decisions and exports complete the evidence chain. A decision records what a responsible user concluded after inspecting the available evidence. An export records which version was released, where it was sent and whether the destination accepted it. Neither should be inferred merely from opening a screen or downloading a file.

Proposed reviewer decision contract

The record should identify the reviewer, role, time, input record versions, disposition and reasons for changes or rejection. It should preserve the distinction between accepting a measurement, accepting an interpretation and authorizing release under the laboratory workflow.

Table III.5. Review correction and export interfaces

Disposition	Proposed consequence
Accept	Preserve the reviewed versions and permit the next authorized workflow step.
Correct	Create linked revised values and retain the original evidence and reason.
Reject	Prevent release of the rejected result and record the relevant limitation.
Defer	Keep the record pending with the information or action needed for resolution.
Reacquire	Request a new observation while preserving the failed or insufficient one.

Export the reviewed state

An export package should identify the released record version, schema, destination mapping and integrity information. It should carry the status and correction semantics supported by the receiving workflow. Delivery acknowledgement and clinical-system acceptance may be different events.

If a released record is corrected, the product should issue a linked correction through the agreed interface and track reconciliation. The original export remains part of the audit history. This design makes it possible to investigate what a receiving system or user saw at a particular time, rather than reconstructing history from the latest database values alone.

III / AI SOFTWARE AND DATA

Edge and cloud responsibilities

Edge and cloud responsibilities should follow latency, reliability, privacy and operating needs. Local processing can preserve acquisition and review during a network interruption. Remote services can support authorized collaboration, larger research computations and fleet operations. Neither location automatically guarantees safety or efficiency.

Keep essential local functions explicit

The proposed instrument should retain the functions required to complete or safely interrupt its permitted local workflow. These include physical control, acquisition records, quality checks and access to the qualified local configuration. A remote outage should produce a clear synchronization state rather than a fabricated success or a silent change in analysis behaviour.

Table III.6. Edge and cloud responsibilities

Function	Proposed deployment consideration
Physical control	Local deterministic response and independent limits.
Image acquisition	Local integrity checks and sufficient temporary storage.
Endpoint analysis and review	Qualified local capability where offline operation is required.
Large simulation ensembles	Remote or scheduled computation with reproducible inputs and resource limits.
Model and rules distribution	Signed packages, authorization, staged rollout and rollback.
Cross-site research	Purpose-limited exchange under agreed access and provenance rules.

Synchronization is a workflow

Queued records need unique identities, retry behaviour and conflict handling. A reconnect should not replace a reviewed local result with a different remote analysis silently. The system should preserve both versions and apply the appropriate review policy.

Resource selection should be based on measured workload, power, storage and service needs. The newest processor or largest model is not automatically the best choice. A smaller qualified local model may be preferable for a time-sensitive measurement, while a larger research model can run separately without changing the clinical workflow.

Data collection and annotation strategy

Data collection should be designed around the intended evaluation, not only the convenience of acquisition. A large number of near-duplicate images from a few plates can provide less useful evidence than a smaller, well-characterized set spanning the relevant conditions. The data model should preserve the relationships needed to recognize that difference.

Define the unit of independence

Images can be nested within observations, plates, isolates, patients, batches and sites. Training and test partitions should prevent related examples from crossing the boundary in a way that inflates apparent performance. The appropriate separation depends on the claim: robustness to a new image is different from transfer to a new laboratory.

Table III.7. Data collection and annotation strategy

Data element	Why it is needed
Acquisition context	Identifies camera, optics, lighting, configuration and time.
Assay context	Supports eligibility and assessment of domain coverage.
Annotation provenance	Identifies who labelled the data, under what instructions and with what uncertainty.
Relationship identifiers	Enable plate, isolate, patient, batch and site separation where applicable.
Failure and exclusion records	Prevent evaluation from silently discarding difficult cases.

Collect disagreement deliberately

Annotations should allow uncertainty and disagreement rather than forcing every case into a falsely precise label. Independent review and adjudication can help establish the reference for a specific task. Original annotations should remain available so that the effect of adjudication can be assessed.

The collection programme should also record why data were acquired and what reuse is permitted. Data suitable for service troubleshooting are not automatically authorized for model training. Representative data, appropriate reference labels and a controlled evaluation design are central to medical AI development principles. [\[R21\]](#) The proposed implementation should make those requirements part of routine data operations.

Genomic data access and provenance

Genomic and scientific repositories can provide valuable evidence, but they differ in curation, scope, licensing, update practices and uncertainty. The proposed platform should use selected resources through lawful, purpose-defined access and preserve the versions used in each analysis. It should not promise to ingest every repository or treat all deposited records as equally reliable.

Separate sequence evidence from image inference

A measured sequence can be analysed for features supported by a qualified pipeline and database. An image-based prediction of a genotype is a different research task. The interface should identify whether a molecular feature was measured, computationally detected from sequence data, inferred from another phenotype or simply retrieved from literature.

Table III.8. Genomic data access and provenance

Resource type	Proposed handling
Curated resistance knowledge	Preserve ontology, evidence model, database release and applicable use terms.
Sequence repositories	Retain accession, version, sample context and quality limitations.
Reference genomes	Identify assembly and annotation versions and avoid treating one reference as universal.
Literature	Link to the original study and retain the supported claim and limitations.
Partner datasets	Apply the agreed purpose, access, retention and publication conditions.

CARD and NCBI Pathogen Detection illustrate complementary curated and sequence-based resources. Their documentation should guide interpretation of their outputs; database access does not establish clinical validity for a new product workflow. [\[R24\]](#) [\[R25\]](#)

The proposed service should record query parameters, tool versions and the exact evidence used. When a database changes, a new analysis should remain distinguishable from the earlier result. A disagreement between phenotype and genotype should trigger investigation of method, context and uncertainty rather than an automatic assumption that one source must be correct.

Using the AI capability catalogue

The forty capabilities that follow are candidates for task-specific AI across the platform. They span acquisition, measurement, quality, modelling, bioinformatics, workflow and service. Some may prove useful with conventional algorithms alone. The catalogue treats that outcome as a valid engineering decision.

Evaluate the complete task

Each entry identifies inputs, outputs, why learning might help, a conventional baseline, data needs, failure modes and validation. A capability is useful only if it improves the intended decision or process under realistic conditions. A high model score on a convenient dataset is insufficient.

Table III.9. Using the AI capability catalogue

Capability range	Principal function
A01 to A15	Capture, geometry, measurement and image-quality assistance.
A16 to A20	Temporal analysis and early-response or MIC research.
A21 to A28	Fusion, uncertainty, missingness and dataset quality.
A29 to A34	Bioinformatics, evidence retrieval and rule-consistency support.
A35 to A40	Workflow, maintenance, fleet and operator assistance.

Shared evidence requirements

Inputs should be representative of the intended use, evaluation partitions should respect data relationships and performance should include failed or abstained cases. Confidence needs empirical calibration. Changes in acquisition, population or workflow can alter performance and should be monitored. [\[R21\]](#) [\[R23\]](#)

The catalogue does not prescribe model generations. Candidate families include geometric methods, convolutional or transformer-based vision models, probabilistic models, sequence methods and task-specific language systems. Custom development should be driven by evidence and efficiency requirements. A model that cannot identify when its inputs are unsuitable should not be trusted merely because it performs well on routine examples.

III / AI SOFTWARE AND DATA

A01 Capture readiness assessment

A01 · Proposed acquisition assistance

Capture readiness assessment asks whether the next observation is likely to be usable and correctly associated. AI may recognize combinations of visual conditions that are difficult to express as a small set of thresholds. It should complement explicit checks for identity, calibration and hardware state, not override them.

Table III.10. A01 Capture readiness assessment

Capability element	Proposed definition
Inputs	Preview image, plate association, configuration, calibration validity and instrument status.
Outputs	Specific readiness findings, affected region and a recommended permitted action.
Conventional baseline	Deterministic identity, presence, calibration and exposure checks.
Why AI may help	Recognition of variable visual obstructions or placement patterns that recur across captures.
Data requirements	Representative ready and not-ready examples with independently labelled causes and acquisition context.
Failure modes	Learning site backgrounds, overlooking unfamiliar obstructions or treating a plausible image as proof of identity.
Validation	Cause-specific detection, false rejection, usable capture yield and recovery performance on held-out sessions and sites.

Keep hard prerequisites explicit

An expired calibration or unresolved identifier is not a visual classification problem. Such conditions should prevent the relevant workflow step regardless of the model's confidence. Conversely, a visual warning should explain what the operator needs to inspect rather than simply displaying a red readiness score.

The qualification should compare the complete acquisition workflow with the deterministic baseline. A useful model should reduce avoidable failed captures without creating excessive operator interruptions. Cases that remain uncertain should be retained for review and future analysis, including whether the warning was justified after a successful reacquisition.

A02 Exposure and illumination selection

A02 · Proposed acquisition optimization

Exposure and illumination selection can adapt acquisition to the appearance of a permitted plate while keeping the measurement pipeline within a qualified range. AI may predict which allowed setting will produce the most useful evidence, especially when global brightness is a poor guide to the feature of interest.

Table III.11. A02 Exposure and illumination selection

Capability element	Proposed definition
Inputs	Preview images, permitted lighting modes, exposure limits and the intended measurement task.
Outputs	A selected qualified setting and the reason or expected quality improvement.
Conventional baseline	Fixed settings or histogram-based exposure control with saturation limits.
Why AI may help	Task-aware selection where local glare, medium appearance and boundary contrast interact.
Data requirements	Paired observations across allowed settings, with independent measurement-quality assessment.
Failure modes	Optimizing visual appeal, hiding weak growth, selecting an unqualified mode or creating inconsistent time-series appearance.
Validation	Measurement quality, saturation, repeatability and setting-selection stability against fixed-setting baselines.

Optimize the evidence rather than the picture

The objective should be a useful, reproducible measurement image, not maximum contrast or aesthetic enhancement. A setting that makes one boundary conspicuous may suppress another relevant feature. Evaluation should therefore assess the complete permitted image domain.

The instrument should record the selected settings and preserve limits enforced independently of the model. If the model cannot find an eligible setting, it should request review or declare acquisition unsuccessful. For time-series research, any setting change should remain explicit so that intensity differences are not mistaken for biological dynamics.

A03 Focus quality assessment

A03 · Proposed image-quality assessment

Focus assessment determines whether the acquired optical detail is sufficient for a specified task. A model may distinguish genuinely informative edges from texture, noise or printed labels that can make a simple sharpness score appear favourable. The required focus quality depends on the feature being measured.

Table III.12. A03 Focus quality assessment

Capability element	Proposed definition
Inputs	Image regions, optical configuration, expected feature scale and optional reference-target observations.
Outputs	Local focus-quality map, task suitability and regions requiring attention.
Conventional baseline	Gradient, Laplacian or frequency-based sharpness measures on selected regions.
Why AI may help	Separating useful biological detail from high-contrast artifacts and variable medium appearance.
Data requirements	Controlled focus variation, realistic plate examples and independent assessments of measurement usability.
Failure modes	Scoring labels instead of the measurement plane, confusing noise with detail or missing spatially uneven focus.
Validation	Task-level measurement error across focus conditions, local defect detection and false rejection.

Evaluate the relevant plane

A sharp lid inscription does not establish that the agar surface is in focus. The assessment should therefore examine the regions and plane that support the intended measurement. A global score may hide a blurred edge at the periphery of the field.

The model should be compared with conventional metrics under the same acquisition conditions. Its output should identify the affected region and support a permitted recovery action. If focus adjustment changes the qualified optical configuration, the system should apply the relevant calibration rules rather than assume that a sharper image preserves the previous measurement scale.

A04 Glare and condensation detection

A04 · Proposed artifact assessment

Glare and condensation detection identifies optical conditions that obscure or distort useful evidence. AI may recognize varied droplet patterns, reflections and local veiling that are difficult to capture with a single intensity threshold. The output should describe the observation problem, not promise that another spectral mode can always repair it.

Table III.13. A04 Glare and condensation detection

Capability element	Proposed definition
Inputs	Original image, optional paired illumination modes and permitted vessel context.
Outputs	Artifact type or uncertainty, spatial mask and affected measurement regions.
Conventional baseline	Saturation masks, local contrast checks and paired-image difference rules.
Why AI may help	Recognition of diverse artifact shapes and interactions with relevant boundaries.
Data requirements	Representative vessels, media, lighting conditions and independently labelled artifact effects.
Failure modes	Confusing biological texture with droplets, missing transparent distortion or relying on one lid material.
Validation	Region-level detection and the effect on measurement eligibility, including novel artifact conditions.

Assess consequence as well as presence

An artifact outside the measurement region may be harmless, while a small obstruction at a critical edge can invalidate the result. The system should connect the artifact finding to the affected task rather than rejecting every imperfect image indiscriminately.

Evaluation should include cases in which the model is uncertain and cases where a different imaging mode improves appearance without restoring a valid measurement. Original observations remain available for review. Any reacquisition creates a new record, allowing investigators to assess whether the proposed recovery action actually resolved the limitation.

A05 Plate pose estimation

A05 · Proposed geometric estimation

Plate pose estimation determines the position and orientation needed to relate an image to the measurement geometry. A learned model may help locate partially obscured boundaries or variable vessel shapes. The resulting transform must still be checked against the qualified optical model.

Table III.14. A05 Plate pose estimation

Capability element	Proposed definition
Inputs	Plate image, permitted geometry and optional fiducial observations.
Outputs	Position, rotation, tilt-related indicators, transform and uncertainty.
Conventional baseline	Circle or ellipse fitting, edge detection and calibrated fiducial geometry.
Why AI may help	Robust feature localization when labels, reflections or partial occlusion challenge simple fitting.
Data requirements	Known poses across the permitted range, vessel variation and independent geometric references.
Failure modes	Fitting the lid instead of the measurement plane, accepting unsupported vessel shapes or overcorrecting distorted images.
Validation	Pose residuals, downstream length error and detection of out-of-range configurations.

Geometry needs an independent reference

A visually centred plate may still be tilted or displaced in height. The model should not infer a precise three-dimensional state from an image that does not contain enough information. Where pose is not identifiable, the output should express that limitation.

Any image transform used for measurement should preserve the relationship to original coordinates. The reviewer should be able to inspect the source image and the transformation applied. Qualification should include loading and positioning variations that are independent of the images used to train the estimator, with acceptance based on their effect on the actual measurement.

A06 Plate and vessel identification

A06 · Proposed vessel classification

Plate and vessel identification determines whether the observed container belongs to a permitted format and which configuration rules apply. AI may assist with visual classification when external appearance varies, but it should not infer missing assay identity merely from the vessel’s shape or colour.

Table III.15. A06 Plate and vessel identification

Capability element	Proposed definition
Inputs	Overview image, declared vessel metadata and permitted format catalogue.
Outputs	Candidate vessel class, consistency findings and unsupported-format state.
Conventional baseline	Barcode metadata, dimensions and explicit operator selection checked against geometric rules.
Why AI may help	Recognition of unexpected formats, damaged vessels or discrepancies between metadata and appearance.
Data requirements	Representative permitted and non-permitted vessels, manufacturing variation and realistic labels.
Failure modes	Treating similar geometry as equivalent, relying on supplier-specific markings or confusing lid and base dimensions.
Validation	Format classification, unknown-format rejection and effects of incorrect classification on measurement eligibility.

Classification is not assay confirmation

The same container can hold different media or support different workflows. The software should therefore use visual classification as a consistency check rather than an authoritative source for all assay fields.

A useful evaluation includes unfamiliar but visually similar vessels and cases in which the declared metadata are intentionally inconsistent. The system should request resolution when the conflict matters to calibration or interpretation. It should preserve the operator’s decision and the observed evidence so that recurring supplier or workflow differences can be investigated without silently broadening the permitted domain.

A07 Barcode and label association

A07 · Proposed identity-association assistance

Barcode and label association connects a visible identifier with the correct plate record. Conventional decoding and explicit reconciliation should remain the primary path. AI may help locate a label, distinguish overlapping text or flag a mismatch, but it should never invent a missing identifier.

Table III.16. A07 Barcode and label association

Capability element	Proposed definition
Inputs	Label image, decoded candidate values and authorized order or study context.
Outputs	Located label regions, candidate associations and conflicts requiring resolution.
Conventional baseline	Standard barcode decoding, check digits and exact identifier matching.
Why AI may help	Locating damaged, rotated or partially obscured labels and detecting inconsistent visible text.
Data requirements	Diverse label formats and difficult cases with independently verified associations.
Failure modes	Plausible transcription of an unreadable label, matching to a nearby case or hiding a duplicate identifier.
Validation	Association error, unresolved-case handling and performance on damaged or unfamiliar labels.

Identity errors deserve a separate metric

A correctly measured zone attached to the wrong case is not a successful result. Evaluation should therefore measure association integrity independently of image-analysis accuracy. A high barcode decoding rate is insufficient if the workflow can attach a valid code to the wrong physical object.

Uncertain associations should remain unresolved until an authorized user or qualified process supplies the missing evidence. The record should retain the original label image, decoded values and decision history. AI assistance is justified only if it improves this process without converting uncertainty into a confident but unsupported match.

A08 Antimicrobial disk localization

A08 · Proposed object localization

Antimicrobial disk localization identifies candidate disk centres and boundaries within an eligible plate image. Learned object detectors may handle appearance variation, partial occlusion and clutter better than a purely geometric method. Their detections still need consistency checks and a relationship to the permitted assay layout.

Table III.17. A08 Antimicrobial disk localization

Capability element	Proposed definition
Inputs	Original plate image, spatial calibration and permitted disk or layout context.
Outputs	Candidate disk locations, extents, confidence and unresolved detections.
Conventional baseline	Circular-feature detection, template matching and geometric consistency rules.
Why AI may help	Handling varied printing, illumination, overlapping features and nonuniform backgrounds.
Data requirements	Independently annotated disks across sites, suppliers, plate appearances and difficult cases.
Failure modes	Treating labels, bubbles or colonies as disks, missing low-contrast disks or merging nearby objects.
Validation	Localization error, missed and false detections, and downstream effects on zone measurement.

Evaluate the complete disk map

Per-object detection scores can conceal a plate-level error that matters to the workflow. The evaluation should therefore assess whether the complete disk map is correct and whether uncertain objects are flagged before interpretation.

The software should preserve detection geometry and link any reviewer correction to the original output. A detector should not determine antimicrobial identity from position unless the layout is independently established and validated for that workflow. Object localization supplies a geometric input; it does not by itself establish the drug, disk content or eligible interpretation rule.

A09 Disk label recognition

A09 · Proposed text recognition assistance

Disk label recognition attempts to read visible markings and associate them with the correct antimicrobial metadata. AI may assist when characters are small, rotated or partially obscured. The consequence of a plausible but incorrect reading can be substantial, so the system should favour unresolved status over guessing.

Table III.18. A09 Disk label recognition

Capability element	Proposed definition
Inputs	Disk-region images, permitted terminology and independently supplied panel or layout information.
Outputs	Candidate text, recognition uncertainty and consistency findings.
Conventional baseline	Optical character recognition with controlled dictionaries and exact metadata checks.
Why AI may help	Recognition under variable printing, orientation and limited image detail.
Data requirements	Label examples spanning suppliers, abbreviations, quality conditions and confusable characters.
Failure modes	Hallucinating characters, choosing a common drug name or confusing concentration markings.
Validation	Exact-field accuracy, confusable-label errors, abstention and downstream interpretation integrity.

Use dictionaries carefully

A restricted vocabulary can reduce impossible outputs, but it can also force an incorrect valid label when the true label is absent. Unknown and unreadable states should therefore remain available. The model should not use a likely susceptibility pattern to decide which drug label it thinks it sees.

The preferred workflow combines visible evidence with independently established assay metadata and review. Corrections should preserve both the original recognition output and the final association. Qualification should examine rare but consequential confusions, rather than reporting only an average character accuracy that hides a wrong antimicrobial identity.

A10 Zone boundary segmentation

A10 · Proposed boundary estimation

Zone boundary segmentation estimates the image region corresponding to a method-defined inhibition boundary. Learned segmentation can model appearance variation and ambiguous local texture, but it must be evaluated against the actual reading definition. A smooth or visually attractive contour is not necessarily the correct contour.

Table III.19. A10 Zone boundary segmentation

Capability element	Proposed definition
Inputs	Eligible image, disk geometry, calibration and relevant reading context.
Outputs	Boundary representation, calibrated measurement inputs and local uncertainty.
Conventional baseline	Radial intensity profiles, edge detection, thresholding and geometric fitting.
Why AI may help	Complex edge appearance, uneven backgrounds and contextual separation of boundary from artifacts.
Data requirements	Independent expert annotations, disagreement records and representative difficult boundaries.
Failure modes	Smoothing away meaningful irregularity, following glare, ignoring faint growth or importing training annotation bias.
Validation	Diameter error, spatial residuals, agreement, readability coverage and performance near consequential interpretation boundaries.

Preserve the boundary definition

Different biological appearances may require different reading rules. The model should not silently standardize every edge to the same visual pattern. Method and organism context must remain available to the qualified workflow. [\[R03\]](#)

Evaluation should include comparison with conventional algorithms and independent human measurements. Where readers disagree, the reference process and uncertainty should be documented. The product should preserve the original image, estimated boundary and any reviewer adjustment so that an error can be traced to acquisition, segmentation, calibration or interpretation rather than attributed vaguely to “the AI.”

A11 Spatial measurement uncertainty

A11 · Proposed uncertainty estimation

Spatial measurement uncertainty describes how precisely a physical quantity can be supported by the observation and measurement chain. AI may estimate local ambiguity or predict when an image is likely to produce a large measurement error. It should complement, not replace, a physically grounded calibration uncertainty budget.

Table III.20. A11 Spatial measurement uncertainty

Capability element	Proposed definition
Inputs	Image, boundary candidates, spatial calibration, configuration and repeatability evidence.
Outputs	Uncertainty components, interval or distribution, and the basis of the estimate.
Conventional baseline	Propagation of calibration and localization uncertainty with repeated-measurement analysis.
Why AI may help	Learning how complex edge appearance relates to measurement variability.
Data requirements	Independent reference measurements, repeated observations and difficult cases with documented reader disagreement.
Failure modes	Confidently extrapolating beyond the data, omitting systematic bias or treating model confidence as physical uncertainty.
Validation	Empirical interval coverage, interval width, bias and performance across image-quality and site strata.

Combine evidence without double counting

If the model’s uncertainty estimate already reflects a calibration effect, adding the same effect again can overstate uncertainty. Omitting it can understate uncertainty. The proposed budget should therefore identify the source and dependence of each component. [\[R16\]](#)

A useful uncertainty output should support a decision: whether a measurement is suitable, whether more evidence is needed and how a reviewer should interpret a near-boundary result. Reporting a narrow interval is not an objective in itself. The interval must behave as claimed on independent data and remain meaningful under the declared operating conditions.

A12 Lawn quality assessment

A12 · Proposed plate-quality assessment

Lawn quality assessment identifies visible patterns that may make a disk diffusion observation unsuitable or difficult to interpret. AI may recognize irregular coverage, strong spatial variation or other appearance problems. The output should be a quality finding tied to the image, not an unsupported diagnosis of the cause.

Table III.21. A12 Lawn quality assessment

Capability element	Proposed definition
Inputs	Whole-plate image, local texture, assay context and optional paired observations.
Outputs	Quality regions, observed pattern description and measurement-eligibility recommendation.
Conventional baseline	Spatial intensity and texture uniformity measures with explicit review rules.
Why AI may help	Recognition of complex patterns that vary with medium and organism appearance.
Data requirements	Representative quality variation and independent expert labels with reasons and uncertainty.
Failure modes	Rejecting legitimate morphology, missing localized defects or learning one laboratory's preparation style.
Validation	Quality-defect detection, false rejection and the effect on downstream measurement and review.

Describe the observation before its cause

An uneven appearance may have several explanations. The interface should identify what is visible and why it affects the task. It should avoid claiming that an image proves a specific preparation error when that conclusion requires additional records.

The qualification should include the permitted organism and medium range and preserve cases on which experts disagree. A model that correctly flags obvious failures but rejects many useful plates may add workload without improving the laboratory result. Evaluate usable coverage and consequential missed defects together, with the laboratory's established alternative workflow available for unresolved cases. [\[R03\]](#)

A13 Mixed morphology detection

A13 · Proposed morphology review assistance

Mixed morphology detection flags visually distinct growth patterns that may deserve expert review. AI can assist with locating differences in colour, texture, size or spatial organization. A visual difference is not sufficient to confirm a mixed culture, contamination or species identity.

Table III.22. A13 Mixed morphology detection

Capability element	Proposed definition
Inputs	Plate or colony-region images, acquisition context and optional temporal observations.
Outputs	Candidate regions with differing appearance and an explicit review flag.
Conventional baseline	Colour and texture clustering, size distributions and manual comparison.
Why AI may help	Combining several appearance features and recognizing context-dependent differences.
Data requirements	Independently annotated regions with relevant confirmation data where a biological conclusion is studied.
Failure modes	Confusing illumination or medium variation with morphology, or missing subtle differences between biological populations.
Validation	Region detection and review usefulness; biological conclusions evaluated separately against appropriate confirmation.

Keep visual and biological labels separate

The dataset should distinguish “different appearance observed” from “mixed culture confirmed.” Combining those labels would teach a model to make a stronger claim than the image supports. The same caution applies to colony size variation that can arise within one population.

The initial proposed function is triage for review. It should direct attention to the relevant original evidence and preserve the reviewer’s disposition. Evaluation should ask whether the flag improves detection of important cases without creating excessive false alarms. Any later biological classification would require a separately defined task, representative reference data and an explicit intended-use boundary.

A14 Within zone colony flagging

A14 · Proposed within-zone observation assistance

Within-zone colony flagging locates candidate growth-like features inside an inhibition region. AI may help detect small or visually subtle objects and preserve their location for review. The feature can have several explanations, including artifacts, mixed populations or biological phenomena that require confirmation.

Table III.23. A14 Within zone colony flagging

Capability element	Proposed definition
Inputs	Eligible high-detail image, zone geometry and optional repeated observations.
Outputs	Candidate feature locations, appearance evidence, size estimates where calibrated and uncertainty.
Conventional baseline	Local contrast detection, connected components and expert inspection.
Why AI may help	Distinguishing variable small features from background structure across image conditions.
Data requirements	Region annotations, artifact examples, temporal correspondence and separately recorded confirmation results.
Failure modes	Missing low-contrast growth, labelling dust or droplets as colonies, or equating a visual feature with heteroresistance.
Validation	Object-level detection, plate-level review burden and biological confirmation performance as separate outcomes.

A flag is a prompt to investigate

Heteroresistance research provides reasons to take population heterogeneity seriously, but an image flag does not establish a resistant subpopulation. The dedicated application chapter develops the alternative explanations and confirmation boundary. [\[R07\]](#) [\[R08\]](#)

The proposed workflow should preserve both flagged and unflagged original images so that missed features can be investigated. If the image resolution cannot support the requested observation, the system should say so. Upscaling or generating a sharper-looking feature must not be used as evidence that a colony was actually observed.

A15 Readability and abstention

A15 · Proposed task-level readability decision

Readability assessment combines relevant quality findings to decide whether the available evidence supports a particular measurement. It is broader than detecting blur or glare alone. AI may help model interactions among defects, but the final eligibility logic should remain explicit and reviewable.

Table III.24. A15 Readability and abstention

Capability element	Proposed definition
Inputs	Image, task, calibration state, artifact findings, boundary uncertainty and required metadata.
Outputs	Readable, unreadable or uncertain state with specific reasons and permitted next actions.
Conventional baseline	Prespecified quality rules and expert review.
Why AI may help	Combining weak quality indicators whose joint effect predicts measurement failure.
Data requirements	Independent task-usability labels, including difficult and rejected observations.
Failure modes	Hiding errors by abstaining on inconvenient cases or accepting familiar-looking but unsupported images.
Validation	Error among accepted cases, coverage, failure detection and review workload, reported together.

Report the denominator

In an illustrative set of 100 observations, accepting 90 and abstaining on 10 gives 90 percent coverage. Accuracy among the accepted observations does not describe the full set by itself. This example is a reporting illustration, not AMReye.AI performance.

The proposed evaluation should prespecify how abstentions enter the analysis and what happens operationally to an unreadable case. A useful system should reduce consequential accepted errors while keeping the fallback workload manageable. It should not optimize a headline agreement rate by excluding precisely the cases that matter most to laboratory users.

A16 Temporal image registration

A16 · Proposed temporal geometry alignment

Temporal image registration aligns repeated observations so that apparent movement can be distinguished from a change in the object. AI may assist where conventional feature matching struggles with changing biological appearance. Registration must preserve the physical meaning of the measurement rather than warp every image into superficial agreement.

Table III.25. A16 Temporal image registration

Capability element	Proposed definition
Inputs	Original image series, timestamps, configuration and stable reference features.
Outputs	Transforms, alignment residuals, valid regions and registration uncertainty.
Conventional baseline	Fiducial alignment, phase correlation and constrained geometric feature matching.
Why AI may help	Robust matching under appearance change, partial obstruction or weak stable features.
Data requirements	Known or independently assessed transformations and realistic temporal changes.
Failure modes	Removing real growth through excessive warping, matching different regions or masking a changed optical configuration.
Validation	Landmark residuals, measurement preservation and detection of unregistrable intervals.

Retain original coordinates

Every registered feature should remain traceable to the original observation. A researcher should be able to inspect the transform and determine whether a reported growth displacement exceeds the registration uncertainty.

Rigid or constrained transforms may be preferable when they match the physical motion of the plate. More flexible models should be justified by the acquisition geometry and evaluated for distortion of biological features. A failed registration should not be repaired by silently interpolating or generating an image. The temporal model can handle an explicit gap more honestly than a convincing but unsupported alignment.

A17 Colony tracking

A17 · Proposed object tracking research

Colony tracking links candidate objects across time to study persistence, growth and spatial relationships. AI may combine appearance and motion information when objects change shape or become difficult to separate. The tracking identity is a computational hypothesis, not a permanent biological identity established by the image alone.

Table III.26. A17 Colony tracking

Capability element	Proposed definition
Inputs	Registered observations, candidate colony regions, timestamps and quality states.
Outputs	Track associations, split or merge events, confidence and unresolved links.
Conventional baseline	Nearest-neighbour association with geometric constraints and manual review.
Why AI may help	Handling appearance changes, temporary occlusion and crowded regions.
Data requirements	Independently reviewed tracks with explicit ambiguous, merged and lost-object states.
Failure modes	Identity swaps, false splits, missed merges and treating disappearance as biological death.
Validation	Association accuracy, event detection, track completeness and effects on derived growth features.

Preserve ambiguous events

When two regions merge visually, the software may no longer be able to attribute subsequent area to the original objects. It should mark that limitation rather than fabricate two precise tracks. A missing observation should likewise remain distinguishable from a true disappearance.

The proposed initial role is research analysis. Evaluation should focus on the quantities investigators intend to derive, such as growth trajectories or spatial interaction measures. A tracker with visually smooth paths can still bias those quantities. Original segmentations, association decisions and manual corrections should therefore remain available for reproducibility and error analysis.

A18 Growth curve estimation

A18 · Proposed temporal feature estimation

Growth-curve estimation fits a temporal representation to observed features such as area, intensity or another qualified signal. AI may model nonlinear relationships or combine observations with different quality levels. The fitted curve should identify what quantity it describes; an image-area trajectory is not automatically a viable-cell growth curve.

Table III.27. A18 Growth curve estimation

Capability element	Proposed definition
Inputs	Time-stamped measured features, observation uncertainty, quality states and assay context.
Outputs	Fitted trajectory, parameters, residuals and uncertainty over the observed interval.
Conventional baseline	Prespecified parametric growth curves, smoothing or state-space estimation.
Why AI may help	Complex temporal patterns and interactions among multiple qualified features.
Data requirements	Representative time series, independent reference measurements where biological quantities are inferred and realistic missingness.
Failure modes	Confusing exposure changes with growth, overfitting sparse data or reporting parameters that are not identifiable.
Validation	Fit residuals, parameter recovery where assessable and performance on independent trajectories.

Distinguish fitting from forecasting

A curve fitted through existing observations summarizes those observations. Extending it into the future is a forecast that needs additional evaluation. The software should keep the observed interval and forecast interval visually and numerically distinct.

The research record should preserve units, preprocessing and model assumptions. If a parameter such as a rate cannot be estimated reliably from the available data, it should remain uncertain or unavailable. A smooth curve should not create the impression that a sparse or interrupted observation series contains more biological evidence than was actually acquired.

A19 Early response research classification

A19 · Research target for selected early-response workflows

Early-response classification investigates whether a defined observation window can support a reliable susceptibility-related conclusion in a selected assay. AI may combine temporal and morphological evidence, but the biological response must be present and measurable. Computation cannot recover a response that has not yet become informative.

Table III.28. A19 Early response research classification

Capability element	Proposed definition
Inputs	Qualified early observations, declared assay clock, organism and drug context, and quality records.
Outputs	Research classification or uncertainty, observation window and eligibility state.
Conventional baseline	Prespecified feature rules or a simple time-dependent classifier.
Why AI may help	Combining several early features whose relationships vary within a defined assay domain.
Data requirements	Paired early observations and appropriate reference results, with site and isolate separation.
Failure modes	Reporting only easy cases, confusing specimen-to-result with assay time or extrapolating across unsupported drugs.
Validation	Consequential errors, coverage, uncertainty and time-to-eligible-result within the prespecified scope.

Define the sub-two-hour target

The target should state whether the clock begins at sample loading, another assay event or a different documented origin. The specimen context, organism-drug combinations, comparator and success criteria must be fixed before evaluation. Upstream preparation time should be reported separately when it is outside the target clock.

Published single-cell and electrical studies demonstrate that short windows are scientifically investigable in particular methods. They do not establish a universal limit or AMReye.AI performance. [\[R11\]](#) [\[R38\]](#) Endpoint Reader V1 and this research track should therefore retain separate eligibility rules and claims.

A20 MIC estimation research

A20 · Research target for quantitative inference

MIC estimation research asks whether a defined set of observations can support a useful prediction of a method-specific MIC. The output is a prediction, not a direct broth microdilution measurement. A disk-zone diameter does not contain a universally invertible mapping to a precise MIC across organisms, drugs and conditions.

Table III.29. A20 MIC estimation research

Capability element	Proposed definition
Inputs	Qualified phenotypic observations, assay context and paired reference MIC data.
Outputs	Predicted MIC range or distribution, units, uncertainty and domain eligibility.
Conventional baseline	Simple calibrated regression or ordinal models within a narrowly defined organism-drug domain.
Why AI may help	Modelling nonlinear relationships and combining additional validated phenotypic features.
Data requirements	Representative paired reference measurements, repeated results and independent site or isolate partitions.
Failure modes	False numerical precision, leakage from reference labels, poor transfer and confusion of categorical with quantitative agreement.
Validation	Appropriate quantitative agreement, bias, uncertainty calibration and clinically consequential interpretation errors.

Preserve the reference method

Reference MIC data should retain method, dilution structure, off-scale results and quality context. A model should not turn an interval-censored or off-scale reference into an invented exact value. Units such as milligrams per litre must remain explicit.

The proposed output should be visibly labelled predicted and should not overwrite a measured result. Any future clinical use would require a bounded intended use and its own validation. Direct MIC testing through a qualified assay remains a different product function, even when both outputs can be stored in the same PIMS environment.

A21 Multimodal representation learning

A21 · Proposed multimodal research representation

Multimodal representation learning combines information from different observation channels into features useful for a defined task. It can model relationships that are difficult to express manually. Its value depends on whether the channels add relevant information and whether their timing, calibration and missingness are understood.

Table III.30. A21 Multimodal representation learning

Capability element	Proposed definition
Inputs	Qualified images and sensor observations with units, intervals, calibration and quality states.
Outputs	Task-specific representation, contributing channels and a bounded downstream prediction.
Conventional baseline	Explicit feature concatenation, simple statistical models and single-modality comparisons.
Why AI may help	Nonlinear relationships and interactions among complementary signals.
Data requirements	Synchronized representative data, controlled confounders and enough independent cases for the model complexity.
Failure modes	Learning site or batch identity, relying on a shortcut channel or failing when a modality is missing.
Validation	Held-out task performance, modality ablation, missing-channel tests and cross-site transfer.

Demonstrate complementarity

A combined model should be compared with its strongest single-channel baseline. Removing each channel can reveal whether it contributes useful information or merely increases complexity. The evaluation should also challenge common-cause disturbances that can affect several channels at once.

The representation itself is not a biological result. Its downstream use must be defined and evaluated. A fused score should retain the original observations and their quality states so that a reviewer can investigate why the result changed after a sensor fault, configuration change or reprocessing. Unqualified signals should not gain credibility merely by entering a larger model.

A22 Sensor disagreement detection

A22 · Proposed disagreement and fault assessment

Sensor disagreement detection identifies observations that are inconsistent with their expected relationship. AI may learn normal cross-channel patterns and highlight unusual residuals. The detected discrepancy can have several explanations, including a sensor fault, timing error, changed environment or an inadequate model.

Table III.31. A22 Sensor disagreement detection

Capability element	Proposed definition
Inputs	Sensor observations, independent quality checks, synchronization information and expected relationships.
Outputs	Disagreement event, affected channels, supporting residuals and uncertainty.
Conventional baseline	Physical consistency rules, calibrated residual thresholds and independent reference checks.
Why AI may help	Detecting complex patterns of inconsistency across many operating conditions.
Data requirements	Normal variation and realistic faults, including shared disturbances and incomplete observations.
Failure modes	Missing common-cause errors, overflagging legitimate differences or identifying the wrong channel as faulty.
Validation	Fault-specific detection, false alarms, detection delay and correct escalation or abstention.

Disagreement does not identify the truth

If two channels disagree, the model should not automatically choose the one that resembles its training data. The investigation should consider calibration, timing, location and what each channel measures. A third channel helps only if it contributes sufficiently independent information.

The proposed output should support a defined recovery path. Some events may justify excluding a failed auxiliary sensor; others may invalidate the whole observation. These paths need separate evaluation. A system that detects a discrepancy but still releases the same unsupported conclusion has not achieved the intended error-reduction function.

A23 Missing modality handling

A23 · Proposed missing-data handling

Missing-modality handling determines what analysis remains justified when one or more expected channels are unavailable. AI may support a qualified reduced-input model or a research imputation, but neither should be represented as if the missing sensor produced a real observation.

Table III.32. A23 Missing modality handling

Capability element	Proposed definition
Inputs	Available channels, explicit missingness reasons, quality states and intended task.
Outputs	Eligible reduced-input result, imputed research value or abstention with provenance.
Conventional baseline	Complete-case rules, prespecified fallback models and explicit unavailable status.
Why AI may help	Learning robust representations when missingness patterns are realistic and sufficiently represented.
Data requirements	Natural and deliberately simulated missingness, with reasons and independent reference outcomes.
Failure modes	Treating informative missingness as random, hiding a sensor failure or promoting imputation to measured evidence.
Validation	Performance by missingness pattern, uncertainty, coverage and detection of unsupported combinations.

Missingness has a cause

A channel absent because a sample is outside its range may contain different information from a channel absent because a cable disconnected. The model should not assume those events are equivalent. The original reason should remain part of the record.

For a clinical workflow, only separately qualified degraded modes should be eligible. Research imputation can be useful for model exploration or sensitivity analysis, but it should carry an imputed value type and a link to the method. It must not fill gaps in the original measured series or conceal that the biological observation was never acquired.

A24 Out of distribution detection

A24 · Proposed domain suitability assessment

Out-of-distribution detection asks whether an input differs materially from the domain in which a model was developed and evaluated. No detector can guarantee recognition of every unfamiliar case. Its role is to provide an additional signal for review and abstention, alongside explicit eligibility checks.

Table III.33. A24 Out of distribution detection

Capability element	Proposed definition
Inputs	Images, metadata, model representations and declared training or validation domain.
Outputs	Domain-suitability indicators, relevant differences and an uncertain or unsupported state.
Conventional baseline	Metadata eligibility rules, range checks and distance measures on explicit features.
Why AI may help	Recognizing complex appearance shifts that are not captured by a few metadata fields.
Data requirements	Representative in-domain data and diverse held-out shifts with known context.
Failure modes	High confidence on unfamiliar inputs, rejecting harmless variation or using arbitrary novelty scores.
Validation	Shift-specific detection, accepted-case error, coverage and transfer to unseen shift types.

Domain is more than appearance

A familiar-looking plate can still have an unsupported organism, drug, medium or reading time. Those conditions require explicit context checks. Conversely, an unfamiliar background may not affect the measurement if the relevant evidence remains qualified.

The proposed system should combine domain indicators with task-specific quality assessment. Evaluation should include new sites, instruments, vessel types and workflow conditions without using them to tune the detector after seeing the final test results. A novelty score should be explained through its tested operational meaning, not presented as a universal measure of biological uncertainty.

A25 Confidence calibration

A25 · Proposed confidence calibration

Confidence calibration examines whether a model’s stated confidence corresponds to observed performance. A model can rank easy and difficult cases well while still assigning misleading probabilities. Calibration should be evaluated separately from discrimination and should remain specific to the task and population.

Table III.34. A25 Confidence calibration

Capability element	Proposed definition
Inputs	Model outputs, independent reference outcomes and relevant population or quality strata.
Outputs	Calibrated probabilities or intervals, calibration diagnostics and applicability limits.
Conventional baseline	Held-out empirical reliability analysis and simple post-processing calibration methods.
Why AI may help	Conditional uncertainty models where error behaviour varies with image or workflow features.
Data requirements	Independent calibration data and a separate final evaluation set with sufficient event counts.
Failure modes	Reusing test data, hiding subgroup miscalibration or assuming calibration survives domain shift.
Validation	Reliability curves, suitable scoring rules, interval coverage and subgroup analysis.

Confidence is not a permission rule by itself

A high calibrated probability still needs to be interpreted against the consequence of error, the intended use and the availability of review. A low-confidence output may require another observation or an established alternative workflow.

Calibration should be reassessed when the model, acquisition pipeline or data distribution changes materially. The record should identify the calibration method and dataset version. A display should avoid false precision: a percentage with several decimal places does not communicate more useful certainty if the supporting evaluation is limited. The purpose is an empirically meaningful uncertainty statement that helps the user make an appropriate decision.

III / AI SOFTWARE AND DATA

A26 Active data selection

A26 · Proposed data acquisition support

Active data selection prioritizes examples that may be useful for improving a model or resolving a known evidence gap. AI can identify uncertainty, diversity or disagreement patterns, but its selection can also bias the resulting dataset. A representative evaluation set must remain independent of that process.

Table III.35. A26 Active data selection

Capability element	Proposed definition
Inputs	Existing model outputs, data coverage, uncertainty, disagreement and permitted collection context.
Outputs	Ranked candidates for annotation or additional collection with selection reasons.
Conventional baseline	Stratified sampling, random sampling and explicit gap-based collection.
Why AI may help	Finding informative edge cases within a large pool of routine observations.
Data requirements	A broad eligible pool, relationship identifiers and records of selection and non-selection.
Failure modes	Repeatedly selecting the same artifact, neglecting important rare groups or contaminating the final test set.
Validation	Improvement per reviewed case, coverage of evidence gaps and performance on an untouched representative test set.

Preserve the sampling story

A dataset enriched for difficult cases is useful for development but may not estimate routine prevalence or workflow performance. Its selection policy should therefore accompany the data. Researchers should be able to reconstruct why a record entered the training set.

The proposed function should support expert-led collection decisions rather than autonomously expand data use. Privacy, consent and partner permissions still apply. A model should not request patient-linked data merely because an uncertainty score is high. The collection plan should identify the minimum information needed to answer the development question and maintain separate evaluation data for the final claim.

A27 Annotation quality assistance

A27 · Proposed annotation quality assistance

Annotation assistance identifies potentially inconsistent, incomplete or difficult labels for human review. AI may compare annotations with image features or with related records. Its role is to surface questions, not to replace independent reference labels with the model's preferred answer.

Table III.36. A27 Annotation quality assistance

Capability element	Proposed definition
Inputs	Original observations, annotation versions, instructions and reviewer disagreement records.
Outputs	Candidate inconsistencies, missing fields and examples for adjudication.
Conventional baseline	Schema checks, double annotation, inter-reader comparison and random quality audits.
Why AI may help	Finding unusual patterns across large annotation collections.
Data requirements	Known annotation histories, independent reviews and examples of legitimate ambiguity.
Failure modes	Enforcing a shared bias, treating minority expert judgement as error or leaking model predictions into reference creation.
Validation	Confirmed issue yield, missed inconsistencies and preservation of independent adjudication.

Avoid circular ground truth

If a model's prediction becomes the label used to evaluate the same model, apparent agreement can improve without any improvement in truth. The annotation process should preserve who saw which information and when. Blinded review is important where it supports the selected evaluation design.

The proposed interface should present the original evidence, competing annotations and the relevant instructions. An adjudicator should be able to retain uncertainty rather than being forced to choose a precise value. Quality assistance succeeds when it improves the reliability and transparency of the reference process, not when it makes the dataset look more consistent with the current model.

A28 Dataset shift monitoring

A28 · Proposed distribution monitoring

Dataset shift monitoring tracks changes in the observations or populations encountered by a deployed or research system. AI may summarize complex image or feature distributions, while explicit metadata can reveal changes in instruments, sites or workflows. A detected shift does not automatically mean performance has degraded.

Table III.37. A28 Dataset shift monitoring

Capability element	Proposed definition
Inputs	Authorized operational metadata, quality indicators, model outputs and delayed reference information where available.
Outputs	Shift alerts, affected strata and evidence for investigation.
Conventional baseline	Statistical process monitoring, metadata counts and prespecified distribution comparisons.
Why AI may help	Detecting appearance changes not captured by simple operational fields.
Data requirements	Baseline distributions, sufficient ongoing samples and known configuration changes.
Failure modes	Confusing case-mix change with instrument drift, alert fatigue or missing a harmful shift in a rare group.
Validation	Detection delay, false alerts and association between monitored signals and actual performance changes.

Connect an alert to a review process

The proposed response should identify who investigates, what additional evidence is needed and whether any workflow restriction is justified. Automatic retraining should not be the default reaction, because it can introduce new unvalidated behaviour.

Monitoring should preserve privacy and site context. Aggregated statistics may support an investigation without transferring patient-linked records. Changes in data availability or review practice can also alter monitored distributions and should be recorded. The objective is early recognition of a meaningful change and a controlled response, not an uninterrupted stream of novelty alerts.

A29 Sequence quality triage

A29 · Proposed sequence-data triage

Sequence quality triage helps determine whether molecular data are suitable for a proposed downstream analysis. AI may identify complex quality patterns, but established pipeline checks and explicit metadata should remain central. The task concerns the quality of measured sequence data, not inference of a sequence from a plate image.

Table III.38. A29 Sequence quality triage

Capability element	Proposed definition
Inputs	Authorized reads or assemblies, sequencing metadata and established quality metrics.
Outputs	Quality findings, limitations and eligibility for specified downstream tasks.
Conventional baseline	Standard read, assembly, coverage and contamination quality checks.
Why AI may help	Recognizing combinations of quality signals associated with problematic analyses.
Data requirements	Representative sequencing technologies, known quality issues and independently assessed outcomes.
Failure modes	Treating database similarity as sample purity, overlooking mixed material or transferring thresholds between incompatible workflows.
Validation	Detection of relevant quality failures and effects on the accuracy of downstream analyses.

Preserve the molecular provenance

The record should retain accession or file identity, tool versions, reference data and the quality decisions made. A later database or pipeline update should produce a new analysis version. NCBI documentation provides a basis for understanding the context and limitations of its sequence-based resources. [\[R25\]](#)

This proposed capability belongs to the bioinformatics product track. It should not silently reject or alter measured molecular data. Where quality is insufficient, the output should identify the limitation and the analysis it affects, allowing a qualified user to decide whether additional evidence or a different workflow is required.

A30 Resistance gene evidence retrieval

A30 · Proposed resistance-evidence analysis

Resistance-gene evidence retrieval connects qualified molecular observations with curated knowledge about resistance determinants. The function may combine sequence-analysis outputs with structured evidence retrieval. It should preserve the distinction between a detected sequence feature, a database annotation and a clinically expressed phenotype.

Table III.39. A30 Resistance gene evidence retrieval

Capability element	Proposed definition
Inputs	Qualified sequence-analysis results, accessions, database releases and the permitted research question.
Outputs	Source-linked determinant evidence, match context, limitations and provenance.
Conventional baseline	Established sequence-analysis pipelines and curated database queries.
Why AI may help	Organizing heterogeneous evidence and identifying relevant contextual qualifications.
Data requirements	Curated annotations, representative sequence quality and independently reviewed evidence associations.
Failure modes	Equating gene presence with expression, missing novel mechanisms or mixing database releases.
Validation	Correct evidence linkage, reproducibility, retrieval completeness and expert review of interpretation.

Keep the evidence chain visible

CARD and NCBI resources offer different forms of resistance-related information. The proposed system should identify the resource and version used, the analysis method and the exact molecular input. It should not describe a database hit as a new laboratory measurement. [\[R24\]](#) [\[R25\]](#)

The initial role is research and qualified molecular-result support. An absent match does not prove susceptibility, and a present determinant does not by itself establish every clinical consequence. The interface should preserve these limits and direct discrepancies to review rather than resolving them through an unsupported single confidence score.

A31 Phenotype genotype discrepancy review

A31 · Proposed phenotype and genotype reconciliation

A measured susceptibility phenotype and a molecular finding answer different questions. A discrepancy may reflect expression, an unrepresented mechanism, assay conditions, mixed populations, sample identity or an error in either workflow. The proposed capability organizes these explanations for expert review; it does not decide that one evidence class always overrides the other.

Table III.40. A31 Phenotype genotype discrepancy review

Capability element	Proposed definition
Inputs	Reviewed phenotypic result, molecular assay or sequence evidence, organism identification and matched specimen lineage.
Outputs	Discrepancy summary, plausible explanations and links to the underlying records.
Conventional baseline	Curated organism–mechanism rules and manual microbiologist review.
Why AI may help	Ranking complex combinations of evidence and retrieving relevant contextual observations.
Data requirements	Paired, independently assessed records with assay limits, database versions and adjudicated discrepancies.
Failure modes	Treating an absent gene call as susceptibility, linking different isolates or confusing predicted and measured results.
Validation	Explanation fidelity, linkage accuracy, reviewer usefulness and errors in proposed escalation.

Preserve the direction of inference

A sequence-derived mechanism prediction should identify the detection method and reference knowledge used. A phenotype-derived hypothesis about a gene remains an inference even when it resembles a known association. Database evidence can support interpretation, but it cannot supply a molecular measurement that was never performed. [\[R24\]](#) [\[R25\]](#)

The review interface should display both original records, not a merged value that conceals disagreement. Proposed actions may include checking sample identity, reviewing assay quality or obtaining an appropriate confirmatory investigation. Clinical reporting rules remain separately controlled. Performance assessment should include discrepancies that the system cannot explain, because honest unresolved cases are preferable to convincing invented mechanisms.

A32 Genomic similarity research

A32 · Proposed genomic similarity research

Genomic similarity analysis can support research into related isolates and population structure. Similarity depends on the chosen representation, sequence coverage, organism and comparison method. It is not by itself proof of transmission, common exposure or identical susceptibility.

Table III.41. A32 Genomic similarity research

Capability element	Proposed definition
Inputs	Qualified sequences or authorized sequence-derived features, organism metadata and analysis versions.
Outputs	Similarity relationships, method-specific distances and contextual limitations.
Conventional baseline	Established sequence alignment, allele comparison and phylogenetic methods.
Why AI may help	Efficient retrieval or representation learning for large, heterogeneous research collections.
Data requirements	Diverse, quality-controlled populations with independent comparison results and realistic sampling structure.
Failure modes	Site leakage, contamination, incomplete assemblies, oversampling and unjustified transmission claims.
Validation	Agreement with appropriate genomic methods, retrieval accuracy and robustness to coverage and sampling changes.

Similarity requires context

An embedding distance is a model output, not a universal genetic distance. Its interpretation must be established for the organism group and use case. The platform should retain the sequence-analysis pipeline, reference collection snapshot and thresholds used to generate a relationship. Public repository access must follow the repository's terms and data-use conditions. [\[R25\]](#)

An authorized research workspace could place phenotypic observations beside genomic relationships to explore hypotheses. Patient-level epidemiological conclusions require additional information, appropriate governance and expert assessment. The proposed software should therefore distinguish a computational cluster from an investigated outbreak, and it should avoid presenting either as a demonstrated AMReye.AI clinical capability.

A33 Literature evidence retrieval

A33 · Proposed literature evidence retrieval

Evidence retrieval helps a reviewer find relevant studies, methods and limitations. AI may expand terminology, rank passages and organize contradictory findings. It should operate over identifiable sources and show what was actually retrieved, rather than generating citations from linguistic plausibility.

Table III.42. A33 Literature evidence retrieval

Capability element	Proposed definition
Inputs	A bounded research question, approved source collections and search constraints.
Outputs	Source-linked evidence map with population, method, outcome and limitation fields.
Conventional baseline	Structured database searches, citation chaining and manual critical appraisal.
Why AI may help	Reconciling terminology and locating relevant material across large text collections.
Data requirements	Licensed source access, reliable metadata and independently assessed relevance examples.
Failure modes	Fabricated citations, missed contrary evidence, abstract-only overinterpretation and obsolete guidance.
Validation	Citation existence, passage support, retrieval recall and faithful treatment of negative findings.

Separate retrieval from authority

A search result is a candidate source. Study design, specimen context, comparator, sample size and limitations determine what the source can support. A vendor performance statement, an original clinical study and a laboratory proof of concept should not receive the same evidentiary weight merely because they share keywords.

The interface should record the search date and query, distinguish full-text review from abstract access and expose unresolved contradictions. Generated summaries require source checking before external use. Standards questions should point to the controlled standards service, because a language model's recollection is unsuitable for selecting clinical interpretation rules. The retrieval system is a research aid with its own measurable failure rate.

A34 Standards rule consistency checking

A34 · Proposed standards consistency assistance

Standards interpretation should be implemented through a versioned, testable rules service. AI can assist with finding inconsistent mappings, missing metadata or unexpected changes during expert review. It should not improvise breakpoints or silently reinterpret a result when a new document appears. [\[R01\]](#) [\[R02\]](#)

Table III.43. A34 Standards rule consistency checking

Capability element	Proposed definition
Inputs	Authorized standards content, structured rules, method scope and curated test cases.
Outputs	Suspected inconsistencies with exact rule and source locations.
Conventional baseline	Schema validation, deterministic rule tests and expert double checking.
Why AI may help	Identifying semantic mismatches and unusual relationships in large rule collections.
Data requirements	Licensed content, correction notices, expert mappings and examples of known rule defects.
Failure modes	Invented thresholds, missed footnotes, conflating methods or treating an advisory as a breakpoint.
Validation	Defect detection, false alarms and expert confirmation against the applicable source version.

The rules service remains authoritative

An assistance output should be a review item, not an executable change. Promotion of a correction requires an authorized reviewer, a controlled source and regression tests. Historical reports retain their original interpretation and rule version; a later reinterpretation becomes a linked event with a reason.

Rapid methods deserve separate namespaces and eligibility checks. A routine endpoint threshold must not be substituted into an early reading because the organism and antimicrobial names match. The evaluation should deliberately include missing organism information, unsupported combinations, unit errors and superseded corrections. An explicit unavailable interpretation is the appropriate result when required scope information cannot be established. [\[R04\]](#) [\[R05\]](#)

A35 Reviewer workload prioritization

A35 · Proposed reviewer workload prioritization

Review prioritization organizes work according to defined laboratory policies, technical uncertainty and operational urgency. Its purpose is to help qualified staff attend to cases efficiently while preserving every case's required review. It should not create an invisible queue in which difficult or unfamiliar samples are indefinitely deferred.

Table III.44. A35 Reviewer workload prioritization

Capability element	Proposed definition
Inputs	Queue age, requested workflow, capture quality, disagreement flags and authorized urgency metadata.
Outputs	Explainable queue ordering and visible reasons for escalation.
Conventional baseline	First-in-first-out processing with explicit priority classes and age limits.
Why AI may help	Estimating review effort and recognizing combinations associated with delayed resolution.
Data requirements	Representative workflow timestamps, adjudicated reasons and staffing context.
Failure modes	Starvation of complex cases, proxy bias, excessive alerts and learning historical staffing shortages.
Validation	Waiting-time distribution, overdue cases, priority compliance and reviewer override outcomes.

Operational advice needs boundaries

A predicted review duration is an operational estimate. It is not a measure of clinical severity. Any use of clinical urgency must follow the laboratory's authorized policy and data permissions. The queue should make aging visible and provide a deterministic maximum-delay escalation independent of the model.

Evaluation should compare the proposed ordering with a simple baseline under realistic arrival patterns and staff availability. Average turnaround alone can conceal poor service for a smaller case group. Report tail delays and unresolved counts as well. If the model becomes unavailable, staff should retain a complete, intelligible queue and a predictable manual ordering method.

A36 Inventory and consumable forecasting

A36 · Proposed inventory and consumable forecasting

Consumable forecasting estimates future demand to support laboratory planning. The useful unit may be a disk, plate, calibration material, replacement component or service kit. The forecast must account for expiry and compatible use, because an apparently adequate stock count can still be unusable for the scheduled work.

Table III.45. A36 Inventory and consumable forecasting

Capability element	Proposed definition
Inputs	Consumption, stock movements, expiry dates, planned workload and supplier lead-time observations.
Outputs	Demand intervals, stock-risk warnings and suggested planning quantities.
Conventional baseline	Reorder points, moving averages and minimum–maximum stock policies.
Why AI may help	Modelling irregular demand and interactions among workload, seasonality and replenishment delay.
Data requirements	Reconciled inventory events, complete stockout records and changing procurement constraints.
Failure modes	Confusing unmet demand with low demand, ignoring expiry or recommending incompatible substitutes.
Validation	Stockouts, expiry waste, forecast calibration and performance during demand changes.

Recommendations are not purchases

The software should expose the assumptions behind a planning quantity, including the service period and uncertainty. Approval, supplier qualification and purchasing remain controlled business processes. No automatic order should arise merely because an AI forecast crosses a threshold.

Laboratories with limited history need a simple, inspectable baseline and conservative uncertainty. Data from another site may help only when workload and procurement conditions are comparable. A forecast should be revised when a method changes its consumable mix. The original forecast and subsequent actual consumption should remain available so that apparent improvements can be assessed without retrospectively rewriting the prediction.

A37 Predictive maintenance

A37 · Proposed predictive maintenance

Predictive maintenance estimates whether instrument condition is deteriorating before a functional failure occurs. Candidate inputs include illumination stability, focus actuator behavior, temperature excursions, fan performance and repeated acquisition faults. The model should support service decisions without overriding mandatory calibration or safety checks.

Table III.46. A37 Predictive maintenance

Capability element	Proposed definition
Inputs	Instrument telemetry, calibration trends, operating hours, faults and verified service findings.
Outputs	Condition indicators, inspection recommendations and bounded failure-risk estimates.
Conventional baseline	Scheduled service, threshold alarms and statistical trend monitoring.
Why AI may help	Recognizing multivariable degradation patterns that precede certain known failures.
Data requirements	Traceable maintenance histories, confirmed failure modes and representative healthy operation.
Failure modes	Rare-failure overfitting, confusing ambient changes with damage or falsely reassuring staff.
Validation	Failure-specific detection, useful warning time, missed events and unnecessary service visits.

A service label must be trustworthy

Replacing a component does not prove that the component caused the original fault. Service records should distinguish suspected causes from findings confirmed by inspection or testing. Models trained on ambiguous repair labels can learn misleading associations.

The proposed system should retain deterministic limits for conditions that require immediate action. A low predicted risk cannot waive an overdue calibration or an electrical fault. For Reader V1, basic fault logging and stable maintenance procedures are more credible initial functions than precise remaining-life estimates. Advanced forecasts become useful only after sufficient field evidence establishes their reliability across instrument revisions and operating environments.

A38 Energy and acquisition scheduling

A38 · Proposed energy and acquisition scheduling

Acquisition scheduling chooses when and how to collect observations within a qualified workflow. AI may allocate computation or suggest research sampling times that capture informative changes. The optimization must respect biological timing, thermal stability, illumination exposure and required quality checks.

Table III.47. A38 Energy and acquisition scheduling

Capability element	Proposed definition
Inputs	Workflow constraints, power state, resource availability and current observation uncertainty.
Outputs	Eligible acquisition schedule, compute allocation and estimated resource demand.
Conventional baseline	Fixed sampling intervals and deterministic resource scheduling.
Why AI may help	Balancing changing information needs against constrained power and processing capacity.
Data requirements	Measured power profiles, realistic workloads and independently evaluated observation schedules.
Failure modes	Missing transient behavior, altering the assay through illumination or postponing a required measurement.
Validation	Information retained, timing compliance, energy use and performance under resource shortages.

Optimize within the method

For endpoint Reader V1, an optimization may concern processing and queue management after a valid acquisition. It should not modify the biological method to save energy. Adaptive temporal sampling belongs to a separately evaluated research workflow, especially when the timing of growth observations affects the conclusion.

The scheduler must record the observations it intentionally omitted and why. A forecast cannot retroactively replace those measurements. On loss of cloud connectivity, the device needs a defined local schedule that preserves required operations. Evaluation should include workload bursts and degraded power conditions, because a policy optimized only for average conditions may fail at the moment reliability matters most.

A39 Fleet quality surveillance

A39 · Proposed fleet quality surveillance

Fleet surveillance compares instrument quality indicators across deployments to identify emerging problems. A coordinated view may reveal a drifting illumination batch, a software regression or an unusual calibration pattern. Cross-site comparison is meaningful only after accounting for configuration, workload and environmental differences.

Table III.48. A39 Fleet quality surveillance

Capability element	Proposed definition
Inputs	Authorized quality summaries, hardware and software versions, calibration results and service events.
Outputs	Investigable quality signals with affected configurations and supporting trends.
Conventional baseline	Stratified control charts, release monitoring and scheduled quality review.
Why AI may help	Finding interacting patterns across a large and diverse instrument population.
Data requirements	Stable metric definitions, complete version lineage and comparable reference observations.
Failure modes	Pooling incompatible metrics, exposing sensitive data or confusing case mix with instrument drift.
Validation	Detection delay, false investigations, root-cause confirmation and privacy-preserving utility.

Aggregation does not remove governance

The smallest useful data set should be selected for the surveillance question. A service engineer may need a reference-image metric and instrument version without needing patient identifiers or biological images. Access rights and retention should follow the purpose of each data flow.

A fleet alert should trigger investigation through the quality system. It should not silently change local measurement algorithms. Corrective actions require assessment of affected records, release controls and deployment verification. Independent sites should retain essential local functions during a central outage, while the central record later reconciles event timing and versions without overwriting the original local history.

A40 Natural language workflow assistance

A40 · Proposed natural language workflow assistance

A language interface can help authorized users locate records, understand software messages and navigate documented procedures. Its practical value is reduced interaction burden. It should distinguish information retrieval from actions that alter records, interpretations or access permissions.

Table III.49. A40 Natural language workflow assistance

Capability element	Proposed definition
Inputs	User request, role permissions and a bounded set of approved records or instructions.
Outputs	Source-linked explanations, navigation or a reviewable action proposal.
Conventional baseline	Search, structured forms, contextual help and deterministic shortcuts.
Why AI may help	Translating varied phrasing into appropriate information requests and interface actions.
Data requirements	Authorized documentation, representative tasks and examples of ambiguous or hostile content.
Failure modes	Invented advice, wrong-record actions, prompt injection, privacy leakage and excessive authority.
Validation	Task success, source fidelity, permission enforcement and resistance to misleading retrieved content.

Keep consequential actions explicit

The assistant should identify the intended record and show the exact proposed change before a consequential action. Retrieved text is data, not permission to execute instructions. Approval should come from the authenticated user with the appropriate role, through the same controls as the conventional interface.

Clinical interpretation should remain in the qualified measurement and standards workflow. The assistant may explain why a record is awaiting review, but it should not invent a susceptibility category or approve its own explanation. Voice, mobile and touchless access can use this interaction layer only after usability, environmental noise, authentication and error-recovery behavior are evaluated for the intended setting. [\[R22\]](#) [\[R23\]](#)

IV / DIGITAL TWINS AND SIMULATIONS

The digital plate record

The durable representation of an observed plate

The digital plate record is the foundation of the proposed simulation system. It stores observations, identity, timing, calibration, handling events and reviewer decisions. A record can be valuable without a predictive model: it allows a laboratory to revisit what was observed and to understand how a reported measurement was produced.

What belongs in the record

Each acquisition should retain its original file, a content integrity value, device configuration and the time interval over which the signal was acquired. Measurements derived from an image should link to that image and to the calibration and algorithm versions used. A corrected boundary becomes a new measurement event associated with the reviewer and reason.

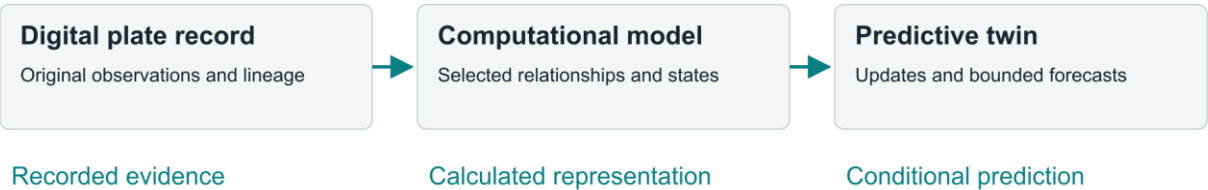
The record should distinguish physical identity, direct observations, derived measurements, qualified interpretations and simulations. These layers answer different questions: which object was observed, what was recorded, what analysis calculated, what the reviewed workflow concluded and what a model estimates beyond observation.

Preservation is a product capability

The stored record should remain intelligible if the original instrument is replaced or a model is retired. This requires documented formats, units, reference identifiers and an export path. Retaining only a screenshot of a final result would lose information needed for investigation and reanalysis.

The record does not become a predictive twin merely because it contains many sensors or a three-dimensional visualization. Prediction requires an explicit model, a defined question and evidence that the model can answer that question within a stated domain. Those additional requirements are developed throughout this part.

Three linked objects, three different evidentiary roles



More detail does not change simulated values into measurements.

AMReye.AI · conceptual diagram

Figure IV.1. Three linked representations with distinct evidentiary roles. Original conceptual diagram.

From record to computational model

A model turns a question into a calculation

A computational model represents selected features of a physical system using mathematical rules or learned relationships. It may describe an optical measurement, the expansion of a colony or the distribution of an antimicrobial in a medium. Its scope should follow the question; a detailed rendering of an enclosure does not necessarily improve a biological forecast.

Define the quantity of interest

For AMReye.AI research, candidate quantities include visible colony area, a spatial growth probability, a time to a defined observation threshold or a forecast distribution of an image-derived measurement. These quantities differ from susceptibility categories and require separate validation before any clinical interpretation is considered.

Table IV.1. From record to computational model

Model specification	Required statement
System boundary	Plate, region, population or instrument process being represented.
State variables	Quantities that evolve, with units or explicitly defined dimensionless scales.
Inputs	Environmental conditions, assay context and observed interventions.
Parameters	Values estimated from data or fixed from justified evidence.
Output	The exact quantity the model is intended to predict.
Domain	Conditions in which the model has been evaluated.

The same image may support several models, but each needs a separate output identity. An estimated biomass should not be relabelled as a measured viable count. A model can fit the observed past while being unsuitable for extrapolation, particularly when multiple parameter combinations explain the same early trajectory.

Start with a baseline

A constant-value forecast or a simple growth curve establishes whether a more complex model adds information. Complexity should be justified by improved predictive performance, useful uncertainty and practical interpretability. The proposed platform should preserve baseline results alongside experimental models during development, making improvement testable rather than assumed.

IV / DIGITAL TWINS AND SIMULATIONS

Requirements for a predictive twin

A predictive twin must remain connected to evidence

The proposed predictive twin is a computational representation linked to a particular physical plate, updated when qualified observations arrive and used to answer a bounded forecasting question. A general probabilistic twin framework has been demonstrated in engineering research; transferring that idea to microbiological plates requires new biological evidence. [\[R59\]](#)

Minimum functional contract

The twin should identify its physical counterpart, current model version, last accepted observation, estimated state and permitted prediction horizon. It should also describe the conditions under which updating is suspended. An image rejected for identity or calibration failure should not automatically adjust the biological state.

Table IV.2. Requirements for a predictive twin

Requirement	Acceptance question
Identity linkage	Can every update be traced to the correct plate and acquisition?
State update	Does the algorithm incorporate observations using a documented method?
Forecast	Is the output a dated prediction with an uncertainty statement?
Suitability	Are unsupported conditions detected or explicitly excluded?
Decision boundary	Is the permitted research or operational use clearly defined?

Bidirectional use needs an additional gate

A twin might eventually recommend a further observation or a virtual experiment. Any action that changes the physical assay, such as adjusting incubation or exposure, needs its own validation and control path. A simulation service should not gain authority over an instrument simply because the two exchange data.

Reader V1 can establish the record and measurement foundations before a predictive twin is available. This sequence supports future research while keeping the initial product proposition credible. The twin's maturity should be assessed by its validated forecasting domain, not the number of simulated images it can generate or the visual realism of its interface.

IV / DIGITAL TWINS AND SIMULATIONS

A state space view of plate development

State evolution and observation are different processes

A state-space formulation separates biological development from the process used to observe it. Let x_k represent the latent state at time t_k , u_k the recorded inputs and θ the model parameters. A proposed discrete representation is:

$$x_{k+1} = f(x_k, u_k, \Delta t_k; \theta) + w_k$$

$$y_k = h(x_k, c_k) + v_k$$

Here y_k is an observation, c_k describes calibration or capture conditions, w_k represents unresolved process variation and v_k represents observation error. The equations are a modelling scaffold, not an implemented AMReye.AI algorithm. Vector components may have different units and therefore need explicit scaling and covariance definitions.

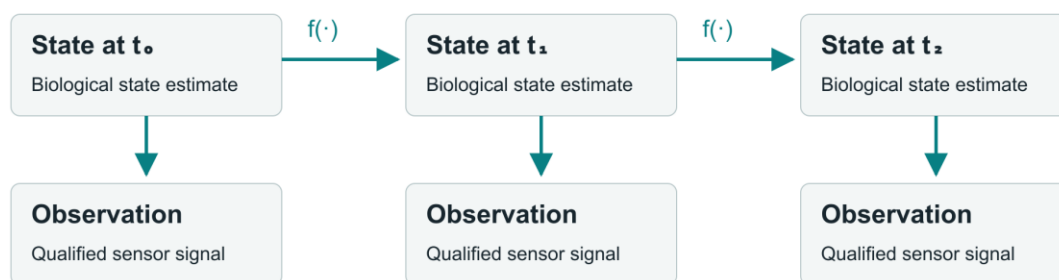
Choose interpretable states where possible

A simple research state might contain visible occupied area and a growth-rate parameter. A richer state could represent local population density, nutrient availability or antimicrobial concentration. Including an unobservable state adds assumptions; it does not make that quantity measured.

State variables must remain within their defined ranges. Missing temperature intervals cannot be assumed constant without an explicit assumption. Segmented area in mm² retains image and segmentation uncertainty. Process variation should describe unresolved biological changes rather than absorb every instrument defect.

The model should use actual elapsed time, especially when acquisition intervals vary. Numerical propagation over a long gap may require smaller internal steps than the observation interval. Testing should challenge irregular timing, abrupt input changes and boundary conditions before the representation is used to support forecasts.

State evolution and sensor observation



$h(\cdot)$ maps state to signal; calibration and noise remain explicit

AMReye.AI · conceptual diagram

Figure IV.2. State propagation and observation mapping. Arrows represent model dependencies, not fixed hardware connections.

Observation models and sensor likelihoods

The observation model connects biology to a sensor

An observation likelihood describes the probability of obtaining a signal for a given model state and capture condition. This connection is essential because cameras, gas sensors and electrical probes measure different physical effects. Their outputs should not be treated as interchangeable estimates of bacterial abundance.

Examples of observation relationships

For an image, a forward model may include illumination, object geometry, optical blur, sampling and sensor noise. A simplified model might instead relate segmented area to a latent occupancy state. For a gas channel, the relationship may additionally depend on chamber mixing, background atmosphere and transport delay.

Table IV.3. Observation models and sensor likelihoods

Question	Design implication
Is the signal saturated?	A bounded or censored likelihood may be more suitable than an unrestricted error model.
Is noise signal-dependent?	Estimate variation across the relevant intensity or concentration range.
Are channels correlated?	Include dependence or justify a conditional-independence approximation.
Is there a delay?	Align the observation to the state interval that could have produced it.
Is the value missing?	Record why it is absent; do not assign a zero likelihood measurement.

Avoid double counting

Two features extracted from the same image often share errors. Treating them as independent can produce an unjustifiably narrow posterior distribution. A model should account for shared image quality, calibration and preprocessing where these materially affect the output.

Observation models need calibration data that distinguish sensor behavior from biological variation. A stable optical target can help assess image formation, while it cannot establish the relation between image appearance and viability. The proposed system should retain both types of evidence and identify which relationship each experiment actually tests.

IV / DIGITAL TWINS AND SIMULATIONS

Time origins and biological age

Clock time is not automatically biological age

A forecast needs an explicit time origin. Specimen collection, inoculation, antimicrobial exposure, incubation entry and first image are different events. A trajectory indexed only by file creation time can conceal a long and variable period of biological development before the first observation.

Proposed temporal record

Each event should carry a timestamp, time zone or UTC representation, precision and provenance. Device clocks also need a synchronization status. If an operator enters an approximate time, the record should preserve that approximation rather than displaying false second-level precision.

Table IV.4. Time origins and biological age

Time concept	Example use
Calendar timestamp	Relating an acquisition to the laboratory workflow.
Elapsed assay time	Comparing observations from a declared method start.
Exposure duration	Representing time under a defined antimicrobial condition.
Observation interval	Accounting for signals integrated over several seconds or minutes.
Biological age estimate	Research state inferred from prior conditions and observations.

Timing uncertainty propagates

In a rapidly changing phase, uncertainty about when growth began may dominate uncertainty about an image-derived area. A model fitted to a single early image cannot always distinguish a larger initial population from a longer unobserved growth period. The resulting ambiguity should widen the forecast or limit the permitted question.

For a sub-two-hour research target, the clock origin must be stated before evaluation. Reporting time from first image when the intended claim concerns specimen-to-result time would omit preparation and prior incubation. The platform should preserve multiple workflow intervals so that speed can be assessed honestly for each intended use, including unsuccessful and delayed cases.

IV / DIGITAL TWINS AND SIMULATIONS

Spatial representations of plate growth

Represent the spatial feature that matters

Plate growth can be represented as a continuous field, a grid of local measurements, a set of segmented objects or a graph of tracked colonies. These representations trade spatial detail against data requirements and computational cost. A whole-plate average can conceal a small localized event that matters to heteroresistance research.

Candidate representations

An occupancy map describes where visible growth is present. An object representation records individual boundaries, locations and tracking identities. A radial profile can summarize an approximately circular inhibition pattern, but its validity decreases when the boundary is irregular or neighboring zones interact.

Table IV.5. Spatial representations of plate growth

Representation	Useful question	Main limitation
Global summary	Is overall visible coverage changing?	Loses local structure.
Spatial grid	Where does growth emerge?	Resolution and registration affect the field.
Object set	How do individual colonies evolve?	Merging and splitting complicate identity.
Boundary curve	How does a visible zone change?	A boundary alone omits internal features.

Coordinate systems are part of the evidence

Spatial observations should refer to a plate coordinate system with units and a transform from image coordinates. Registration uncertainty should accompany comparisons across time. If the plate rotates or shifts, a stationary pixel does not represent a stationary biological location.

The proposed model should retain access to original images even when a reduced representation is used for computation. A reviewer may need to inspect a location excluded by the segmentation. Spatial compression should therefore be evaluated against the intended research question, including rare local features, rather than judged only by average reconstruction quality.

IV / DIGITAL TWINS AND SIMULATIONS

Population growth model families

Growth curves are hypotheses with domains

Population models provide compact descriptions of change. An exponential model, $dN/dt = \mu N$, represents growth at a constant specific rate μ . A logistic form, $dN/dt = \mu N(1 - N/K)$, adds a limiting level K . These equations are illustrative mathematical baselines; neither automatically describes the visible area, viability or antimicrobial response of a real plate.

Map the variable before fitting

If N denotes cell number, its units and measurement basis differ from a normalized image intensity. An image-derived proxy may have a nonlinear relationship to population size, especially after objects overlap or the detector saturates. Fitting the same equation to different proxies does not give the fitted parameters the same biological meaning.

Table IV.6. Population growth model families

Model family	Potential role	Assumption to challenge
Exponential	Short interval of approximately constant rate	Rate stability and absence of saturation.
Logistic or sigmoidal	Describing bounded trajectories	Identifiability of the plateau from early data.
Lag-aware	Separating delay from subsequent expansion	Whether the delay is observable.
Stochastic population	Representing heterogeneous growth histories	Adequacy of the assumed variation.

Early agreement can hide later divergence

Several curves may fit the first observations almost equally well but imply very different later behavior. A one-day record therefore does not establish that the remaining days can be reconstructed reliably. The useful forecast horizon depends on the organism, assay, conditions and output.

The research programme should compare model families prospectively on withheld future observations. Reporting only goodness of fit to the observed past would not test forecasting. Parameters should retain uncertainty and correlation, particularly when the observed period covers only a small part of the proposed growth trajectory.

IV / DIGITAL TWINS AND SIMULATIONS

Diffusion and spatial transport models

Transport models connect location, time and exposure

Spatial antimicrobial exposure can be explored with transport equations. An illustrative reaction–diffusion form is:

$$\partial c/\partial t = D\nabla^2 c - r(c, \mathbf{x}, t)$$

Here c is concentration, D is an effective diffusion coefficient and r represents a specified loss or reaction term. If distance is expressed in mm and time in hours, D has units mm²/h. The equation is a research starting point, not a calibrated disk-diffusion model for AMReye.AI.

Boundary and initial conditions matter

A simulation requires a description of the initial distribution and the behavior at the domain boundaries. Disk release, agar properties, geometry and environmental history may influence the modeled exposure. A visually plausible gradient is insufficient evidence that the concentration field is correct.

Table IV.7. Diffusion and spatial transport models

Model element	Evidence needed
Source term or release boundary	Justification for the assumed delivery over time.
Effective transport coefficient	Relevant material and condition measurements or defensible bounds.
Medium geometry	Dimensions and assumptions about spatial uniformity.
Biological response	Independently evaluated relation between exposure and the chosen output.
Numerical solution	Convergence and conservation checks appropriate to the equation.

Do not reverse a clinical method casually

An observed zone diameter does not uniquely identify the underlying concentration history or a minimum inhibitory concentration. Different parameter combinations can produce similar visible boundaries. Inferring an MIC from a simulated field is therefore a separate research claim requiring a suitable reference method and validation.

A useful first application may be sensitivity analysis: determining which uncertain transport assumptions most affect a proposed observation schedule. Such a study can inform experiment design without claiming that the simulation replaces an established susceptibility method.

Mechanistic and learned hybrid models

Hybrid models combine constraints and learned relationships

A hybrid model uses mechanistic structure for some processes and learned components for others. For example, a transport calculation might describe a spatial input while a learned observation model relates local state to image features. Another design could use a simple growth equation with a bounded, data-driven residual term.

Define responsibility at each boundary

The mechanistic component should declare its assumptions and parameter units. The learned component should declare its training domain and output constraints. Their combined uncertainty is not necessarily the sum of two independent errors, because the components may have been fitted to the same data.

Table IV.8. Mechanistic and learned hybrid models

Hybrid pattern	Proposed benefit	Failure to investigate
Learned observation model	Flexible mapping from state to sensor signal	Learning illumination artifacts as biology.
Learned parameter map	Context-sensitive parameter estimates	Extrapolating outside represented conditions.
Residual correction	Capturing systematic model mismatch	Concealing a structurally incorrect mechanism.
Reduced-order surrogate	Faster evaluation of an expensive model	Failing near unrepresented boundaries.

Preserve meaningful constraints

Nonnegative quantities, bounded occupancy and causal timing can be enforced where the underlying definition requires them. Constraints reduce some implausible behavior but do not prove that the model is biologically correct. A perfectly smooth, nonnegative trajectory can still make an inaccurate forecast.

Evaluation should compare the hybrid with both its mechanistic baseline and a suitable purely empirical model. The comparison needs held-out conditions and an explicit computational cost measure. A more elaborate model is justified when it offers a demonstrated advantage for the intended question, such as better calibrated forecasts or useful transfer with limited data, while preserving an intelligible route for investigating failure.

Parameter identifiability

A fitted parameter is not always identifiable

Identifiability concerns whether the available observations can determine a model's parameters. Structural identifiability asks what could be determined with ideal data under the model. Practical identifiability concerns the actual sampling, noise and observed time span. Both matter before a fitted parameter is given a biological interpretation.

An illustrative ambiguity

In $N(t) = N_0 \exp(\mu t)$, a single value of N at a known time cannot determine both the initial level N_0 and growth rate μ . Multiple pairs can produce the same observation. Additional time points may help, but an uncertain time origin introduces another source of ambiguity.

Table IV.9. Parameter identifiability

Symptom	Possible implication	Response
Many fits with similar error	Parameters are weakly constrained	Report ranges and compare forecasts.
Large parameter changes after one observation	Estimation is unstable	Reassess model complexity and sampling.
Strong parameter correlation	One quantity compensates for another	Use independent measurements or simplify.
Boundary-valued estimates	Bounds may be driving the result	Examine assumptions and residuals.

Design observations to distinguish explanations

A new sensor is valuable when it helps resolve a relevant ambiguity. Measuring an independent environmental input may be more informative than adding another feature from the same image. Experimental design should therefore consider parameter sensitivity and expected information, not just the number of channels.

For the proposed twin, parameter estimates should retain uncertainty and the data interval used to estimate them. A precise-looking best fit should not conceal a broad set of plausible alternatives. If different parameter sets lead to incompatible future decisions, the appropriate outcome is a shorter horizon, an additional observation or an unresolved research result.

State estimation from sparse observations

Sparse data require disciplined updating

State estimation combines prior information, a process model and new observations. Candidate methods include deterministic optimization, filtering and sequential sampling. The choice depends on nonlinearity, the form of uncertainty and computational constraints. No estimation method can recover information that the data and assumptions do not contain.

Proposed update sequence

The platform should first validate identity, timing and observation quality. It then predicts the state to the observation time, evaluates the new signal under the observation model and updates the state distribution. Finally, it records diagnostics and determines whether forecasting remains eligible.

Table IV.10. State estimation from sparse observations

Update stage	Required diagnostic
Propagate	Time step, inputs and any unobserved interval.
Compare	Residual between expected and observed signal.
Assimilate	Method, parameter restrictions and accepted channels.
Assess	Remaining uncertainty and evidence of model mismatch.
Preserve	Previous state, new state and the observations used.

A large residual needs investigation

An unexpected observation may reveal new biological behavior, a changed environment, an incorrect identity or an instrument fault. Automatically forcing the model through every observation can hide these problems. Conversely, rejecting every surprising point can prevent the model from recognizing a real change.

The proposed update policy should make these alternatives explicit. A quarantine state can preserve a disputed observation while stopping its use in downstream predictions. Research comparisons should include sparse and irregular schedules that resemble actual operation. Randomly deleting isolated images from an otherwise complete series is only one missing-data scenario and does not represent every real interruption.

Forecasting beyond the last image

A future trajectory is a conditional prediction

Forecasting propagates a model beyond the last accepted observation. The output is conditional on the estimated state, parameters and assumed future inputs. If temperature, atmosphere or handling changes after observation stops, a forecast based on unchanged conditions may no longer apply.

Preserve the forecast origin

Each prediction should identify the last real observation and the exact time at which the forecast was issued. Later observations can evaluate that prediction, but they must not be used to revise its historical score. A revised forecast is a new record.

Table IV.11. Forecasting beyond the last image

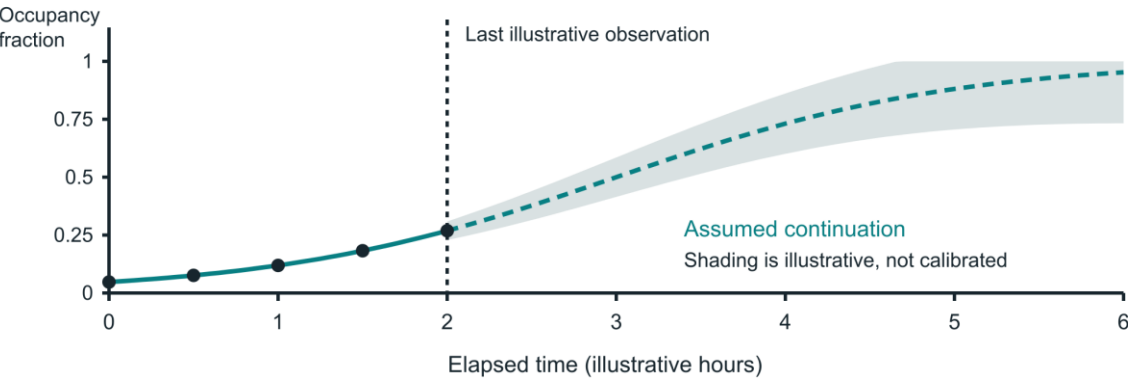
Forecast field	Example meaning
Origin	Last accepted biological observation and model state.
Target	Quantity and future time being predicted.
Conditions	Assumed future environment and handling.
Distribution	Central estimate and an evaluated uncertainty interval.
Status	Research eligible, unsupported, expired or interrupted.

One day observed does not guarantee later days

The proposed capability may investigate whether a limited observation period supports useful forecasts over a longer period. It should test each horizon against genuinely withheld future measurements. Performance can deteriorate after a growth transition, a previously unseen morphology or a change in the relevant environment.

Generated images may illustrate a predicted state, but visual realism is not a forecast metric. Evaluation should use the defined biological or image-derived quantity and its uncertainty. The interface should show measured observations with a different visual encoding from simulated continuation, and exports should retain the distinction in machine-readable fields as well as captions.

Illustrative forecast: synthetic values only



AMReye.AI · conceptual diagram

Figure IV.3. Synthetic illustration of a forecast origin. All plotted values and shading are constructed examples, not experimental data or calibrated performance.

The prediction horizon

A horizon is a validated limit, not a display preference

The prediction horizon is the future interval over which a forecast is permitted for a particular use. Its useful length depends on the output, acceptable error, input uncertainty and model domain. Different quantities from the same twin may have different horizons.

Define a horizon policy

A research programme should prespecify performance requirements at several future intervals. These may include a maximum error, interval calibration, a minimum useful coverage and an eligibility domain. The permitted horizon should be the longest interval supported by those requirements, not the furthest point that the solver can calculate.

Table IV.12. The prediction horizon

Limiting factor	Effect on eligibility
Sparse recent observations	State uncertainty may remain too large.
Unknown future environment	Only explicitly conditional scenarios are justified.
New morphology or assay condition	The established forecasting domain may not apply.
Excessive interval width	The prediction may be too imprecise for the question.
Failed model diagnostics	Suspend or shorten the forecast.

Horizon evaluation must include failures

It is misleading to report long-horizon accuracy only for cases where the model happened to remain confident. The evaluation should show the proportion of all eligible cases receiving a forecast at each horizon and the error among those accepted. Rejected and unavailable cases remain in the accounting.

A forecast should expire when its intended use or input assumptions cease to apply. Expiry does not delete the simulation; it changes what the result is allowed to support. For continuity after a lost plate, an exploratory trajectory may remain available beyond the qualified horizon, but it must be visibly outside validated predictive use and excluded from laboratory result release.

Uncertainty propagation

Uncertainty passes through the entire calculation

Measurement noise, calibration, time origin, model parameters and future inputs all contribute to forecast uncertainty. Propagating only the final model's statistical variance can omit dominant sources. Shared errors also matter: two measurements using the same spatial calibration are not independent simply because they came from different images. [\[R16\]](#)

An illustrative measurement calculation

If a diameter is calculated as $d = sp$, where s is scale in mm/pixel and p is a pixel diameter, a first-order variance approximation is:

$$\text{Var}(d) \approx p^2\text{Var}(s) + s^2\text{Var}(p) + 2sp \text{Cov}(s, p)$$

The covariance term vanishes only when independence is justified. This equation illustrates propagation for one measurement; it does not quantify an AMReye.AI instrument's uncertainty.

Table IV.13. Uncertainty propagation

Source	Candidate treatment
Calibration uncertainty	Propagate a shared scale distribution across linked observations.
Segmentation uncertainty	Evaluate plausible boundaries and their measurement effect.
Timing uncertainty	Sample or bound the possible observation origin.
Parameter uncertainty	Propagate joint parameter distributions, preserving correlation.
Model discrepancy	Assess systematic mismatch using independent observations.

Use intervals that answer the question

A confidence interval for an estimated mean differs from a prediction interval for a future observation. The latter must account for future variation relevant to the defined outcome. Interval labels should state their meaning and nominal probability.

Monte Carlo propagation can handle nonlinear relationships when the input distributions are defensible. More simulation draws reduce numerical sampling error; they do not correct missing uncertainty sources or invalid assumptions. The report should identify which uncertainties were included, which were bounded and which remain unquantified.

IV / DIGITAL TWINS AND SIMULATIONS

Aleatoric and epistemic uncertainty

Different uncertainties call for different remedies

Aleatoric uncertainty describes variation represented as inherent randomness in the chosen model, such as variability among otherwise comparable observations. Epistemic uncertainty concerns limited knowledge, including poorly estimated parameters or an inadequate model. The distinction depends on the model and available information; it is not a permanent label attached to a phenomenon.

Decide what additional information could help

Repeated measurements can characterize some observation variation. A better calibration may reduce uncertainty about scale. Additional laboratories can reveal a domain limitation that repeated measurements at one site would miss. An extra sensor helps only if it addresses an uncertainty relevant to the prediction.

Table IV.14. Aleatoric and epistemic uncertainty

Uncertainty question	Useful research action
Is the signal repeatable?	Replicate acquisition under controlled conditions.
Is the parameter constrained?	Collect observations that distinguish plausible parameter values.
Is the model structure adequate?	Compare alternative explanations on withheld conditions.
Is future input unknown?	Report conditional scenarios or obtain input monitoring.
Is the deployment unfamiliar?	Conduct domain-specific evaluation before extension.

Do not interpret confidence as certainty

A model can produce a narrow interval when its assumptions are wrong. Agreement among several models can also be misleading if they share training data, features or structural omissions. Confidence should therefore be assessed against independent outcomes and challenged under plausible shifts.

For the proposed twin, the user interface should explain the dominant reason for uncertainty where that reason is known. “Insufficient recent observation” suggests a different response from “unsupported organism–assay combination.” A generic confidence percentage without those distinctions would provide less useful guidance and could encourage overinterpretation.

IV / DIGITAL TWINS AND SIMULATIONS

Forecast calibration and scoring

Score the forecast that existed before the outcome

Forecast evaluation compares predictions issued at a defined origin with later independent observations. A rolling-origin design repeats this process at several prespecified points in a trajectory. The training process must not see the future observations used for scoring, including through normalization, parameter tuning or selection of an apparently successful horizon.

Evaluate accuracy and uncertainty together

Point metrics such as absolute error describe a central prediction. Interval coverage checks how often an interval contains the defined future outcome. Interval width describes precision. A very broad interval can achieve high coverage while being operationally unhelpful, so both should be reported.

Table IV.15. Forecast calibration and scoring

Evaluation view	Proposed report
Horizon	Error and interval behavior at each future interval.
Case domain	Results by organism, assay, site and relevant condition.
Eligibility	Fraction forecast, rejected, failed and unavailable.
Calibration	Empirical coverage relative to the stated probability.
Baseline comparison	Difference from persistence and simple growth models.

Avoid a favorable average concealing failure

Repeated predictions from the same plate are correlated. Treating them as independent samples can overstate precision. Statistical analysis should preserve the plate or isolate grouping and the study's sampling structure. Results should also include cases with interrupted observation, because continuity is an explicit product research objective.

Model selection and final evaluation need separate data. A model that was repeatedly adjusted after viewing the test trajectories has effectively trained on them. The proposed programme should preserve a frozen evaluation set or a prospective evaluation period, with the analysis plan and model version recorded before outcomes are examined.

Missing observation intervals

A gap remains a gap in the measured series

Missing observations can arise from scheduled omission, failed acquisition, network delay, physical handling or loss of the sample. These causes carry different information. The proposed record should preserve the expected observation, its status and the reason it is absent wherever that reason is known.

Distinguish delayed data from absent data

A locally acquired image may arrive at the server late. That is different from an image never captured. The server should use the acquisition time for biological analysis and the receipt time for operational auditing. Backdated ingestion must not make a historical forecast appear to have used information that arrived later.

Table IV.16. Missing observation intervals

Gap type	Record behavior
Acquisition failed	Store the failure and available technical diagnostics.
Upload delayed	Preserve the local acquisition identity and reconcile on receipt.
Observation intentionally omitted	Record the schedule decision and its authorizing policy.
Value imputed	Store a separate imputed object linked to the gap.
State simulated	Preserve the model and assumptions used during the interval.

Evaluate the return of observation

When a real observation resumes, compare it with the earlier forecast before updating the model. A large discrepancy may require investigation rather than immediate assimilation. The historical prediction remains available for scoring even if the new observation produces a better state estimate.

Missingness tests should include long contiguous outages, failures associated with poor image quality and unrecorded environmental changes. An algorithm that performs well when random frames are removed may still fail when all missing observations occur during condensation or rapid biological change. The proposed validation should reflect those mechanisms explicitly.

IV / DIGITAL TWINS AND SIMULATIONS

Plate loss and continuity of the record

Plate loss ends observation, not the historical record

If a physical plate is lost, damaged or no longer usable, the stored observations can remain available for review and research. A computational model may continue a conditional simulation from its last accepted state. That continuation cannot establish what the missing plate would actually have shown, nor can it confirm an unobserved biological outcome.

Proposed loss event

The record should identify the last confirmed physical observation, the time and nature of the loss if known, and the consequences for the workflow. Uncertainty about the loss time should be preserved. The plate's operational state becomes unavailable for further physical confirmation.

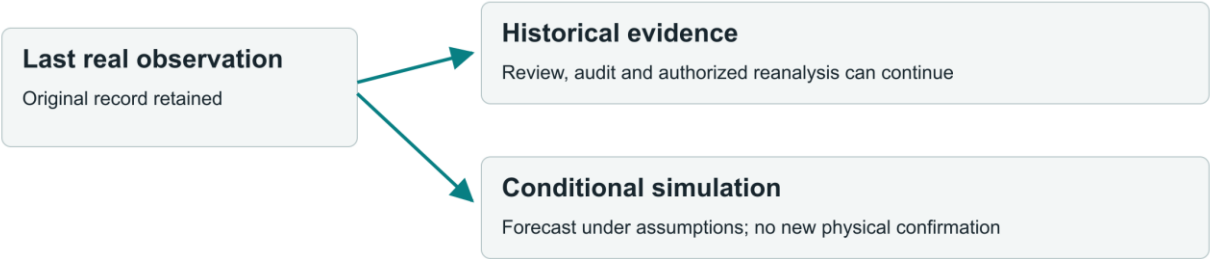
Original observations support historical review. A model supports conditional propagation. A prior forecast can be evaluated only if an independent outcome becomes available; its own generated trajectory cannot validate it. The laboratory workflow can escalate or repeat according to policy, but must not complete a missing physical result using synthetic growth.

Continuity is a resilience benefit

Preserving the record can prevent the loss of already captured information and help investigate an incident. It may support planning a repeat observation or reviewing an earlier endpoint. Whether an existing measured result remains reportable depends on the qualified method and laboratory policy, not on the availability of a simulated continuation.

The interface should make the loss state prominent on every subsequent simulation view and export. A later record from a replacement plate needs a new physical identity and lineage link. It must not be spliced into the original series as though the same biological object had resumed observation.

What survives when a physical plate is lost?



Loss event and unknown intervals remain visible.

AMReye.AI · conceptual diagram

Figure IV.4. Continuity after plate loss preserves historical evidence and separately labelled conditional simulation.

Instrument failure and recovery

Recovery must reconcile both instrument and biology

Instrument failure may affect acquisition, environmental control, data storage or several functions at once. Restoring the computer does not necessarily restore the assay's validity. A power interruption, for example, can leave a complete digital log while the physical conditions during the outage remain uncertain.

Proposed recovery sequence

The instrument should identify the last durable event, verify stored data integrity and determine which components require a readiness check. The workflow then assesses whether the physical plate experienced an excursion relevant to the method. Only eligible observations resume the original series.

Table IV.17. Instrument failure and recovery

Failure domain	Recovery evidence
Compute or software	Controlled restart, version identity and job reconciliation.
Storage	Integrity verification and explicit accounting for incomplete writes.
Timekeeping	Clock status and reconstruction of uncertain event intervals.
Optical system	Readiness and calibration checks appropriate to the disturbance.
Environment	Recorded excursion or an explicit unknown interval.

Prevent duplicate or invented events

An interrupted upload may be retried without creating a second biological observation. Stable event identifiers and idempotent ingestion are proposed safeguards. Conversely, a queued acquisition command should not be marked completed unless evidence shows that capture actually occurred.

A twin may propagate through the outage under stated assumptions, but its state should be quarantined if the physical environment became unsupported. On recovery, compare new observations with the pre-existing forecast and preserve the discrepancy. The system should support a clear decision to resume, re-estimate, repeat the assay or close the series as incomplete. These outcomes require validation through realistic fault and recovery exercises.

Simulation ensembles

An ensemble explores a defined set of possibilities

A simulation ensemble contains multiple runs that vary initial state, parameters, inputs, stochastic events or model structure. It can reveal how uncertainty affects a forecast. The ensemble is informative only to the extent that its variations represent plausible uncertainty for the intended question.

Design the ensemble explicitly

Parameter draws should preserve known dependencies. Sampling growth rate and lag independently when the fitted estimates are strongly correlated can create implausible combinations. Structural alternatives should be documented separately from repeated stochastic runs of one model.

Table IV.18. Simulation ensembles

Ensemble dimension	Example research question
Initial state	How does uncertain starting occupancy affect later area?
Parameters	Which plausible rate values change the forecast materially?
Future inputs	What follows under alternative recorded-condition scenarios?
Stochastic events	How variable are trajectories under the same model assumptions?
Model structure	Do different plausible mechanisms imply different outcomes?

Agreement needs interpretation

If every ensemble member uses the same incorrect observation model, a narrow distribution may still be wrong. An ensemble is not a substitute for independent validation. Weighting models by past performance also requires care when the deployment conditions differ from the evaluation data.

The proposed platform should retain the ensemble specification, random seeds where applicable and a summary of failed runs. Excluding numerical failures without explanation could bias the result toward easier cases. Reports should distinguish a range of scenarios from a calibrated probability interval. A scenario range becomes probabilistic only when the weighting and uncertainty assumptions justify that interpretation and have been evaluated for the intended use.

Virtual experiments and counterfactuals

Virtual experiments generate hypotheses for physical testing

A virtual experiment changes an input or model assumption and examines the calculated consequence. Candidate uses include comparing observation schedules, exploring sensitivity to temperature histories or evaluating alternative optical configurations. These calculations can prioritize physical experiments, but they do not themselves demonstrate biological effectiveness.

Counterfactuals require causal assumptions

A model trained on observational associations may not predict what happens when an input is deliberately changed. For example, an association between a site and a growth pattern does not imply that moving a plate to that site would cause the pattern. An intervention model needs a defensible account of the relevant mechanism and confounders.

Table IV.19. Virtual experiments and counterfactuals

Virtual question	Appropriate initial use
Would another sampling interval capture a transition?	Plan a paired acquisition study.
Which uncertain input dominates the forecast?	Prioritize a measurement or calibration experiment.
Could a simpler sensor configuration suffice?	Define a modality-ablation comparison.
How might a transport assumption change a boundary?	Select conditions for model discrimination.

Keep scenarios tied to their assumptions

The report should identify what was changed, what remained fixed and whether the change lies within an evaluated range. A hypothetical environment should not be described as an observed exposure. An apparent benefit should be reported in the model's output units and accompanied by sensitivity to alternative assumptions.

For AMReye.AI, virtual experiments are a proposed research service and engineering aid. They should not recommend patient treatment or replace laboratory confirmation. A successful virtual study ends with a testable physical question, an experimental comparison or a clear statement that the available model cannot resolve the decision.

IV / DIGITAL TWINS AND SIMULATIONS

Synthetic data for software development

Synthetic data can test software behavior safely and reproducibly

Synthetic records are useful for exercising interfaces, permissions, export formats and failure handling without exposing patient data. Artificial plate images can also test geometry and boundary measurement against known constructed shapes. The validity of such tests depends on the property being assessed.

Match the generator to the test

A geometric image with a known circle can test a diameter calculation. It cannot establish how the algorithm handles genuine biological zone edges. A simulated missing observation can test an unavailable-value export, while it cannot reproduce every cause of a real acquisition failure.

Table IV.20. Synthetic data for software development

Synthetic asset	Suitable verification target	Insufficient evidence for
Known geometric boundary	Coordinate transforms and units	Clinical measurement accuracy.
Artificial event sequence	Audit and recovery logic	Real workflow usability.
Generated sensor fault	Detection-path behavior	Field fault prevalence.
Mock laboratory record	Interface and access controls	Clinical data representativeness.

Make synthetic origin durable

Synthetic status should be stored in the record schema and preserved in every export. A visible label alone can disappear when an image is copied or a table is transformed. Dataset lineage should identify the generator, parameters and source material used to construct the asset.

The proposed public demonstration should use synthetic or appropriately authorized nonclinical examples with clear labels. Production and demonstration records require separate namespaces or equivalent controls so that a demonstration result cannot enter a clinical workflow accidentally. Reproducible synthetic tests complement physical and clinical evidence; their scope should be stated in the verification report.

IV / DIGITAL TWINS AND SIMULATIONS

Synthetic data and training contamination

Synthetic training data can introduce hidden dependence

Synthetic augmentation may improve robustness when it represents plausible variation. It can also amplify a generator's biases or leak information from evaluation data. A large synthetic dataset derived from a few source plates does not create an equally large number of independent biological examples.

Split by the originating biological unit

Training and evaluation separation should account for parent images, isolates, plates and acquisition sessions. If synthetic variants of a test image enter training, visual differences do not restore independence. The same concern applies when a generator was fitted using the intended evaluation collection.

Table IV.21. Synthetic data and training contamination

Contamination route	Proposed control
Augmented copies across splits	Assign the parent biological record before augmentation.
Generator trained on test material	Record generator training lineage and exclude overlap.
Simulated labels treated as truth	Retain the model and assumptions producing each label.
Repeated tuning on test outcomes	Reserve an untouched or prospective evaluation set.
Site-specific synthetic style	Test transfer using independent real observations.

Measure the contribution on real data

An augmentation experiment should compare otherwise comparable models with and without the synthetic material. The decisive evaluation should include real held-out observations and relevant rare or difficult conditions. Improved performance on another synthetic collection may simply reflect shared generator assumptions.

The proposed data service should retain a synthetic fraction and origin report for each training release. A model trained with generated growth should not cite that growth as experimental confirmation of its own forecast. Synthetic data may support engineering efficiency, but claims about biological performance require an independent evidentiary basis.

IV / DIGITAL TWINS AND SIMULATIONS

Model verification

Verification asks whether the calculation was implemented correctly

Model verification concerns the software and numerical solution. A solver can be implemented correctly while representing an inadequate biological model. Keeping verification distinct from validation makes failures easier to investigate and prevents a passing software test from being interpreted as proof of biological predictiveness.

Proposed numerical checks

Simple cases with known solutions can test the implementation. For exponential growth with constant μ , a numerical solver can be compared with $N_0 \exp(\mu t)$. Spatial transport solvers can be challenged with relevant conservation properties, boundary conditions and grid refinement. The exact checks should follow the equations and numerical method selected.

Table IV.22. Model verification

Verification area	Example acceptance evidence
Units and dimensions	Consistent conversion and dimensional analysis.
Analytic benchmark	Error against a known special-case solution.
Numerical convergence	Stable quantity of interest as step size or mesh changes.
Boundary behavior	Correct handling of limits and domain conditions.
Reproducibility	Recorded environment, seed and tolerances where relevant.

A failed run is a result state

The simulation service should detect nonconvergence, impossible values and exhausted resources. It should return a failed or incomplete status with diagnostics, rather than a partially calculated trajectory labelled successful. Recovery must preserve the original failure and identify any revised configuration.

Verification evidence should accompany each released model implementation. Changes to a solver, precision mode or acceleration library may require targeted rechecking even when the biological equations are unchanged. FDA's computational-model guidance provides a risk-informed credibility framework for relevant mechanistic models; it does not establish the credibility of this proposed platform. [\[R58\]](#)

IV / DIGITAL TWINS AND SIMULATIONS

Biological model validation

Validation asks whether the model is useful for the stated biology

Biological model validation compares predictions with observations that can test the intended use. It should establish the conditions under which the model is sufficiently accurate and informative. A model validated for visible colony area is not thereby validated for viability, susceptibility category or patient outcome.

Design the comparison before collecting the outcome

The study should specify organisms, assay conditions, observation schedules, forecast origins, horizons and reference quantities. It should also define how missing outcomes and failed simulations enter the analysis. Blinded or otherwise independent measurement can reduce the risk that reference assessment is influenced by the model's prediction.

Table IV.23. Biological model validation

Validation question	Required design feature
Does the forecast generalize?	Independent plates, isolates, time periods and relevant sites.
Is uncertainty credible?	Evaluation of interval calibration and width.
Does the model add value?	Comparison with prespecified simple baselines.
When does it fail?	Deliberate coverage of boundary and difficult conditions.
Is the output useful?	A defined decision or research question with measurable criteria.

Preserve unsuccessful hypotheses

A model that fails under a particular condition can still provide useful knowledge about its domain. The appropriate response is to restrict the claim, improve the model or collect additional evidence. Retrospectively removing difficult cases without a justified eligibility rule would overstate performance.

No AMReye.AI predictive twin is presented here as biologically validated. The proposed programme should progress from controlled measurements to independent forecasting studies and then, only where justified, to a specific operational or clinical investigation. Evidence from unrelated engineering twins demonstrates methods, not microbiological validity.

IV / DIGITAL TWINS AND SIMULATIONS

Sensitivity analysis

Sensitivity analysis identifies what drives the answer

Sensitivity analysis examines how a model output changes when an input or parameter changes. Local analysis studies behavior near a selected point. Global analysis explores variation across a defined range or distribution. The useful approach depends on nonlinearity, interactions and the uncertainty relevant to the decision.

Choose a meaningful output and range

A sensitivity ranking for mean colony area may differ from one for time to a threshold. Parameter ranges should be justified; an arbitrary wide range can make one input appear dominant. Correlated parameters need joint treatment or a clearly stated approximation.

Table IV.24. Sensitivity analysis

Analysis	Useful role	Limitation
One-at-a-time perturbation	Initial debugging and local intuition	Can miss interactions and nonlinear regions.
Derivative-based analysis	Efficient local responsiveness	Depends on the chosen operating point.
Sample-based global analysis	Exploring broad uncertainty and interactions	Requires a defensible sampling distribution.
Structural comparison	Assessing alternative model assumptions	Results depend on which alternatives are included.

Connect sensitivity to a research action

If timing uncertainty dominates an early forecast, improving clock records may be more useful than increasing image resolution. If a model is highly sensitive to an unobserved parameter, the next experiment should seek information about that parameter or test whether the model can be simplified.

Sensitivity is not the same as uncertainty: a highly influential parameter known precisely may contribute little predictive uncertainty. The proposed report should therefore present both input uncertainty and output sensitivity. This combination supports a rational choice of additional sensors, calibration effort and experimental conditions without assuming that more measurements always improve the result.

IV / DIGITAL TWINS AND SIMULATIONS

Model transfer between laboratories

Transfer requires evidence about the new laboratory

A model developed at one laboratory may encounter different media, workflows, environmental conditions, imaging systems or populations elsewhere. Some differences can be corrected through calibration; others change the relationship between biological state and observation. The proposed transfer process should identify which type of change is present.

Separate instrument calibration from biological adaptation

A reference target can help align spatial scale or illumination response. It cannot prove that an assay's biological behavior is equivalent across laboratories. Conversely, retraining a biological model should not be used to conceal an unstable optical system.

Table IV.25. Model transfer between laboratories

Transfer layer	Proposed evidence
Measurement	Comparable calibration, quality metrics and reference observations.
Workflow	Matching time origins, handling and eligibility definitions.
Population	Relevant organism, specimen and condition coverage.
Model	Independent forecast evaluation in the new domain.
Operations	Local review, outage and service behavior.

Adaptation changes the model

If parameters or learned components are updated using local data, the adapted system needs a new version and an evaluation appropriate to the change. Performance measured before adaptation should not be presented as evidence for the revised system. A holdout or prospective period is needed to assess the local update fairly.

The platform should support a site-specific eligibility state. A capability may remain available for research while being unavailable for a more consequential use at a new site. This approach allows collaborative development without treating every deployment as automatically covered by evidence from the first laboratory.

Computational infrastructure and scheduling

Cloud computation should preserve scientific traceability

The proposed cloud or Hive Mind layer can schedule simulations, retain model versions and support authorized collaboration. Compute scale may allow more ensemble members or more expensive models. It does not remove uncertainty in the input data or establish the biological validity of a model.

A simulation is a reproducible job

Each job should record its inputs, model package, numerical settings, random seeds where applicable and execution status. Input references should resolve to immutable versions or content-addressed objects. The job result should include diagnostics and the scope of any approximation used to meet resource limits.

Table IV.26. Computational infrastructure and scheduling

Infrastructure function	Proposed requirement
Scheduling	Explicit priority, resource limits and cancellation behavior.
Isolation	Separation of users, projects and execution permissions.
Reproducibility	Versioned software environment and dependency information.
Storage	Durable inputs, outputs and provenance with retention controls.
Recovery	Safe retry without duplicate or silently changed results.

Local operation needs a defined fallback

An instrument should retain essential capture and record functions during a cloud outage when its qualified workflow supports local operation. A queued simulation can remain pending; the interface should not suggest that a missing result is a negative biological finding.

Resource optimization should be evaluated against scientific accuracy. Reduced precision, a smaller ensemble or a surrogate model can change the output and should be recorded. The research service should show when a job was completed, partially completed or failed. A successful server response alone is insufficient evidence that the requested calculation met its numerical and scientific requirements.

Human interpretation of a twin

A useful twin interface makes uncertainty actionable

The twin interface should help a reviewer understand what was measured, what was inferred and what remains unknown. Visual continuity is attractive, but a seamless animation can conceal the point at which observation stopped. The proposed display therefore uses explicit visual and textual distinctions between evidence classes.

Proposed viewing behavior

Measured observations should appear at their actual acquisition times. Forecasts begin at a visible origin and use a different line style or shaded region. The last accepted observation, horizon and assumed conditions should remain visible when the user changes the time view or exports an image.

Table IV.27. Human interpretation of a twin

Reviewer question	Interface response
What is real?	Direct access to original observations and their quality states.
What is predicted?	A separately labelled trajectory and interval.
Why is it uncertain?	Dominant known limitations and missing information.
Is it still eligible?	Current domain, horizon and interruption status.
What changed?	Comparison of successive state updates and forecast versions.

Test understanding, not visual preference alone

Usability evaluation should ask users to identify the last real observation, distinguish an imputed value and recognize an expired forecast. It should include plate loss, delayed upload and disagreement scenarios. A visually appealing display that causes users to mistake simulation for measurement has failed an essential requirement.

The interface should offer a compact explanation with access to deeper technical detail. Experts may need model diagnostics, while routine reviewers need the permitted action and reason for abstention. Both views should derive from the same traceable record and preserve the same evidence boundaries.

Simulation reports and provenance

A simulation report must survive outside the application

A report may be downloaded, printed or shared with an authorized collaborator. It therefore needs enough context to prevent the simulated output being mistaken for an experimental result after it leaves the original interface. A watermark alone is insufficient because the essential scientific meaning belongs in the content and structured data.

Proposed report contract

The report should identify the plate record, simulation job, model version and issue time. It should state the question, output units, last actual observation, forecast horizon, assumptions and uncertainty definition. Figures should identify measured points and simulated regions in their captions.

Table IV.28. Simulation reports and provenance

Report section	Minimum content
Identity and scope	Physical record, authorized research purpose and result status.
Evidence inputs	Observation identifiers, quality states and excluded data.
Model	Version, parameters, numerical settings and domain.
Results	Defined quantities, intervals, diagnostics and failed outputs.
Limitations	Missing information, untested assumptions and prohibited inference.
Reproduction	Input versions and an authorized route to the computation record.

Corrections create linked reports

If an input record is corrected, a new simulation can be issued with a link to the superseded result. The original report should remain auditable and clearly marked when no longer suitable for use. A correction should not silently rewrite the evidence that a collaborator previously reviewed.

Machine-readable exports should carry the same status fields as the human-readable report. Empty, unavailable, imputed and simulated values need distinct representations. This design supports later analysis without requiring downstream users to infer scientific status from a color, filename or free-text footnote.

IV / DIGITAL TWINS AND SIMULATIONS

Research gates for predictive twins

Advance the twin through questions that can be answered

The predictive-twin programme should progress through evidence gates tied to specific functions. A functioning data pipeline is an important engineering achievement, but it is separate from a validated forecast. Each gate should define the decision, required evidence and consequence of failure.

Table IV.29. Research gates for predictive twins

Gate	Evidence needed to advance
Record integrity	Correct identity, timing, units, provenance and recovery behavior.
Measurement foundation	Qualified observation and calibration performance for the chosen output.
Model verification	Correct implementation and numerical behavior within specified limits.
Controlled forecasting	Independent future observations and credible uncertainty in a bounded setting.
Transfer	Evidence under additional relevant sites, populations or conditions.
Intended use	Appropriate study, risk review and regulatory assessment for the proposed decision.

The first useful milestone

A credible initial research milestone could be a reproducible forecast of a narrowly defined image-derived quantity over a short horizon under controlled conditions. The study should include a simple baseline, unsuccessful predictions and a clear distinction between measured and simulated values. Its success would support that specific capability, not a universal biological twin.

Subsequent work can investigate longer horizons, additional modalities and interrupted observation. Each extension changes the evidence question. A plate-loss simulation remains a conditional research continuation even if the same model performs well while observations are available.

The platform vision is a coordinated system of records, measurements, models and expert decisions. Its strength should come from making each connection testable. AMReye.AI can pursue ambitious simulation capabilities while keeping the current endpoint reader, proposed engineering and future predictive research clearly distinguishable.

Applications and intended use boundaries

Each application needs its own intended use

AMReye.AI is conceived as a platform with several product and research families. Shared imaging, data and service infrastructure can support different applications, but a common platform does not make their evidence interchangeable. The intended use should state the user, specimen or material, method, output and decision supported.

Organize applications by the evidence transition

Endpoint reading begins with a prepared, incubated plate and seeks a traceable measurement within an established workflow. Direct-from-specimen testing changes preparation and biological interpretation. A research observation system may investigate growth patterns without producing a clinical result. Each transition changes the study design and risk assessment.

Table V.1. Applications and intended use boundaries

Application class	Proposed contribution	Principal evidence boundary
Endpoint reader	Calibrated imaging, measurement and review	Agreement within a specified method and domain.
Temporal research	Repeated observations and growth features	Validity of the measured or predicted quantity.
Integrated instrument	Coordinated handling and observation	Whole-workflow performance and safety.
Molecular integration	Linked external assay evidence	Correct identity, semantics and assay scope.
Nonclinical research	Environmental, industrial or teaching records	Sector-specific methods and permitted decisions.

Shared infrastructure creates practical value

PIMS can preserve identity and provenance across these workflows. Calibration services, software quality controls and training materials can also be reused where their scope remains applicable. Biological models and clinical interpretation rules require more careful qualification.

The applications in this part are product intentions and research pathways. They are not a list of current cleared indications or deployed capabilities. The proposed portfolio should advance through explicit evidence gates, with the endpoint Reader V1 remaining the near-term priority and broader functions evaluated individually before external performance claims are made.

Routine endpoint disk diffusion

Routine endpoint reading is the first measurement priority

The Reader V1 concept supports imaging and measurement of eligible disk-diffusion plates at the appropriate endpoint. It aims to reduce avoidable variability in capture, measurement transcription and record retrieval while retaining mandatory expert review. The biological method and applicable interpretation rules remain independently controlled. [\[R01\]](#) [\[R02\]](#) [\[R03\]](#)

Proposed operational sequence

The operator associates the plate with its authorized record and confirms workflow eligibility. The reader checks capture readiness and image quality, applies a qualified spatial calibration and proposes disk and zone measurements. The reviewer inspects the image, resolves flags and approves or corrects the result through a traceable record.

Table V.2. Routine endpoint disk diffusion

Stage	Required outcome
Identity	Correct plate and case association before analysis.
Capture	Complete, readable image with preserved acquisition context.
Measurement	Diameter in mm with calibration and quality provenance.
Interpretation	Eligible organism–drug–method mapping and rule version.
Review	Authorized decision with any correction reason.
Export	Reviewed result and status transferred without semantic loss.

Measure the actual benefit

Evaluation should compare the proposed workflow with an appropriate reference reading process. Relevant outcomes include measurement bias, agreement, consequential category errors, unreadable cases, review effort and transcription failures. Faster image processing alone does not establish faster specimen-to-result service.

The study should retain difficult plate appearances and document why a case was excluded or repeated. An endpoint reader should not infer that an unreadable boundary has zero diameter or assign a susceptibility category from an incomplete record. Its reliability depends on recognizing when the available observation cannot support the intended result.

Clinical laboratory review workflows

Review is a designed part of the product

A clinical laboratory workflow requires clear responsibility for the result. Mandatory human review should be supported by an interface that makes important evidence visible and enables efficient correction. A nominal approval button is insufficient if the reviewer cannot inspect the original image, measurement basis or unresolved flags.

Proposed review workspace

The workspace should present plate identity, method context, image quality and the proposed measurement together. Relevant standards information should be available through the controlled rules service. Reviewers need a way to mark a result unreadable, request a repeat or escalate a discrepancy without fabricating a numerical value.

Table V.3. Clinical laboratory review workflows

Review action	Required trace
Accept a measurement	Reviewer identity, time and exact output version.
Correct a boundary	Original and revised geometry with reason.
Resolve an identity issue	Authorized reassociation and affected analyses.
Reject a capture	Failure reason and subsequent workflow state.
Release an export	Approved record version and destination mapping.

Corrections must invalidate dependent approval

If a measurement or relevant context changes after approval, the prior decision should not automatically apply to the revised result. The system needs a defined re-review path. A receiving laboratory system should be able to distinguish a replacement report from a new independent test.

Usability studies should include experienced and less-experienced intended users under realistic workload. They should measure whether users detect deliberately introduced identity, calibration and interpretation problems. Review time and satisfaction matter, but the central question is whether the complete workflow produces an accurate, understandable and accountable result under the conditions in which it will be used.

Urinary tract infection workflows

UTI workflows contain several distinct testing questions

Urinary tract infection is an important proposed application area, but “UTI testing” does not describe a single assay. Specimen assessment, growth detection, organism identification and susceptibility testing answer different questions. Reader V1's endpoint scope begins after an eligible isolate and plate preparation; it does not currently establish a direct urine diagnosis.

Separate the proposed pathways

A laboratory may use an endpoint reader within its established isolate-based workflow. A future direct-from-urine research pathway would need additional evidence about specimen variability, organism burden, mixed populations, preparation and reference comparison. Research using selected isolates cannot establish performance across all urine specimens.

Table V.4. Urinary tract infection workflows

Pathway	Proposed role	Feasibility boundary
Isolate-based endpoint	Traceable zone measurement and review	Qualified disk-diffusion workflow.
Temporal imaging research	Study early visible growth or response	Independent future and reference outcomes.
Direct specimen research	Investigate reduced preparation delay	Specimen-level eligibility and failure accounting.
Result connectivity	Link laboratory findings to the authorized case	Correct identity and reviewed status.

Define the clock before discussing speed

A research target below two hours should state whether timing begins at specimen receipt, preparation completion, exposure or first observation. It should include all required steps within that claim and report the proportion of specimens producing an eligible result.

The product should preserve the laboratory's distinction between a biological observation and the clinical diagnosis made in context. A growth pattern or susceptibility result is not, by itself, a complete assessment of a patient's symptoms. The proposed benefit is better laboratory information and workflow traceability, with patient-level impact requiring separate clinical evaluation.

Women and access to timely diagnostics

Diagnostic access should be evaluated through the whole pathway

Bladder infections are more common in women than in men, making the accessibility of appropriate assessment and laboratory services a relevant design concern. NIDDK describes this difference in its clinical information; it does not establish an AMReye.AI benefit or a local prevalence estimate. [\[R62\]](#)

Design for practical access

The proposed platform should be evaluated in the context of specimen collection, transport, laboratory availability, result communication and follow-up. A rapid analytical step has limited value if a result is delayed in a queue or cannot be communicated securely to the authorized care team.

Table V.5. Women and access to timely diagnostics

Access consideration	Proposed product response
Travel and repeat visits	Study whether reliable reporting reduces avoidable return journeys.
Privacy	Limit displayed identifiers and protect authorized result access.
Language and understanding	Test clear instructions and explanations with intended users.
Uneven connectivity	Preserve eligible local work and reconcile later uploads.
Affordability	Assess total workflow cost, maintenance and service availability.

Avoid assuming the same pathway for everyone

Pregnancy, age, comorbidity and clinical presentation can change the relevant care pathway. The product should not collapse these contexts into a universal automated recommendation. Clinical decisions remain with qualified professionals using the appropriate evidence and guidance.

A future impact study should assess who receives an eligible result, who experiences delays and who requires repeat testing. Differences across groups should be investigated with adequate sample sizes and privacy safeguards. Equity is a measurable design objective: the platform should demonstrate that an improvement in average service does not conceal poorer access or performance for a less-represented group.

Blood culture and rapid AST pathways

Blood culture and rapid AST require a separate method pathway

Rapid susceptibility testing from positive blood cultures is distinct from routine endpoint disk-diffusion reading. EUCAST provides a dedicated RAST method with time-specific interpretation and quality-control material. Those conditions cannot be replaced by routine endpoint thresholds simply because the same antimicrobial appears in both workflows. [\[R04\]](#) [\[R05\]](#)

Proposed role of the platform

AMReye.AI could investigate controlled timed imaging, capture completeness, traceable reading and integration of a separately qualified rapid method. The initial research should define the positive-culture starting point, organism and drug scope, preparation process, reading times and comparator. Direct testing before culture positivity would be another intended use.

Table V.6. Blood culture and rapid AST pathways

Timing interval	Why it must remain visible
Specimen collection to positivity	May dominate the overall diagnostic timeline.
Positivity to method start	Includes preparation and operational delay.
Method start to eligible reading	The analytical interval of a timed rapid method.
Reading to reviewed communication	Determines when the information becomes available for use.

Sub-two-hour work remains a research target

A selected rapid research workflow may aim for an eligible result in less than two hours from a declared origin. The study must measure errors, abstention, organism–drug coverage and unsuccessful cases at that time. It should not imply that every bloodstream infection can receive a complete result within two hours of specimen collection.

Clinical value depends on how reviewed information enters care. A future study should examine communication and decision processes alongside analytical performance. The current compendium proposes the measurement and data foundations; it does not present a validated rapid blood-culture AST product.

Heteroresistance and population diversity

A population can contain more than one susceptibility phenotype

Heteroresistance describes a population in which a subpopulation exhibits greater resistance than the main population under the conditions being assessed. The term is used with differing operational definitions, so a study must state its organism, antimicrobial, population relationship and detection criteria. A single whole-population result may not describe that internal diversity adequately. [\[R07\]](#) [\[R08\]](#)

Why the distinction matters for observation

AMReye.AI's research interest is whether spatial and temporal records can help identify features that merit further investigation. Candidate features include localized growth, differences in emergence time and inconsistent responses across regions. These are observations or flags; none is a universal visual signature of heteroresistance.

Table V.7. Heteroresistance and population diversity

Evidentiary level	Appropriate statement
Image feature	A localized object or growth pattern was observed.
Expert assessment	The feature is compatible with a biological explanation requiring investigation.
Population evidence	A defined method supports susceptibility heterogeneity.
Mechanistic evidence	Additional work supports a particular explanation for the heterogeneity.
Clinical association	A study evaluates relevance to outcomes in a specified setting.

Retain population and isolate identity

Mixed species, contamination and within-isolate heterogeneity should not be conflated. The proposed record should preserve the level at which identity was established and whether a subpopulation was independently characterized. Repeated appearance alone does not prove a clonal relationship.

The product research opportunity is to make unusual observations visible, traceable and available for qualified follow-up. Demonstrating that opportunity requires paired image and confirmation records, an appropriate negative comparison group and accounting for missed features. The system should not use the heteroresistance label merely because an AI model finds a plate visually unusual.

Resistance tolerance and persistence

Resistance, tolerance and persistence describe different behavior

Resistance concerns growth at antimicrobial concentrations that inhibit a more susceptible comparator. Tolerance concerns prolonged survival during exposure without the corresponding increase in MIC. Persistence is a population phenomenon in which a subpopulation survives substantially better than the remainder. The consensus terminology emphasizes that persistence and a clinically persistent infection are not interchangeable concepts. [\[R10\]](#)

Table V.8. Resistance tolerance and persistence

Concept	Main distinction	What a still image cannot establish
Resistance	Ability to grow under a defined inhibitory challenge	The complete concentration–response relationship.
Heteroresistance	Resistance differs within the population	Frequency, identity and stability of the subpopulation.
Tolerance	Survival duration differs without increased MIC	Killing kinetics and subsequent recoverability.
Persistence	A surviving subpopulation produces heterogeneous survival	Whether survival reflects persistence or another mechanism.

Build different evidence objects

The proposed platform should avoid a single “abnormal growth” label that obscures the biological question. A temporal morphology record, a population susceptibility assessment and a survival study should retain separate methods and outputs. Their relationships can be investigated without merging them into one categorical result.

The interface can help by showing whether the evidence concerns growth, survival or a computational proxy. A fluorescent or metabolic signal also needs a defined relationship to the quantity being inferred. Absence of visible change is not sufficient to establish death or susceptibility.

For AMReye.AI development, these distinctions guide annotation and evaluation. Experts should label directly observable image features separately from confirmed biological classifications. That structure allows later models to be assessed against the right reference and prevents a convenient image label from becoming an unsupported mechanistic claim.

Why heteroresistance can be missed

Rare subpopulations create a sampling problem

A minor subpopulation may be absent from the material actually sampled, remain below a detection threshold or become visible only after the observation period. The measurement system must therefore consider both biological sampling and optical detectability. Improving camera resolution cannot recover a subpopulation that was never represented in the observed material.

An illustrative probability model

If each of n independently sampled cells has probability f of belonging to a subpopulation, the probability of sampling none is $(1 - f)^n$. For small f , this is approximately $\exp(-nf)$. This simplified calculation assumes independent, representative sampling; clumping, growth and selection can violate those assumptions.

Table V.9. Why heteroresistance can be missed

Limitation	Consequence for a detection study
Low representation	Include the sampling basis and an appropriate detection-limit analysis.
Delayed emergence	Evaluate observation timing and late-appearing features.
Population instability	Preserve handling history and time between linked assessments.
Limited field or focus	Map where an object could be missed optically.
Automated suppression	Test whether denoising or segmentation removes small real features.

The initial study question should be narrow

The proposed imaging study can ask whether a defined feature is detected when it is visibly present in a qualified image. A broader claim that the instrument detects heteroresistance in a specimen requires additional evidence about sampling, method suitability and confirmation.

A 2025 mechanistic study of an *Enterobacter cloacae* complex isolate linked transient gene amplification with variable resistance and growth behavior. That finding supports attention to changing population states, but it does not establish a universal detection time or a visual threshold for this platform. [\[R60\]](#)

Interpreting growth within inhibition zones

Growth within an inhibition zone is an observation to interpret in context

An apparent object inside a zone may be biological growth, contamination, a mixed population, glare, debris or an imaging artifact. Organism–antimicrobial-specific reading instructions can also determine how particular appearances should be handled. The system must use the applicable method guidance and expert review, rather than a universal “inner colony equals heteroresistance” rule. [\[R03\]](#)

Proposed observation record

The reader should preserve the object's location, size estimate, appearance, image quality and relationship to the proposed zone boundary. A flag should link to the original pixels. The label should describe the observable feature before proposing a biological interpretation.

A visible feature is the start of an investigation

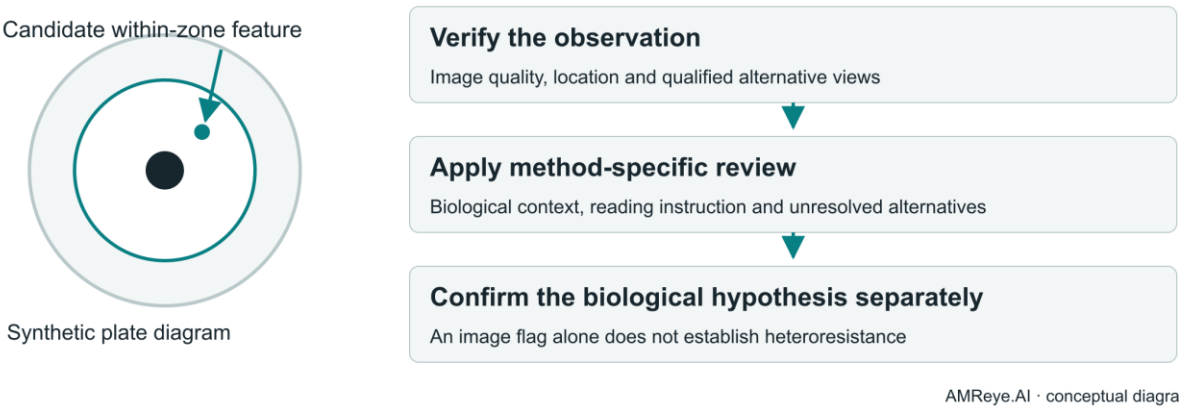


Figure V.1. Conceptual review sequence; all plate geometry is synthetic and no susceptibility result is implied.

Avoid erasing the finding through correction

A reviewer may alter a zone boundary without resolving the nature of an internal feature. These should be separate actions. The feature record should remain available even if the final diameter changes, with its review status and reason.

Evaluation should include visually similar artifacts and confirmed negative cases. Report false flags per plate as well as sensitivity for visible target features. Excessive false alerts can burden review and encourage dismissal of important findings. The desired contribution is reliable attention to an interpretable observation, followed by the laboratory's qualified assessment process.

Imaging contributions to heteroresistance research

Imaging can improve the record available for investigation

The proposed imaging contribution to heteroresistance research is richer spatial and temporal evidence. Repeated observations can show where a feature appeared, how its visible extent changed and whether it remained associated with the same plate location. These records may help experts formulate and test hypotheses about population behavior.

Separate feature detection from biological classification

An initial algorithm can be trained to identify small visible objects or local changes. Its reference should be expert-reviewed image annotations, with disagreement retained. A later biological classification model would require independently confirmed labels and a study designed for that claim.

Table V.10. Imaging contributions to heteroresistance research

Candidate imaging output	Proposed evaluation
Small-object localization	Detection and position error across the usable field.
Emergence-time estimate	Agreement with independently reviewed temporal records.
Growth trajectory	Measurement error and tracking continuity.
Spatial association	Robustness to plate motion and registration uncertainty.
Investigation flag	Confirmation yield, missed confirmed cases and review burden.

Additional modalities need an information test

Spectral, phase or metabolic channels might add complementary information, but their value cannot be assumed. A paired study should compare the combined model with a strong imaging baseline and assess whether gains persist on independent material. Shared confounders, such as condensation or altered illumination, should be challenged explicitly.

The platform should retain observations that the current model cannot classify. Those cases may be especially useful for expert-led research. A curated archive of uncertainty and discordance can support future learning, provided access and consent are appropriate and historical model predictions are not retrospectively treated as confirmed biological labels.

Confirmation and discordant results

Confirmation must answer the actual discrepancy

A suspected heteroresistance finding requires an appropriately designed laboratory investigation. Population analysis profiling is used in original heteroresistance studies, but methods and operational definitions vary. The compendium does not prescribe a universal confirmation protocol or replace organism-specific laboratory expertise. [\[R09\]](#) [\[R61\]](#)

Proposed evidence linkage

The platform should connect the initial image feature with the expert assessment and subsequent laboratory findings. It should preserve whether the follow-up used the same physical material, a related isolate or a later specimen. These relationships affect how a disagreement can be interpreted.

Table V.11. Confirmation and discordant results

Discrepancy	Proposed investigation record
Image flag without confirmation	Feature remains observed; biological classification unresolved or unsupported.
Confirmation without visible feature	Assess sampling, timing, field coverage and optical limits.
Repeated tests disagree	Preserve method conditions, handling history and population lineage.
Molecular and phenotypic evidence differ	Review assay scope and evidence identity without automatic override.
Identity is uncertain	Suspend cross-record conclusions until the linkage is resolved.

Adjudication should be independent

The expert process used to establish reference labels should not simply accept the AI flag as evidence of the target condition. Blinding where feasible, prespecified criteria and retention of unresolved cases improve interpretability. A consensus label should also retain whether it was unanimous or uncertain.

For validation, every eligible case should be accounted for, including missing confirmation and failed follow-up. Treating only successfully confirmed positive flags as the study population would exaggerate usefulness. The proposed product should make this accounting straightforward through explicit investigation states and links, while leaving the biological decision to qualified personnel and the applicable method.

Longitudinal heteroresistance observations

Longitudinal records preserve changes that a single endpoint can hide

Population composition and visible behavior can change over time. A longitudinal research record should preserve observation timing, handling events and the relationship among repeated assessments. It should not assume that a phenotype measured after additional processing represents the earlier state without qualification.

Follow the same object when possible

Tracking a visible colony or region requires stable identity through movement, merging and occlusion. When identity is lost, the trajectory should become uncertain or terminate. Automatically joining the nearest objects can create a convincing but incorrect history.

Table V.12. Longitudinal heteroresistance observations

Longitudinal element	Required distinction
Same plate, later image	Repeated observation of one physical object.
Derived material	A new biological object linked by lineage.
Repeated assay	A separate experiment with its own conditions.
Later specimen	A different sampling event, even for the same patient.
Simulated continuation	A conditional model output with no new observation.

Use temporal evidence to refine questions

Researchers can examine whether a feature emerges consistently, whether its appearance depends on the observation window and whether alternative imaging channels improve detection. Those analyses should preserve unobserved intervals and uncertainty in onset time. A feature first visible at one image may have appeared at any point since the preceding eligible observation.

The proposed twin can summarize trajectories and test forecasting hypotheses, but it should not complete missing biological history with synthetic labels. Longitudinal heteroresistance research is most valuable when the platform makes uncertain lineage, timing and measurement visible enough for experts to evaluate competing explanations.

Heteroresistance research priorities

Prioritize evidence that establishes useful detection boundaries

The heteroresistance programme should begin with a clearly defined observation task and progress toward independently confirmed biological questions. The aim is to determine what the platform adds, where it fails and what follow-up remains necessary. A broad promise to detect all resistant subpopulations would be unsupported.

Table V.13. Heteroresistance research priorities

Priority question	Proposed study output
Which visible features are reproducibly identifiable?	Expert agreement and image-quality-dependent detection limits.
Does time-lapse observation add information?	Paired comparison with a single eligible endpoint image.
Do extra modalities help?	Independent modality-ablation results and added review burden.
Which flags are biologically informative?	Confirmation-linked performance with complete case accounting.
Does the workflow improve practice?	Prospective operational and, where justified, clinical evaluation.

Clinical relevance is a separate evidence question

Band and colleagues demonstrated treatment failure associated with resistant subpopulations in a murine infection model. A 2025 retrospective ECC study did not find significant increases in hospitalization duration or disease progression in the outcomes it assessed. These findings concern different designs and populations; neither justifies a universal prediction for an individual patient. [\[R08\]](#) [\[R61\]](#)

The programme should retain negative and discordant findings and avoid selecting only striking images. Suitable collaborations would combine microbiology, imaging metrology, biostatistics and data governance. No such partnership or study completion is implied here.

A successful early milestone would be a transparent, reproducible assessment of a bounded feature-detection task with an independently linked confirmation pathway. That result could justify further research into broader biological inference. It would still require separate evidence before supporting clinical reporting or claims about improved patient outcomes.

Molecular result integration

Molecular findings should remain linked to the assay that produced them

The proposed platform can integrate authorized molecular results alongside phenotypic observations. Integration means preserving identity and meaning across systems. It does not imply that Reader V1 performs molecular testing or that an image-derived hypothesis is equivalent to a measured molecular result.

A molecular result needs context

The incoming record should identify the specimen or isolate, assay, target, result state, quality status and reporting laboratory. For sequence-derived results, the pipeline and reference database versions also matter. An absent target, an assay failure and an untested target must remain distinguishable.

[\[R24\]](#) [\[R25\]](#)

Table V.14. Molecular result integration

Integration element	Proposed requirement
Identity	Establish the permitted relationship between molecular and phenotypic material.
Assay scope	Record which targets and variants the method can assess.
Result semantics	Preserve detected, not detected, indeterminate, failed and not tested states.
Provenance	Retain the originating system, version and reviewed status.
Reconciliation	Escalate discrepancies through a qualified review path.

Validate the interface as well as the biology

An accurate external assay can be misrepresented by an incorrect mapping. Integration tests should challenge target synonyms, organism names, units, amended reports and duplicate transmissions. The receiving record should retain the original source result so that a mapping error can be investigated.

The proposed user experience places complementary evidence together while showing its origin and limitations. A combined summary should not erase a discordance. Any future model that uses molecular data to improve a phenotypic prediction needs its own evaluation, including performance when molecular data are absent, incomplete or outside the model's training domain.

MRSA related molecular information

MRSA-related information requires organism and assay specificity

MRSA-related workflows may involve phenotypic testing, molecular detection or protein-based evidence. CDC describes the relevance of *mecA* and PBP2a and notes that a *mecA*-only assay does not detect every alternative mechanism, including *mecC*. This illustrates why an integrated platform must preserve the scope of the actual assay rather than treating every negative molecular result as equivalent. [\[R63\]](#)

Proposed product role

Reader V1 may investigate eligible endpoint measurement within a qualified staphylococcal workflow. Molecular-result integration can place an authorized external finding beside that measurement. These functions should retain separate method identities and be interpreted using the applicable current standards and laboratory rules.

Table V.15. MRSA related molecular information

Evidence object	Information to preserve
Phenotypic observation	Organism, method, measured value, quality and interpretation version.
Molecular assay	Target coverage, result state, assay identity and specimen lineage.
Protein-based assay	Method, target, quality status and originating result.
Discrepancy review	Competing findings, expert decision and any follow-up.

Avoid a broad image-to-mechanism claim

A model may identify a visual feature associated with a particular laboratory result in a research dataset. That association does not establish a molecular mechanism in a new isolate. The platform should label such outputs as research predictions and retain uncertainty and domain limits.

Current clinical interpretation must come from controlled standards content, not an unversioned web summary. The CDC page's general testing explanations are useful context, but its numerical tables are not used here as a current breakpoint source. This separation prevents historical or typographically imperfect reference material from becoming executable clinical rules. [\[R01\]](#) [\[R02\]](#)

Phenotype genotype research

Phenotype–genotype research needs paired evidence

The proposed research service can investigate relationships between measured phenotypes and sequence-derived features. A useful dataset requires correct pairing, adequate assay quality and clear definitions of both sides of the relationship. Apparent predictive success can arise from shared lineage, site or batch information rather than a transferable biological association.

Design the prediction direction explicitly

Predicting a phenotype from sequence data differs from generating a hypothesis about a mechanism from an image. Each direction needs its own target, reference and limitations. A phenotype-derived gene hypothesis remains computational until molecular evidence independently supports it.

Table V.16. Phenotype genotype research

Research element	Proposed control
Pairing	Trace the phenotype and sequence to the intended biological material.
Reference quality	Retain assay limits, reviewed status and unresolved discrepancies.
Population structure	Separate related isolates appropriately during evaluation.
Database dependence	Preserve reference releases and annotation methods.
Missing mechanisms	Test performance when the relevant biology is poorly represented.

Use repository knowledge selectively

Curated resources can provide resistance-model definitions and sequence context. They should be accessed under appropriate terms, with versioned retrieval and a documented role in the analysis. A large repository does not guarantee representative labels for the intended clinical or geographic setting. [\[R24\]](#) [\[R25\]](#)

The first research outputs should emphasize reproducible associations and their failure boundaries. External validation should challenge lineage and site transfer, not merely random splits of closely related isolates. Claims about improved treatment decisions would require additional studies beyond predictive agreement in a retrospective dataset.

Microscopy and cell level observations

Cell-level observation is a distinct instrument family

Microscopy can reveal cellular morphology and dynamic behavior at a scale different from whole-plate imaging. The proposed imaging family may investigate bright-field, phase, fluorescence or other qualified modalities. Each requires appropriate sample presentation, optical resolution, focus control and a clearly defined biological output.

Scale changes the engineering problem

A whole-plate reader prioritizes field coverage and calibrated spatial measurement. Cell-level imaging may require a much smaller field, more demanding focus stability and different acquisition timing. A sensor suitable for plate presence or coarse distance measurement cannot be assumed to resolve cellular morphology.

Table V.17. Microscopy and cell level observations

Candidate output	Evidence requirement
Cell or object count	Detection and separation accuracy against an appropriate reference.
Morphology measurement	Calibrated geometry and robust segmentation.
Temporal cell behavior	Tracking identity and timing accuracy.
Fluorescence-derived feature	Assay-specific signal interpretation and photophysical controls.
Early response prediction	Independent biological reference and time-defined evaluation.

Learn from external demonstrations without importing performance

Original rapid-AST research has used single-cell observations in tightly defined experimental systems. Such work motivates investigation, but its reported timing and accuracy belong to the original method and study population. It does not establish that a different AMReye.AI optical configuration will reproduce those results. [\[R11\]](#)

The proposed development path should first establish image and measurement quality on the intended sample format. Biological inference follows as a separate stage. Microscopy data can enrich the platform's research records, while integration with endpoint plate results requires explicit biological lineage and independent evidence that the two observations answer a compatible question.

Robotics for repeatable observation

Robotics should improve repeatability without compromising identity

Robotic handling can position plates, schedule observations and reduce repetitive manual movement. Its value depends on reliable physical handling, traceable custody and recovery from failure. A faster transfer mechanism is not useful if it increases plate association errors, disturbs the sample or complicates cleaning.

Proposed automation functions

Candidate functions include loading, positioning, imaging access, queue management and controlled transfer between qualified stations. More invasive operations, including liquid handling or specimen preparation, belong to separately evaluated workflows with additional safety and contamination requirements.

Table V.18. Robotics for repeatable observation

Function	Engineering question
Plate pickup and placement	Is the permitted vessel range handled without damage or unstable pose?
Identification	Is identity verified at the point where a mix-up could occur?
Positioning	Does repeatability support the measurement's calibration assumptions?
Interlocks	Can a hazardous or unsupported movement be prevented?
Recovery	Can staff locate and reconcile every physical object after interruption?

Human access remains part of the design

Operators need clear indications of instrument state and a safe, documented recovery path. Maintenance should not require bypassing safeguards during normal operation. The system should preserve the distinction between a commanded movement and a confirmed completed transfer.

Validation should include vessel variation, misloading, obstruction, interrupted power and failed identity reads. The proposed acceptance criteria should cover successful throughput and the behavior of unsuccessful transfers. Robotics is an evidence-gated product family; it should not be implied as an installed capability of the simpler Reader V1 reference configuration.

Food and dairy applications

Food and dairy work requires sector-specific methods

The proposed platform may support research and documentation in food and dairy microbiology, including imaging of eligible culture formats and traceable observation records. The relevant decisions differ from clinical susceptibility reporting. A clinical breakpoint or a patient-oriented interpretation should not be transferred to a food-quality decision without an appropriate basis.

Define the material and question

Milk, processed food, production-environment samples and reference cultures can create different matrices and workflows. Imaging may be used to document visible growth or assist a qualified counting task, while the accepted method determines what the observation means for the sample.

Table V.19. Food and dairy applications

Proposed use	Feasibility boundary
Culture documentation	Demonstrate readable records for the intended media and vessels.
Counting assistance	Validate detection, overlap handling and reporting range.
Research susceptibility records	Preserve organism, method and sector-appropriate interpretation.
Production trend analysis	Account for sampling plans, process changes and missing data.
Training	Use clearly labelled examples and qualified reference assessments.

Quality decisions need qualified evidence

An image-derived count is not automatically equivalent to a recognized method result. Performance should be established against the appropriate reference, including difficult matrices and low or high count conditions. Sampling and preparation remain part of the complete method even when the reader performs only the observation step.

The service proposition may include calibration, workflow integration, record retention and competency support. Any claim concerning product release, shelf life or regulatory conformity needs its own method and quality-system assessment. The platform's common data infrastructure can support that work while keeping clinical and food-sector interpretations separate.

Water and environmental surveillance

Environmental observation should retain sampling context

Water and environmental surveillance are proposed research applications for traceable culture imaging and data integration. Interpretation depends on where, when and how a sample was collected, the method used and the decision being considered. A plate image alone cannot establish that drinking water is safe or identify the source of contamination.

Proposed environmental record

The record should include sampling location at an appropriate privacy and security level, material type, collection event, transport context and analytical method. Repeated observations should distinguish a new sample from a repeated measurement of the same material.

Table V.20. Water and environmental surveillance

Application question	Required context
Is an observed count changing?	Consistent sampling and method definitions over time.
Are resistance patterns different across sites?	Comparable organism and method scope with sampling denominators.
Is a local feature unusual?	Relevant baseline and confirmation pathway.
Can results be combined regionally?	Shared metadata, units and data-use permissions.
Does a signal require investigation?	Authorized public-health or environmental decision process.

Avoid overinterpreting a map

Spatial dashboards can make patterns visible, but sampling intensity and site selection can create apparent hotspots. The proposed analytics should show denominators, coverage and missing areas. An untested location should not appear equivalent to a tested location with no detected finding.

Future work could connect environmental, animal and human research records through a One Health framework while preserving their different methods. Such integration should support hypothesis generation and coordinated investigation. It does not demonstrate transmission or a causal pathway without additional evidence and appropriate expert analysis.

Veterinary and agricultural research

Veterinary and agricultural applications need their own interpretation domain

Animal health and agricultural microbiology offer proposed opportunities for imaging, data management and research services. Species, specimen context, antimicrobial use and accepted interpretation can differ from human clinical work. A common organism name is insufficient reason to reuse a human clinical rule without checking applicability.

Proposed areas of investigation

The platform could support qualified veterinary laboratory records, research on isolate populations and comparisons across authorized One Health studies. Agricultural applications may involve environmental or production-system questions that require different sampling and analytical methods.

Table V.21. Veterinary and agricultural research

Product function	Domain-specific requirement
Endpoint observation	Confirm method, media and vessel suitability.
Interpretation	Use the applicable veterinary or research framework.
Identity and metadata	Preserve host, specimen and sampling context at an appropriate level.
Comparative research	Account for population and exposure differences.
Reporting	Separate laboratory findings from treatment or production decisions.

Cross-domain evidence can inform research

A 2025 study of a veterinary clinical *Enterobacter cloacae* complex isolate demonstrated a transient gene-amplification mechanism associated with heteroresistance. Its origin illustrates the relevance of population-level questions beyond human clinical laboratories. The study does not establish prevalence or platform performance across animal species. [\[R60\]](#)

The proposed service should work with qualified domain experts to define study endpoints and reference methods. Shared infrastructure can reduce duplication in record handling and calibration, but each new application needs a documented eligibility domain. Expansion should follow evidence about the intended material and decision, rather than treating One Health as permission to generalize every clinical claim.

Pharmaceutical quality research

Pharmaceutical quality research demands a controlled measurement chain

The platform may investigate imaging and data-management support for pharmaceutical microbiology research and method development. Potential functions include documenting eligible culture observations, assessing imaging repeatability and preserving analytical review. These proposals do not establish a qualified sterility test, a batch-release method or compliance with a pharmacopoeial requirement.

Define the proposed role precisely

An observation aid may improve record retrieval without changing the accepted analytical method. A new detection algorithm or earlier endpoint can change that method's performance and requires a separate qualification strategy. Absence of visible growth within a shorter period cannot automatically support an absence-of-contamination claim.

Table V.22. Pharmaceutical quality research

Proposed function	Evidence needed
Image archive	Identity, completeness, integrity and authorized retrieval.
Measurement assistance	Accuracy and precision for the defined observation.
Automated flagging	Sensitivity, false alarms and handling of difficult appearances.
Earlier research endpoint	Independent evidence across the relevant challenge domain.
Quality-system integration	Controlled review, change management and validated interfaces.

Record integrity is necessary but insufficient

An immutable audit trail can establish what happened in the software. It cannot establish that the biological method was suitable. Both analytical performance and data integrity need assessment appropriate to the intended quality decision.

A practical first collaboration could evaluate a bounded documentation or measurement function within an existing qualified workflow. The study should include deviations, failed acquisitions and reviewer corrections. Any subsequent proposal to alter a regulated release method would require specialist quality and regulatory assessment, with the relevant standards and pharmacopoeial requirements verified at that time.

Teaching and competency development

Teaching tools should make measurement reasoning visible

The proposed education family can use curated images, synthetic demonstrations and structured review exercises to teach observation, calibration and evidence interpretation. The aim is to help learners understand why a result is supported, uncertain or unavailable. A teaching score should not be confused with authorization to perform clinical work independently.

Candidate learning activities

Learners could compare boundary choices, identify image-quality problems, inspect the effect of calibration and distinguish measured values from predictions. Heteroresistance examples should separate visible features from confirmed biological labels. Demonstrations should disclose whether an image is experimental, synthetic or illustrative.

Table V.23. Teaching and competency development

Learning objective	Proposed assessment
Spatial measurement	Correct use of units and calibration context.
Quality assessment	Recognition of unreadable or compromised observations.
Interpretation	Appropriate use of a supplied, versioned method context.
Evidence discipline	Distinguishing observation, inference and confirmation.
Record handling	Correct identity, correction and audit behavior.

Preserve expert disagreement where it is informative

Some image boundaries are ambiguous. A teaching collection can show a justified range or explain why the case should be escalated. Presenting every example as having a single obvious numerical answer can teach false certainty.

The service may include competency support for laboratories, but formal authorization remains with the responsible institution and its qualified process. Training effectiveness should be measured through retained understanding and performance on independent examples. Completion of a module or familiarity with an interface alone does not establish competence for every organism, method or clinical context.

Hospital stewardship interfaces

Stewardship interfaces should deliver reviewed information clearly

The proposed platform can support antimicrobial stewardship by making qualified laboratory results available with their timing, method and review status. It should help the authorized team understand what evidence exists and what remains pending. It does not replace clinical judgment or independently select treatment.

Connect the result to its useful context

A stewardship-facing view may need specimen and isolate identity, collection and reporting times, organism identification, reviewed susceptibility findings and significant amendments. The amount of patient information exposed should be limited to the authorized purpose and role.

Table V.24. Hospital stewardship interfaces

Interface function	Proposed requirement
Result availability	Distinguish preliminary, reviewed, amended and unavailable states.
Timeliness	Preserve the interval from relevant workflow events to communication.
Alerts	Use an authorized policy with escalation and acknowledgment.
Amendments	Make corrected results visible without erasing the original report.
Aggregate review	Provide denominators, eligibility and method context.

Measure whether information reaches a decision

A shorter analytical interval is one part of a longer pathway. A future study should assess whether reviewed information was received, understood and used appropriately by the intended team. Alert volume, missed acknowledgments and after-hours workflow can affect practical value.

The platform should not infer that every delayed action is inappropriate or that a model can judge treatment quality from a laboratory record alone. Clinical context may justify different decisions. Proposed analytics should therefore support expert review of the process, with carefully defined endpoints and access controls. Claims about reduced antimicrobial use, improved outcomes or shorter hospitalization require studies designed to assess those outcomes and their confounders.

Public health surveillance interfaces

Surveillance needs consistent definitions and complete denominators

Public-health surveillance combines laboratory observations across time and place. A proposed AMReye.AI interface should preserve method, organism, interpretation version and sampling context so that aggregates remain interpretable. Changes in testing practice can create an apparent trend even when the underlying population has not changed.

Prepare reviewed data for authorized reporting

The export should distinguish patient, specimen, isolate and test records and apply the receiving programme's approved deduplication and inclusion rules. A software default should not silently decide which repeated isolates count in a surveillance denominator. NCDC provides programme-specific surveillance and reporting resources that require controlled implementation. [\[R33\]](#)

Table V.25. Public health surveillance interfaces

Data issue	Proposed handling
Repeated isolates	Apply a documented rule and preserve the original records.
Changed breakpoints	Retain rule versions and assess comparability across periods.
Missing tests	Distinguish not tested from tested with no detected finding.
Site coverage	Show reporting completeness and participating-site scope.
Amended results	Reconcile corrections through the authorized reporting process.

Quality is more than a larger dataset

Automated aggregation can make errors propagate quickly. The platform should support validation of mappings, exception review and traceability from a summary back to authorized source records. Public outputs should use appropriate aggregation and disclosure controls.

The proposed service could reduce repetitive data preparation and make quality problems easier to investigate. It should not claim participation in a government network, automatic approval of its data or national representativeness without verified agreements and evidence. Surveillance utility should be assessed against the receiving programme's actual requirements and the completeness of the contributing laboratory data.

India and diagnostic access

India provides a relevant setting for practical diagnostic engineering

India's National Action Plan on Antimicrobial Resistance 2.0 covers 2025–2029 and adopts coordinated action across sectors. Laboratory capacity, surveillance, responsible antimicrobial use and research are relevant areas in which a carefully evaluated platform could contribute. Alignment with a national priority does not imply government endorsement or a procurement commitment. [\[R34\]](#)

Design around operational diversity

The proposed product should be evaluated across the laboratory conditions it intends to serve. Power quality, connectivity, staffing, environmental conditions, workload and access to service can vary. A design suitable for one well-supported site may require different operational provisions elsewhere.

Table V.26. India and diagnostic access

Indian deployment consideration	Proposed response
Service reach	Replaceable modules, clear diagnostics and qualified local support.
Connectivity	Eligible local capture and review with controlled later synchronization.
Training	Practical competency materials and evaluated language support.
Procurement	INR cost visibility, supplier qualification and traceable substitutions.
Data exchange	Interfaces matched to authorized laboratory and surveillance workflows.

Build evidence locally without assuming uniformity

Indian evaluation should include relevant clinical material, laboratory practices and intended users. ICMR's AMR diagnostic evaluation guidance provides a national reference for planning appropriate evidence; the exact study must follow the intended product claim and current requirements. [\[R35\]](#)

The long-term contribution could include domestic engineering capability, maintainable instruments, high-quality data and research collaboration. These benefits should be measured through reliability, qualified workflow performance and service outcomes. Local manufacture is valuable when it improves quality, availability or sustainability, not merely when an imported component is replaced by a custom part.

Goa and a regional research pathway

Goa can be a focused setting for a regional research pathway

Goa has relevant public-health and laboratory infrastructure. NCDC lists Goa Medical College and Hospital in its AMR programme material, and the Directorate of Health Services describes a regional laboratory at the college's Department of Microbiology within its surveillance context. Goa's government also documented an FDA-led AMR awareness programme in November 2023. These are contextual facts, not AMReye.AI partnerships. [\[R33\]](#) [\[R36\]](#) [\[R37\]](#)

A proposed regional programme

A bounded research pathway could bring together laboratory users, engineering expertise and data-governance support to evaluate endpoint observation and review. The first objective should be a well-defined measurement study, followed by usability and operational assessment where justified. Any collaboration would require the institutions' own agreement and approvals.

Table V.27. Goa and a regional research pathway

Proposed contribution	Evidence of value
Laboratory measurement	Independent agreement, repeatability and failure accounting.
Technical education	Demonstrated understanding of calibration and evidence boundaries.
Service development	Documented response, repairability and calibration continuity.
Regional research	Reproducible datasets with appropriate permissions and provenance.
Wider transfer	Evaluation beyond the initial local conditions.

Connect regional work to broader relevance

Goa-based development could provide a manageable environment for close interaction between instrument engineers and laboratory users. Its wider value would depend on whether findings transfer to different workloads, climates, populations and service settings. A successful local study should therefore define what was tested and what remains outside its scope.

The intended contribution is scientific and practical: better observation records, a stronger measurement culture and evidence that can support responsible product development. No institution, deployment, endorsement or clinical outcome is claimed as secured by this compendium.

One Health data integration

One Health integration should preserve differences between sectors

WHO describes One Health as an integrated approach to the linked health of people, animals, plants and ecosystems. For AMReye.AI, the proposed relevance is a common foundation for traceable observations and authorized research across those domains. A shared platform should preserve the different sampling and interpretation methods used by each sector. [\[R65\]](#)

A shared schema with domain-specific meaning

Common elements can include sample identity, location at an appropriate granularity, time, organism, method, measured value and provenance. Sector-specific extensions should capture host, material, sampling design and the applicable interpretation framework. Harmonization should not force unlike results into a misleading common category.

Table V.28. One Health data integration

Integration layer	Proposed approach
Identity	Preserve source-system and biological lineage without unnecessary personal data.
Measurement	Retain units, method and calibration context.
Interpretation	Keep human, veterinary and research frameworks explicitly distinct.
Geography and time	Use appropriate resolution and acknowledge sampling gaps.
Governance	Define permitted purposes, access and onward sharing.

Correlation is a starting point for investigation

Similar patterns across sectors may reflect shared pressures, sampling practices or genuine biological relationships. A dashboard cannot distinguish these explanations by itself. Proposed analyses should account for coverage and population structure and identify the additional evidence needed for causal or transmission claims.

The product opportunity includes interoperable records, coordinated research tools and reusable quality controls. Its humanitarian value depends on whether those tools improve understanding and action while respecting local governance. Universal access to every repository is neither necessary nor assumed; curated, licensed and purpose-limited data use is the proposed strategy.

Global partnership and transferability

Global relevance depends on transferability and local ownership

AMR is a global challenge, but laboratories differ in their resources, methods, populations and regulatory settings. A platform developed in India can be relevant elsewhere if its measurement and service evidence transfer to the intended setting. Exporting a device does not automatically export the validity of every associated model or interpretation.

Proposed partnership model

Technical and laboratory partners could contribute domain evaluation, workflow expertise, calibration support or authorized research data. Each collaboration should define responsibilities, access rights, intellectual-property terms and the evidence question. The product should remain usable by its intended laboratory even when a research collaboration ends.

Table V.29. Global partnership and transferability

Transfer requirement	Proposed evidence
Measurement domain	Performance on relevant vessels, media and optical conditions.
Clinical scope	Applicable intended use, standards and regulatory assessment.
User context	Evaluated language, training and review behavior.
Service	Sustainable maintenance, spares and calibration arrangements.
Data governance	Lawful cross-border access and local institutional control.

Avoid extracting data without returning value

Research partnerships should specify what participating laboratories receive, such as analytical support, quality insights, training or co-developed methods. Data access should follow a defined purpose and agreement, not an assumption that central AI development justifies unrestricted collection.

Global performance claims should be based on appropriately diverse evidence and disclose important limitations. A model may need local qualification or remain restricted to research in a new setting. This approach supports ambitious collaboration while keeping the laboratory's operational needs and responsibilities central to deployment decisions.

Humanitarian design and equitable access

Humanitarian value requires dependable operation and fair access

The proposed platform's humanitarian objective is to improve the availability and usefulness of reliable laboratory information. Achieving that objective requires more than a lower component price or a faster algorithm. Maintenance, training, consumables, data access and recovery from failure determine whether a laboratory can depend on the system over time.

Design for the full service life

The instrument should support repair, understandable fault messages and a practical calibration path. Software should preserve access to authorized records and documented exports. A laboratory should not lose its historical evidence merely because a service contract or network connection changes.

Table V.30. Humanitarian design and equitable access

Design objective	Proposed measure
Reliability	Uptime within the qualified environment and complete failure accounting.
Repairability	Time, tools and qualified effort needed to restore service.
Affordability	Total cost of ownership for the actual workflow.
Accessibility	Successful use by the intended staff with evaluated support.
Continuity	Record recovery and safe operation during defined interruptions.

Evaluate distribution, not only averages

An improvement should be examined across sites and user groups. A system that performs well only with ideal infrastructure may widen service differences. Studies should therefore identify conditions under which the product is unavailable, requires additional support or imposes excessive repeat work.

No claim that AMReye.AI has already reduced mortality, improved equity or transformed healthcare is made here. Those are potential downstream goals that require evidence. The immediate engineering responsibility is to produce a trustworthy, maintainable measurement and information system whose benefits and limitations can be assessed by the people expected to use it.

Measuring application impact

Impact should be measured at the level of the claim

The platform's impact framework should connect technical outputs to operational and, where justified, clinical outcomes. These are related but distinct levels of evidence. Better image quality can support better measurement, but the subsequent effects on reporting and patient care must be evaluated rather than assumed.

Table V.31. Measuring application impact

Impact level	Candidate metric	Important limitation
Measurement	Bias, agreement and uncertainty	Does not establish clinical outcome benefit.
Workflow	Reviewed turnaround, repeats and staff effort	Depends on local process and workload.
Service	Uptime, repair time and calibration continuity	Requires representative field observation.
Research	Reproducibility and useful confirmed findings	More data alone is not scientific progress.
Clinical or public health	Prespecified outcome in an appropriate study	Requires control of confounding and adequate evidence.

Establish the baseline before deployment

A prospective evaluation should define how the existing workflow is measured and how changes will be attributed. Seasonal workload, staffing and concurrent interventions can affect results. The analysis should include eligible cases that fail, abstain or require a repeat, because these influence real service performance.

For each metric, record the denominator, time window, data source and uncertainty. A reduction in average review time should be accompanied by information about difficult cases and tail delays. An apparent reduction in errors should identify which error class was assessed and whether the reference process was independent.

The proposed compendium's applications are a research and product agenda. Progress should be reported as specific verified increments: a qualified measurement, a reliable interface, a successful service process or a demonstrated outcome within its study domain. This makes the broader vision accountable to evidence without restricting future exploration to the initial Reader V1 configuration.

VI / VALIDATION SAFETY AND SECURITY

Evidence required before a product claim

A product claim needs evidence matched to its meaning

Validation begins with a precise claim. “Measures an eligible endpoint zone diameter” and “detects resistance within two hours” require different evidence. The claim should identify the material, method, users, environment, output and decision. Its wording should not exceed the evaluated domain.

Proposed evidence ladder

Engineering verification establishes whether specified functions work. Analytical evaluation addresses measurement performance. Workflow and usability studies assess the complete interaction with intended users. Clinical performance or outcome claims require the corresponding study and regulatory assessment. A successful demonstration cannot stand in for all these levels. [\[R26\]](#) [\[R35\]](#)

Table VI.1. Evidence required before a product claim

Claim type	Minimum evidence question
Capture capability	Does the system acquire complete, readable records under specified conditions?
Measurement capability	How do values agree with an appropriate independent reference?
Interpretation capability	Are rules, scope and corrections implemented correctly?
Workflow benefit	Does the complete reviewed process improve the stated operational metric?
Predictive or clinical benefit	Does an appropriately designed independent study support the exact outcome?

Make evidence maturity visible

The proposed product record should distinguish demonstrated functions, qualified functions, proposed engineering and future research. An external paper can establish that an approach is scientifically plausible, but its results belong to that study. Historical prototypes and software demonstrations also need their own scope and verification records.

Before external use, each claim should have an owner, supporting evidence, limitations and a review date. When the product, method or intended use changes, the evidence relationship should be reassessed. This discipline allows ambitious research to remain visible while keeping the released product description accurate.

Intended use and risk analysis

Risk analysis connects failure to the decision it could affect

The proposed risk process should examine hazards arising from incorrect measurement, identity errors, misleading interpretation, unavailable service and unsafe instrument behavior. It should consider normal use, foreseeable misuse and combinations of failures. A low average error rate does not resolve a rare but consequential failure mode. [\[R51\]](#)

Start with a concrete sequence

For example, an incorrect plate association can lead to a technically accurate measurement being attached to the wrong case. Better segmentation would not mitigate that hazard. Identity verification, review context and export checks address different points in the sequence.

Table VI.2. Intended use and risk analysis

Failure sequence	Candidate control	Verification focus
Wrong identity → wrong case result	Positive association and mismatch handling	Realistic misidentification scenarios.
Poor image → incorrect boundary	Quality checks and abstention	Difficult appearances and false acceptance.
Wrong rules → wrong category	Versioned rules and eligibility checks	Scope, corrections and regression cases.
Changed result → stale approval	Dependent approval invalidation	Correction and amendment behavior.

Controls need evidence of effectiveness

A warning is useful only if the intended user notices and understands it in time. An interlock needs testing under the failure conditions it is meant to prevent. The assessment should also consider new risks introduced by the control, such as excessive false alarms or a recovery path that encourages bypassing checks.

The risk record should link hazards, controls, verification and residual-risk decisions. It should evolve with field findings and product changes. This compendium proposes the process and relevant failure classes; it does not claim that a complete product risk-management file or acceptable residual-risk determination already exists.

VI / VALIDATION SAFETY AND SECURITY

Measurement validation design

Measurement validation needs a prespecified comparison

A zone-measurement study should define the measurand, reference process, eligibility criteria and analysis before examining the final results. The measurand includes the physical plane, boundary convention, units and method context. Without these definitions, two accurate observers may appear to disagree because they are measuring different features.

Proposed study structure

The study should include representative plate appearances and independent reference assessment. It should separate development data from evaluation data and account for repeated measurements within plates or isolates. Reference uncertainty and reviewer disagreement should be retained rather than treated as zero.

Table VI.3. Measurement validation design

Design element	Required specification
Population	Organisms, antimicrobials, media and relevant plate appearances.
Measurement	Boundary convention, calibration and reporting resolution.
Reference	Qualified method, assessors and adjudication process.
Endpoints	Bias, agreement, uncertainty and clinically relevant error classes.
Exclusions	Prespecified reasons with complete counts and outcomes.
Analysis	Grouping, confidence intervals and handling of missing values.

Challenge the useful range

The dataset should include measurements near interpretation boundaries, irregular zones and other difficult conditions within the proposed scope. A collection dominated by easy, clearly separated cases may conceal performance weaknesses that matter in practice.

Acceptance criteria should be justified against the intended method and decision, with relevant national evaluation guidance and standards consulted. They should not be copied from an unrelated instrument or selected after viewing the results. The proposed evaluation should report both numerical measurement behavior and the consequences of that behavior in the intended reviewed workflow.

[\[R35\]](#)

Reference methods and adjudication

The reference process also has uncertainty

An independent reference provides the basis for assessing a proposed measurement or classification. Its suitability depends on the question. A manual zone reading can assess a zone measurement, while an MIC prediction requires an appropriate MIC reference. A software model cannot validate itself by comparing with labels generated by the same model.

Proposed reference governance

Reference assessors should use a defined method and have appropriate competency. Blinding to the investigational output should be used where feasible. Disagreements should follow a prespecified adjudication process that retains the original readings and reasons for resolution.

Table VI.4. Reference methods and adjudication

Reference issue	Proposed treatment
Inter-observer variation	Quantify and retain individual readings.
Ambiguous boundary	Apply the method's reading guidance and preserve uncertainty.
Conflicting methods	Investigate scope and conditions before assigning a final label.
Missing reference	Keep the case in the accounting with an explicit status.
Adjudication bias	Avoid allowing the investigational prediction to determine its own reference.

A consensus is a process, not an infallible truth

Consensus can improve consistency, but it may still contain systematic error. Where possible, the study should examine reference repeatability and use independent checks appropriate to the measurand. Difficult cases should not be discarded solely because experts disagree.

The proposed data model should store reference values separately from routine reviewer corrections. A correction made during product use may be useful training material after assessment, but it is not automatically a research reference label. Keeping these roles distinct supports honest evaluation and makes later reanalysis possible when a reference decision is revised.

VI / VALIDATION SAFETY AND SECURITY

Dataset representativeness

Representativeness is a property of the intended domain

A validation collection should reflect the conditions under which the product is expected to work. Relevant dimensions include organisms, antimicrobial combinations, media, plate geometry, image quality, sites, users and clinically important result ranges. A large dataset can still be narrow if most observations come from similar cases.

Build a coverage matrix

The proposed study should map intended-use dimensions to available cases and identify gaps before model training and final evaluation. Rare but consequential conditions may need deliberate enrichment. Their prevalence in the evaluation collection should be reported so that overall metrics are interpreted correctly.

Table VI.5. Dataset representativeness

Coverage dimension	Example question
Biological	Are relevant organisms and difficult phenotypes represented?
Method	Are permitted media, disks and reading conditions covered?
Instrument	Are manufacturing and calibration variations represented?
Site and user	Does the collection reflect intended operational diversity?
Result range	Are boundary cases and consequential categories sufficiently assessed?

More images are not always more independent evidence

Many images of one plate can help study temporal behavior or acquisition repeatability. They do not provide the same population evidence as the same number of independent isolates. The analysis should identify the biological unit and account for dependence among observations.

Data collection should preserve failed captures and unavailable results, because these influence real-world coverage. A model evaluated only on images that passed an expert selection process may appear more capable than it is in routine operation. The proposed product evaluation should therefore distinguish performance on accepted images from the complete workflow's ability to obtain an acceptable result.

VI / VALIDATION SAFETY AND SECURITY

Site and temporal separation

Separate development from evaluation at meaningful boundaries

Training and test separation should prevent information about the evaluation outcome from influencing model development. Random image splitting is often insufficient when several images share a plate, isolate, patient, batch or acquisition session. The relevant grouping depends on the claim being evaluated.

Proposed separation strategy

The development collection supports model fitting and exploratory decisions. A validation collection supports tuning and threshold selection. A final independent collection or prospective period assesses the frozen system. Site and temporal separation can challenge transfer beyond the conditions that shaped development.

Table VI.6. Site and temporal separation

Leakage route	Proposed prevention
Same plate across partitions	Group all linked observations before splitting.
Related isolate records	Account for the biological relationship relevant to the study.
Shared preprocessing statistics	Fit transformations using permitted development data only.
Repeated test-set inspection	Reserve independent evaluation after final choices are frozen.
Synthetic derivatives	Keep each derivative with its originating source unit.

Document what the split actually tests

A held-out site can test some operational transfer, but it does not guarantee coverage of every new organism or instrument revision. A temporal split can reveal drift while still sharing many site-specific features. The report should state which forms of generalization the design supports.

The proposed software should retain dataset release identifiers and partition assignments. If a reference label changes, affected analyses should be traceable. Evaluation data should not quietly migrate into training while the same historical metrics continue to describe the revised model. New training releases require an appropriate new assessment of performance.

VI / VALIDATION SAFETY AND SECURITY

Repeatability and reproducibility

Repeatability and reproducibility test different sources of variation

Repeatability examines variation under closely controlled conditions. Reproducibility examines variation when specified conditions change, such as instrument, operator, day or laboratory. Both are relevant to a measurement product, and neither alone establishes agreement with the correct value.

Proposed variance study

A nested or crossed design can separate contributions from repeated capture, repositioning, operator and instrument. The design should reflect the permitted workflow and avoid attributing biological change to instrument variation when the same plate is observed over a period in which its appearance evolves.

Table VI.7. Repeatability and reproducibility

Variation source	Candidate comparison
Capture noise	Repeated eligible acquisitions without repositioning.
Placement	Removal and replacement within the permitted pose range.
Operator	Different trained users following the same workflow.
Instrument	Multiple qualified units and relevant configurations.
Time or site	Defined changes with biological and environmental context retained.

Use stable targets for instrument questions

Reference artifacts can help isolate optical and geometric variation. Biological material is needed to assess the complete measurement of real appearances. These studies answer complementary questions and should be reported separately.

The analysis should estimate relevant variance components and uncertainty, not merely report that repeated values look similar. A highly repeatable systematic bias is still an accuracy problem. Proposed acceptance should therefore combine repeatability and reproducibility evidence with independent reference agreement and clear operating limits. The resulting calibration and service requirements should follow the observed sources of variation rather than an arbitrary maintenance interval.

VI / VALIDATION SAFETY AND SECURITY

Bias agreement and uncertainty

Agreement needs more than correlation

Two methods can be strongly correlated while differing by a clinically relevant offset or scale error. Measurement validation should examine differences in the reported units, their dependence on magnitude and the uncertainty of the reference comparison. A scatter plot alone can conceal systematic disagreement.

Proposed analysis views

For paired diameters, calculate the difference between the investigational and reference measurements in mm. Examine its average, spread and relationship to diameter, organism, image quality and other relevant factors. Agreement limits require assumptions about the difference distribution and appropriate handling of repeated observations.

Table VI.8. Bias agreement and uncertainty

Analysis	What it addresses	What it does not establish alone
Mean difference	Systematic offset	Acceptable spread for individual cases.
Absolute error	Magnitude of disagreement	Direction or cause of bias.
Difference plot	Range-dependent patterns	Clinical acceptability without criteria.
Repeatability	Consistency under specified conditions	Accuracy against an independent reference.
Uncertainty budget	Contributors to measurement doubt	Biological validity of the interpretation.

Preserve reporting resolution

Rounding can change comparisons near a decision boundary. The system should retain the original measured value and the rule governing its reported form. Analysis should distinguish raw measurement behavior from the effect of rounding and categorical interpretation.

A suitable study should report confidence intervals and case counts for important subgroups. Pooled agreement may conceal a systematic problem in a smaller domain. The proposed product should restrict its qualified scope when evidence identifies such a limitation, rather than allowing a favorable overall average to override the unresolved subgroup result.

Categorical interpretation evaluation

Categorical agreement depends on the applicable interpretation rules

Categorical evaluation compares the investigational and reference interpretations within a defined organism–antimicrobial–method domain. It should use the applicable rule version and retain the underlying measurements. The same diameter can lead to different interpretations when the method or standards context changes. [\[R01\]](#) [\[R02\]](#)

Use correct denominators

Categorical agreement is the proportion of evaluable paired results assigned the same category under the specified framework. For conventional error analyses, a very major error concerns an investigational susceptible result when the reference is resistant; its denominator is the relevant reference-resistant set. A major error concerns an investigational resistant result when the reference is susceptible, with the corresponding reference-susceptible denominator. Exact handling must follow the chosen evaluation framework.

Table VI.9. Categorical interpretation evaluation

Analysis requirement	Proposed report
Category framework	Standards body, version, method and applicable definitions.
Error class	Explicit direction of disagreement and denominator.
Boundary cases	Distribution near relevant interpretation thresholds.
Unavailable results	Separate failed, abstained and unsupported cases.
Uncertainty	Confidence intervals appropriate to the sample and grouping.

Do not merge unlike category meanings

EUCAST's I category denotes susceptibility with increased exposure; it should not be casually equated with another framework's intermediate category. The system should preserve the source vocabulary and any approved mapping rather than translating categories through a generic three-letter lookup.

The proposed evaluation should report measurement and interpretation errors separately where possible. A correct numerical measurement can still receive an incorrect category through a rule-mapping defect. Conversely, a measurement error may remain within the same category while exposing a weakness that matters near another threshold. Both patterns require investigation.

Evaluating rare and consequential errors

Zero observed errors do not establish a zero error rate

Rare consequential errors require deliberate statistical planning. If n independent eligible cases produce zero errors, an exact one-sided 95% binomial upper bound for the underlying error probability is $1 - 0.05^{(1/n)}$. This is a mathematical illustration under independent, identically distributed Bernoulli assumptions, not a product performance result.

Table VI.10. Evaluating rare and consequential errors

Illustrative error-free sample	One-sided 95% upper bound, approximately
100 independent relevant cases	2.95%
300 independent relevant cases	0.994%
1,000 independent relevant cases	0.299%

Plan the relevant denominator

For a particular error class, the relevant sample may be only the reference-resistant or reference-susceptible subset, not every test in the study. Multiple drug results from one isolate may also be dependent. The analysis should reflect that structure rather than treating every output as an independent opportunity to demonstrate safety.

The zero-error ambition is translated into engineering requirements: prevent known failure sequences, detect disagreement, abstain when evidence is inadequate and recover without inventing data. Each control needs measurable performance, including false alarms and missed failures. Multiple sensors cannot guarantee zero error because common causes can make them agree incorrectly.

Interpret the evidence conservatively

An enriched challenge collection can help investigate rare failure modes, while a representative workflow study helps estimate operational coverage. These designs answer different questions. A credible claim should state the assessed error class, domain, denominator and confidence interval. The absence of an observed error is encouraging evidence within that study, not proof that an error is impossible.

VI / VALIDATION SAFETY AND SECURITY

Abstention and coverage evaluation

Abstention is useful only when its cost and coverage are visible

An abstaining system declines to produce an eligible result when evidence is insufficient. This can reduce unsafe outputs, but a model can achieve a low accepted-case error rate by rejecting most difficult cases. Evaluation must therefore report both accuracy among accepted cases and the proportion of the intended workload that receives a result.

Proposed selective-performance report

The study should define eligibility independently of model confidence, then account for every eligible case. Reasons for abstention should be meaningful, such as unreadable image, identity uncertainty, unsupported domain or sensor disagreement. A generic failure category can conceal distinct problems requiring different remedies.

Table VI.11. Abstention and coverage evaluation

Metric	Meaning
Coverage	Accepted results divided by all eligible cases.
Accepted-case error	Errors among outputs the system allowed to proceed.
False rejection	Cases unnecessarily deferred under the defined reference assessment.
Repeat burden	Additional captures, assays or review caused by abstention.
Recovery success	Deferred cases resolved through an eligible follow-up path.

Test thresholds on independent data

Confidence thresholds should be selected using permitted development data and evaluated after being fixed. Reporting the most favorable threshold from the final test set would bias the result. Performance should also be stratified by site, organism and relevant difficult conditions.

The proposed interface should offer a useful next step without disguising abstention as a negative result. “Unavailable because the boundary is unreadable” has different meaning from a measured diameter. A receiving system must preserve this distinction so that an abstained case is not accidentally converted into zero, susceptible or normal during export.

Robustness to imaging disturbances

Robustness testing should reproduce meaningful disturbances

The imaging system should be challenged with disturbances expected within or near its operating limits. Relevant factors include glare, condensation, exposure variation, blur, plate pose, label occlusion, background changes and vessel variation. Tests should assess both measurement degradation and the ability to recognize an unsupported image.

Separate tolerance from detection

Some disturbances may be small enough for qualified measurement to continue. Others should trigger abstention. The study should establish the boundary between these states and verify that the quality check recognizes it. A system that remains numerically stable on a visually corrupted image may be confidently wrong.

Table VI.12. Robustness to imaging disturbances

Disturbance	Proposed evaluation
Illumination change	Bias, contrast and quality-check response.
Focus degradation	Boundary error and false acceptance.
Pose or height variation	Scale and distortion effects within the permitted range.
Glare or condensation	Local feature loss and detection of unreadable regions.
Occlusion	Missing disks, labels or boundaries and appropriate workflow status.

Use physical and synthetic challenges appropriately

Synthetic perturbations can isolate a software property, while physical tests reveal interactions among optics, material and environment. A simulated blur filter may not reproduce all effects of an actual focus change. Both forms of evidence should state their scope.

The proposed report should identify the disturbance level, affected area and reference behavior. Combined disturbances deserve attention because several mild effects can jointly invalidate a measurement. Recovery tests should verify that a successful recapture produces a new linked observation and that the original failed image remains available for investigation.

VI / VALIDATION SAFETY AND SECURITY

Sensor failure and common causes

Multiple sensors can share the same failure

Multimodal sensing can improve error detection when channels provide complementary information. It can also create false agreement when several channels share a calibration error, clock fault, power disturbance or confounding environment. The proposed validation should test the independence assumed by the fusion method.

Build a common-cause map

The engineering team should identify shared power, timing, mounting, calibration, preprocessing and environmental dependencies. A fault in one shared element may affect many apparently separate measurements. Voting across those measurements does not create independent confirmation.

Table VI.13. Sensor failure and common causes

Fault class	Proposed challenge
Single-channel failure	Stuck, missing, saturated or drifting signal.
Shared timing error	Misaligned channels that remain individually plausible.
Shared calibration defect	Consistent but incorrect values across linked observations.
Environmental confounder	Multiple channels responding to a non-target change.
Fusion failure	Overweighting a channel or narrowing uncertainty unjustifiably.

Evaluate the recovery decision

Detection alone is insufficient. The system must choose an appropriate qualified fallback, request review or abstain. Excluding a faulty auxiliary sensor may be acceptable for one task and invalidate another. Reduced-input modes need separate evidence and explicit eligibility rules.

The study should report fault-specific detection, false alarms, detection delay and the correctness of the resulting workflow state. It should also include faults not represented during model training. AMReye.AI's error-reduction strategy is therefore a combination of diverse evidence, explicit dependency analysis and safe handling of uncertainty, with no guarantee that sensor multiplicity eliminates every error.

VI / VALIDATION SAFETY AND SECURITY

Early result validation

An early result must be evaluated at the claimed time

Rapid-AST research should specify the clock origin, specimen context, organism–drug scope and reference method before evaluation. An early output is eligible only if the method and interpretation have been established for that time. Routine endpoint breakpoints are not automatically applicable to a shorter observation interval. [\[R04\]](#) [\[R05\]](#)

Proposed sub-two-hour study definition

The study should state which preparations and observations occur within the two-hour interval and which occur beforehand. It should preserve timestamps for unsuccessful cases and include the time needed for mandatory review when the claim concerns a reviewed result.

Table VI.14. Early result validation

Study element	Required statement
Start	Specimen receipt, positive culture, exposure or another explicit event.
Population	Eligible material, organisms, drugs and relevant exclusions.
Output	Measured response, research prediction or reviewed category.
Comparator	Appropriate independent reference with defined timing.
Success	Prespecified error, coverage and uncertainty criteria at the target time.
Failure	Late, unavailable, abstained and incorrect results included in reporting.

Preserve the time-to-result distribution

A median below two hours can coexist with many much later cases. Report the fraction meeting the target and the distribution of delays. A model should not be scored at its eventual successful time when the claim concerns an earlier fixed deadline.

The initial programme may focus on selected workflows where sufficient signal is plausible. Expansion to other organisms, drugs or specimens requires new evidence. External studies motivate these investigations, but their analytical speed and performance cannot be imported as an AMReye.AI result.

VI / VALIDATION SAFETY AND SECURITY

Human review and usability validation

Human review needs validation as part of the system

The complete product includes the interaction between software and intended users. A correct algorithm can still lead to an incorrect result through confusing presentation, wrong-record selection or ineffective correction controls. Usability evaluation should focus on tasks whose failure could affect safety, data integrity or the laboratory decision.

Proposed critical tasks

Users should be able to verify identity, recognize unreadable observations, inspect measurement context, resolve flags, correct a result and understand the consequences of approval. Simulation views add the task of distinguishing measured evidence from predicted continuation.

Table VI.15. Human review and usability validation

Scenario	Proposed observation
Wrong plate association	Does the user recognize and resolve the mismatch?
Poor-quality image	Is the case appropriately rejected or repeated?
Conflicting sensor evidence	Does the user understand the unresolved state?
Corrected measurement	Is dependent approval renewed appropriately?
Interrupted simulation	Can the user identify the last real observation?

Assess realistic conditions

Participants should represent intended users, and tasks should reflect workload, interruptions and available training. Assistance from the development team during a study can conceal interface weaknesses. Observed difficulties should be analyzed for their causes and addressed through design where feasible.

The evaluation should measure errors, recoveries and understanding as well as task duration. A faster interface that encourages automatic acceptance of uncertain outputs would not satisfy the review objective. The proposed release evidence should show that users can carry out critical tasks reliably within the product's intended context, with remaining limitations reflected in design, training and use restrictions.

VI / VALIDATION SAFETY AND SECURITY

Software verification and release control

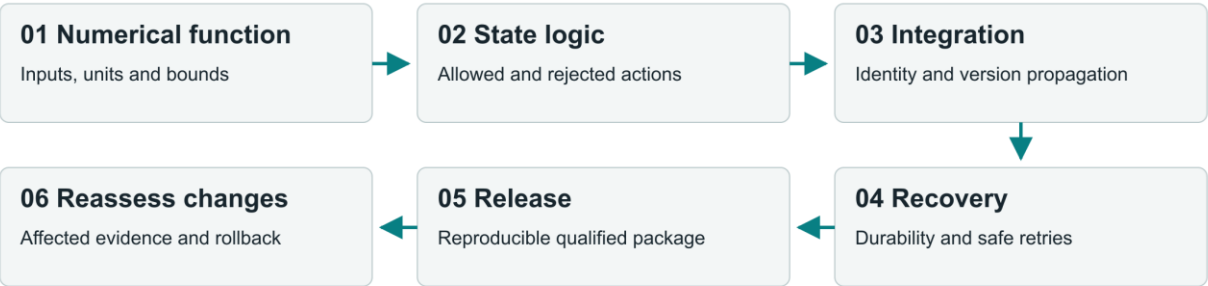
Software verification should follow the consequences of failure

Verification checks that software requirements are implemented correctly and that components interact as specified. The proposed development process should link requirements, risk controls, design and test evidence. Lifecycle standards such as IEC 62304 provide a relevant planning framework; applicability and classification require product-specific assessment. [\[R52\]](#)

Test the complete result path

Unit checks can validate numerical functions and state transitions. Integration tests should verify identity, calibration, interpretation, review and export together. System tests should include interrupted operations and realistic failure states. Passing isolated algorithm tests does not establish a correct end-to-end record.

Verification follows the complete result path



AMReye.AI · conceptual diagram

Figure VI.1. Proposed verification layers and the need to reassess evidence when the configuration changes.

Control what changes after testing

A release should identify software, models, standards rules and relevant dependencies independently. If one changes, the affected evidence should be reassessed. A signed or integrity-checked package supports distribution control, but it does not establish that the contents are clinically correct.

The proposed release process should retain test failures and their resolution, known limitations and rollback instructions. Unresolved issues need a documented impact assessment. A development build, public demonstration and qualified product release should have distinct identities so that evidence or records from one cannot be mistaken for another.

Standards content verification

Standards content is a separately controlled product dependency

The interpretation service should identify the standards body, document version, correction state and applicable method. Measurement algorithms and standards rules need independent version identities. Updating a model should not silently change clinical thresholds, and updating thresholds should not obscure the measurement originally produced. [\[R01\]](#) [\[R02\]](#)

Proposed content verification

Authorized source material should be mapped into structured rules by a controlled process with expert review. Tests should cover organism and antimicrobial mappings, units, method restrictions, footnotes, unavailable combinations and correction notices. Licensed content should be handled under its applicable terms.

Table VI.16. Standards content verification

Verification case	Expected behavior
Supported complete input	Apply the exact eligible rule and preserve its identity.
Missing organism or method	Return an unresolved interpretation state.
Unsupported combination	Do not substitute a nearby or remembered rule.
Superseded correction	Identify the affected release and re-review requirement.
Historical record	Retain the interpretation used at release and link later reassessment.

Check boundary behavior deliberately

Test values at, immediately below and immediately above relevant thresholds, following the method's rounding and reporting conventions. Character encoding, decimal separators and unit conversion can produce defects even when the source table was read correctly.

The proposed release package should include a standards-content manifest and regression evidence. A language model may assist expert review but should not generate executable breakpoints from memory. Before deployment, the team must verify the current source version and corrections again, because a dated compendium cannot guarantee that its reference snapshot remains the applicable rule set indefinitely.

VI / VALIDATION SAFETY AND SECURITY

Interoperability verification

Interoperability is preservation of meaning across a boundary

An interface can transmit data successfully while changing its meaning. A missing value converted to zero, a simulated result treated as measured or a corrected report handled as a duplicate can create serious downstream errors. Verification should test semantics, identity and workflow states as well as network delivery.

Proposed interface contract

Each receiving system should have an approved mapping for identifiers, units, terminology, result states, timestamps and amendments. The mapping should state which information is mandatory and what happens when it is absent. A transport acknowledgment should not be mistaken for clinical acceptance or review.

Table VI.17. Interoperability verification

Interface test	Required outcome
Missing or unavailable value	Preserve the reason and avoid numerical substitution.
Duplicate transmission	Reconcile safely without creating a second test.
Out-of-order amendment	Apply version logic and retain the history.
Unit mismatch	Reject or perform an explicitly authorized conversion.
Wrong identity	Stop association and expose the conflict.
Unauthorized destination	Deny export and record the attempted operation.

Test both sides of the exchange

A local export file that looks correct does not establish that the receiving system interpreted it correctly. End-to-end testing should include readback or equivalent confirmation of the stored meaning. The assessment should also cover retries, timeouts and partial failures.

The proposed platform should preserve the original export payload and mapping version where appropriate for audit and privacy requirements. Changes to a laboratory system or interface profile need impact assessment. Interoperability is a maintained service responsibility, not a one-time claim based on the presence of an API endpoint.

Traceable audit and correction records

An audit trail should explain a result's history

The audit record should capture who performed a relevant action, what changed, when it occurred and which record versions were affected. It should distinguish user actions, automated processing and administrative operations. The purpose is to reconstruct the result pathway and investigate discrepancies without relying on recollection.

Proposed event model

Events should use stable identifiers and reference the exact input and output versions. Original observations remain preserved. Corrections append a new state and reason, while dependent analyses and approvals are assessed for invalidation. Audit access itself should be controlled and monitored.

Table VI.18. Traceable audit and correction records

Event	Essential relationship
Capture	Physical record, device configuration and original observation.
Analysis	Input versions, calibration, algorithm and output.
Correction	Previous state, revised state, author and reason.
Approval	Reviewer and exact record version approved.
Export	Released version, destination and delivery state.
Administrative change	Authorized actor, scope and affected access or configuration.

Integrity and completeness are different

A cryptographic integrity check can reveal changed bytes, but it cannot prove that an event was recorded truthfully or that all relevant events exist. The system needs reliable event generation, durable storage, clock context and reconciliation of interrupted operations.

Verification should include attempts to bypass normal editing paths, interrupted writes and corrections after export. The resulting history should remain understandable to an authorized reviewer. Retention and deletion policies must also account for applicable legal and institutional requirements; “immutable” should describe controlled alteration behavior, not an unqualified promise to retain personal data forever.

VI / VALIDATION SAFETY AND SECURITY

Data minimization and purpose

Collect data for a defined purpose

Data minimization begins by identifying what each product function needs. A service engineer may require a calibration trend and hardware version without needing patient identifiers. A measurement algorithm may require a plate image whose label region contains information irrelevant to analysis. The design should reduce unnecessary exposure at the point of collection and use.

Proposed purpose map

Each data class should have a documented purpose, authorized roles, retention basis and permitted onward use. Clinical operation, service diagnostics, product improvement and research are distinct purposes that may require different permissions and agreements.

Table VI.19. Data minimization and purpose

Data class	Example purpose	Minimization question
Plate image	Measurement and review	Can irrelevant label information be excluded from a research derivative?
Clinical identifier	Correct case association	Which roles need direct identification?
Device telemetry	Maintenance and quality	Can service proceed without biological content?
Research annotation	Model development	Is the dataset authorized for this specific reuse?
Audit event	Accountability and investigation	What detail is necessary and how long is it required?

Preserve useful lineage without excessive disclosure

Pseudonymous identifiers can support linkage within an authorized workflow while limiting routine access to direct identifiers. The reidentification mapping should have its own access controls. Pseudonymization is not the same as anonymization, because the relationship may still be recoverable.

The proposed platform should make purpose restrictions enforceable through permissions and data flows. A broad statement that all available repositories will train an AI would be inconsistent with this approach. Curated, licensed and purpose-specific access can support useful research while preserving the rights and responsibilities attached to each source.

VI / VALIDATION SAFETY AND SECURITY

Consent and lawful data use

Lawful use must be assessed for the actual data flow

The legal and ethical basis for collecting, storing and reusing data depends on jurisdiction, institutional roles, purpose and the nature of the information. A laboratory's permission to process a specimen does not automatically authorize every form of commercial model training or onward sharing. The proposed programme should document these distinctions before data collection.

Current Indian planning context

India's final Digital Personal Data Protection Rules, 2025 provide phased commencement. The Gazette distinguishes provisions commencing at publication, after one year and after eighteen months. Applicable obligations, corrections and other relevant law must therefore be checked for the deployment date and processing arrangement. This compendium describes design intentions, not a legal compliance determination. [\[R54\]](#)

Table VI.20. Consent and lawful data use

Governance question	Required project decision
Who determines the purpose?	Identify institutional and company responsibilities.
What use is authorized?	Separate clinical, service, research and model-development purposes.
What information is provided?	Use understandable notices and appropriate consent or other lawful basis.
Can data be shared?	Define recipients, locations, restrictions and onward use.
What happens later?	Address withdrawal, retention, correction and rights handling as applicable.

Build controls that support the decision

The system should retain evidence of the applicable permission or agreement without exposing unnecessary personal information. Research datasets need approval and access records appropriate to the study. A change in purpose should trigger reassessment rather than being treated as covered by a historical generic permission.

The deployment team should obtain qualified legal and institutional review for the specific arrangement. Technical safeguards support lawful processing, but encryption or pseudonymization alone does not establish that a proposed use is authorized.

VI / VALIDATION SAFETY AND SECURITY

Identity access and authorization

Authorization should be enforced where the data are accessed

Authentication establishes an identity; authorization determines what that identity may do. The proposed platform should enforce permissions on the server or equivalent trusted boundary, including direct requests for records, images, exports and administrative actions. Hiding a button in the interface is not an access control.

Proposed role separation

Laboratory reviewers, operators, service staff, researchers and administrators may need different privileges. Access should follow the minimum scope required for the task and be reviewed when responsibilities change. Shared accounts weaken accountability and should not be the default for consequential actions.

Table VI.21. Identity access and authorization

Access area	Proposed control
Clinical records	Organization and role-scoped authorization.
Review and release	Qualified role with exact-version approval.
Service diagnostics	Restricted technical data and time-bounded support access.
Research exports	Purpose-specific approval and dataset scope.
Administration	Strong authentication and auditable privilege changes.

Test the denied paths

Verification should attempt access through direct object identifiers, stale links, alternate routes and lower-privilege sessions. A successful authorized request does not demonstrate that unauthorized requests are rejected. Signed links and cached files need expiry and scope consistent with the intended access policy.

Session revocation, account recovery and emergency access should have defined behavior. The system should preserve essential audit evidence while avoiding disclosure of credentials or secrets in logs. Controlled external review of a product document or research dataset should use an explicit audience and should not accidentally expose the same material through a public route or search index.

VI / VALIDATION SAFETY AND SECURITY

Encryption and key management

Encryption depends on key management and endpoint controls

The proposed system should protect sensitive data in transit and at rest using established, appropriately configured cryptographic mechanisms. Encryption is one layer of protection. An authorized application with excessive access, a compromised endpoint or an exposed key can still disclose encrypted information after decryption.

Proposed key lifecycle

Keys and credentials should be provisioned through controlled mechanisms, stored outside source code and ordinary logs, rotated where appropriate and revoked when no longer authorized. Recovery arrangements must avoid making a single lost key the cause of permanent loss of required records.

Table VI.22. Encryption and key management

Protection area	Proposed verification
Transport	Correct certificate validation and rejection of insecure connections.
Stored data	Coverage of databases, object storage, backups and local caches.
Secrets	Absence from source, public assets, logs and exports.
Key access	Role separation and audit of sensitive operations.
Recovery	Tested restoration with authorized access to required keys.

Include devices and removable media

An instrument may retain images or queued records during offline operation. Its local storage and service access need protection appropriate to the data. Exported files can also leave the application's controls, so the export workflow should identify the recipient, purpose and permitted handling.

The engineering specification should use current supported security mechanisms and independent review rather than inventing a custom encryption scheme. Verification should include configuration and operational procedures, not only the presence of an encryption library. The product claim should describe the implemented protection accurately without suggesting that encryption makes the system immune to compromise.

VI / VALIDATION SAFETY AND SECURITY

Secure software development

Security engineering should be part of ordinary development

The proposed software lifecycle should include threat modelling, secure design, dependency management, code review, verification and vulnerability response. Security requirements should follow the product's assets and trust boundaries. Medical-device cybersecurity guidance and health-software security standards provide relevant planning references, with applicability assessed for the intended product. [\[R49\]](#) [\[R53\]](#)

Track the software supply chain

A software bill of materials identifies relevant components and versions. It supports vulnerability assessment and change management but does not itself prove security. The team should determine whether a reported vulnerability is reachable in the product and what mitigation or update is needed.

Table VI.23. Secure software development

Development activity	Proposed evidence
Threat assessment	Assets, actors, trust boundaries and misuse scenarios.
Dependency review	Component inventory, support status and vulnerability disposition.
Code review	Assessment of critical data, permission and update paths.
Security verification	Tests of relevant abuse cases and configuration.
Release response	Supported update, rollback and communication process.

Test the AI-specific boundaries

Retrieved documents and user-supplied text should not gain authority to change permissions or execute actions. A language assistant needs constrained tools, source-aware responses and explicit authorization for consequential operations. Training and evaluation should include misleading instructions embedded in otherwise relevant material.

Security findings should enter the same controlled change process as other product defects, with priority based on impact and exposure. The proposed programme should retain unresolved risks and support periods transparently. A clean automated scan is useful evidence for the checks performed, not a guarantee that no exploitable weakness exists.

VI / VALIDATION SAFETY AND SECURITY

Instrument and network security

Instrument security includes physical and network boundaries

An instrument connects software to a physical laboratory workflow. Security assessment should consider local storage, service ports, removable media, update mechanisms and network access. A protected cloud service does not compensate for an instrument that exposes records or accepts unauthorized control locally.

Proposed deployment boundary

The device should expose only the services needed for its qualified functions. Network segmentation, authenticated communication and controlled administration can limit unnecessary reachability. Remote support should be explicitly authorized, scoped and auditable, with a defined end to the session.

Table VI.24. Instrument and network security

Boundary	Proposed control
Local user access	Role-based operation and protected administrative functions.
Service interface	Restricted access with documented maintenance procedures.
Network services	Minimal exposure and authenticated, authorized requests.
Software update	Verified package identity and safe installation behavior.
Removable media	Controlled import and export with malicious-content handling.
Physical storage	Protection appropriate to retained data and theft risk.

Safe operation during isolation

If a security incident requires network isolation, the instrument should enter a defined operating mode. It may retain eligible local capture and review or suspend functions that depend on trusted external services. The choice must follow the qualified workflow and risk assessment, not an improvised response during an incident.

Verification should include unauthorized control attempts, invalid updates and loss of network trust. The product should preserve record integrity and avoid unsafe physical actions when communication fails. Security and biological reliability interact: an incident that changes timing, calibration or environmental control may require assessment of affected laboratory records as well as restoration of the software.

VI / VALIDATION SAFETY AND SECURITY

Incident response and recovery

Incident response should restore trustworthy service

The proposed incident process should identify, contain, investigate and recover from events affecting confidentiality, integrity or availability. Recovery is complete only when the system and relevant records are trustworthy for their intended use. Restarting a service without assessing altered data or configuration can leave the underlying problem unresolved.

Proposed response record

The team should preserve the incident timeline, affected systems, observed indicators and actions taken. Access to evidence should be controlled, and sensitive information should not be copied into broad support channels. Notification obligations depend on the applicable law, agreement and incident context.

Table VI.25. Incident response and recovery

Response stage	Required decision
Detection	Is the event a security, reliability or combined incident?
Containment	Which functions or connections must be restricted?
Assessment	Which records, versions and physical workflows may be affected?
Recovery	What evidence supports safe restoration?
Communication	Who must be informed through authorized channels?
Improvement	Which corrective actions prevent recurrence or improve detection?

Test restoration before it is needed

Backups should be tested for completeness, integrity and usability, including required keys and configuration. Recovery objectives should reflect the laboratory workflow and retention requirements. A backup that exists but cannot be restored within the needed context is not an effective continuity measure.

Exercises should include corrupted records, lost credentials, unavailable cloud services and instrument replacement. After restoration, the system should reconcile queued events and identify uncertain intervals. Affected approvals or interpretations may require review if their inputs or integrity cannot be established. Incident closure should retain the evidence and residual limitations rather than merely marking the service online.

VI / VALIDATION SAFETY AND SECURITY

Privacy in shared research datasets

Research sharing should preserve utility while limiting disclosure

A research dataset may contain direct identifiers, indirect identifiers, labels embedded in images and sensitive institutional information. Removing a name field is not sufficient to establish anonymity. The proposed sharing process should assess what can be inferred from the combination of fields and the recipient's access to other information.

Proposed dataset release process

The release should have a defined purpose, authorized recipients, permitted analyses and retention conditions. Data minimization should be applied to images, metadata and free text. The team should verify that transformation has preserved the scientific variables required for the study.

Table VI.26. Privacy in shared research datasets

Dataset element	Proposed review
Images	Inspect labels, overlays and metadata for unnecessary identifiers.
Dates and locations	Use only the resolution needed for the research question.
Rare combinations	Assess potential identification from unusual linked attributes.
Genomic material	Review source permissions, sensitive content and intended sharing scope.
Free-text annotations	Remove unrelated personal or institutional information.
Linkage keys	Restrict access and separate from the research release.

Treat transformed data as a new release

The dataset should retain a version, transformation description and relationship to the authorized source. Researchers need to know whether values were removed, generalized, imputed or simulated. Privacy transformations can affect analytical validity and should be considered in interpretation.

The proposed programme should use institutional and legal review appropriate to the data and jurisdiction. A public repository's availability does not automatically authorize every reuse or combination. Shared research should return useful scientific value while respecting the original permissions and avoiding claims that a technical de-identification step guarantees zero reidentification risk.

VI / VALIDATION SAFETY AND SECURITY

AI assurance and change management

AI assurance continues after the first evaluation

An AI capability needs a defined intended use, data lineage, performance assessment and change process. Assurance also includes behavior under uncertainty, shift and misuse. General AI risk-management and good machine-learning-practice resources provide useful principles, but they do not certify a particular model or replace product-specific evidence. [\[R21\]](#) [\[R23\]](#)

Proposed model release record

The record should identify the model, training data release, preprocessing, decision thresholds and evaluated domain. It should retain conventional baselines and known limitations. The interface and downstream rules that use the model are part of the evaluated system configuration.

Table VI.27. AI assurance and change management

Change	Proposed impact assessment
New training data	Population coverage, label quality and evaluation independence.
New model structure	Performance, uncertainty and computational behavior.
Changed threshold	Error–coverage tradeoff and workflow burden.
New preprocessing	Measurement meaning and compatibility with prior calibration.
New deployment domain	Transfer evidence and revised eligibility.

Monitor without uncontrolled learning

Field monitoring can identify drift, unexpected failures and review patterns. It should not automatically convert every user correction into a training label or update the deployed model continuously. Candidate changes need curation, evaluation and controlled release.

The proposed system should support rollback to a qualified version while preserving which version produced each historical output. Monitoring should include accepted-case errors, abstention, data quality and important subgroups. A stable average score can conceal changing case mix or a local failure. Assurance therefore requires interpretable evidence and accountable decisions throughout the model's service life.

VI / VALIDATION SAFETY AND SECURITY

Clinical study preparation

Clinical study preparation starts before data collection

A clinical performance investigation should be designed around the intended product claim and conducted through the applicable ethical, institutional and regulatory processes. The protocol should define responsibilities, material handling, reference methods, data governance and analysis. A retrospective convenience collection may answer an exploratory question while remaining insufficient for a broader product claim. [\[R35\]](#)

Proposed preparation package

The study team should specify eligibility, recruitment or specimen-selection methods, endpoints, sample-size rationale and handling of missing or indeterminate results. Training and instrument configuration should be controlled. The investigational output's role in clinical care must be explicitly determined and authorized.

Table VI.28. Clinical study preparation

Preparation element	Required outcome
Protocol	Clear question, population, method and prespecified analysis.
Approvals	Applicable institutional, ethical and regulatory permissions.
Reference process	Independent assessment and adjudication rules.
Data plan	Purpose, access, retention and quality checks.
Operational plan	Training, calibration, monitoring and deviation handling.
Reporting plan	Complete accounting of outcomes and limitations.

Preserve deviations and unsuccessful cases

Protocol deviations should be recorded with their effect on interpretation. Exclusions should follow prespecified rules and be reported transparently. A study should not remove difficult cases merely because they reduce apparent performance.

The proposed programme should involve appropriate microbiology, clinical, statistical, engineering and governance expertise. No clinical study completion, recruitment, approval or patient benefit is claimed here. The product's next evidence step should be selected from the actual maturity of its measurement and software functions, with exploratory work separated from evidence intended to support release or clinical claims.

India regulatory assessment

India regulatory assessment follows intended use and product function

The proposed reader, interpretation software, integrated instrument and predictive research functions may raise different regulatory questions. Assessment should begin with what each function does, the material it processes and the medical decision it supports. Calling software an assistant or labelling a device research-only does not by itself resolve its regulatory status.

Current reference and its limits

CDSCO's 2026 medical-device software guidance discusses intended use, risk classification, quality systems and regulatory pathways under the Medical Devices Rules, 2017. The document states that it is an awareness resource and directs professional use to the statutory provisions and applicable clarifications. It is a planning reference here, not a product-specific legal determination. [\[R26\]](#)

Table VI.29. India regulatory assessment

Assessment topic	Required product-specific work
Device boundary	Identify hardware, software, accessories and connected functions.
Intended use	Define users, specimens, outputs and decision significance.
Classification	Determine applicable rules and competent authority interpretation.
Evidence pathway	Establish required evaluation, permissions and submission material.
Lifecycle duties	Plan quality, change, vigilance and post-market responsibilities.

Keep endpoint and predictive functions separate

A qualified endpoint measurement function should not be used as evidence that a rapid classification or simulated biological result is authorized. Each extension needs an assessment of its intended use and evidence consequences. Public copy, demonstrations and instructions should remain consistent with that scope.

Before regulated evaluation or commercialization, the programme should obtain appropriate specialist review and verify current statutory requirements. No device classification, licence, clearance or certification is asserted by this compendium. The immediate deliverable is a coherent technical and evidence plan from which that assessment can proceed.

Quality management and standards planning

Quality management connects design, evidence and delivered units

The proposed quality system should ensure that requirements, risk controls, design decisions, suppliers, production and service remain traceable. ISO 13485 is a relevant medical-device quality-management reference; ISO 14971 addresses risk management. Their applicability and implementation should be assessed for the product and organization, with no certification implied. [\[R50\]](#) [\[R51\]](#)

Select standards by function

Software lifecycle, health-software security, laboratory competence and instrument safety address different aspects of the programme. The standards plan should identify the exact edition, applicable scope and required evidence. A list of standards is not evidence that the product complies with them.

Table VI.30. Quality management and standards planning

Quality activity	Proposed controlled output
Design and development	Requirements, reviews, verification and validation records.
Supplier management	Qualification, incoming criteria and substitution assessment.
Production	Assembly instructions, acceptance checks and unit configuration.
Calibration	Traceability, intervals and response to out-of-tolerance findings.
Service	Fault investigation, repair verification and field feedback.
Corrective action	Root-cause assessment and evidence of effective resolution.

Laboratory competence is a separate consideration

Testing and calibration services may require competence arrangements appropriate to their work. ISO/IEC 17025 provides a relevant framework for testing and calibration laboratories, but it does not replace medical-device product requirements or establish an accreditation for AMReye.AI. [\[R64\]](#)

The quality system should scale with development maturity while preserving the evidence needed for later decisions. Early prototypes can be flexible, but their configurations and test conditions must still be recorded if the findings are to inform subsequent engineering or external claims.

Release gates and residual risk review

Release requires a defined decision on the remaining risk

A release gate should assess the complete configuration intended for use: instrument, software, model, calibration, standards content, interfaces, instructions and service arrangements. Each component may pass its own tests while their combination remains inadequately evaluated. The release decision should therefore review the integrated evidence and unresolved limitations.

Table VI.31. Release gates and residual risk review

Gate question	Required evidence
Is the intended use stable?	Approved scope and consistent product description.
Are measurements qualified?	Relevant analytical and robustness results.
Are critical workflows reliable?	Identity, review, correction, export and recovery verification.
Are users prepared?	Usability evidence and appropriate training.
Are security and privacy addressed?	Implemented controls and assessed residual risks.
Can the product be supported?	Installation, calibration, service and update arrangements.

Make unresolved matters explicit

Known issues should identify affected conditions, consequences and controls. A limitation should not be hidden in a technical file while marketing implies unrestricted capability. Where evidence is insufficient, the release may be restricted, deferred or retained for a defined research purpose.

The zero-error aspiration informs this process through prevention, detection, abstention and recovery. It is not a release criterion that can be proven universally. The responsible decision is whether the evidence and residual risks support the exact intended use under the applicable requirements.

AMReye.AI remains at research and prototype stage in this compendium. The proposed gates provide a route toward a dependable product and a basis for partner review. Future claims should be revised only when the corresponding engineering, analytical, clinical or operational evidence has actually been obtained and assessed.

VII / SCIENTIFIC EVIDENCE

How to read the evidence synthesis

Evidence supports a particular proposition under particular conditions

This synthesis evaluates the scientific basis and unresolved questions behind the proposed platform. It is a focused critical review, not a systematic review with exhaustive database coverage or a meta-analysis. Original studies, standards, agency guidance and manufacturer documentation serve different purposes and should not be treated as interchangeable levels of proof.

Four questions for each source

The relevant questions are what was measured, in which population, against which comparator and under which operating conditions. A result can be credible within its study while remaining insufficient for a different specimen, organism, instrument or clinical decision. Publication establishes an accessible account of work; it does not independently certify every claim in that account.

Table VII.1. How to read the evidence synthesis

Evidence class	Appropriate use in this compendium
Original experiment	Assess a measured finding and its design limitations.
Method or standard	Establish applicable procedural and interpretive requirements.
Regulatory document	Understand scope, responsibilities and evidence expectations.
Manufacturer source	Identify a product's described function and stated conditions.
AMReye.AI proposal	Define an engineering hypothesis and the evidence still needed.

Reading quantitative results

Agreement, precision, accuracy, sensitivity and time to result answer different questions. A high correlation can coexist with systematic measurement bias. A small average error can conceal clinically important outliers. A rapid answer among selected successful runs can coexist with a substantial unresolved fraction.

Where the available evidence does not establish a denominator, independent validation or local applicability, the conclusion remains bounded. The practical outcome of this review is a set of testable design decisions and research questions. No external study is represented as an AMReye.AI experiment, and no aggregate performance score is assigned to the proposed platform.

VII / SCIENTIFIC EVIDENCE

Disk diffusion standards and reader evidence

A calibrated image does not replace the biological method

CLSI M100 Edition 36 and EUCAST clinical breakpoint tables version 16.1 were the current editions verified for this publication date. Their applicability depends on the relevant method, organism, antimicrobial and conditions. A breakpoint value detached from those conditions is not a complete interpretation rule. [\[R01\]](#) [\[R02\]](#)

The CLSI correction of 9 July 2026 restores the sulfamethoxazole concentrations in combination MIC notation in Table 2H-1. It illustrates why an edition number alone is insufficient version control. The correction does not create a disk-diffusion indication where the table's method restrictions do not permit one. [\[R74\]](#)

Reader precision is one component

Hombach and colleagues' variance analysis distinguished biological variation from technical contributions, including preparation, streaking, materials and reading. Their estimated contributions varied substantially across the tested organism–drug combinations. The findings support evaluating the entire preparation-to-reading process rather than assuming that image precision determines total assay precision. [\[R67\]](#)

Table VII.2. Disk diffusion standards and reader evidence

Proposed AMReye.AI question	Evidence needed
Is the image geometrically reliable?	Traceable scale and distortion evaluation.
Is the plate eligible for interpretation?	Documented method and quality conditions.
Is the boundary reproducible?	Independent readings and controlled repeat captures.
Is the category appropriate?	Eligible, corrected standards content and reviewer decision.

Consequence for development

Reader V1 should preserve the distinction between measurement uncertainty and method variability. Its interface should expose missing method context before category assignment. Validation should include plates near relevant decision boundaries and difficult reading conditions, with a predefined route for unresolved cases. Greater numerical precision should never be presented as proof that an unsuitable biological preparation has become valid.

VII / SCIENTIFIC EVIDENCE

External evidence on automated zone measurement

Automated measurement evidence must disclose human intervention

Hombach et al. compared on-screen adjusted Sirscan readings with manual measurements and separately assessed repeated automated readings. The study reported improved repeatability in the automated condition. The distinction between adjusted clinical readings and the precision experiment matters: their results do not establish that every autonomous reading was correct without review. [\[R66\]](#)

A recent AI study requires additional clarification

The 2026 study by Ciftci and colleagues reports categorical agreement of 94.2% and mean absolute diameter error of 0.42 mm. Its methods describe 207 original photographs in one passage and 1,024 baseline images elsewhere, expanded to 5,120. The detection metric also differs between a table and the ablation narrative. These counts and the independence of plate/isolate partitions require clarification before the headline metrics can serve as a robust benchmark. [\[R12\]](#)

Proposed evaluation response

AMReye.AI should report the unedited model output and the final reviewed result separately. Every evaluated plate should retain its split assignment, source image lineage and correction history. An augmented image must remain in the same partition as its source and associated biological unit.

Table VII.3. External evidence on automated zone measurement

Reporting element	Why it changes interpretation
Original independent plates	Determines the biological information available.
Images and augmentations	Describes views and transformations, not new isolates.
Unassisted measurement	Evaluates the algorithm's actual starting performance.
Reviewer changes	Quantifies the work required to reach the final result.
Unreadable and failed cases	Prevents favorable exclusion of difficult inputs.

Evidence should make these distinctions auditable. A model explanation or attention map can support investigation, but it cannot substitute for independent performance evaluation or establish that a model has learned a biologically valid mechanism.

VII / SCIENTIFIC EVIDENCE

External evidence on rapid single cell AST

Rapid single-cell AST demonstrates a route, not a universal timescale

Baltekin et al. demonstrated direct single-cell imaging in a microfluidic setting and reported susceptibility classification for 49 clinical uropathogenic E. coli isolates against ciprofloxacin within ten minutes of the measurement sequence. The overall method was described as requiring less than thirty minutes from sample loading. The narrow organism–drug evaluation and the timing origin are essential parts of that result. [\[R11\]](#)

Translation questions for AMReye.AI

The study motivates investigation of early phenotypic responses where sufficient cells can be observed under a controlled antimicrobial exposure. It does not demonstrate that an ordinary endpoint plate photograph can deliver the same answer. Differences in specimen preparation, cell capture, exposure, optical sampling and decision criteria must be evaluated explicitly.

Table VII.4. External evidence on rapid single cell AST

Translation question	Proposed test
Are enough relevant cells observed?	Capture yield and eligibility across intended specimen conditions.
Does early response discriminate susceptibility?	Blinded comparison by organism and drug.
Are slow or heterogeneous responses retained?	Time-dependent unresolved and error analysis.
Does the workflow remain fast outside the bench?	End-to-end timing including preparation and review.

Sub-two-hour research target

The proposed target should begin with a defined specimen and clock origin, not a platform-wide promise. A feasibility study should prespecify the proportion of eligible samples expected to receive a reportable answer, the comparator method and acceptable uncertainty. Time distributions should include failed runs and repeats.

A useful early answer is one whose correctness and limits are understood. If a sample requires longer observation, abstention preserves the distinction between a target and an established capability. Faster inference cannot compensate for insufficient biological information.

VII / SCIENTIFIC EVIDENCE

External evidence on heteroresistance in infection

Heteroresistance has mechanistic and context-dependent clinical evidence

Band et al. linked an initially susceptible classification of *Enterobacter cloacae* with a resistant subpopulation and treatment failure in a murine infection model. This establishes a biologically consequential example under the tested conditions, rather than a universal human outcome or an imaging diagnostic rule. [\[R08\]](#)

Doijad and colleagues examined mechanistic determinants of colistin heteroresistance in the *Enterobacter cloacae* complex. Kupke and colleagues later described transient amplification events in a veterinary clinical isolate. Together these studies motivate attention to population structure and unstable resistance phenotypes without implying a single mechanism for all organisms and drugs. [\[R09\]](#) [\[R60\]](#)

Clinical association is not uniform

Wei et al. identified heteroresistance in 40 of 164 non-resistant isolates within a 212-isolate collection. The study did not find a significant increase in the assessed hospitalization or progression outcomes. Its retrospective context and denominators should accompany interpretation. This finding does not negate the murine mechanism; it limits a universal clinical-outcome inference. [\[R61\]](#)

Proposed contribution of imaging

AMReye.AI can investigate whether longitudinal spatial observations improve recognition of candidate minority growth. The first validation question is detection of a reproducible visual event. Confirmation of heteroresistance is a separate biological question requiring appropriate reference characterization.

Table VII.5. External evidence on heteroresistance in infection

Observation	Defensible initial interpretation
Small within-zone object	Candidate feature requiring review.
Repeated growth at that location	Temporal evidence of a biological event, subject to confounders.
Confirmed resistant subpopulation	A method-supported heteroresistance finding within defined scope.

The product should preserve the chain between these stages. It should not label every satellite colony as resistant or infer clinical treatment failure from an image alone.

VII / SCIENTIFIC EVIDENCE

External evidence on heteroresistance and combinations

Combination findings generate hypotheses that require their own evaluation

Band et al.'s 2019 work investigated the exploitation of heteroresistance to identify effective antibiotic combinations. The study provides a mechanistic reason to examine how differently susceptible subpopulations respond to combined exposures. It does not authorize a general combination recommendation or establish a treatment-selection function for AMReye.AI. [\[R07\]](#)

A proposed research use

The platform could support controlled experiments by recording spatial growth, exposure context and time-dependent responses under a prespecified design. Its contribution would be measurement and traceability. Interpretation of interaction, mechanism and therapeutic significance would require the appropriate experimental and clinical expertise.

Table VII.6. External evidence on heteroresistance and combinations

Research element	Proposed evidence requirement
Population composition	Establish the relevant heterogeneity before the comparison.
Exposure definition	Preserve concentrations, timing and the experimental context.
Response endpoint	Define growth, survival or another measured outcome in advance.
Interaction assessment	Use a suitable comparator and prespecified analysis.
Clinical translation	Evaluate separately from laboratory interaction evidence.

Avoid a false shortcut from image to therapy

An apparent reduction in visible growth can result from different biological processes. A single image cannot establish killing, durable suppression, mechanism or patient benefit. Even a reproducible laboratory interaction remains conditional on the tested system.

The proposed software should therefore present research measurements and uncertainty without generating autonomous treatment instructions. A useful capability would be the ability to compare hypotheses and retrieve the complete experimental record. The development objective is to make combination research more inspectable, while keeping therapeutic claims behind their own evidence and authorization requirements.

VII / SCIENTIFIC EVIDENCE

Definitions of persistence and tolerance

Resistance, tolerance and persistence describe different response patterns

The consensus statement by Balaban and colleagues distinguishes resistance, which concerns growth under antimicrobial exposure, from tolerance and persistence, which concern survival during killing. Persistence refers to a surviving subpopulation rather than uniformly slower killing across the population. These distinctions require suitable experimental endpoints and cannot be established by vocabulary alone. [\[R10\]](#)

Implication for a measurement platform

An endpoint zone image primarily records spatial growth inhibition under its method conditions. A time series may reveal additional dynamics, but the observable signal still has to match the biological question. An unchanged object might be alive, dead or an imaging artifact. Absence of visible expansion does not establish eradication.

Table VII.7. Definitions of persistence and tolerance

Proposed output	Necessary boundary
Growth inhibition measurement	Does not itself quantify killing.
Delayed visible growth	Does not uniquely identify persistence.
Surviving subpopulation hypothesis	Requires an appropriate survival-based investigation.
Resistance interpretation	Requires the applicable assay and interpretive context.

Design the record around the actual endpoint

The proposed data model should store whether an experiment measures growth, viable survival, metabolic response or another signal. Analysis should preserve the observation period and exposure definition. A generic field called “resistance score” would obscure distinctions that matter scientifically.

For research comparisons, the programme should agree on terminology before data collection. Reviewers should be able to identify whether a claim follows from the measured endpoint or remains a hypothesis. This is especially relevant when multimodal measurements disagree: a metabolic signal and an optical growth signal may answer different questions rather than indicate that one sensor has failed. Biological interpretation should precede automatic reconciliation.

VII / SCIENTIFIC EVIDENCE

Evidence for spatial and temporal growth models

Growth models explain variation without guaranteeing an individual future

Thomas et al. developed a stochastic cellular model linking gene expression, metabolism and replication. The work connects fluctuations in cellular processes with heterogeneous growth and evaluates model behavior against reported physiological observations. It provides a basis for investigating variable responses, not a validated forecasting model for a particular AMReye.AI plate. [\[R13\]](#)

Kapteyn, Pretorius and Willcox describe probabilistic graphical modelling for digital twins in an engineering setting. Its relevance here is methodological: distinguish latent state, observation, uncertainty and decision. Biological calibration and validation remain separate work. [\[R59\]](#)

Evidence needed at plate scale

The proposed twin must connect its latent variables to actual measurable quantities. A plausible growth equation is insufficient if several parameter combinations fit the same short observation period but predict incompatible futures. That situation is an identifiability limitation, not a reason to select the most visually convincing trajectory.

Table VII.8. Evidence for spatial and temporal growth models

Model question	Proposed evaluation
Can parameters be estimated?	Sensitivity and identifiability analysis using the intended observations.
Does forecasting beat a simple baseline?	Prospective time-split comparison against persistence and simple growth fits.
Is uncertainty informative?	Horizon-specific coverage and calibration assessment.
Does interruption change reliability?	Withheld-observation experiments with known later measurements.

Scientific consequence

Simulation continuity preserves computation after physical observation ends. It cannot preserve knowledge of unobserved contamination, environmental changes or new biological events. The proposed publication and interface should retain the final measured state and show how uncertainty evolves beyond it. Model-based exploration remains valuable precisely when its assumptions and limits can be inspected.

VII / SCIENTIFIC EVIDENCE

Optical metrology evidence

Sensor specifications and system metrology answer different questions

EMVA 1288 provides a framework for characterizing camera behavior. Optical documentation explains how geometry, perspective and lens choice affect imaging. The JCGM uncertainty framework supports an explicit account of contributions to a measurement result. None supplies an automatic accuracy specification for an assembled plate reader. [\[R14\]](#) [\[R15\]](#) [\[R16\]](#)

The measurement chain is the object of validation

A camera can have a small pixel pitch while the final system has substantial error from distortion, defocus, plate height, reflections or boundary interpretation. Similarly, a lens chosen for reduced perspective sensitivity still requires alignment and verification across the actual object field.

Table VII.9. Optical metrology evidence

Evidence source	What AMReye.AI must establish independently
Camera characterization	Image quality under the selected exposure and illumination.
Lens specification	Usable field, focus, distortion and mounting stability.
Calibration artifact	Traceability and fitness for the spatial scale of interest.
Uncertainty framework	A defensible budget for the assembled measurement procedure.

Proposed metrology experiment

The optical bench should compare known geometric targets at multiple positions and relevant heights. Repeat captures should vary remounting, illumination and qualified environmental conditions. The analysis should separate scale error, spatial distortion and repeatability, then examine biological boundaries using independent reference readings.

The resulting evidence should state where measurements are eligible and when recalibration or rejection is necessary. Object sampling in micrometres per pixel is a useful design calculation, but it should be reported separately from diameter uncertainty in millimetres. Keeping those quantities distinct prevents a component specification from becoming an unsupported biological measurement claim.

VII / SCIENTIFIC EVIDENCE

Gas and environmental sensor evidence

Environmental sensing supports context before it supports biological inference

Manufacturer documentation for gas and environmental sensors establishes their physical signals, operating ranges and integration requirements. A broad gas response or a temperature channel does not identify a bacterial species, resistance mechanism or metabolite without additional evidence. The platform's candidate catalogue therefore separates environmental monitoring from exploratory biological sensing. [\[R17\]](#) [\[R18\]](#)

A controlled question for each signal

Temperature, humidity and gas measurements can help explain whether acquisition occurred under qualified conditions. Using the same channels as biological predictors is a separate research programme. It requires demonstrating that the signal reflects the intended biology rather than media, cleaning agents, room conditions or sensor history.

Table VII.10. Gas and environmental sensor evidence

Potential confounder	Proposed study response
Medium or vessel background	Include appropriate non-biological controls and lots.
Room-air variation	Record the ambient and enclosed measurement contexts.
Humidity and temperature coupling	Characterize cross-sensitivity within the intended range.
Sensor drift or carryover	Assess repeated exposure, recovery and recalibration.
Shared preparation effect	Evaluate whether multiple channels fail together.

Evidence threshold for inclusion

A channel should earn its place through incremental information, reliability or control value. A modality that improves apparent training accuracy but fails across sites should not be described as robust biological sensing. Candidate selection should compare its benefit with the added calibration, maintenance and contamination-control burden.

The strongest near-term use may be detecting a condition that makes the optical result unreliable. That is a meaningful contribution even when the channel cannot support a standalone diagnostic inference. The product should express this role directly, rather than convert every installed sensor into a separate clinical claim.

VII / SCIENTIFIC EVIDENCE

Sequence based resistance evidence

Sequence evidence does not remove phenotype uncertainty

CARD and NCBI Pathogen Detection provide curated resistance-related resources and documented analytical contexts. Their content supports structured investigation of genetic determinants and isolate relationships. A database match is not equivalent to a measured susceptibility result, and an absent match does not prove susceptibility. Version, coverage and the underlying sequence quality remain material. [\[R24\]](#) [\[R25\]](#)

Separate three evidence paths

The proposed platform should distinguish externally measured molecular results, computational interpretation of sequence data and predictions from images or other phenotypes. These paths can be compared, but they should not be merged into a field that hides how a conclusion was obtained.

Table VII.11. Sequence based resistance evidence

Evidence path	Proposed interpretation limit
Validated molecular assay result	Limited to the assay's target and intended use.
Sequence-based determinant call	Conditional on sequence quality, pipeline and database coverage.
Phenotype-to-genotype prediction	A research inference requiring independent molecular confirmation.

A useful disagreement study

The programme can examine cases where a measured phenotype and a molecular result diverge. The investigation should consider assay coverage, expression, regulatory context, mixed populations, measurement quality and record linkage before assigning an explanation. The aim is to identify resolvable causes and persistent uncertainty.

Model training should avoid learning site, lineage or specimen shortcuts that merely correlate with resistance in the available collection. Evaluation should separate related isolates and examine performance when the distribution changes. A useful genomic integration service preserves evidence provenance and explains discordance. It does not promise to process every repository or infer every resistance mechanism from a plate image.

VII / SCIENTIFIC EVIDENCE

AI assurance evidence

AI assurance is a lifecycle argument supported by measurements

Good machine learning practice, transparency principles and the NIST AI Risk Management Framework emphasize structured development and evaluation of AI systems. These resources help organize evidence and responsibilities; they do not supply an automatic approval or a universal performance threshold. [\[R21\]](#) [\[R22\]](#) [\[R23\]](#)

Test the proposed decision behavior

For AMReye.AI, an assurance case should connect intended use to data, models, interfaces and human decisions. The evidence should show how the system behaves when an input is outside its training experience, a sensor disagrees or a reviewer changes a result. Performance on a convenient retrospective test set is only one component.

Table VII.12. AI assurance evidence

Assurance question	Evidence expected from the programme
Is the task represented?	Documented population, workflow and data coverage.
Is evaluation independent?	Patient, isolate, plate and site lineage controls as applicable.
Does uncertainty support action?	Tested abstention and review behavior.
Are updates controlled?	Versioned models, change assessment and rollback evidence.
Can failures be investigated?	Preserved inputs, outputs and decision lineage.

Meaningful baselines

Each proposed AI capability should be compared with an appropriate conventional method. Fixed image-quality rules, geometric fitting, simple trend models and explicit workflow logic may outperform a learned approach when data are limited or the task is well constrained. The choice should follow measured benefit.

The assurance programme should report both capability and burden: correction frequency, unresolved cases, reviewer workload and operational failures. A system that reaches high agreement by deferring nearly every difficult case has a different value proposition from one that handles those cases reliably. Both accuracy and coverage belong in the product evidence.

VII / SCIENTIFIC EVIDENCE

Manual and digital reader comparators

Manual and digital readers occupy a defined part of the workflow

Manual disk-diffusion reading combines method-specific visual judgment with measurement and transcription. A digital reader can support imaging, zone measurement and result handling while retaining operator responsibility for ambiguous growth. BIOMIC V3 is an example of a commercial digital reading system; the Sirscan study provides historical experimental evidence relevant to this category. [\[R29\]](#) [\[R66\]](#)

Compare equivalent tasks

Reader V1's appropriate comparison begins with the endpoint reading workflow. The evaluation should use the same eligible plates and distinguish preparation, incubation, capture, analysis, review and export. Incubation time should not disappear from a total-turnaround claim merely because it occurs before loading the reader.

Table VII.13. Manual and digital reader comparators

Comparison dimension	Proposed AMReye.AI evidence
Measurement	Bias, spread and outliers against independent reference readings.
Readability	Detection of invalid or ambiguous inputs, including false acceptance.
Review	Correction frequency, decision time and unresolved fraction.
Records	Traceability of source image, scale, interpretation and correction.
Integration	Export accuracy and resilience to interrupted transfer.

Potential differentiation to investigate

AMReye.AI proposes a tightly linked record of capture quality, measurement uncertainty and reviewer actions. The question is whether that linkage improves reproducibility and reduces avoidable work in the intended laboratory. This requires a measured comparison; it is not established by including more software functions.

Competitive assessment should account for local service, consumables, training, maintenance and existing laboratory systems. An institution may value interoperability and dependable support more than an expanded feature list. The development programme should therefore compare the complete endpoint service offered to a user, with current regional specifications checked before any procurement decision.

VII / SCIENTIFIC EVIDENCE

Integrated disk diffusion comparators

Integrated disk-diffusion automation extends beyond image analysis

COPAN describes RADIAN within an automated microbiology workflow that includes disk application, incubation-related handling, digital reading and supporting software functions. Its scope is broader than a standalone endpoint reader. The existence of this category rules out a general claim that automated plate-based AST or AI-assisted disk reading is unique to AMReye.AI. [\[R27\]](#) [\[R28\]](#)

Cherkaoui et al. evaluated a Colibri–RADIAN workflow against VITEK 2 in routine laboratory conditions. The study is relevant because it assesses an integrated process and categorical agreement, rather than only segmentation on stored photographs. Its findings remain specific to the evaluated configuration and comparison. [\[R68\]](#)

Proposed comparison design

The integrated AMReye.AI track should be evaluated independently from Reader V1. It introduces additional responsibilities for sample identity, movement, preparation, timing and recovery. These functions change the risk profile and the amount of verification required.

Table VII.14. Integrated disk diffusion comparators

Integrated function	Evidence needed beyond reader validation
Automated preparation	Reproducibility and conformity of the biological preparation.
Disk handling	Correct identity, placement and exception management.
Plate movement	Tracking, collision prevention and recovery without identity loss.
Controlled observation	Qualified timing and environmental conditions.
Workflow coordination	Queue behavior, failed-run handling and human intervention.

Distinguishing ambition from demonstrated advantage

The proposed platform's modular sensing and longitudinal data could create research value, but adding these functions does not establish superior clinical results or lower total cost. A credible comparison should examine defined workflows, throughput distributions, quality outcomes and service requirements. The integrated track should progress only when each added automation function improves a verified process without obscuring its failure modes.

VII / SCIENTIFIC EVIDENCE

Automated ID and AST comparators

Identification and susceptibility systems should be compared by output

VITEK 2, BD Phoenix M50 and DxM MicroScan WalkAway represent established automated ID/AST system families. Their cards or panels, supported organisms and reporting functions differ. Current manufacturer documentation is useful for identifying those functions, while the applicable regional labeling and panel-specific evidence determine actual use. [\[R30\]](#) [\[R31\]](#) [\[R69\]](#)

MALDI-TOF systems such as VITEK MS PRIME primarily support organism identification through mass-spectral analysis. Identification can inform an AST workflow, but it is not the same measurement as phenotypic antimicrobial susceptibility. [\[R72\]](#)

Table VII.15. Automated ID and AST comparators

Comparator category	Primary comparison question
Automated identification	How is organism identity established and linked to the specimen?
Automated phenotypic AST	What susceptibility output is supported for the relevant panel?
Combined ID/AST workflow	How are identification, susceptibility and review coordinated?
Endpoint disk reader	How reliably is a prepared plate measured and interpreted?

Implication for the portfolio

Reader V1 should support reliable integration of independently established identity where required. It should not imply that a disk image performs the same identification function as a validated biochemical, molecular or mass-spectral method. Phenotype-to-genotype and morphology-based identification remain separately evaluated research capabilities.

Competitive differentiation should be assessed at the level of the user's decision. A laboratory may use an integrated ID/AST system for routine panels and retain disk diffusion for selected requirements. The proposed product should explain where it fits, how records connect and what evidence supports that use. A list of instrument names cannot substitute for a comparison of specimen requirements, reportable scope, exception handling and operational fit.

VII / SCIENTIFIC EVIDENCE

Rapid AST and microfluidic comparators

Rapid AST is a family of methods with different clocks and specimens

Manufacturer documentation describes dRAST as a positive-blood-culture phenotypic AST workflow with results in as little as four hours. VITEK REVEAL describes rapid phenotypic AST for Gram-negative positive blood cultures, with average timings around 5.5–6 hours. Q-linea describes ASTar reports in approximately six hours from its relevant workflow. These are manufacturer-described timings, not a direct independent comparison or time from patient sampling. [\[R70\]](#) [\[R71\]](#) [\[R75\]](#)

EUCAST RAST has its own eligibility, timing and interpretation documents. A research single-cell method, a commercial positive-blood-culture system and an early disk-diffusion method should therefore remain distinct comparator classes. [\[R04\]](#) [\[R05\]](#)

Table VII.16. Rapid AST and microfluidic comparators

Method class	Question to resolve before comparing speed
Positive-blood-culture AST	Is time before culture positivity included?
Direct-specimen research method	Which preparation and eligibility steps precede measurement?
Early disk reading	Is a validated time-specific interpretation available?
Endpoint reader	Has the required biological incubation already occurred?

AMReye.AI research position

Sub-two-hour performance remains a target for selected, explicitly defined workflows. The proposed studies should compare reportable-answer distributions, not the fastest successful example. Organism–drug coverage, invalid runs, repeat testing and reviewer time belong beside the timing result.

Microfluidics may shorten diffusion distances or permit observation of individual cells, but it also introduces loading, material, exposure and sampling constraints. Its value must be tested in the chosen biological workflow. The platform should pursue the mechanism that best supports reliable early information, while maintaining endpoint Reader V1 as a distinct development priority.

VII / SCIENTIFIC EVIDENCE

Research imaging and software comparators

Research imaging platforms compete on experimental information

Reshape Biotech describes automated endpoint and time-lapse capture, structured analysis and data-sharing workflows for microbiology research and quality-control applications. This is a relevant comparator for longitudinal imaging and research data services. Manufacturer workflow descriptions do not establish clinical AST performance or equivalence with a diagnostic system. [\[R73\]](#)

Research modalities such as speckle imaging, Raman-based analysis, electrical microfluidics and calorimetry provide additional examples of measurable microbial responses. Their existence supports a broad technology search, while each has its own sampling and validation boundaries. [\[R38\]](#) [\[R40\]](#) [\[R42\]](#) [\[R44\]](#) [\[R45\]](#)

Proposed research-platform comparison

For the AMReye.AI research family, useful outcomes include experimental reproducibility, observation completeness, analysis transparency and the ability to compare conditions without losing provenance. Clinical categories may not be the appropriate primary endpoint for these applications.

Table VII.17. Research imaging and software comparators

Research capability	Evidence of practical value
Longitudinal imaging	Stable registration and interpretable temporal measurements.
Experiment metadata	Reproducible grouping and retention of conditions and exceptions.
Analysis software	Transparent transformations and exportable measurements.
Multimodal experiments	Incremental information beyond a simpler measurement set.
Collaboration	Authorized access and reproducible analysis across participating teams.

A disciplined breadth of scope

The proposed portfolio should offer distinct research functions rather than imply that one device performs every method. Microscopy, spectroscopy and automated plate handling may belong to different configurations or collaborations. The decision to integrate a modality should follow a demonstrated user question and measurable benefit.

An effective research service can help laboratories preserve and analyze experiments even before a predictive twin becomes useful. That near-term value should be evaluated on its own terms, while predictive and diagnostic extensions remain subject to additional evidence.

VII / SCIENTIFIC EVIDENCE

Regulatory evidence and label specificity

Regulatory evidence belongs to a specific device and intended use

The CDSCO medical-device software guidance provides context for assessing software under India's Medical Devices Rules. It does not, by itself, assign AMReye.AI a classification or establish permission to market. Intended use, software function, hardware relationship and applicable statutory requirements require a product-specific assessment. [\[R26\]](#)

The FDA decision identified in reference R32 illustrates another boundary: a regulatory record supports the device and indications covered by that record. Similarity in imaging or AI terminology cannot transfer another device's authorization to the proposed platform. [\[R32\]](#)

Evidence that should accompany a regulatory comparison

The programme should retain the exact device, version, jurisdiction, indication and limitations when discussing an authorization. A manufacturer's broad product-family page may not identify the particular configuration relevant to an institution's intended workflow.

Table VII.18. Regulatory evidence and label specificity

Proposed statement	Required supporting material
Device is authorized for a use	Applicable regulator record and labeling.
Method is recognized	Exact method, scope and current conditions.
Software change is acceptable	Product-specific assessment and required evidence.
Security expectations are addressed	Implemented controls and verification appropriate to the device.

AMReye.AI publication boundary

This compendium describes proposed engineering and research requirements. It is not a regulatory submission, clinical-use instruction or declaration of conformity. The validation and quality sections identify work that the development programme must complete.

Current FDA cybersecurity guidance and software lifecycle standards can inform planning, but a citation to them does not demonstrate implementation. [\[R49\]](#) [\[R52\]](#) [\[R53\]](#) The product's eventual regulatory case should connect its actual design and evidence to the applicable requirements, with qualified review of the intended market and use.

VII / SCIENTIFIC EVIDENCE

Contradictions and unverified propositions

Contradictions should change the strength of a claim

The literature and product landscape support meaningful opportunities, but several attractive propositions remain unverified. The proposed platform should retain these as explicit research questions. A stronger vocabulary cannot resolve an evidence gap, and a larger sensor catalogue does not supply validation by itself.

Table VII.19. Contradictions and unverified propositions

Proposition	Evidence-based disposition
Multiple sensors guarantee zero error	Unsupported; common causes and false agreement require evaluation.
A day of observations predicts all later growth	Unsupported; forecasting depends on identifiability, conditions and horizon.
Within-zone growth proves heteroresistance	Unsupported without biological confirmation.
A published AI accuracy is a transferable benchmark	Conditional on dataset integrity, independence and matching task.
A rapid method works for every specimen	Unsupported outside its studied and authorized scope.
The latest chip is the best design choice	An engineering question requiring workload and lifecycle evidence.

Contradiction does not always mean one source is wrong

The murine heteroresistance study and the retrospective clinical collection answer different questions. An effect under a controlled infection model can coexist with an absence of a significant association in a particular clinical analysis. Both should inform the research design without being collapsed into a universal conclusion. [\[R08\]](#) [\[R61\]](#)

Proposed response to unresolved evidence

AMReye.AI should identify the decision affected by each uncertainty. Some gaps prevent a claim; others determine an experiment or a configuration choice. The response should be proportionate: preserve an unresolved status, obtain independent characterization or revise the intended use.

The same discipline applies to negative findings. A failed exploratory modality can be valuable if it reveals a confounder or an integration cost. Development should reward informative evidence, including evidence that narrows the portfolio, rather than treating every proposed capability as an inevitable product feature.

VII / SCIENTIFIC EVIDENCE

Priority questions for original AMReye.AI evidence

The next decisive evidence should come from AMReye.AI's own defined tasks

External work establishes scientific plausibility, known alternatives and important limitations. The development programme now needs original evidence that connects the proposed system to its intended users. The highest-value studies are those that resolve a specific release or research decision.

Priority evidence sequence

Reader V1 should first demonstrate a dependable measurement chain and review workflow. Advanced sensing and forecasting can then be investigated against that foundation, using independently defined outcomes rather than a single broad “platform accuracy” measure.

Table VII.20. Priority questions for original AMReye.AI evidence

Priority question	Proposed decision enabled
Can eligible endpoint plates be captured consistently?	Qualify the acquisition configuration.
Are calibrated diameters reproducible across conditions?	Define measurement scope and uncertainty.
Are difficult inputs detected and reviewed effectively?	Establish safe exception and correction behavior.
Does an added modality contribute independent information?	Select or reject the candidate integration.
Can a twin forecast withheld observations reliably?	Define a qualified prediction horizon or retain research status.
Can selected early workflows meet the sub-two-hour target?	Establish scope-specific feasibility and next validation needs.

What a publishable result should include

The proposed reports should describe configuration, biological population, sampling, reference method, exclusions, missing data, uncertainty and failures. Data splits should respect biological and capture lineage. Where sharing is permitted, analysis definitions and appropriately protected data should support independent scrutiny.

The programme should also measure practical consequences: review effort, instrument downtime, calibration burden and integration errors. A technically impressive measurement can have limited value if it creates an unreliable workflow. Original AMReye.AI evidence should therefore show both what the system measures and how that measurement supports a dependable laboratory process.

VIII / DEVELOPMENT AND DELIVERY

From requirements to verified increments

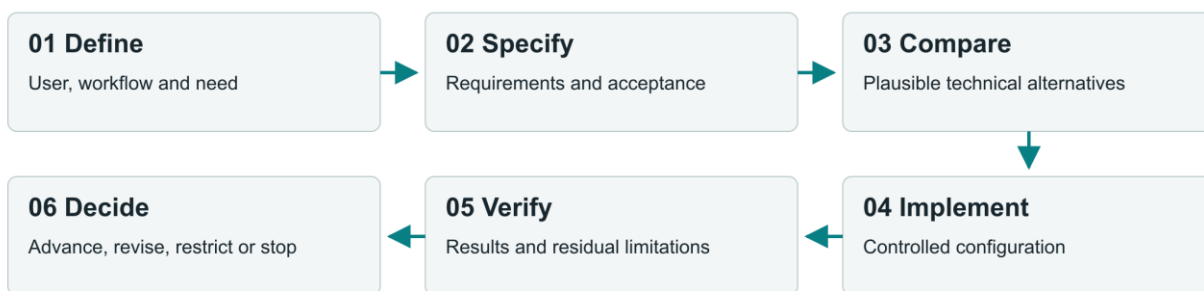
Development should produce evidence-bearing increments

The proposed programme should convert the platform vision into bounded engineering questions. Each increment should deliver a function and the evidence needed to assess it. This approach allows configuration ideas to remain flexible while preventing an untested concept from becoming a fixed product promise.

Begin with observable requirements

A requirement should describe a behavior, condition and acceptance method. “Reliable imaging” is too broad to test directly. A more useful requirement identifies the permitted plate range, capture conditions, quality decision and retained record. Numerical thresholds should follow justified design targets and evidence, not arbitrary precision.

A development increment ends with an evidence-based decision



AMReye.AI · conceptual diagram

Figure VIII.1. Proposed sequence for one bounded development increment; an adverse finding can justify restriction or stopping.

Preserve learning across changes

An unsuccessful prototype can establish a useful limit, such as insufficient optical contrast or an impractical service requirement. The relevant conditions and measurements should be retained so that the same assumption is not repeatedly rediscovered. Historical evidence must still identify the configuration that produced it.

For AMReye.AI, the first increments should establish the endpoint reader's capture, calibration, measurement, review and export chain. Advanced sensing, automation and predictive twins can develop through separate research questions that use those foundations. Their progress should be assessed individually, without making the initial product depend on every element of the long-term vision.

VIII / DEVELOPMENT AND DELIVERY

Reader V1 engineering sequence

Reader V1 should establish a complete, narrow workflow

The near-term engineering sequence should prioritize controlled endpoint imaging and a traceable reviewed measurement. A small complete workflow is more informative than a collection of disconnected demonstrations because it exposes identity, calibration and state-management failures at their interfaces.

Proposed sequence

First establish a repeatable optical arrangement and qualified capture records. Then develop spatial calibration and image-quality assessment. Add measurement with independent comparison, followed by controlled interpretation, review and export. Integrate service and recovery behavior before evaluating the workflow in a broader setting.

Table VIII.1. Reader V1 engineering sequence

Increment	Exit evidence
Capture bench	Readable field coverage and stable acquisition records.
Calibration	Traceable scale and characterized spatial residuals.
Quality checks	Detection of relevant unreadable or unsupported images.
Measurement	Independent agreement within the proposed domain.
Review and interpretation	Correct rules, corrections and approval behavior.
Integrated workflow	End-to-end identity, export and recovery verification.

Keep the biological method controlled

The reader should not change incubation or preparation conditions merely to simplify imaging. Any proposed method change requires a separate assessment. The exact reference components in the concluding BOM are a dated procurement configuration, not a commitment to a permanent chip generation or enclosure shape.

The programme should use explicit go/no-go decisions after each major uncertainty is tested. If the optical arrangement cannot meet the required field and measurement performance, software complexity should not be used to conceal the limitation. The appropriate response is to revise the arrangement or narrow the intended scope before proceeding.

VIII / DEVELOPMENT AND DELIVERY

Optical bench development

The optical bench should measure its own limitations

Optical development should establish field coverage, sampling, distortion, focus behavior, illumination uniformity and stability. The bench should allow controlled comparison of alternatives without implying that the final product must use one fixed enclosure or station arrangement.

Proposed bench programme

Use appropriate reference artifacts to characterize geometry and image formation, then evaluate representative biological appearances within authorized laboratory work. Record lens settings, distances, illumination, exposure and processing. A photograph of a clear plate is insufficient evidence of calibrated measurement performance.

Table VIII.2. Optical bench development

Bench question	Proposed observation
Is the full field usable?	Focus and distortion behavior across the intended region.
Does scale change with pose?	Measurement effect of permitted height and tilt variation.
Is illumination stable?	Spatial and temporal reference-image metrics.
Are boundaries visible?	Contrast and ambiguity across relevant plate appearances.
Can the system be recalibrated?	Repeatable setup and independent verification after adjustment.

Compare alternatives on the same task

Conventional and telecentric optics, different illumination geometries and alternate acquisition settings should be compared using the intended measurement and service constraints. Telecentricity may reduce some magnification variation, but field size, cost and integration burden remain relevant. [\[R15\]](#)

The development report should separate object-plane sampling from achieved accuracy. A small nominal pixel pitch does not account for blur, distortion, calibration or boundary ambiguity. The selected arrangement should be justified by a measured uncertainty budget and practical reproducibility, with unresolved limitations carried into the next prototype rather than hidden by a high-resolution specification.

VIII / DEVELOPMENT AND DELIVERY

Mechanical design and service access

Mechanical design should support measurement and maintenance

The mechanical system should hold relevant optical and sample relationships within their permitted ranges while allowing cleaning, service and safe operation. Plate loading, access and thermal behavior are part of the measurement design. A rigid-looking enclosure does not establish stability under actual use.

Proposed design considerations

Candidate configurations should be assessed for vessel compatibility, repeatable placement, access to components, cable routing and contamination-sensitive surfaces. The design should avoid unnecessary crevices and provide materials and finishes appropriate to the cleaning process being qualified.

Table VIII.3. Mechanical design and service access

Mechanical function	Verification question
Plate positioning	Does placement variation remain within the calibrated domain?
Optical support	Are alignment and focus stable during ordinary handling?
Light exclusion	Is ambient-light influence controlled over the intended environment?
Service access	Can components be replaced without uncontrolled misalignment?
Cleaning	Can the intended procedure be performed without damage or hidden contamination paths?

Design for recovery from ordinary mistakes

Misloading should be detectable before measurement. Access panels and moving parts need appropriate safeguards. The instrument should make it clear when a service action requires recalibration or readiness checks before returning to use.

Early prototypes may use adjustable fixtures to explore geometry. A later production design should reduce unnecessary adjustment while retaining the controls needed for calibration and service. The transition should be supported by tolerance analysis and measured assembly variation. No particular multi-zone layout or final industrial form is fixed by this compendium; alternative configurations remain open to evidence-led selection.

VIII / DEVELOPMENT AND DELIVERY

Electronics and power development

Electronics should provide stable, observable power and signal paths

Electronics development should define power domains, interfaces, protection, grounding and timing requirements before selecting components. Compute capability is only one requirement. Stable acquisition, manageable heat, serviceability and predictable failure behavior are equally relevant to a measurement instrument.

Proposed electrical design record

The design should document expected and peak loads, supply tolerances, connection ratings and behavior during startup, shutdown and interruption. Sensor signals should retain appropriate conditioning and calibration. Loads such as illumination or motors should not be driven directly from interfaces that are not rated for them.

Table VIII.4. Electronics and power development

Design area	Required engineering evidence
Power budget	Measured demand under representative and worst relevant workloads.
Protection	Appropriate response to overload, incorrect connection and supply disturbance.
Signal integrity	Reliable communication and measurement under expected interference.
Thermal behavior	Stable operation within qualified component and system limits.
Serviceability	Identifiable connections and controlled replacement procedures.

Evaluate substitutions as system changes

A newer processor may offer more performance but introduce thermal, software or availability disadvantages. Selection should compare the actual workload, power envelope, support and validation burden. The concluding BOM records current candidates for one reference configuration; the main design remains generation-independent.

Custom electronics can later reduce wiring complexity or improve efficiency when the required interfaces and performance are stable. Before that transition, modular procurement can support rapid learning. Both stages require configuration records and verification of the complete signal and power path, especially where an electrical fault could produce a plausible but incorrect observation.

VIII / DEVELOPMENT AND DELIVERY

Embedded software and control development

Embedded control should make instrument state explicit

The embedded layer coordinates acquisition, readiness checks, local storage and hardware status. Its design should distinguish commanded actions from confirmed outcomes. An image request, for example, is not a completed observation until the acquired data and associated record have been verified and stored.

Proposed state model

Useful states include initialization, ready, acquiring, processing, awaiting review, degraded and faulted, with transitions defined by the relevant workflow. A physical-control subsystem may require additional states. The exact implementation can vary, but unsupported transitions should be rejected consistently.

Table VIII.5. Embedded software and control development

Control responsibility	Proposed behavior
Startup	Verify configuration, essential components and storage readiness.
Acquisition	Associate data with the correct plate and timing context.
Local persistence	Commit complete records and identify interrupted writes.
Fault handling	Preserve diagnostics and stop affected functions appropriately.
Recovery	Reconcile pending jobs without inventing completed actions.

Keep timing and identity close to acquisition

The capture record should be created from the actual instrument event rather than reconstructed from a later filename or upload time. Clock status, sequence identifiers and configuration should remain available during offline operation.

Verification should challenge rapid repeated commands, power loss, full storage, unavailable sensors and invalid configuration. A responsive interface is not enough if background acquisition silently fails. The embedded layer should provide reliable evidence of what the instrument did, allowing higher-level AI and workflow software to reason from trustworthy observations rather than optimistic command status.

Measurement software development

Measurement software should preserve the numerical chain

The measurement pipeline should connect the original image to calibrated geometry, quality assessment and a reviewable output. Each transformation needs a defined purpose and version. Preprocessing that improves visual appearance can change edge location or suppress small features, so its effect on the measurand must be evaluated.

Proposed pipeline development

Begin with explicit geometric baselines and independently assessed boundaries. Compare candidate segmentation approaches on representative data, including ambiguous cases. Preserve calibration and image-coordinate transforms so that a reviewer can understand how a numerical diameter was obtained.

Table VIII.6. Measurement software development

Pipeline stage	Development evidence
Input qualification	Correct format, identity, completeness and image-quality state.
Calibration	Valid transform and uncertainty for the current configuration.
Feature analysis	Detection and segmentation performance in the intended domain.
Measurement	Correct units, geometry and reporting conventions.
Review overlay	Faithful display of the measured feature on the original image.

Compare custom models with conventional methods

Custom models should be selected because they improve the defined task, not because a particular architecture is fashionable. A simpler algorithm may offer adequate performance, easier verification or lower compute demand. The comparison should include error, coverage, uncertainty and operational cost.

The pipeline should return explicit unavailable states when required evidence is missing. It should not manufacture a diameter from a low-confidence boundary or treat a failed calculation as zero. Later AI improvements can be introduced through controlled releases while historical records retain the exact measurement pipeline that produced them.

VIII / DEVELOPMENT AND DELIVERY

PIMS and interface development

PIMS development should begin with identity and record integrity

PIMS is the proposed Plate Information Management System linking physical objects, observations, analyses and reviewer decisions. Its initial implementation should make those relationships reliable before adding advanced analytics or conversational features. A complete record path supports both product use and later research.

Proposed implementation sequence

Define stable identifiers and the core event schema. Implement capture ingestion, calibration references and measurement records. Add review, correction and export with explicit version relationships. Then test recovery, retention and access behavior across the complete workflow.

Table VIII.7. PIMS and interface development

PIMS increment	Acceptance evidence
Identity model	Correct case, plate and observation relationships.
Ingestion	Complete, idempotent records with integrity checks.
Analysis linkage	Exact input, calibration and software versions retained.
Review	Authorized decisions and correction-driven re-review.
Export	Preserved meaning and receiving-system verification.
Governance	Enforced permissions, retention and audit behavior.

Interfaces should remain explicit

The field dictionaries in the appendices are proposed contracts. They identify the information that must survive between components, not a claim that every interface has been implemented. Implementation should validate required fields, units and allowed value states at the boundary.

Data migrations need their own verification. A new schema should preserve historical meaning and identify transformations rather than silently replacing old records. PIMS should make it possible to retrieve an original observation and reproduce or explain its result even after models, rules and interfaces have changed. That capability is central to the platform's long-term scientific and service value.

VIII / DEVELOPMENT AND DELIVERY

Dataset operations and model development

Dataset operations are an ongoing product capability

Useful model development depends on consistent collection, annotation, quality assessment and release of authorized data. These activities should be treated as a managed operation rather than an occasional export from routine records. The process must preserve biological identity and the distinction between observation, reviewer judgment and confirmed reference evidence.

Proposed dataset workflow

Collection plans should identify the evidence gaps they address. Incoming data are checked for completeness, permission, duplication and technical quality. Annotation uses defined instructions and expert review appropriate to the task. A release records its composition, lineage, known limitations and permitted use.

Table VIII.8. Dataset operations and model development

Operational stage	Controlled output
Collection planning	Target domain, permissions and gap coverage.
Ingestion review	Identity, completeness and quality disposition.
Annotation	Versioned labels, assessor context and disagreement.
Release	Dataset manifest, partitions and use restrictions.
Model development	Reproducible training and baseline comparison.
Evaluation	Frozen configuration and independent results.

Prioritize information, not volume alone

Active data selection may help identify uncertain or underrepresented cases for expert review. It should not distort the final evaluation population or turn model confidence into a biological reference. Collection priorities should consider rare failure modes and the practical cost of confirmation.

The proposed operation should support corrections to labels and permissions with clear downstream impact tracking. A revised annotation may affect a training release, while a withdrawn authorization may affect future use. Model development should remain connected to those governance events. This makes the data service useful for sustained research without assuming that every stored observation is automatically available for every future AI purpose.

VIII / DEVELOPMENT AND DELIVERY

Design reviews and configuration control

Design reviews should resolve specific technical decisions

A design review should assess whether evidence supports the next development step. It should bring together the people responsible for measurement, software, biological interpretation, safety, manufacturing and service as appropriate. Its value lies in resolving assumptions and interfaces before they become expensive or difficult to change.

Proposed review inputs

The review should include requirements, alternative designs, test results, unresolved issues and the configuration being assessed. A rendered concept or demonstration can communicate an idea, but it should not be mistaken for verified engineering evidence.

Table VIII.9. Design reviews and configuration control

Review decision	Required rationale
Select an approach	Measured or justified advantage for the intended requirement.
Accept a limitation	Defined affected domain and consequence for use.
Request more evidence	Specific uncertainty and the test that could resolve it.
Change a requirement	Reason, affected interfaces and evidence impact.
Stop a track	Clear failure boundary or insufficient value for further work.

Configuration control preserves comparability

Each prototype and test should identify the hardware, software, calibration and method context. A result from one configuration should not automatically support another. Changes to optics, mounting or preprocessing may alter measurement behavior even when the product name remains the same.

The proposed process should retain the decision and its evidence, with follow-up actions tied to a clear owner. This is product development discipline, not a requirement to freeze the platform's form prematurely. Flexible exploration remains possible when the conditions and consequences of each alternative are recorded well enough to compare them fairly.

VIII / DEVELOPMENT AND DELIVERY

Supplier qualification and substitution

Supplier qualification should address quality and continuity

Component selection should consider specification, authenticity, availability, support and the consequences of substitution. A low public price is useful procurement information, but it does not establish domestic stock, delivery to the intended location or suitability for a measurement product. The dated BOM therefore separates observed listings from quotations and unresolved procurement checks.

Proposed supplier assessment

For critical components, the programme should identify authorized supply routes, revision control, documentation and incoming acceptance criteria. Supplier changes should trigger an assessment of whether the component's relevant behavior remains equivalent.

Table VIII.10. Supplier qualification and substitution

Supplier question	Proposed evidence
Is the exact part identified?	Manufacturer, part number, variant and revision where relevant.
Is the source reliable?	Traceable supply route and authenticity checks.
Is availability confirmed?	Current stock, dispatch location and destination-specific quotation.
Can changes be detected?	Revision and discontinuation information.
Is substitution acceptable?	Compatibility and performance verification for affected functions.

Define critical characteristics

For a lens, focal length alone may be insufficient; mount, image circle, focus range and distortion can matter. For a power supply, connector fit does not establish electrical compatibility. Incoming checks should target the properties that affect the system rather than merely confirming that the item powers on.

The proposed programme should maintain alternatives where practical, but avoid claiming interchangeability before assessment. A substitution may require recalibration, software changes or additional validation. Procurement resilience comes from understanding these dependencies and planning qualification, not from listing many nominally similar products in a catalogue.

VIII / DEVELOPMENT AND DELIVERY

Prototyping and design for manufacture

Prototyping should answer questions about production as well as function

An adjustable laboratory prototype can establish a measurement principle. A manufacturable design must also tolerate component variation, repeatable assembly, inspection, cleaning and service. The transition should preserve the function demonstrated on the bench while reducing dependence on individual craftsmanship or undocumented adjustment.

Proposed design-for-manufacture review

The team should examine part count, tolerances, assembly sequence, fasteners, cable routing and access for inspection. Critical optical relationships should have a defined assembly and verification method. A design that requires repeated subjective alignment may be difficult to reproduce consistently.

Table VIII.11. Prototyping and design for manufacture

Production concern	Development response
Tolerance accumulation	Analyze and measure its effect on calibration and field coverage.
Assembly variation	Use fixtures or references for critical relationships.
Part identification	Prevent visually similar but incompatible components being confused.
Inspection access	Make important joints, settings and serial identities observable.
Repair	Support replacement without unnecessary damage or recalibration burden.

Pilot builds reveal different problems

A small series of controlled builds can expose variation that one prototype does not show. Record assembly time, rework, failed checks and measurement performance across units. These findings should inform design changes before claims about scalable manufacture are made.

The proposed vertical-integration strategy should retain this evidence-led approach. Custom parts can improve consistency when requirements are stable, while premature tooling can lock in unresolved assumptions. Manufacturing readiness should be demonstrated through repeatable builds and acceptance evidence, not inferred from a detailed rendering or a completed bill of materials.

Assembly and production verification

Production verification should establish the identity and readiness of each unit

Each delivered instrument should have a traceable configuration and evidence that it meets the applicable production acceptance requirements. Assembly completion is not equivalent to measurement readiness. The acceptance process should verify the functions and critical characteristics that can vary between units.

Proposed unit record

The record should identify the instrument, critical components, software and model versions, calibration, inspection results and disposition. Rework should remain visible, with the affected checks repeated as appropriate. A failed unit should not disappear from production quality statistics after repair.

Table VIII.12. Assembly and production verification

Acceptance area	Proposed check
Identity and configuration	Correct components, revisions and software package.
Electrical readiness	Relevant power, protection and thermal behavior.
Optical performance	Field, focus, illumination and calibration suitability.
Functional workflow	Capture, storage, review and export behavior.
Safety and security	Applicable safeguards and deployment configuration.
Documentation	Instructions, service information and unit-specific records.

Acceptance criteria need a validated basis

Production checks should relate to design verification and validation. A convenient test is insufficient if it does not detect variation that can affect the result. The programme should establish whether a reference-artifact check predicts the relevant instrument behavior and what additional checks remain necessary.

Trend analysis across production can identify drift, supplier variation or an assembly problem. Such findings should enter corrective action and configuration review. The proposed process supports consistent delivery while preserving the distinction between unit acceptance and the broader clinical or analytical evidence supporting the product's intended use.

Calibration transfer to production

Calibration must transfer from development to routine production

The calibration method should be reproducible across instruments and service locations. It needs appropriate reference artifacts, environmental conditions, calculation methods and acceptance criteria. A calibration performed successfully by the development team does not establish that the same result can be obtained reliably during production or field service.

Proposed transfer package

The package should define the reference target, placement, acquisition settings, computation and independent verification. It should specify what invalidates calibration and how out-of-tolerance findings affect prior measurements. Traceability should be appropriate to the measurand and intended uncertainty. [\[R16\]](#)

Table VIII.13. Calibration transfer to production

Transfer element	Required evidence
Reference artifact	Identity, uncertainty or verification basis and condition.
Procedure	Reproducible setup and controlled acquisition.
Computation	Versioned method and correct units.
Verification	Independent check not simply reusing the fitted target points.
Recalibration trigger	Defined changes, damage or drift conditions.

Assess the calibration operator and fixture

Placement variation, target wear and interpretation of instructions can affect results. A transfer study should include the people and equipment expected to perform calibration. It should measure repeatability and identify where fixtures or clearer instructions reduce variability.

The instrument should store calibration identity and validity with each measurement. Replacing a lens, altering focus or changing a relevant image pipeline can require requalification. Service should therefore include both the physical replacement and the evidence needed to restore measurement readiness. Calibration is a continuing product responsibility rather than a one-time factory certificate detached from subsequent changes.

VIII / DEVELOPMENT AND DELIVERY

Installation and site acceptance

Installation should verify the actual operating environment

Site acceptance establishes that the delivered instrument and workflow are ready for the intended local use. It should assess the environment, configuration, interfaces and trained users. A successful factory test does not establish that power, networking, space and laboratory procedures at the destination are suitable.

Proposed installation assessment

The site should confirm placement, relevant environmental limits, electrical arrangements and authorized network access. The instrument configuration and calibration should be checked after transport. Interfaces should be verified with the receiving laboratory system using appropriate test records.

Table VIII.14. Installation and site acceptance

Site acceptance area	Proposed evidence
Environment	Conditions within the qualified operating range.
Instrument	Correct identity, configuration and post-transport readiness.
Data flow	Authorized access, export mapping and receiving-system readback.
Users	Completion of relevant training and critical-task assessment.
Continuity	Defined response to outages, faults and support needs.

Agree on responsibility before routine use

The laboratory and service team should understand who performs readiness checks, calibration review, software updates and incident escalation. The installation record should identify limitations or local restrictions. Unresolved issues should lead to a defined disposition rather than an assumption that ordinary use will reveal whether they matter.

The proposed acceptance process should preserve evidence of unsuccessful checks and their resolution. Where the product remains investigational, site activities must follow the authorized study and regulatory conditions. Installation of a prototype does not itself establish clinical approval or qualification for routine diagnostic reporting.

Training and operational handover

Training should prepare users for normal work and exceptions

Operational handover should ensure that intended users understand the product's scope, critical tasks and limits. Training should cover ordinary capture and review as well as identity conflicts, unreadable images, corrections and recovery. Users should know when to stop and seek support rather than improvise around a failed control.

Proposed training structure

Role-specific instruction can separate operator, reviewer, administrator and service responsibilities. Practical exercises should use appropriate examples with known teaching objectives. Synthetic examples are useful for rare software states, while real authorized images are needed for relevant visual interpretation skills.

Table VIII.15. Training and operational handover

Training area	Demonstrated competency
Identity and capture	Correct association and recognition of incomplete acquisition.
Measurement review	Understanding calibration, boundaries and uncertainty.
Interpretation	Use of the applicable method and versioned rules.
Correction	Preserving history and renewing affected approval.
Failure handling	Appropriate abstention, repeat or escalation.
Data protection	Authorized access, export and incident reporting.

Evaluate understanding after instruction

Attendance does not establish competency. Assessment should use independent examples and critical scenarios appropriate to the role. Refresher training may be needed after significant interface or workflow changes. Records should identify what was assessed and any limitations in authorization.

The proposed handover should include usable instructions, service contacts and a clear support process. Language and accessibility should be evaluated with intended users. The objective is dependable operation within a qualified scope, not reliance on the development team's continued presence to explain ambiguous behavior or resolve every routine exception.

VIII / DEVELOPMENT AND DELIVERY

Maintenance and support delivery

Service delivery should restore a qualified function

Maintenance and support are part of the proposed product portfolio. Their purpose is to keep instruments and records dependable over the service life. A repair is complete when the affected function has been verified, not merely when a replacement part has been installed or an error message has disappeared.

Proposed service workflow

Support begins with a clear description of the fault, instrument identity and affected workflow. Remote diagnostics should use only authorized data. On-site or depot work should preserve configuration, record replaced components and identify any need for recalibration or software requalification.

Table VIII.16. Maintenance and support delivery

Service activity	Required output
Triage	Affected function, urgency and safe operating disposition.
Diagnosis	Evidence supporting the suspected cause.
Repair	Controlled action and component or configuration changes.
Verification	Relevant readiness, calibration and functional checks.
Return to use	Authorized disposition and remaining limitations.
Feedback	Quality trend and corrective-action input.

Service terms should match evidence and capacity

Response times, availability commitments and maintenance intervals should be established from the actual delivery model and field evidence. They are not fixed promises in this compendium. The programme should measure time to diagnosis, restoration and recurrence alongside user burden.

The service portfolio can include calibration, preventive maintenance, training, interface support and research assistance. Each requires a defined scope and qualified personnel. Predictive maintenance may later improve planning, but it should complement mandatory checks and confirmed service findings. A model's low failure-risk estimate cannot override evidence that an instrument is outside its qualified operating condition.

VIII / DEVELOPMENT AND DELIVERY

Fleet updates and rollback

Fleet updates should be reversible and traceable

Software, models and standards content may change during the product's life. The proposed update process should establish what is changing, which units are eligible and what evidence supports deployment. Historical records must retain the versions that produced their results, even after the instrument is updated.

Proposed staged deployment

An update should be verified in a controlled environment, then introduced through a defined deployment sequence with monitoring and rollback criteria. Compatibility checks should include hardware, calibration, data schema and interfaces. The update should not interrupt a critical operation without a qualified transition plan.

Table VIII.17. Fleet updates and rollback

Update concern	Proposed control
Package integrity	Verify the authorized release and reject altered content.
Compatibility	Check required versions and configuration before installation.
Incomplete installation	Restore a defined state without corrupting records.
Performance regression	Monitor prespecified quality and workflow indicators.
Rollback	Preserve data meaning and return to an eligible configuration.

A rollback is more than replacing an executable

Database migrations, changed calibration and new result semantics can make reversal difficult. The design should specify which changes are reversible and how records created under a newer version remain interpretable. A rollback must not erase the audit trail of the attempted update.

Fleet monitoring should identify affected configurations and sites without collecting unnecessary patient information. If a defect can affect prior results, the quality process should assess the scope and appropriate communication. The proposed system should support that investigation through version lineage, rather than relying on users to remember when their instrument changed.

VIII / DEVELOPMENT AND DELIVERY

Custom electronics strategy

Custom electronics should follow stable measurement requirements

Vertical integration can improve power efficiency, signal quality, assembly consistency or serviceability when the product's requirements are sufficiently understood. The proposed strategy begins with modular, available components for learning and moves toward custom boards where measured benefits justify the engineering and qualification burden.

Candidate integration stages

An early custom board might organize power, illumination control and sensor interfaces while retaining a replaceable compute module. Later revisions could integrate acquisition or specialized processing. The sequence should follow bottlenecks demonstrated in the actual workload.

Table VIII.18. Custom electronics strategy

Integration decision	Evidence needed
Combine interfaces	Reduced wiring or improved reliability without unacceptable coupling.
Customize power	Measured efficiency, stability and protection benefits.
Integrate acquisition	Timing or signal-quality improvement for the defined measurement.
Reduce component count	Reproducible assembly and maintainable service path.
Change compute	Performance per watt, support and validation advantage.

Custom does not automatically mean better

Integration can create shared failure modes and make replacement more difficult. It can also increase dependence on a specific supplier or fabrication process. The comparison should include lifecycle support, test access and recovery from component obsolescence.

Each custom revision should retain verification points for critical signals and power domains. Production testing and calibration must be planned alongside circuit design. The long-term objective is an efficient and dependable platform whose custom elements serve demonstrated needs. No custom electronic assembly or manufacturing readiness is claimed as completed merely because this strategy is described.

VIII / DEVELOPMENT AND DELIVERY

Custom model and silicon strategy

Custom models and eventual silicon require different maturity gates

AMReye.AI intends to develop task-specific models and may eventually investigate specialized silicon for efficient inference or acquisition. Custom model development can begin while hardware remains modular. Custom silicon requires a much more stable workload, interface and support strategy because its physical implementation is harder to revise.

Establish the workload before specializing it

The programme should measure inference latency, memory use, energy, accuracy and uncertainty on representative tasks. Model compression or acceleration must preserve the behavior required by the validated workflow. A smaller model that changes abstention or rare-error behavior needs appropriate reassessment.

Table VIII.19. Custom model and silicon strategy

Stage	Proposed exit evidence
Custom software models	Independent task performance and reproducible training.
Software optimization	Measured efficiency without unacceptable functional change.
Programmable acceleration	Stable workload and qualified numerical behavior.
Specialized electronics	System-level benefit and maintainable interfaces.
Custom silicon research	Justified scale, stable requirements and verification capability.

Preserve an update and service path

Clinical knowledge, standards and model requirements evolve. A specialized design should keep appropriate functions updateable and avoid hard-coding assumptions that are likely to change. Standards interpretation, in particular, should remain independently versioned rather than embedded as an immutable hardware truth.

The ambition for deep customization is a staged strategy, not a claim that every chip is currently designed in-house. Selection should follow measured performance, power, quality, availability and lifecycle needs. The newest commercial component is not automatically the best choice, and a custom component is not automatically the most efficient solution for a still-changing research workload.

VIII / DEVELOPMENT AND DELIVERY

Partnerships and technology transfer

Partnerships should define contributions and evidence ownership

The proposed platform can benefit from laboratory, engineering, manufacturing and research partnerships. Each partnership should identify the problem being addressed, the work each party performs and the outputs each receives. A broad expression of interest should not be represented as a secured collaboration or completed validation.

Proposed collaboration structure

A laboratory partner may contribute workflow expertise and authorized evaluation material. An engineering partner may support optics, electronics or manufacturing. A research partner may develop methods or independent analysis. These contributions need compatible quality, data and intellectual-property arrangements.

Table VIII.20. Partnerships and technology transfer

Partnership element	Required agreement
Scope	Specific function, study or service to be delivered.
Responsibilities	Technical, clinical, statistical and governance roles.
Data	Access, permitted use, retention and publication conditions.
Intellectual property	Background rights and treatment of new developments.
Evidence	Reference methods, review and reporting of unsuccessful findings.
Continuity	Support, termination and access to required records.

Technology transfer needs reproducibility

A design or model is transferable when another qualified team can reproduce the relevant function using the supplied specifications, procedures and evidence. Source code or a drawing alone may be insufficient. Calibration, test fixtures, training and quality controls can be equally important.

The programme should assess whether a partner can maintain the intended quality and service responsibilities. Transfer should preserve configuration and evidence lineage so that local changes do not inherit unsupported claims. The product compendium describes the proposed collaboration model; it does not announce any institution, supplier or regulator as an appointed or committed partner.

Development decisions and exit criteria

Exit criteria should decide what the evidence permits next

Development decisions should connect the current evidence to a bounded next step. A successful prototype may justify a more representative evaluation, while a failed assumption may justify redesign or restriction. The decision should state what has been demonstrated and which uncertainties remain material.

Table VIII.21. Development decisions and exit criteria

Development track	Proposed exit criterion
Reader V1	Complete qualified capture-to-review chain in the defined endpoint domain.
Advanced sensing	Added information demonstrated against a strong baseline.
Automation	Reliable handling, identity and recovery within a specified workflow.
Predictive twin	Independent forecasting evidence with calibrated uncertainty and bounded horizon.
Data services	Authorized, reproducible and useful information delivery.
Vertical integration	Measured system benefit with production and service feasibility.

Decide on evidence rather than accumulated features

A larger sensor catalogue, more AI functions or a more elaborate enclosure does not establish a better product. The relevant question is whether each addition improves the defined measurement, workflow or research capability enough to justify its complexity and risk.

The programme should maintain alternative paths where major uncertainties remain. For example, an imaging-only reader can progress while a gas-sensing hypothesis is tested separately. Failure of an exploratory modality should not invalidate a qualified function that does not depend on it.

The long-term AMReye.AI vision combines instruments, software, data and services around traceable observation. Its execution should produce evidence that partners can inspect and users can rely on. Each released capability should carry a clear scope, support model and maturity statement, while future research remains visible as an opportunity requiring its own proof.

Interface catalogue and record relationships

Proposed interfaces preserve evidence across transformations

The following dictionaries document proposed record contracts. They are design specifications for review, not a claim that an API, database or device interface has been implemented. Field names are illustrative and can be revised through schema governance without changing the underlying provenance requirements.

Record relationships

A specimen or research sample can produce multiple preparations and plates. Each plate can have multiple capture attempts. Observations and measurements reference the exact capture or biological context that produced them. Interpretations and reviewer decisions reference immutable versions of those results.

Table A.1. Interface catalogue and record relationships

Record	Principal links
Capture	Plate, acquisition attempt, instrument and configuration.
Sensor observation	Capture or observation session, channel and calibration.
Calibration	Artifact, procedure, instrument and validity conditions.
Measurement	Source image, region, calibration and algorithm.
AI output	Input set, model, inference configuration and uncertainty.
Interpretation	Measurement, organism context and standards package.
Reviewer decision	Reviewed versions, identity, action and reason.
Export	Approved snapshot, destination mapping and transfer outcome.

Shared semantics

Every record should have a unique identifier, schema version, creation time and originating system. A revision should identify the prior version without overwriting it. Missing data require a reason rather than a plausible default.

The value state should distinguish **measured, derived, imputed, simulated and unavailable**. These states describe origin, not quality. A measured value can be unreliable; a simulated value can be useful within its research scope. Quality, eligibility and review disposition therefore require separate fields. The export contract should preserve these distinctions so that another system cannot silently convert an estimate into an observation.

Capture record field dictionary

A capture record identifies an acquisition attempt, including failure

The proposed capture contract separates the planned observation from the image actually obtained. An unsuccessful attempt still produces an auditable event. A later successful capture has its own identifier and timestamp; it does not replace the missing earlier observation.

Table A.2. Capture record field dictionary

Field	Proposed meaning and constraint
capture_id	Unique immutable identifier for this attempt.
plate_id / session_id	Links to the physical plate and observation sequence.
captured_at / clock_state	Timestamp, time zone or UTC basis, and synchronization status.
instrument_id / config_id	Qualified hardware and acquisition configuration.
image_uri / image_hash	Controlled storage reference and content integrity digest.
geometry / image_format	Width, height, channels, encoding and orientation.
acquisition_settings	Exposure, illumination and relevant capture parameters.
calibration_ids	Applicable geometric and radiometric calibration references.
quality_flags	Focus, clipping, glare, obstruction and other evaluated conditions.
capture_status / reason	Succeeded, failed or incomplete, with a specific reason.

Validation rules

An image hash cannot be supplied when no image exists. A failed capture must not receive a fabricated zero-size image or a copied prior frame. If the clock is unreliable, the system should retain that uncertainty and avoid unsupported temporal analysis.

Image orientation and coordinate conventions should be explicit so that a measurement region remains reproducible after export. Storage references should not expose unrestricted access to confidential images. The proposed consumer should verify integrity and eligibility before analysis, then retain the original bytes alongside any normalized derivative. A recapture can resolve an image-quality problem while leaving a gap in a time series that still requires disclosure.

Sensor observation field dictionary

A sensor observation retains its raw signal and interpretation state

The proposed contract accommodates biological signals, environmental measurements and instrument-control channels. A common format should not imply that these channels have the same biological meaning. Each observation needs a defined quantity and context.

Table A.3. Sensor observation field dictionary

Field	Proposed meaning and constraint
observation_id / channel_id	Unique record and identified sensing channel.
observed_at / interval	Acquisition time and integration or averaging duration.
context_id	Plate, capture, enclosure or instrument context.
quantity / unit	Defined measurand and machine-readable unit.
raw_value / raw_unit	Original output before conversion or compensation.
value / value_state	Interpreted value and measured/derived/imputed/simulated status.
calibration_id	Applicable conversion and traceability record.
uncertainty	Value, coverage or method, and stated interpretation.
quality / range_state	Validity, saturation, below-range or above-range condition.
missing_reason	Required when the intended observation is unavailable.

Preserve different kinds of absence

A disconnected channel, a value below a detection limit and a channel not installed are different states. The record should not encode all three as zero. A below-range observation should retain the bound and method context where known.

Compensation should produce a derived value linked to the raw signal and conversion version. If several channels share a compensation input, that dependency should remain inspectable. Downstream fusion should receive both the value and its quality state, including any timing mismatch. This allows a model or reviewer to distinguish biological disagreement from a sensor that was never eligible to contribute.

Calibration record field dictionary

Calibration records define where a conversion is valid

A calibration is a relationship between a known reference and an instrument response under stated conditions. The proposed record should preserve the artifact, procedure and results needed to understand that relationship. A calibration identifier alone is insufficient if its scope is inaccessible.

Table A.4. Calibration record field dictionary

Field	Proposed meaning and constraint
calibration_id / revision	Unique calibration and immutable revision.
instrument_id / channel	Instrument, optical path or sensor channel covered.
procedure_id / version	Controlled procedure and analysis version.
artifact_id / certificate	Reference identity, certificate and traceability where available.
performed_at / operator	Time and authorized person or qualified automated process.
conditions	Temperature, geometry and other relevant conditions.
coefficients / model	Conversion relationship and coordinate applicability.
residuals / uncertainty	Fit residuals, uncertainty basis and unresolved effects.
valid_domain	Spatial field, range, configuration and other eligibility limits.
disposition / triggers	Accepted or rejected, and recheck or recalibration triggers.

Consumer responsibilities

The proposed measurement service should verify that a calibration matches the actual configuration and operating domain. An expired, superseded or incompatible record should not be selected merely because it is the most recent entry.

Changes to lens position, camera orientation, mechanical alignment or relevant firmware can require reassessment. The record should support those decisions without automatically assuming every change has the same impact. Historical measurements retain their original calibration reference. If a later investigation changes their interpretation, the correction should identify the affected records and preserve the original result rather than silently rewriting its metrological history.

Measurement record field dictionary

A measurement record links a quantity to the evidence used to estimate it

The proposed measurement contract supports zone diameters and other defined image-derived quantities. It separates an algorithm's geometric output from standards interpretation and final review. A numeric diameter should not implicitly carry a susceptibility category.

Table A.5. Measurement record field dictionary

Field	Proposed meaning and constraint
measurement_id / revision	Immutable measurement version.
capture_id / image_hash	Exact source image and integrity link.
object_id / region	Disk or feature identity and image coordinates.
measurand / unit	Defined quantity, such as diameter in mm.
value / value_state	Numeric result and origin, normally derived from the image.
boundary_geometry	Contour or geometric representation used for measurement.
calibration_id	Applicable scale and distortion relationship.
algorithm_id / version	Method and configuration used to produce the result.
uncertainty / quality	Quantified uncertainty where established and quality flags.
eligibility / reason	Whether the measurement can support the intended next step.

Corrections and unavailable values

A reviewer-adjusted boundary should create a new measurement version linked to the original and the associated decision. A missing or unreadable zone should use an unavailable state with a reason. It should not use a conventional numeric value that another system might interpret as a real diameter.

The record should retain sufficient precision for analysis while applying any required reporting convention only at the appropriate stage. If rounding affects interpretation, the selected rule should be explicit and reproducible. Geometry should remain tied to the source coordinate system so that a reviewer can inspect the exact boundary behind the number.

AI output field dictionary

An AI output is a versioned inference with a defined task

The proposed interface should make an inference reproducible and distinguish it from a reviewed conclusion. Different capabilities may return geometry, a quality flag, a forecast or a ranked research hypothesis. Their outputs should declare their semantics instead of sharing an ambiguous confidence field.

Table A.6. AI output field dictionary

Field	Proposed meaning and constraint
inference_id / task_id	Unique execution and defined analytical task.
input_manifest	Exact input records, hashes and value states.
model_id / version	Controlled model artifact and integrity reference.
inference_config	Preprocessing, thresholds and relevant runtime configuration.
output / output_schema	Typed result with units and task-specific meaning.
uncertainty_method	Method and interpretation of any reported uncertainty.
domain_checks	Evidence of input eligibility and distribution checks.
abstention / reason	Whether the model declined to supply a usable result.
dependencies	Calibration, reference data and other model outputs used.
runtime_status	Completion, failure, partial output and timing.

Confidence is not automatically probability

A raw score, calibrated probability and prediction interval are different objects. The contract should identify which is being returned and under which validation conditions it has meaning. A percentage should not be displayed simply because it looks intuitive.

The system should preserve outputs from failed or abstaining runs for investigation where appropriate, while preventing them from becoming reportable conclusions. A later model version may produce a different answer to the same image. That new inference should be linked as a separate analysis, retaining the earlier output and the reviewer decisions that relied on it. Retrospective reanalysis does not silently revise a previously authorized report.

Standards interpretation field dictionary

Standards interpretation is a traceable transformation of eligible evidence

The proposed interpretation record should bind a measurement to the exact standards package and biological context used. Standards content requires independent version control, including corrections. It should not be embedded invisibly in a model artifact or inferred from the current date.

Table A.7. Standards interpretation field dictionary

Field	Proposed meaning and constraint
interpretation_id	Unique immutable interpretation record.
measurement_revision	Exact measurement version being interpreted.
organism_context	Identity, identification source and unresolved ambiguity.
antimicrobial_context	Agent, disk content or exposure definition as applicable.
method_context	Method, medium, incubation and specimen conditions required.
standards_package	Authority, edition, correction set and package integrity.
rule_id	Applicable rule and relevant comments or restrictions.
rounding_rule	Transformation between retained precision and reported input.
category / eligibility	Result only when all necessary conditions are met.
unresolved_reason	Missing context, absent rule or another blocking condition.

No cross-method fallback

The absence of an eligible rule should result in an unresolved interpretation. An endpoint rule should not be substituted for an early reading, and a rule for one organism should not be applied to another because their measurements appear similar. [\[R01\]](#) [\[R02\]](#) [\[R04\]](#)

The proposed interface should preserve a readable explanation of the rule selection. When standards change, historical reports retain their original interpretation basis. Any authorized re-interpretation should create a new version with a reason and a clear relationship to the earlier record. This supports clinical governance without erasing the conditions under which an earlier decision was made.

Reviewer decision field dictionary

Reviewer decisions identify exactly what was accepted or changed

Mandatory review in Reader V1 requires more than a generic approval flag. The proposed decision record should identify the evidence shown, the versions reviewed and the action taken. Changes to material inputs should invalidate approval of the affected result until the revised evidence is reviewed.

Table A.8. Reviewer decision field dictionary

Field	Proposed meaning and constraint
decision_id / reviewed_at	Unique event and time.
reviewer_id / role	Authorized identity and role at the time of review.
reviewed_manifest	Exact capture, measurement and interpretation versions.
action	Accept, correct, reject, request recapture or retain unresolved.
reason / annotation	Structured reason and necessary explanatory note.
corrected_record_ids	New versions resulting from a correction.
scope	Specific result or defined group covered by the decision.
approval_state	Current validity relative to the reviewed evidence.
supersedes	Prior decision replaced by a later authorized action.
authorization_context	Policy or workflow governing release.

A review must be informed

The interface should expose the original image, relevant quality flags, calibration context and proposed boundary. It should identify simulated or unavailable data clearly. A reviewer should not have to infer which version is being approved from a mutable screen.

Bulk review may be useful for eligible routine cases, but the workflow must preserve the ability to inspect exceptions and define what the bulk action covers. A correction should retain both the initial and revised measurements. The service should prevent export of an obsolete approval after an upstream change, and should show the specific reason that another review is required.

Export and provenance field dictionary

An export is an authorized snapshot with a verifiable delivery outcome

The proposed export contract should preserve the meaning of the reviewed result across system boundaries. A successful network connection does not prove that the receiving system accepted the correct record or interpreted its units and states properly.

Table A.9. Export and provenance field dictionary

Field	Proposed meaning and constraint
export_id / created_at	Unique transfer event and time.
snapshot_id / manifest	Immutable set of approved source versions.
destination_id	Authorized receiving system and route.
mapping_version	Field, unit and terminology transformation used.
payload_hash / schema	Integrity and expected structure.
approval_reference	Decision authorizing the exported content.
value_states	Retained measured, derived, imputed, simulated or unavailable status.
delivery_state	Pending, acknowledged, rejected or uncertain.
acknowledgement	Receiving reference, time and validation response.
retry_lineage	Prior attempts and duplicate-prevention identity.

Transfer and correction behavior

Retries should use an idempotency mechanism so that an uncertain acknowledgement does not create duplicate reports. A receiver rejection should remain visible, with a reason and an authorized recovery path. The instrument should retain the source record even if a destination is temporarily unavailable.

If an approved result changes after export, the workflow should issue a traceable correction or replacement according to the receiving system's conventions. It should not silently resend altered content under the same immutable identity. Research exports must preserve prediction labels and access restrictions. Public publication of a summary is a separate decision from authorized transfer of detailed laboratory records.

Worked example of a reviewed endpoint record

Synthetic example: a reviewed endpoint measurement

This example illustrates proposed record relationships using invented identifiers and values. It is not a patient result, validation observation or clinical interpretation. No susceptibility breakpoint is applied.

Example event sequence

A qualified capture of synthetic plate EX-PLATE-001 produces image EX-IMG-01. Calibration EX-CAL-04 is eligible for the illustrated configuration. The initial derived diameter is 18.6 mm. A reviewer identifies an incorrectly placed boundary and creates a revised measurement of 18.2 mm, retaining both versions.

Table A.10. Worked example of a reviewed endpoint record

Record	Illustrative content
Capture EX-CAP-01	Successful; source image and integrity digest retained.
Measurement EX-M-01	Derived diameter 18.6 mm; algorithm version recorded.
Decision EX-D-01	Correction requested because the boundary needs adjustment.
Measurement EX-M-02	Derived diameter 18.2 mm; supersedes EX-M-01 for review.
Interpretation EX-I-01	Unavailable: no eligible biological and standards context supplied.
Decision EX-D-02	Measurement reviewed; no clinical category authorized.

What the example demonstrates

The corrected number is not substituted into the original record. The reviewer action explains why a new measurement exists, and the absent interpretation remains absent. A measurement review can be complete while a clinical result remains unavailable.

An export for a permitted research purpose could include the reviewed diameter, its derivation and the unresolved interpretation state. It should not insert a susceptible or resistant category merely to satisfy a required field in another system. If the receiving schema cannot represent this state, the export should fail with a meaningful mapping error rather than invent a clinical conclusion.

Worked example of interrupted observation

Synthetic example: observation stops while simulation continues

This invented sequence illustrates the proposed digital-twin contract. Times are relative to an explicitly defined research observation start. Values are illustrative and do not describe measured AMReye.AI performance or biological growth.

Evidence and forecast branches

The final successful observation occurs at 24 hours. A planned 26-hour capture fails, and the physical plate is subsequently unavailable. A previously qualified research model can be run conditionally from the last observed state. Its outputs remain simulations and cannot be relabeled as recovered measurements.

Table A.11. Worked example of interrupted observation

Event	Required record behavior
Observation at 24 h	Preserve the actual source and associated quality state.
Failed capture at 26 h	Record unavailable observation with the failure reason.
Plate unavailable	Record the event and stop expecting physical continuity.
Simulation to 30 h	Identify model, inputs, assumptions and a 6 h horizon.
Simulation to 36 h	Identify a 12 h horizon and its uncertainty or lack of qualification.

When continuation should stop

If the model has no validated horizon for the new conditions, the result should remain exploratory or be withheld. A user request for a longer animation should not extend the qualified prediction domain. The record should distinguish computation completed from prediction eligible for use.

If later observations become available from the same verified physical plate, they form new evidence. They can update the model state and support a comparison with earlier forecasts, but should not overwrite the historical predictions. If the replacement is a different plate, it needs a different physical identity even when prepared from related material.

Worked example of disagreement and recovery

Synthetic example: disagreement triggers investigation and recapture

This invented case illustrates a proposed recovery workflow. A plate image passes a simple brightness check, but the focus assessment fails. An environmental channel also reports an out-of-range condition. Neither a majority vote among checks nor a plausible-looking diameter is sufficient to authorize the result.

Proposed sequence

The system retains the initial capture, the failed checks and the attempted measurement as ineligible. A reviewer investigates the setup and identifies a positioning problem. The environmental condition is assessed separately; it is not assumed to share the same cause merely because it occurred at the same time.

Table A.12. Worked example of disagreement and recovery

Step	Record outcome
Initial capture	Image retained; quality conflict visible.
Measurement attempt	Ineligible output preserved for investigation.
Reviewer investigation	Suspected positioning cause and separate environmental assessment.
Corrective action	Adjustment recorded with any calibration implications.
New capture	New identifier, timestamp and complete quality evaluation.
Final disposition	Review based on the new evidence; earlier approval cannot be reused.

Recovery has a biological boundary

A successful recapture can resolve an optical defect. It cannot automatically undo an inappropriate incubation condition or recreate a missed observation time. The reviewer must decide whether the underlying plate remains eligible for the intended method.

The proposed audit should retain both resolved and unresolved causes. This allows later analysis of common faults and prevents the system from learning that any recapture is an acceptable remedy. Recovery is complete when the relevant measurement and workflow are qualified again, with a documented reason for using or rejecting the resulting evidence.

Glossary of biological terms

Biological terms should identify the phenomenon being discussed

These concise definitions support reading the compendium. Operational definitions for a study must additionally specify the assay, conditions and endpoint. The glossary does not replace current laboratory methods or clinical standards.

Table A.13. Glossary of biological terms

Term	Meaning in this publication
Antimicrobial resistance	Ability to grow despite an antimicrobial exposure under a defined assessment.
AST	Antimicrobial susceptibility testing within an identified method and interpretive context.
Phenotype	Observable properties under specified biological and environmental conditions.
Genotype	Genetic constitution or identified sequence features; distinct from measured phenotype.
Heteroresistance	Coexistence of subpopulations with different susceptibility within a population, established by an appropriate method.
Tolerance	Increased survival duration during antimicrobial killing without necessarily increased resistance.
Persistence	Survival of a subpopulation during killing, distinguished through suitable experimental analysis.
MIC	Minimum inhibitory concentration determined under a specified dilution-based or other validated method.
Inhibition zone	Spatial region of inhibited visible growth around an antimicrobial source in an eligible assay.
Isolate	A recovered microbial population used for characterization; not interchangeable with every image of it.
Inoculum	Microbial material introduced into an assay, with preparation and quantity relevant to results.
One Health	An approach linking human, animal, plant and ecosystem health.

Language and inference

“Candidate minority growth” describes an observation for investigation. “Confirmed heteroresistance” describes a biological conclusion supported by the relevant method. “Predicted resistance determinant” describes a computational inference, while a measured molecular result has a different evidence path.

Using these distinctions consistently helps the reader see where the platform measures a signal and where further characterization is required. Tolerance and persistence terminology follows the cited consensus; the applications chapters explain its limits for imaging. [\[R10\]](#) [\[R65\]](#)

Glossary of measurement terms

Measurement terms separate resolution, uncertainty and reliability

The quantity being measured should always be stated. A specification in pixels, micrometres or milliseconds does not automatically express the quality of the final biological result. The same term can otherwise be misread across optics, software and clinical evaluation.

Table A.14. Glossary of measurement terms

Term	Meaning in this publication
Measurand	The quantity intended to be measured, defined with sufficient context.
Calibration	Establishing a relationship between reference values and instrument response.
Traceability	A documented connection of a measurement to an appropriate reference through a calibration chain.
Bias	Systematic difference relative to the defined reference or accepted value.
Repeatability	Agreement under specified repeat conditions.
Reproducibility	Agreement when stated conditions, such as site or operator, change.
Uncertainty	A characterized range or dispersion associated with a measurement under a defined method.
Resolution	Smallest distinguishable increment in the relevant representation or measurement context.
Object sampling	Physical object distance represented by an image pixel under the calibrated geometry.
Confounder	An influence that can distort the relationship under investigation.
Common-cause failure	One cause affecting multiple channels or functions.
False agreement	Consistent outputs that share an incorrect basis or unrecognized failure.

Units and reporting

Zone diameters are expressed in millimetres. Optical object sampling may be expressed in micrometres per pixel. Relative humidity is expressed as % RH, while temperatures require an explicit scale. Time-to-result claims require both a unit and a starting event.

A confidence interval for a study estimate is not the same as uncertainty assigned to a single instrument measurement. The validation protocol should state which quantity an interval describes, how it was obtained and which conditions it covers. The sensing and calibration chapters apply these distinctions to candidate technologies. [\[R16\]](#)

Glossary of software and twin terms

Software and twin terms retain the origin of information

These definitions distinguish stored evidence, transformations and predictions. They are intended to support a shared vocabulary across scientific, engineering and laboratory review.

Table A.15. Glossary of software and twin terms

Term	Meaning in this publication
Digital plate record	Preserved observations, context and decisions associated with a physical plate.
Computational model	A mathematical or learned representation of a defined process.
Predictive twin	A model linked to observations and evaluated for a defined forecasting task.
State estimation	Inference of a system's current latent condition from available evidence.
Forecast horizon	Time between the last relevant observation and the predicted future state.
Imputation	Estimation of a missing value; its origin remains distinct from measurement.
Simulation	Computation of a possible system evolution under stated assumptions.
Data fusion	Combining information with explicit treatment of compatibility, quality and dependence.
Abstention	A defined decision to withhold an otherwise possible model output.
PIMS	Plate Information Management System; proposed functions linking identity, capture, analysis, review and records.
Edge processing	Computation near the instrument or acquisition environment.
Cloud processing	Authorized remote computation or storage with defined security and continuity controls.
Audit trail	Preserved sequence of relevant actions, versions and decisions.
Idempotency	A retry behavior that avoids duplicating the intended logical action.

Terms for platform scope

“Lab-in-a-box” denotes an integrated workflow concept; it does not imply that every laboratory function fits within one enclosure. “Lab-on-a-chip” denotes miniaturized functions implemented on an appropriate chip or microfluidic substrate. Neither term establishes clinical completeness.

The proposed “hive mind” is a coordinated, governed learning and analysis concept across authorized data sources. It does not imply unrestricted access to laboratory records or automatic trust in outputs from other instruments.

Cross reference index

Cross-reference index for common review questions

Use these links to move directly to the principal discussion. The contents pages provide the complete chapter sequence. Candidate catalogues use individual entries so that a reviewer can assess a technology or AI function without assuming that it is included in every product.

[Platform proposition and product boundaries](#)

[Reader V1 and the endpoint priority](#)

[Engineering and sensing strategy](#)

[Forty-entry sensing catalogue](#)

[Software and data architecture principles](#)

[PIMS and plate information management](#)

[Forty-entry AI capability catalogue](#)

[Digital record, model and predictive twin](#)

[Continuity after physical plate loss](#)

[Heteroresistance and its detection boundaries](#)

[Validation framework and evidence requirements](#)

[Data minimization and purpose](#)

[India regulatory assessment](#)

[Critical assessment of automated reader studies](#)

[Competitor classes and endpoint readers](#)

[Staged custom electronics and silicon strategy](#)

[Proposed interface dictionaries and worked examples](#)

[Dated Reader V1 reference BOM](#)

Scope of a cross-reference

An entry points to the primary explanation, not to proof that the function is available. The relevant chapter identifies proposed engineering, research questions and evidence requirements. Use the validation section to assess what would be needed before a capability can support a stronger claim.

References for microbiology and standards

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

[R01] CLSI (2026). Performance Standards for Antimicrobial Susceptibility Testing M100 Edition 36.

[R02] EUCAST (2026). Clinical breakpoint tables version 16.1.

[R03] EUCAST (2026). Disk diffusion and quality control.

[R04] EUCAST (2026). Breakpoints for rapid AST.

[R05] EUCAST (2026). Rapid AST methods and quality control.

[R07] Band VI, Hufnagel DA, Jaggavarapu S, et al. (2019). Antibiotic combinations that exploit heteroresistance to multiple drugs effectively control infection. DOI 10.1038/s41564-019-0480-z.

[R08] Band VI, Crispell EK, Napier BA, et al. (2016). Antibiotic failure mediated by a resistant subpopulation in *Enterobacter cloacae*. DOI 10.1038/nmicrobiol.2016.53.

[R09] Dojjad SP, Gisch N, Frantz R, et al. (2023). Resolving colistin resistance and heteroresistance in *Enterobacter* species.

[R10] Balaban NQ, Helaine S, Lewis K, et al. (2019). Definitions and guidelines for research on antibiotic persistence.

[R11] Baltekin Ö, Boucharin A, Tano E, Andersson DI, Elf J (2017). Antibiotic susceptibility testing in less than 30 min using direct single-cell imaging. *Proc Natl Acad Sci U S A* 114:9170-9175. DOI 10.1073/pnas.1708558114.

[R12] Ciftci F, Erarslan A, Rahebi J (2026). Automatic detection and quantification of antimicrobial inhibition zones using YOLO11n with post-hoc interpretability validation. *Front Microbiol* 17:1810754. DOI 10.3389/fmicb.2026.1810754.

[R13] Thomas P, Terradot G, Danos V, Weiße AY (2018). Sources propagation and consequences of stochasticity in cellular growth. DOI 10.1038/s41467-018-06912-9.

[R60] Kupke J, Brombach J, Fang Y, et al. (2025). Heteroresistance in *Enterobacter cloacae* complex caused by variation in transient gene amplification events. DOI 10.1038/s44259-025-00082-7.

[R61] Wei C, Wu J, Zhang J, et al. (2025). Clinical characteristics, molecular epidemiology and mechanisms of colistin heteroresistance in *Enterobacter cloacae* complex. DOI 10.3389/fcimb.2025.1536058.

[R66] Hombach M, Zbinden R, Böttger EC (2013). Standardisation of disk diffusion results for antibiotic susceptibility testing using the sirscan automated zone reader. DOI 10.1186/1471-2180-13-225.

[R67] Hombach M, Ochoa C, Maurer FP, et al. (2016). Relative contribution of biological variation and technical variables to zone diameter variations of disc diffusion susceptibility testing. DOI 10.1093/jac/dkv309.

[R74] CLSI (2026). M100 Edition 36 correction notice, Table 2H-1, 9 July 2026.

References for engineering and sensing

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

[R14] European Machine Vision Association (2021). EMVA 1288 Standard for characterization of image sensors and cameras.

[R15] Edmund Optics (undated). Advantages of telecentricity.

[R16] JCGM BIPM (2008). Evaluation of measurement data Guide to the expression of uncertainty in measurement.

[R17] Bosch Sensortec (undated). Gas sensors and application software.

[R18] Bosch Sensortec (undated). BME690 data sheet.

[R19] STMicroelectronics (undated). VL53L8CX multizone time of flight ranging sensor.

[R20] Analog Devices (undated). Analog front end design considerations for RTD ratiometric measurements.

[R38] Spencer DC, Paton TF, Mulrone KT, Inglis TJJ, Sutton JM, Morgan H (2020). A fast impedance-based antimicrobial susceptibility test. *Nat Commun* 11:5328. DOI 10.1038/s41467-020-18902-x.

[R39] Yang Y, Gupta K, Ekinici KL (2020). All-electrical monitoring of bacterial antibiotic susceptibility in a microfluidic device. *Proc Natl Acad Sci U S A* 117:10639-10644. DOI 10.1073/pnas.1922172117.

[R40] Swami P, Verma G, Holani A, Kamaraju S, Manchanda V, Sriharan V, Gupta S (2022). Rapid antimicrobial susceptibility profiling using impedance spectroscopy. *Biosens Bioelectron* 200:113876. DOI 10.1016/j.bios.2021.113876.

[R41] Pitruzzello G, Johnson S, Krauss TF (2023). Exploring the fundamental limit of antimicrobial susceptibility by near-single-cell electrical impedance spectroscopy. *Biosens Bioelectron* 224:115056. DOI 10.1016/j.bios.2022.115056.

[R42] Balmages I, Reinis A, Kistkins S, Bliznuks D, Plorina EV, Lihachev A, Lihacova I (2023). Laser speckle imaging for visualization of hidden effects for early detection of antibacterial susceptibility in disc diffusion tests. *Front Microbiol* 14:1221134. DOI 10.3389/fmicb.2023.1221134.

[R43] Kim D, Moon S, Lee J, Cho K, Lee C, Yoon J (2025). Light scattering-based screening method for rapid evaluating antibiotic effects on bacteria using laser speckle imaging. *J Biol Eng* 19:68. DOI 10.1186/s13036-025-00542-8.

[R44] Lu W, Li H, Qiu H, Wang L, Feng J, Fu YV (2023). Identification of pathogens and detection of antibiotic susceptibility at single-cell resolution by Raman spectroscopy combined with machine learning. *Front Microbiol* 13:1076965. DOI 10.3389/fmicb.2022.1076965. Published 4 January 2023 in volume13 (2022).

[R45] Christensen MH, Jakobsen TH, Lichtenberg M, Hertz FB, Dahl B, Bjarnsholt T (2024). Antimicrobial susceptibility testing of bone and joint pathogens using isothermal microcalorimetry. *APMIS* 132:814-823. DOI 10.1111/apm.13470.

[R55] Edmund Optics (undated). Hyperspectral and multispectral imaging.

[R56] Hamamatsu Photonics (undated). Fluorescence lifetime imaging microscopy.

[R57] Thorlabs (undated). Optical coherence tomography tutorial.

A / APPENDICES AND REFERENCES

References for AI data and simulation

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

[\[R21\]](#) US FDA and IMDRF (2025). Good machine learning practice for medical device development Guiding principles.

[\[R22\]](#) US FDA (2024). Transparency for machine learning enabled medical devices Guiding principles.

[\[R23\]](#) NIST (2023). Artificial Intelligence Risk Management Framework AI RMF 1.0.

[\[R24\]](#) CARD McMaster University (undated). Comprehensive Antibiotic Resistance Database.

[\[R25\]](#) NCBI (undated). Pathogen Detection help documentation.

[\[R49\]](#) US FDA (2026). Cybersecurity in Medical Devices Quality Management System Considerations and Content of Premarket Submissions.

[\[R52\]](#) IEC (2015). IEC 62304 2006 with Amendment 1 2015 Medical device software Software life cycle processes.

[\[R53\]](#) IEC (2021 corrected 2025). IEC 81001 5 1 2021 Health software and health IT systems safety effectiveness and security Security activities in the product life cycle.

[\[R58\]](#) US Food and Drug Administration (2023). Assessing the Credibility of Computational Modeling and Simulation in Medical Device Submissions.

[\[R59\]](#) Kapteyn MG, Pretorius JVR, Willcox KE (2021). A probabilistic graphical model foundation for enabling predictive digital twins at scale.

Interpretation of these sources

Frameworks and guidance inform proposed requirements; citations do not establish implementation or conformity. Database content and models require independently preserved versions. The digital-twin engineering study supports a modelling approach, not a validated biological forecast.

References for comparators and regulation

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

[R26] CDSCO (2026). Guidance document on Medical Device Software under MDR 2017.

[R27] COPAN Diagnostics (undated). RADIANT disk diffusion automation.

[R28] COPAN Diagnostics (2026). Laboratory automation brochure.

[R29] Giles Scientific (undated). BIOMIC V3.

[R30] bioMérieux (undated). VITEK 2.

[R31] BD (undated). BD Phoenix M50 instrument.

[R32] US FDA (2026). 510 k review K251511.

[R68] Cherkaoui A, Renzi G, Vuilleumier N, Schrenzel J (2021). Performance of Fully Automated Antimicrobial Disk Diffusion Susceptibility Testing Using Copan WASP Colibri Coupled to the Radian In-Line Carousel and Expert System. DOI 10.1128/JCM.00777-21.

[R69] Beckman Coulter (n.d.). DxM MicroScan WalkAway System.

[R70] QuantaMatrix (n.d.). dRAST.

[R71] bioMérieux (n.d.). VITEK REVEAL.

[R72] bioMérieux (n.d.). VITEK MS PRIME.

[R73] Reshape Biotech (2026). AI-driven operating system for microbiology.

[R75] Q-linea (n.d.). ASTar in the lab.

A / APPENDICES AND REFERENCES

References for India and implementation

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

[R06] World Health Organization (2026). Antimicrobial resistance.

[R33] NCDC Ministry of Health and Family Welfare (undated). National programme on AMR containment.

[R34] Government of India NCDC (2025). National Action Plan on Antimicrobial Resistance 2.0.

[R35] ICMR (2025). Guidance document for evaluation of antimicrobial resistance diagnostics.

[R36] Government of Goa Department of Information and Publicity (2023). Awareness program on antimicrobial resistance by FDA.

[R37] Directorate of Health Services Goa (undated). Integrated Disease Surveillance Programme.

[R50] ISO (2016). ISO 13485 2016 Medical devices Quality management systems Requirements for regulatory purposes.

[R51] ISO (2019). ISO 14971 2019 Medical devices Application of risk management to medical devices.

[R54] Ministry of Electronics and Information Technology India (2025). Digital Personal Data Protection Rules 2025 and enforcement timeline.

[R62] US National Institute of Diabetes and Digestive and Kidney Diseases (n.d.). Definition and Facts of Bladder Infection in Adults.

[R63] US Centers for Disease Control and Prevention (2025). Laboratory Testing for Methicillin (oxacillin)-resistant *Staphylococcus aureus* (MRSA).

[R64] International Organization for Standardization (2017). ISO/IEC 17025:2017 General requirements for the competence of testing and calibration laboratories.

[R65] World Health Organization (2026). One Health.

Geographic and legal scope

India and Goa sources establish public programmes and context, not a partnership or institutional commitment to AMReye.AI. Regulatory and data-protection implementation requires product-specific review of current requirements and commencement provisions.

A / APPENDICES AND REFERENCES

References for Reader V1 component selection

Linked source bibliography

Sources were inspected on 13–14 September 2026. Links open the original study, official documentation or supplier source. Titles retain their published terminology and product names.

- [\[R46\]](#) Raspberry Pi (undated). Camera hardware documentation.
- [\[R47\]](#) Raspberry Pi (undated). Camera software documentation.
- [\[R48\]](#) Raspberry Pi (undated). Raspberry Pi Active Cooler.
- [\[B01\]](#) Robocraze (2026 snapshot). Official Raspberry Pi 5 Model 8GB RAM Development Board.
- [\[B02\]](#) Robocraze (2026 snapshot). Raspberry Pi 16mm High Quality Camera Telephoto Lens.
- [\[B03\]](#) Robocraze (2026 snapshot). Official Raspberry Pi 5 27W USB C PD Power Supply Black.
- [\[B04\]](#) Silverline Electronics (2026 snapshot). Official Raspberry Pi 5 White 27W USB C PD Power Supply.
- [\[B05\]](#) Thingbits (2026 snapshot). Official Raspberry Pi 5 Active Cooler.
- [\[B06\]](#) Robocraze (2026 snapshot). Raspberry Pi High Quality Camera with Interchangeable Lens Base.
- [\[B07\]](#) Robocraze (2026 snapshot). Raspberry Pi 5 Camera Cable Standard Mini 300mm.
- [\[B08\]](#) Thingbits (2026). SanDisk MicroSD Card Class 10, 64 GB, No OS, TH1980.
- [\[B09\]](#) Thingbits (2026). Adafruit SHT40 breakout, TH2066.
- [\[B10\]](#) element14 India (2026). MEAN WELL GST25A12-P1J, order2815879.
- [\[B11\]](#) Thingbits (2026). Raspberry Pi 5 Computer, 8 GB, TH0489.
- [\[B12\]](#) Thingbits (2026). Raspberry Pi HQ Camera, C/CS mount, camera only, TH0267.
- [\[B13\]](#) Adafruit (2026). SHT40 Temperature and Humidity Sensor, product4885.
- [\[B14\]](#) MEAN WELL (2026). GST25A series data sheet.

Reader V1 BOM configuration and price basis

Reader V1 reference configuration, priced 13–14 September 2026

This eight-page BOM describes one coherent endpoint imaging reference configuration. It uses a manually loaded plate, controlled visible illumination, a fixed camera, local computation and mandatory human review. It is a prototype procurement basis, not a released manufacturing drawing or a clinically qualified instrument.

Configuration boundary

The reference uses a Raspberry Pi 5 with 8 GB RAM, an HQ C/CS-mount camera and a 16 mm C-mount lens. A separate 12 V adapter supplies the proposed illumination assembly. The computer uses its official USB-C supply. The reference assumes an existing monitor, keyboard and mouse for local review; these are not included in the hardware total.

Table A.16. Reader V1 BOM configuration and price basis

Price code	Meaning throughout these eight pages
L	Displayed supplier price including GST; availability is a listing observation.
U	Displayed supplier price with tax treatment unresolved.
E	Author's engineering allowance excluding GST and shipping; not a supplier quotation.
I	Included in another line; no additional amount.

Procurement status

Forty line items are specified. Named candidate parts and merchant SKUs are retained where verified. Custom assemblies and unresolved commodity selections have explicit quotation gaps. No order, stock reservation or destination-specific checkout was completed. Formal quotation dates are therefore unavailable; the stated dates are public-listing observation dates.

The lens and camera cable listings were sold out at observation. Several other items were listed in stock, but neither an Indian storefront nor a dispatch claim establishes confirmed delivery to the project. Before purchase, verify exact variants, invoice treatment, regional plugs, lead times and the complete assembled compatibility.

A / APPENDICES AND REFERENCES

Reader V1 BOM compute and storage

Local capture and review without a separate accelerator

The selected board is a current candidate for a modest endpoint workload. Its inclusion is not a measured throughput claim. Thermal, storage and power behavior require verification under the actual capture and analysis software. [\[R47\]](#) [\[R48\]](#)

Table A.17. Reader V1 BOM compute and storage

ID	Candidate component or assembly	Qty	Unit INR / basis
01	Pi 5, 8 GB; Thingbits TH0489 [B11]	1	19,617.50 L
02	Official Pi 5 Active Cooler; TH0490 [B05]	1	531.00 L
03	SanDisk 64 GB Class 10, No OS; TH1980 [B08]	1	1,475.00 L
04	Official 27 W USB-C PSU, black; TIFPS0447 [B03]	1	1,204.00 L
05	USB-A microSD reader, SDXC compatible	1	300.00 E
06	Cat6 patch lead, 2 m, RJ45 to RJ45	1	250.00 E
07	Micro-HDMI to HDMI lead, 1 m	1	550.00 E
08	Official Pi 5 RTC battery assembly	1	650.00 E

Compatibility and sourcing

Lines 01–03 use Thingbits listings observed on 14 September 2026. Line 04 uses the Robocraze listing observed on 13 September. The remaining amounts are engineering allowances; exact commodity SKUs and the RTC assembly quotation remain open.

The active cooler should use the board's intended connector and mounting arrangement. The microSD selection is 64 GB with no preinstalled operating system; the manufacturer's full card suffix, authenticity and write endurance require confirmation. A sustained-write and abrupt-power-loss test should precede use with important records.

The micro-HDMI cable supports an existing review display. The RTC battery supports timekeeping and is not a power backup. Network access should use an authorized local connection. No cloud subscription, GPU accelerator, display, keyboard or mouse is included in this configuration.

A / APPENDICES AND REFERENCES

Reader V1 BOM camera and optics

A C/CS optical path with an explicitly qualified field of view

The selected camera is the C/CS-mount, camera-only variant. The M12 variant is not interchangeable with this lens plan. The included adapter is counted once. [\[R46\]](#) [\[B12\]](#)

Table A.18. Reader V1 BOM camera and optics

ID	Candidate component or assembly	Qty	Unit INR / basis
09	HQ Camera C/CS, IMX477; TH0267 [B12]	1	5,900.00 L
10	16 mm C-mount HQ lens; TIFCM0016 [B02]	1	6,723.00 L
11	Standard-mini camera FFC, 300 mm; TIFCW0078 [B07]	1	213.00 L
12	C-to-CS adapter supplied with HQ camera [B12]	1	0.00 I
13	Custom adjustable camera mount, 1/4-20 interface	1	850.00 E
14	Custom lens support and focus-lock bracket	1	450.00 E

Geometric feasibility calculation

Using a 4.712 mm sensor height and a 110 mm illustrative object field gives magnification about 0.0428. A 16 mm thin-lens approximation then places the object about 390 mm from the principal plane. This is an optical design estimate, not a final enclosure dimension. Nominal object sampling is about 36.2 micrometres per pixel, not measurement accuracy.

The lens listing's minimum object distance is 0.2 m; actual focus, field coverage, distortion and edge quality still require a bench check. The frame should allow adjustment and stable locking. The camera cable must be the 15-pin standard to 22-pin mini camera type for the selected board, not a display cable. Place the compute board near the camera so a 300 mm cable is sufficient.

The lens and cable were listed sold out on 13 September. Obtain equivalent exact-part quotations before procurement; do not silently replace the focal length or cable pinout.

A / APPENDICES AND REFERENCES

Reader V1 BOM illumination and calibration

An engineered light path and a traceable geometric reference

All eight amounts below are engineering allowances, not commercial quotations. The proposed visible-light assembly uses 12 V DC with a design ceiling of 12 W. Spectral selection and geometry remain open until the optical bench establishes adequate contrast and uniformity.

Table A.19. Reader V1 BOM illumination and calibration

ID	Candidate component or assembly	Qty	Unit INR / basis
15	Custom white LED assembly, 12 V, ≤ 12 W	1	1,200.00 E
16	Opal diffuser panel, cut to optical enclosure	1	600.00 E
17	White diffuse reflector sheet and mounting	1	900.00 E
18	Matte black stray-light baffle set	1	350.00 E
19	Aluminium LED heat-spreader carrier	1	450.00 E
20	Custom LED enable board, 3.3 V logic, DC drive	1	650.00 E
21	Traceable planar geometric calibration artifact	1	12,000.00 E
22	Neutral diffuse intensity reference target	1	1,500.00 E

Selection and validation requirements

The LED assembly should operate at a stable acquisition condition. The enable board is proposed to switch the low-voltage load using a qualified 3.3 V control interface; it must not draw LED current from a GPIO pin. Continuous DC illumination is the reference mode. Any later dimming or pulsed operation requires exposure-synchronization evaluation.

The calibration artifact allowance is a quotation gap, not a guarantee that INR 12,000 buys the required certificate or uncertainty. Select the artifact after defining the measurand, field and uncertainty budget. The neutral target is for relative intensity checks unless independently characterized. A printed ruler or uncalibrated gray card cannot substitute for traceable geometric calibration.

Diffuser, reflector, baffle and heat-spreader materials require cleanability, stability and optical verification. Their costs must be replaced by supplier quotations after dimensions and material grades are fixed.

Reader V1 BOM mechanics and identification

A manually loaded, serviceable reference instrument

The mechanical items are proposed fabrication and commodity allowances. They do not constitute completed CAD, a fixed product appearance or manufacturing readiness. One frame and one enclosure set are costed for the optical reference configuration.

Table A.20. Reader V1 BOM mechanics and identification

ID	Candidate component or assembly	Qty	Unit INR / basis
23	2020 extrusion frame, cut lengths and corners	1	3,200.00 E
24	Opaque enclosure panel and access-door set	1	2,200.00 E
25	Removable plate holder for reference plate size	1	900.00 E
26	Removable catch tray, cleanable material	1	700.00 E
27	Hinge and latch assembly	1	500.00 E
28	Anti-slip instrument feet	4	75.00 E
29	Fastener and standoff kit	1	600.00 E
30	Durable instrument and plate barcode label kit	1	300.00 E

Fit and serviceability

The frame must accommodate the optical working distance established on the bench, the selected plate field and stable camera mounting. The holder should locate the intended plate family reproducibly without obscuring relevant growth or labels. A different plate size requires a separate fit and field-of-view check.

The removable tray and enclosure should support the laboratory's qualified cleaning procedure. Hinges, latches and feet should preserve stability during normal loading. Fasteners and standoffs are a single kit allowance; final quantities belong in controlled assembly drawings.

The initial identification path uses image-readable labels or manually verified identifiers within the proposed review workflow. A separate barcode scanner and printer are not included. Label media must remain legible under the intended handling and cleaning conditions. Supplier selection, material grades, tolerances and fabrication lead times remain quotation and engineering gaps for every item on this page.

Reader V1 BOM power wiring and environmental checks

Separated power domains and basic environmental context

The 12 V adapter powers illumination; the computer has its separate USB-C supply. The SHT40 is a candidate ambient temperature/humidity monitor, not a resistance detector. [\[B13\]](#) [\[B14\]](#)

Table A.21. Reader V1 BOM power wiring and environmental checks

ID	Candidate component or assembly	Qty	Unit INR / basis
31	MEAN WELL GST25A12-P1J; element14 2815879 [B10]	1	1,333.13 U
32	India earthed plug to IEC C13 mains lead	1	450.00 E
33	Panel DC jack, 5.5 x 2.1 mm, centre positive	1	120.00 E
34	DC illumination master switch	1	150.00 E
35	DC fuse holder and fuse set	1	250.00 E
36	Low-voltage wiring harness and strain relief	1	500.00 E
37	Adafruit SHT40 4885; Thingbits TH2066 [B09]	1	695.02 L
38	JST-SH 4-pin QT to GPIO interface cable	1	180.00 E
39	Lid-position switch and 3.3 V input assembly	1	250.00 E
40	Protective bonding hardware kit	1	250.00 E

Electrical compatibility and quotation gaps

The MEAN WELL adapter is 12 V, 2.08 A with a centre-positive 5.5 × 2.1 mm output. Its inspected listing shows UK stock and INR 1,333.13; tax and destination delivery remain unresolved. The matching earthed IEC lead is a separate allowance. Product-level electrical safety still requires assessment.

Use the SHT40 interface at a qualified 3.3 V logic configuration. Verify the QT cable pinout and mounting away from heat sources. The lid input supports workflow state; it is not a certified safety interlock. Fuse rating, wire gauge, bonding and protection require design review and load tests before assembly release. No mains wiring should be improvised inside the optical enclosure.

A / APPENDICES AND REFERENCES

Reader V1 BOM alternatives compatibility and procurement

Substitution requires a compatibility decision and a new price check

The reference is coherent at the interface level, but assembly performance has not been demonstrated. An available alternative should be selected only after checking its electrical, optical, mechanical and software consequences. A lower listing price does not establish an equivalent measurement system.

Table A.22. Reader V1 BOM alternatives compatibility and procurement

Candidate alternative	Required decision before substitution
Pi 5 8 GB, Robocraze TIFC00186	INR 19,999 including GST; sold out on 13 September. Same memory variant required. [B01]
HQ camera, Robocraze TIFCM0021	INR 6,723 including GST; sold out on 13 September. Confirm C/CS mount and included adapter. [B06]
Official white 27 W PSU, Silverline	INR 1,020 displayed; tax treatment unresolved. Confirm regional plug and exact assembly. [B04]
Another C-mount lens	Re-evaluate focal length, sensor coverage, focus and distortion; current quotation required.
Another storage medium	Requalify write reliability, boot, power-loss behavior and retention.
Industrial camera or compute module	Treat as a configuration change with new interfaces and calibration.

Supplier evidence to obtain

Before ordering, request the full manufacturer part number where a merchant SKU is insufficient, the selected variant, invoice GST treatment, shipping and any import charges. Obtain actual dispatch origin and destination-specific delivery confirmation. A listing's inventory counter is an observation, not reserved stock.

For custom parts, issue drawings or functional specifications before requesting a formal quotation. Record revision, material, quantity, tolerance, finish and acceptance criteria. The procurement decision should preserve the source and date of the accepted quotation, while retaining this dated reference as an earlier planning basis.

Reader V1 BOM totals exclusions and verification gaps

The planning sum is INR 69,241.65 before unresolved charges

This figure combines displayed prices and explicit engineering allowances. It is not a fully taxed delivered quotation, a manufacturing cost or a selling price. The basis is intentionally separated below so that uncertain amounts do not appear commercially verified.

Table A.23. Reader V1 BOM totals exclusions and verification gaps

Cost basis	INR
L: displayed supplier amounts, GST included	36,358.52
U: displayed amount, GST treatment unresolved	1,333.13
E: engineering allowances, excluding GST and shipping	31,550.00
I: adapter included in camera line	0.00
Mixed-basis planning sum	69,241.65

Charges and work excluded

Additional tax on U/E items, shipping, customs where applicable, quote revisions and fabrication overruns are unresolved. The sum excludes an existing review workstation's peripherals, laboratory consumables, reference strains, biological validation, calibration services beyond the artifact allowance, engineering labor, software development, regulatory work, cloud services and production tooling. It is a reference hardware basket, not the total programme budget.

Verification needed before procurement

Resolve the sold-out lens and cable, obtain custom-part quotations, confirm exact commodity SKUs and complete an electrical/optical compatibility review. Verify invoice and delivery terms for each supplier. The geometric artifact's certificate and uncertainty are a particularly material gap.

The BOM is complete as a dated, transparent reference configuration with 40 line items. Procurement readiness remains conditional on these listed checks and quotations. No fundraising, revenue forecast or selling-price assumption is included. Future component changes should be evaluated against the measurement and service requirements rather than selected solely because they are newer.